

The Second Monopoly

The Patent Expired. The Exclusion Didn't — and It Doubled the Cost of American Living.

(formerly titled *The Forgotten Half of the Patent Clause*)

How a Constitutional Bargain Was Cancelled — Restoring It Frees Buyers from Surveillance, Stops Robots Replacing America's Drivers & Operators, and Returns the Joy of Operating Real Machines

Roleigh Martin

Founder, the Make American Living Affordable (MALA) movement on X

Foreword by Gord Magill, author of End of the Road: Inside the War on Truckers

This is the complete back-matter apparatus to the book above — its Endnotes, Endnote Commentary, Supreme Court case notes, Glossary, and Key Terms, published online so that every source link is live and the whole section is searchable.

Endnotes, Glossary & Key Terms

The complete back-matter apparatus, online

CONTENTS

1. Apparatus Convention
2. Endnotes Ordering
3. Endnotes
4. Endnote Commentary
5. Appendix A: SCOTUS Case Master Table
6. Appendix B: Note on AI-Assisted Authorship
7. Appendix C: LLM Responses to Prompt 7 in the Exam
8. Glossary
9. Key Terms

Apparatus Convention

The book uses no superscript reference numbers in the running text. Citations of sources, court cases, and statutes appear in the Endnotes section at the back of the book. The Endnotes, Endnote Commentary, Glossary, and Key Terms follow the Chicago Manual of Style, 17th edition (author-date system), with the house exceptions noted below.

The order of entries is set out in full in the Endnotes Ordering section, which opens the Endnotes. Two departures from Chicago apply throughout the apparatus. Every entry ends with a source URL. And same-year works by one author take no a, b, or c suffix; an in-text citation such as “Martin (2025)” is disambiguated by context, the running text making the specific work plain.

The Endnote Commentary section, which collects in alphabetical order — keyed to the same author surname or case name as the Endnotes section — substantive notes on those sources: full passages a chapter quotes only in part, additional context the running text omits for the sake of pace, related material the curious reader will want and the impatient reader can skip. Not every Endnote has a corresponding Commentary entry; only those that

genuinely reward a longer note. An Endnote that does have a Commentary entry ends with the italicized notice *"See Endnote Commentary."*

The two sections are designed to be read together by a reader who wants depth, and skipped together by a reader who does not.

The Endnote Commentary is followed by a Glossary section, which collects in alphabetical order by abbreviation the acronyms used throughout the book — primarily federal agencies, statutes, and regulatory frameworks defined parenthetically at first use in the running text. The Glossary is followed by a Key Terms section, which collects in alphabetical order by term the book's specific terms of art (such as bootleggers-and-Baptists, regulatory ratchet, anti-backsliding provision, three-tier breakdown, public domain in the patent-law sense, the Expired Patent Commons, remanufacturing in the book's specific sense, grandfather exemption, expired-patent commerce, constitutional forgetting, and two suspenders), each with a brief definition.

Each Endnote entry concludes with a URL — Justia for U.S. Supreme Court cases, Cornell University Legal Information Institute for federal statutes, Founders Online for Founding-era documents, the publisher's canonical URL for periodicals, books, and reports, with Amazon as a last resort for older works without stable publisher URLs — so that readers using a digital edition of the book can follow the source directly.

The reader who wants to verify a claim in the running text consults the Endnote entry alphabetized by the author or case name as they appear in the text. There are no chapter-specific endnote groupings; the Endnotes section is a single book-wide alphabetical list. The reader who wants additional context on a source — the full passage a chapter quotes only in part, or material the running text omits for the sake of pace — consults the parallel entry in the Endnote Commentary section, which is keyed to the same anchor and ends with the italicized notice *"See Endnote for full citation."*

The reader who wants the meaning of an acronym consults the Glossary; the reader who wants to anchor a recurring concept consults the Key Terms section. Each section does one job.

In-text citations follow Chicago author-date style: integrated into the sentence as “Yandle (1983) named the pattern,” with full names and titles reserved for the Endnote entry. Acronyms are spelled out in full at first appearance in the book with the abbreviation following in parentheses; thereafter only the abbreviation is used.

Endnotes Ordering

Sort order. Entries run by headword — the first significant element of the entry. The ten rules below govern placement throughout this section. Rules 2 through 5 also govern the Glossary and the Key Terms sections, each alphabetized by its own headwords.

1. Headword. The headword is the author’s surname, the organization’s name, the case name, the statute’s name, or — for an entry with none of these — the entry’s own leading words.

2. Leading articles. A leading “The,” “A,” or “An” is disregarded for placement. An article anywhere else in the headword counts as an ordinary word.

3. Word by word, not letter by letter. Comparison proceeds one word at a time, and a word boundary sorts ahead of any letter. New Civil Liberties Alliance therefore precedes Newbold, Charles, though a letter-by-letter sort would reverse them. The abbreviation “v.” counts as a word, so United States Code precedes United States v. Ballin.

4. Case and punctuation. Comparison ignores capitalization, commas, periods, apostrophes, ampersands, and hyphens. Sears, Roebuck & Co. v. Stiffel Co. files under Sears; D’Angelo files as DAngelo.

5. Numerals. A numeral inside a headword sorts by value and not as a string, so U.S. Patent No. 39,285 precedes 139,668, which precedes 5,357,751, which precedes 12,196,640. A headword beginning with a numeral sorts ahead of every headword beginning with a letter, so United States v. 50 Acres of Land precedes United States v. Ballin.

6. Court cases. A case files under the first party named in its style. United States v. Moghadam files under United States, never under Moghadam. In re and Ex parte file under In and Ex.

7. Statutes and code sections. A statute files under its own name — National Appliance Energy Conservation Act of 1987. A codified statutory section files under United States Code, and the sections there run by title number, then by section number. A codified regulation files under Code of Federal Regulations, and the entries there run by title number, then by part number, then by section number.

8. Names as written. A name files as the source writes it, not as it would read expanded. United States Code files under United; U.S. Department of Energy files under US. Spelled-out names therefore precede their abbreviated counterparts, and the two form separate runs.

9. Several entries under one headword. Entries sharing a headword run by date of publication, earliest first. An entry carrying no publication date — a continuing series such as the Consumer Price Index, or an organization's standing web page — goes last among them, and those entries run alphabetically by title.

10. Topical entries. An entry standing for a body of material rather than a single document — Self-certification and certification by simulation; Nichia Corporation and the LED-to-laser lighting succession — files under its own leading word like any other entry.

Endnotes

Adarand Constructors, Inc. v. Peña, 515 U.S. 200 (1995). https://supreme.justia.com/cases/federal/us/515/200/.

Advanced Impaired Driving Prevention Technology Rulemaking. Advance Notice of Proposed Rulemaking, "Advanced Impaired Driving Prevention Technology," 89 Fed. Reg. 830 (January 5, 2024). National Highway Traffic Safety Administration. Authorized by Section 24220 of the Infrastructure Investment and Jobs Act, Pub. L. No. 117-58, codified as a statutory note to 49 U.S.C. § 30111 (distinct from the AEB authority at § 24208,

codified at 49 U.S.C. §§ 30129–30130). The statutory deadline of November 15, 2024 was missed; Section 24220(e) permits extension of up to three years on a determination that compliant technology is not yet available. NHTSA Docket NHTSA-2022-0079; rulemaking ongoing.
https://www.federalregister.gov/documents/2024/01/05/2023-27665/advanced-impaired-driving-prevention-technology.

Aetna Life Insurance Co. v. Haworth, 300 U.S. 227 (1937).
https://supreme.justia.com/cases/federal/us/300/227/.

Ahlborn, Richard E., ed. *Man Made Mobile: Early Saddles of Western North America*. Smithsonian Studies in History and Technology, no. 39. Washington, DC: Smithsonian Institution Press, 1980. The standard scholarly reference on the trans-Atlantic and trans-continental development of the American working saddle, drawn from the Smithsonian's Buckaroos in Paradise field project.
https://repository.si.edu/handle/10088/2470.

Aircraft Owners and Pilots Association. "Last of the Old vs. First of the New." *AOPA Pilot*, June 2013. Account of Cessna's 1996 return to single-engine production: the FAA wanted the restarted Cessna 172 certified to current Part 23, but because the design could not meet every later requirement, Cessna and the FAA agreed that it would be certified in part to older Part 23 amendments issued decades earlier. Cited in How a Remake Is Built and Certified for the 1996 restart under the original certificate, with the restarted design certified partly to older Part 23 amendments.
https://www.aopa.org/news-and-media/all-news/2013/june/pilot/last-of-the-old-vs-first-of-the-new.

Akorn, Inc. v. Provepharm, Inc., No. 14-CV-04106 (N.D. Ill. 2019). Federal antitrust litigation involving alleged monopolization of methylene blue through U.S. Pharmacopeia monograph revision; counterclaims survived motion to dismiss.
https://web.archive.org/web/20191114101913/https://lrus.wolterskluwer.com/news/antitrust-law-daily/monopolization-counterclaims-in-methylene-blue-drug-case-go-forward/81090/.

Alfred L. Snapp & Son, Inc. v. Puerto Rico, 458 U.S. 592 (1982). https://supreme.justia.com/cases/federal/us/458/592/.

Allegheny Pittsburgh Coal Co. v. County Commission of Webster County, 488 U.S. 336 (1989). The unanimous Supreme Court decision holding that a county tax-assessment system that valued recently-acquired coal properties at full current value while leaving long-held properties at decades-old assessments violated the Equal Protection Clause. Cited in the chapter on the Takings Clause prong as the doctrinal cousin to the book's argument that the regulatory wall imposes a constitutionally suspect inequality on similarly situated property — the original manufacturer held to his patent-year standards, the remanufacturer of the same now-expired patent held to ratcheted current-year standards. https://supreme.justia.com/cases/federal/us/488/336/.

AllMachines. "Combine Harvesters." AllMachines aggregator listing. Source for the U.S. retail price reference of approximately $646,000 for the John Deere S7 600 cited in the farm-equipment chapter alongside the IndiaMART listing for the India-market John Deere W70. https://www.allmachines.com/combine-harvesters/view-all.

AllMachines. "Tractors — Brand: Mahindra." AllMachines aggregator listing. Cross-reference for U.S.-market Mahindra tractor pricing cited in the farm-equipment chapter against India-market Mahindra pricing. https://www.allmachines.com/tractors/view-all?brand=mahindra.

Alon, Noga, et al. "Remarks on the Disproof of the Unit Distance Conjecture." arXiv preprint, May 2026. The companion paper released the same day as OpenAI's announcement, in which nine mathematicians present a short, human-verified and somewhat generalized rendering of the machine-generated counterexample to Paul Erdős's 1946 unit distance conjecture, together with their reflections on what the result does and does not establish about artificial intelligence in mathematics. Cited in the preface as the human verification that followed the machine's origination. https://arxiv.org/abs/2605.20695

American Farm Bureau Federation. “Farm Bankruptcies Continued to Climb in 2025.” Market Intel report by AFBF economist Samantha Ayoub, February 2026. Source for the 46 percent year-over-year increase in Chapter 12 farm bankruptcy filings cited in the farm-equipment chapter (315 filings in calendar year 2025 against 216 in 2024), drawn from U.S. Courts data. The figure was reported during National Farmers Union President Rob Larew’s keynote at the NFU 124th Anniversary Convention in New Orleans, March 7–9, 2026.
https://www.fb.org/market-intel/farm-bankruptcies-continued-to-climb-in-2025.

American Hospital Association v. Becerra, 596 U.S. 724 (2022).
https://supreme.justia.com/cases/federal/us/596/20-1114/.

American Innovation and Manufacturing Act of 2020, Pub. L. No. 116-260, Div. S, Title I, Subtitle B, 134 Stat. 2253. Codified at 42 U.S.C. § 7675. Authorizes EPA to phase down hydrofluorocarbons by 85 percent over 15 years.
https://www.law.cornell.edu/uscode/text/42/7675.

American Transportation Research Institute. *Issues and Opportunities with Driver-Facing Cameras*. Arlington, VA: ATRI, April 2023. Survey of approximately 2,100 commercial truck drivers documenting that current driver-facing-camera users rate the technology 2.24 on a zero-to-ten approval scale and that female drivers rate the cameras’ privacy protection 24 percent lower than male drivers, with some reporting voyeurism, unwanted comments about their appearance, and sexual harassment from employees tasked with reviewing the footage. Cited in the trucking-section discussion of driver-facing-camera intrusion in the surveillance chapter.
https://truckingresearch.org/2023/04/new-atri-research-identifies-strategies-for-improving-driver-facing-camera-approval-and-utilization/.

American Transportation Research Institute. *An Analysis of the Operational Costs of Trucking: 2025 Update*. Arlington, VA: American Transportation Research Institute, 2025. The trucking industry’s standard annual accounting of marginal operating costs; the 2025 edition reports a fully loaded average operating cost of $2.26 per mile for 2024, of which

driver wages and benefits account for roughly $0.995 per mile. Cited in the autonomous-vehicle chapter for the per-mile economics of the human-driven truck. https://truckingresearch.org/2025/07/an-analysis-of-the-operational-costs-of-trucking-2025-update/.

American Trucking Associations. “Economics and Industry Data.” American Trucking Associations. The trade association’s compilation of industry statistics, reporting roughly 3.5 million professional truck drivers employed in the United States. Cited in the autonomous-vehicle chapter for the scale of the workforce the autonomous transition would displace. https://www.trucking.org/economics-and-industry-data.

American Trucking Associations. “Maintaining Aftertreatment Systems: Operational and Cost Considerations.” ATA Technology and Maintenance Council report; cross-referenced through industry coverage including *OEM Off-Highway* and *Heavy Duty Trucking*. Source for the figure that ongoing aftertreatment maintenance represents approximately thirteen percent of total operating maintenance costs over the life of a Class 8 truck, cited in the trucking chapter. https://www.trucking.org/.

American Waterways Operators. Organizational homepage. The national trade association for the U.S. tugboat, towboat, and barge industry, organized in Washington, D.C. in 1944 and representing more than 300 member companies operating on the nation’s inland rivers, the Great Lakes, and the Atlantic, Pacific, and Gulf coasts. Identified in the plaintiff-categories chapter as the trade body that would carry the associational claim under *Hunt v. Washington State Apple Advertising Commission* for the independent tug and barge operator whose vessel is built around a marine diesel governed by EPA’s Part 1042 standards. https://www.americanwaterways.com/.

Andrus v. Allard, 444 U.S. 51 (1979). The unanimous Supreme Court decision holding that a federal regulation banning the commercial sale of eagle feathers lawfully obtained before the Eagle Protection Act’s effective date was not a Fifth Amendment taking, because

possession and non-commercial transfer were preserved. Cited in the chapter on objections considered as the leading adverse precedent for the Takings Clause prong of the book's argument.
https://supreme.justia.com/cases/federal/us/444/51/.

Anthropic. "Claude Mythos Preview and Project Glasswing." Announcement, April 7, 2026, as reported in Ravie Lakshmanan, "Anthropic's Claude Mythos Finds Thousands of Zero-Day Flaws Across Major Systems," *The Hacker News*, April 8, 2026. A limited-access frontier model that, with minimal human steering, identified thousands of previously unknown high-severity vulnerabilities across every major operating system and web browser, including a twenty-seven-year-old remote-crash flaw in the TCP stack of OpenBSD — the operating system with the field's strongest reputation for line-by-line code audit — and a sixteen-year-old flaw in FFmpeg's H.264 decoder that had survived five million fuzzing runs. Cited in the preface as one of three cases in which a machine found, by reading a system's rules, a flaw its expert defenders had not.
https://thehackernews.com/2026/04/anthropics-claude-mythos-finds.html.

Aristotle. *The Athenian Constitution (Athenaion Politeia)*. Translated by Frederic G. Kenyon. c. 350 BC. Perseus Digital Library, Tufts University. Reconstructs the procedural mechanisms of classical Athenian deliberative bodies, including the two-urn secret-ballot (*kyrios kadiskos* and *akyros kadiskos*) used in the dikasteria and the boule. The mechanical design is the historical antecedent for the secret-ballot procedure used by the Italian parliament for 119 years and proposed in this chapter for Hybrid Voting in the U.S. Congress.
https://www.perseus.tufts.edu/hopper/text?doc=Perseus:text:1999.01.0046.

Arkansas Game & Fish Commission v. United States, 568 U.S. 23 (2012).
https://supreme.justia.com/cases/federal/us/568/23/.

Aro Manufacturing Co. v. Convertible Top Replacement Co., 365 U.S. 336 (1961). The Supreme Court's repair-and-reconstruction decision, arising from replacement tops for Ford and General Motors convertibles. The Court held that replacing a worn, unpatented element

of a patented combination is permissible repair, not infringing reconstruction, and that no unpatented element of a combination is entitled to the patent monopoly, however essential to the combination it may be. Cited for the principle that a combination patent reaches only the totality of its claimed elements.
https://supreme.justia.com/cases/federal/us/365/336/.

Aronson v. Quick Point Pencil Co., 440 U.S. 257 (1979). The modern Supreme Court restatement of the Patterson v. Kentucky principle that states may regulate the use of intellectual property in any manner not inconsistent with federal patent law. Cited in the chapter on objections considered to establish the long-standing rule that patents do not confer immunity from generally applicable regulation.
https://supreme.justia.com/cases/federal/us/440/257/.

Asch, Solomon E. “Studies of Independence and Conformity: A Minority of One Against a Unanimous Majority.” *Psychological Monographs: General and Applied* 70, no. 9 (1956): 1–70.
https://psycnet.apa.org/doiLanding?doi=10.1037%2Fh0093718.

Association for Accessible Medicines. *The U.S. Generic & Biosimilar Medicines Savings Report*. Annual report. Source for the cumulative consumer-savings figure attributed to the Drug Price Competition and Patent Term Restoration Act of 1984 (the Hatch-Waxman Act), reported in the trillion-dollar range cumulatively since the statute’s passage. Cited in the policy-framework chapter for the Hatch-Waxman precedent.
https://accessiblemeds.org/sites/default/files/2023-09/AAM-2023-Generic-Biosimilar-Medicines-Savings-Report-web.pdf.

Association of Home Appliance Manufacturers (AHAM). Federal Lobbying Disclosure, Calendar Year 2019. U.S. Senate Office of Public Records, Lobbying Disclosure Act Section 5 reports (LD-2). Aggregated via OpenSecrets.org. The trade association representing the major appliance manufacturers selling into the United States market

reported total federal lobbying expenditures of $1,104,386 in calendar year 2019; comparable annual totals have been reported in adjacent years. https://www.opensecrets.org/orgs/association-of-home-appliance-mfrs/summary?id=D000024172.

Attia, Zachi I., et al. “Screening for Cardiac Contractile Dysfunction Using an Artificial Intelligence-Enabled Electrocardiogram.” *Nature Medicine* 25 (2019): 70-74. The derivation study behind the Mayo Clinic electrocardiogram work discussed in the chapter on outsiders and blind doctrine.

Aurora Innovation, Inc. Corporate page and commercial-operations record. Aurora Innovation is the autonomous-freight company that began hauling commercial loads without a driver in the cab on the Dallas–Houston corridor of Interstate 45 in 2025; trade reporting places its Driver-as-a-Service fee near $0.85 per mile. Cited in the autonomous-vehicle chapter for the structure of the autonomous-trucking fee. https://aurora.tech/.

Automatic Emergency Braking Rule (FMVSS No. 127). Final Rule, “Federal Motor Vehicle Safety Standards; Automatic Emergency Braking Systems for Light Vehicles,” 89 Fed. Reg. 39686 (May 9, 2024). National Highway Traffic Safety Administration. Compliance required by September 1, 2029, under the mandate at 49 U.S.C. § 30129 added by the Infrastructure Investment and Jobs Act of 2021. https://www.federalregister.gov/documents/2024/05/09/2024-09462/federal-motor-vehicle-safety-standards-automatic-emergency-braking-systems-for-light-vehicles.

Axios. “The Math Is Improving for Autonomous Trucks.” Axios, May 6, 2026. Trade reporting on Goldman Sachs Research projections for autonomous-versus-human trucking costs, including the projected 2028 cost crossover, early single-route claims of sub-$2.00-per-mile driverless operation, the projected fall in the autonomous hardware premium to roughly $35,000 to $40,000 by 2035, and the American Transportation Research Institute

figure of about $2.26 per mile for a human-driven truck. Cited in the autonomous-vehicle chapter on the crossover projection.
https://www.axios.com/2026/05/06/autonomous-trucks-av-freight.

Azoulay, Pierre, Christian Fons-Rosen, and Joshua S. Graff Zivin. “Does Science Advance One Funeral at a Time?” *American Economic Review* 109, no. 8 (August 2019): 2889-2920. Four hundred fifty-two academic life scientists who died prematurely; publication by scientists who had never collaborated with the deceased rose 8.6 percent on average, and entry was most suppressed where the deceased scientist’s collaborators held gatekeeping positions.
https://doi.org/10.1257/aer.20161574.

Backup Camera Rule (FMVSS No. 111). Final Rule, “Federal Motor Vehicle Safety Standards; Rear Visibility,” 79 Fed. Reg. 19178 (April 7, 2014). National Highway Traffic Safety Administration.
https://www.federalregister.gov/documents/2014/04/07/2014-07469/federal-motor-vehicle-safety-standards-rear-visibility.

Baker v. Selden, 101 U.S. 99 (1879). Justice Bradley, for the Court, held that the copyright in a book explaining a system of bookkeeping gave its author no exclusive right to the system itself; copyright protects the author’s explanation, and the method or art the book describes is open to the public unless secured by patent. The foundational statement that copyright reaches expression and not function.
https://supreme.justia.com/cases/federal/us/101/99/.

Bandura, Albert. *Moral Disengagement: How People Do Harm and Live with Themselves*. New York: Worth Publishers, 2016.
https://albertbandura.com/albert-bandura-moral-disengagement.html.

Beck, Andrew H., Ankur R. Sangoi, Samuel Leung, Robert J. Marinelli, Torsten O. Nielsen, Marc J. van de Vijver, Robert B. West, Matt van de Rijn, and Daphne Koller. “Systematic Analysis of Breast Cancer Morphology Uncovers Stromal Features Associated with Survival.” Science Translational Medicine 3, no. 108 (November 9, 2011): 108ra113. The

C-Path (Computational Pathologist) study. Breast cancer had been graded on a small set of features of the cancer epithelium under a scheme largely unchanged since the 1920s. The system measured 6,642 features of each slide across both epithelium and stroma, was trained on the Netherlands Cancer Institute cohort (n = 248) and evaluated on the independent Vancouver General Hospital cohort (n = 328), and found three stromal features whose association with overall survival was stronger than that of the epithelial characteristics in the model — identifying stromal morphology as a prognostic determinant not previously recognized. Cited in the preface as the case in which a machine did not solve what specialists could not, but looked where specialists had not thought to look.
https://pubmed.ncbi.nlm.nih.gov/22072638/

Bellan, Rebecca. “OpenAI Claims It Solved an 80-Year-Old Math Problem — For Real This Time.” *TechCrunch*, May 20, 2026. Reports OpenAI’s announcement that its reasoning model produced an original mathematical proof disproving Paul Erdős’s 1946 planar-unit-distance conjecture, with companion remarks supporting the disproof from mathematicians Noga Alon, Melanie Wood, and Thomas Bloom. OpenAI characterized the result as “the first time AI has autonomously solved a prominent open problem” in mathematics. Source for the preface’s reference to the moment AI tools began solving problems unsolved by human effort alone.
https://techcrunch.com/2026/05/20/openai-claims-it-solved-an-80-year-old-math-problem-for-real-this-time/.

Bharat NCAP. “AIS 197.” Notification page, Bharat New Car Assessment Programme, Ministry of Road Transport and Highways, Government of India. Bharat NCAP star ratings are awarded on cars that manufacturers voluntarily submit, assessed against the AIS-197 laboratory test protocol by an agency authorized under Rule 126 of the Central Motor Vehicles Rules, 1989. Cited in the international-comparison chapter to distinguish a consumer safety assessment from a statutory type-approval certificate: a Bharat NCAP fact sheet is the former and not the latter, and the chapter’s claim about the Thar ROXX is stated as an independent assessment rather than as a certification.
https://www.bncap.in/notifications/ais-197/.

Bharat NCAP. “Mahindra Thar ROXX.” Vehicle test report, Bharat New Car Assessment Programme, Ministry of Road Transport and Highways, Government of India, November 2024. The official BNCAP page documents the five-star ratings for adult and child occupant protection awarded under the BNCAP testing protocol.
https://www.bncap.in/vehicle/mahindra-thar-roxx/.

Bicycle Blue Book. “2005 Trek Madone 5.2.” Value Guide, Bicycle Blue Book, model-year 2005 listing. Pricing reference for the 2005 Trek Madone 5.2 ($3,400–$3,629 new in 2005) cited in the bicycles section of the modern-bargain chapter as the patented-era counterpart to the 2026 SAVA Warwind V3.
https://www.bicyclebluebook.com/value-guide/2005%20Trek%20Madone%205.2.

Bizzozero, Giulio. “Ueber die schlauchförmigen Drüsen des Magendarmkanals und die Beziehungen ihres Epithels zu dem Oberflächenepithel der Schleimhaut.” *Archiv für mikroskopische Anatomie* 42 (1893): 82–152. The organisms Bizzozero described were in the stomachs of dogs, not humans; the chapter on outsiders and blind doctrine states this, because the popular retelling does not. Known here through later histories of *Helicobacter pylori* and not consulted in the original.

Black & Decker portable electric drill, 1916. S. Duncan Black and Alonzo G. Decker devised a pistol grip and trigger switch in 1914 that let one hand control the tool, paired it with a universal motor able to run on alternating or direct current, and began manufacturing their half-inch portable drill in 1916; the patent issued as U.S. Patent No. 1,245,860. Electric drills existed well before then, but as large stationary machines confined to industrial plants. Cited in the preface for the date the portable power tool became available, as against the much later date at which the building trades took it up. National Inventors Hall of Fame, “S. Duncan Black,” inducted 2019.
https://www.invent.org/inductees/s-duncan-black.

Bolling v. Sharpe, 347 U.S. 497 (1954). The federal companion to Brown v. Board of Education, decided the same day, holding that the Fifth Amendment’s Due Process Clause incorporates an equal-protection guarantee against the federal government, since the

Fourteenth Amendment binds only states. Cited in the chapter on objections considered as the doctrinal foundation that makes any federal-equal-protection framing of the regulatory wall possible at all.
https://supreme.justia.com/cases/federal/us/347/497/.

Bonito Boats, Inc. v. Thunder Craft Boats, Inc., 489 U.S. 141 (1989).
https://supreme.justia.com/cases/federal/us/489/141/.

Boston Consulting Group. “Is There a Future for Service Stations?” July 2019. Scenario analysis of fuel-retail network economics under combined electrification, shared mobility, and vehicle autonomy. States that the refueling or recharging of shared-mobility autonomous vehicles will commonly occur while the vehicles are empty of passengers, at dedicated parking areas outside urban areas, and that the result will be a decline in customer traffic at service stations and lower fuel and convenience-store sales. In its faster environments the report puts station throughput down 30 to 50 percent, gasoline demand for light-duty vehicles down 50 to 70 percent by 2035, and up to 80 percent of an unadapted fuel-retail network unprofitable within about fifteen years. Cited in The Last Human Driver for the effect of hired autonomous fleets on the roadside economy. The report does not break out a separate figure for in-store merchandise or foodservice sales; searches in September 2026 across consultancies, oil majors, trade associations, and transport journals located no published study that does.
https://web-assets.bcg.com/img-src/BCG-Is-There-a-Future-for-Service-Stations-July-2019_tcm9-223783.pdf

Boulware, David R., Matthew F. Pullen, Ananta S. Bangdiwala, et al. “A Randomized Trial of Hydroxychloroquine as Postexposure Prophylaxis for Covid-19.” New England Journal of Medicine 383, no. 6 (August 6, 2020): 517–25. The trial most often cited as establishing that hydroxychloroquine does not work as post-exposure prophylaxis. Cited in the hindered-sector chapter as the exhibit for how a null result can be produced by trial design rather than by the molecule.
https://doi.org/10.1056/NEJMoa2016638. See Endnote Commentary.

Boyle, Willard S., and George E. Smith. “Buried Channel Charge Coupled Devices.” U.S. Patent 3,792,322, filed April 19, 1973, issued February 12, 1974. Assigned to Bell Telephone Laboratories, Inc. Foundational charge-coupled-device imaging patent in the corpus underlying automotive backup-camera image sensors; expired in the 1990s under the pre-1995 seventeen-year-from-issue patent term. Cited in the surveillance chapter as one of the foundational expired imaging patents from which a remanufactured vehicle’s backup camera can be built.
https://patents.google.com/patent/US3792322.

Betts v. Brady, 316 U.S. 455 (1942). The Supreme Court’s holding that due process did not require a State to appoint counsel for an indigent defendant in a noncapital case. Justice Black dissented, joined by Justices Douglas and Murphy. Overruled twenty-one years later in *Gideon v. Wainwright* (1963).
https://supreme.justia.com/cases/federal/us/316/455/.

Bowers v. Hardwick, 478 U.S. 186 (1986). The Supreme Court’s decision upholding a Georgia sodomy statute, stating that proscriptions against such conduct had ancient roots. Overruled in *Lawrence v. Texas* (2003) after a brief filed by historians established that the sources had been in print throughout and did not say what the opinion said they said.
https://supreme.justia.com/cases/federal/us/478/186/.

bp p.l.c. “Energy Outlook: Transport.” bp Energy Outlook, annual; transport section consulted September 2026. Models the share of passenger-car vehicle-kilometres powered by electricity that is driven by robotaxis, reaching roughly 40 percent by 2035 in the Rapid and Net Zero scenarios and about 20 percent in the business-as-usual case, with robotaxis used up to nine times as intensively as private cars. Cited in The Last Human Driver for the scale of hired-fleet mileage. bp owns TravelCenters of America; the Outlook models the demand side of this shift and publishes no companion estimate of its effect on the company’s retail business.
https://www.bp.com/en/global/corporate/energy-economics/energy-outlook/demand-by-sector/transport.html

Bracha, Oren. *Owning Ideas: The Intellectual Origins of American Intellectual Property, 1790–1909*. New York: Cambridge University Press, 2016. https://www.cambridge.org/core/books/owning-ideas/7DCA50A24972FB2535CCC306420C51A3.

Brandeis, Louis D., and Josephine Goldmark. Brief for Defendant in Error, *Muller v. Oregon*, 208 U.S. 412 (1908). Of one hundred thirteen pages, roughly ninety-eight consisted of factory inspection reports, medical testimony and foreign labour statutes compiled by Goldmark and her researchers at the National Consumers League. Oregon's attorney general filed a separate conventional brief supplying the legal authorities.

Bretz, J Harlen. "The Channeled Scablands of the Columbia Plateau." *Journal of Geology* 31, no. 8 (1923): 617-649.

BrightFocus Foundation. "New Research Links Blue Light to Macular Degeneration Risk, Antioxidants May Offer Hope." BrightFocus Foundation, news article. Position summary by the BrightFocus Foundation (a U.S. medical-research nonprofit funding research on Alzheimer's disease, macular degeneration, and glaucoma) reviewing peer-reviewed evidence that blue-light exposure reaches the retina and generates oxidative stress contributing to age-related macular degeneration risk. Cited in the healthy-lighting chapter as an authoritative coalition-member source on the retinal-injury dimension of LED-only residential lighting environments. https://www.brightfocus.org/news/new-research-links-blue-light-to-macular-degeneration-risk-antioxidants-may-offer-hope/.

Brown v. Board of Education, 347 U.S. 483 (1954). Chief Justice Warren, for a unanimous Court, held that separate educational facilities are inherently unequal. The doll studies of Kenneth and Mamie Clark appear in footnote eleven, in a list of social-science authorities; the Clarks were psychologists and neither was a lawyer. https://supreme.justia.com/cases/federal/us/347/483/.

Brulotte v. Thys Co., 379 U.S. 29 (1964). The Supreme Court's holding that a patentee may not collect royalties for use of a patented invention after the patent has expired, even where

the parties agreed to the arrangement by contract. The patents in the case expired on schedule and no term was extended; the Court struck the royalty provision because the practical consequence of expiration had been withheld by another instrument. Reaffirmed on stare decisis grounds in *Kimble v. Marvel Entertainment* (2015). https://supreme.justia.com/cases/federal/us/379/29/.

Bryant, Andrew, Theresa A. Lawrie, Therese Dowswell, et al. “Ivermectin for Prevention and Treatment of COVID-19 Infection: A Systematic Review, Meta-Analysis, and Trial Sequential Analysis to Inform Clinical Guidelines.” American Journal of Therapeutics 28, no. 4 (July/August 2021): e434–e460. Meta-analysis of twenty-four randomized controlled trials involving 3,406 participants, concluding in favor of clinical effect. https://doi.org/10.1097/MJT.0000000000001402.

Buffalo Soldiers National Museum. “The Buffalo Soldiers: A Brief History.” Houston, TX. Accessed May 9, 2026. Reference for the all-Black cavalry regiments organized in 1866 from veterans of the United States Colored Troops — the Ninth and Tenth Cavalry and the Twenty-Fourth and Twenty-Fifth Infantry — and their primary assignment to frontier patrol duty in the post-Civil War American West. https://apps.abacus.ai/chatllm/share/conversation/1418155a2b:13f78811d8 ; https://www.thenmusa.org/buffalo-soldiers/ .

Bureau of Economic Analysis. Industry Economic Accounts. U.S. Department of Commerce. The BEA produces the Gross Domestic Product by Industry accounts and related Industry Economic Accounts that disaggregate national consumer and producer spending by industry sector. https://www.bea.gov/data/industry.

Bureau of Labor Statistics. *War and Postwar Wages, Prices, and Hours, 1914–23 and 1939–44*. By Witt Bowden. Bulletin of the United States Bureau of Labor Statistics, No. 852. Washington, DC: U.S. Government Printing Office, 1946. Source for the $1,428 average annual salary-wage in 1923 for American full-time non-governmental, non-agricultural workers cited in the cars-and-trucks chapter as the closest published

contemporary figure to working-man wages a 1925 Model T Runabout buyer would have earned. The BLS did not collect median-income data in the 1920s; the published average is the best available period statistic, and it covers a narrower population (excluding agricultural and governmental employment) than "all working men." Archived by the Federal Reserve Bank of St. Louis, FRASER digital library. https://fraser.stlouisfed.org/title/war-postwar-wages-prices-hours-1914-23-1939-44-4318.

Bureau of Labor Statistics. "Occupational Outlook Handbook: Heavy and Tractor-Trailer Truck Drivers." U.S. Department of Labor, updated August 2025. The 2024 employment estimate for heavy and tractor-trailer truck drivers (SOC 53-3032) is 2,235,100; the broader BLS combined estimate covering heavy and tractor-trailer drivers, light truck drivers (SOC 53-3033), and driver/sales workers (SOC 53-3031) is approximately 3,766,400 in 2024. Source for the trucking chapter's figure of roughly three and a half million U.S. truck drivers. https://www.bls.gov/ooh/transportation-and-material-moving/heavy-and-tractor-trailer-truck-drivers.htm.

Bureau of Labor Statistics. Consumer Expenditure Survey. U.S. Department of Labor. The CES has been conducted continuously and provides detailed data on how American households allocate spending across more than sixty product and service categories. https://www.bls.gov/cex/.

Bureau of Labor Statistics. Consumer Price Index. U.S. Department of Labor. The CPI is the standard federal measure of price change over time for a representative basket of consumer goods and services. The 1925-to-2025 CPI conversion factor used in the cars-and-trucks chapter is approximately 18.37 (CPI 17.5 in 1925 against CPI 321.465 in 2025). https://www.bls.gov/cpi/.

Bureau of Labor Statistics. Median Weekly Earnings of Wage and Salary Workers. U.S. Department of Labor, quarterly news release. The first-quarter 2025 release reports median weekly earnings of $1,194 for the 120.9 million full-time wage and salary workers in the United States, with the men's median at $1,307 weekly and the women's median at $1,096

weekly. The 2025 figure of approximately $67,964 per year used in the cars-and-trucks chapter is the annualized men's median ($1,307 × 52), consistent with the chapter's use of the phrase "American working man's median income."
https://www.bls.gov/news.release/wkyeng.toc.htm.

Bureau of the Census. Annual Survey of Manufactures. U.S. Department of Commerce. The federal government's principal annual data series on U.S. manufacturing activity — value of shipments, value added, capital expenditures, and employment, disaggregated by NAICS industry code — for establishments with one or more paid employees. The ASM was conducted annually (except in years ending in 2 and 7, when its data appeared in the Economic Census of Manufacturing) through the 2021 survey year, after which the data formerly collected for the ASM transitioned to the Annual Integrated Economic Survey (AIES) beginning with the 2023 data year. Used together with the BLS Consumer Expenditure Survey and the BEA Industry Economic Accounts as one of three federal data sources for the three-tier breakdown of expired-patent commerce discussed in Chapter 2.
https://www.census.gov/programs-surveys/asm.html.

Bureau of Transportation Statistics. "Counting the Transportation Workforce: Nearly 1 Million Self-Employed." Data Spotlight, U.S. Department of Transportation. Based on Federal Motor Carrier Safety Administration registration data through November 2023, identifies approximately 922,854 independent owner-operator truck drivers, which BTS endorses as more comprehensive than the lower survey-derived BLS Current Population Survey estimate of approximately 294,000 self-employed truckers. Source for the trucking chapter's owner-operator range of approximately 587,000 to 923,000 (the lower figure reflecting BLS survey methodology, the upper reflecting FMCSA registration data).
https://www.bts.gov/data-spotlight/counting-transportation-workforce-nearly-1-million-self-employed.

Caffey, John. "Multiple Fractures in the Long Bones of Infants Suffering from Chronic Subdural Hematoma." *American Journal of Roentgenology* 56 (1946): 163-173. The radiological pattern described sixteen years before Kempe named it.

California Air Resources Board. Executive Order A-10-1008, Relating to Certification of New Motor Vehicles. Ford Motor Company, model year 2002, test group 2FMXT05.4RF8 (5.4-liter V8; dual three-way catalytic converters, dual heated oxygen sensors, exhaust-gas recirculation, sequential multiport fuel injection). California Air Resources Board. https://arb.ca.gov/msprog/onroad/cert/pcldtmdv/2002/ford_mdv_a0101008_5d4_l1.pdf.

California Air Resources Board. Truck and Bus Regulation. California Air Resources Board. The California regulation barring most heavy-duty diesel trucks with pre-2010 engines from operating on California roads; a number of other states have adopted some or all of California's heavy-duty rules. Cited in the autonomous-vehicle chapter as a state-level operating restriction that, after a favorable ruling, is answered by the Supremacy Clause. https://ww2.arb.ca.gov/our-work/programs/truck-and-bus-regulation.

California Code of Regulations, 13 C.C.R. § 1968.1. Malfunction and Diagnostic System Requirements—Light-Duty Vehicles (OBD II). California Office of Administrative Law. https://govt.westlaw.com/calregs/Document/I0AC5D330D48411DEBC02831C6D6C108E.

California Code of Regulations, 13 C.C.R. § 1968.2. Malfunction and Diagnostic System Requirements—2004 and Subsequent Model-Year Passenger Cars, Light-Duty Trucks, and Medium-Duty Vehicles and Engines (OBD II). Adopted by the Board in 2002; final regulation order codified 2003; applicable to model year 2004 and subsequent. California Office of Administrative Law. https://govt.westlaw.com/calregs/Document/I5A6C8E70D48411DEBC02831C6D6C108E.

Cannon, Jason. "EPA Makes Biggest Push to Date to Deregulate Diesel Emissions." *Commercial Carrier Journal*, March 12, 2025. Trade-press account of EPA Administrator Lee Zeldin's March 12, 2025 announcement of thirty-one deregulatory actions, including formal reconsideration of the Greenhouse Gas Phase 3 rule and the 2022 Heavy-Duty Engine and Vehicle NOx rule. Quotes American Trucking Associations and Owner-

Operator Independent Drivers Association leadership in support. Source for the trucking chapter's account of the EPA-path activity.
https://www.ccjdigital.com/regulations/emissions/article/15739763/epa-makes-biggest-push-to-date-to-deregulate-diesel-emissions.

Car and Driver. "Where's the New DeLorean We Were Promised?" 2020. Documents the multi-year delay in implementation of the Low Volume Motor Vehicle Manufacturers Act of 2015. The Act, signed by President Obama in December 2015, gave the National Highway Traffic Safety Administration and the Environmental Protection Agency one year to establish the registration and certification process required for low-volume replica manufacturers to begin production. Six years passed before the rules were finalized. The article documents the lawsuit filed by the Specialty Equipment Market Association (SEMA) that forced NHTSA's January 2020 publication of the proposed rule. The DeLorean Motor Company of Texas, which had championed the legislation and announced plans for spring-2017 production of new DMC-12s, ultimately placed the project on indefinite hold in October 2022 because of supplier-relationship collapse and engine-certification difficulties produced by the regulatory paralysis. Source for the cars-and-trucks chapter's DeLorean reference.
https://www.caranddriver.com/features/a34673012/where-is-the-new-delorean/.

CarEdge. "The Death of the Affordable Car: How America Got Priced Out." CarEdge, 2026. Automotive-market analysis reporting that the number of new vehicle models priced under $25,000 has fallen from roughly a dozen in 2012 to four, and attributing the decline in part to a regulatory structure that disfavors small, inexpensive vehicles. Cited in the autonomous-vehicle chapter on the disappearance of the affordable new car.
https://caredge.com/guides/the-death-of-the-affordable-car.

Cars.com Editorial Staff. "2025 Ford F-150: Review, Pricing, Trims, and Specs." Cars.com Research, model-year 2025 listing. Manufacturer-pricing reference for the 2025 Ford F-150 base XL Regular Cab 2WD MSRP of $37,250 cited in the cars-and-trucks chapter.
https://www.cars.com/research/ford-f_150-2025/.

Carscoops. “A Third of Americans Are Priced Out of New Cars, and It’s Getting Worse.” Carscoops, February 2026. Reporting, drawing on a Plante Moran analysis, that roughly a third of American households can no longer afford any new vehicle and that the share of new-car buyers earning under $100,000 fell from 50 percent in 2020 to 37 percent by 2026. Cited in the autonomous-vehicle chapter on new-car affordability. https://www.carscoops.com/2026/02/new-car-affordability-crisis/.

Cedar Point Nursery v. Hassid, 594 U.S. 139 (2021). https://supreme.justia.com/cases/federal/us/594/20-107/.

Celgene Corp. v. Peter, 931 F.3d 1342 (Fed. Cir. 2019). The Federal Circuit’s holding that applying inter partes review to patents issued before the Leahy-Smith America Invents Act is not an unconstitutional taking, on the ground that patent owners have always held their patents subject to the possibility of a validity challenge; the court took up the retroactivity question that *Oil States Energy Services v. Greene’s Energy Group* had expressly reserved. Certiorari denied, 140 S. Ct. 2675 (2020). Cited in the chapter on objections considered as the first of two Federal Circuit decisions applying the question Oil States left open; the decision concerns cancellation of a live patent and does not reach the public’s right to build a design after its patent has expired. https://law.justia.com/cases/federal/appellate-courts/cafc/18-1167/18-1167-2019-07-30.html.

Centers for Disease Control and Prevention. “Epilepsy: Facts and Stats.” National Center for Chronic Disease Prevention and Health Promotion, U.S. Department of Health and Human Services, updated May 15, 2024. The current authoritative federal statistical compilation on epilepsy in the United States. Reports approximately 2.9 million U.S. adults aged eighteen and over and 456,000 children seventeen and under with active epilepsy, for a total of roughly 3.4 million Americans. Source for the healthy-lighting chapter’s photosensitive-epilepsy population baseline. https://www.cdc.gov/epilepsy/data-research/facts-stats/index.html.

Centers for Medicare and Medicaid Services. Local Coverage Determination L39058: *Platelet Rich Plasma Injections for Non-Wound Injections*. Effective 2023, current. https://www.cms.gov/medicare-coverage-database/view/lcd.aspx?lcdid=39058.

Charlotin, Damien. “AI Hallucination Cases.” Database, continuously updated. A public, continuously updated database of court decisions worldwide in which a party was found to have relied on citations fabricated by an artificial-intelligence system. As of early July 2026 the database recorded 1,668 such decisions, 1,163 of them in the United States; of that total, 653 involved a practising attorney and 975 involved a self-represented litigant. The database counts only decisions in which a court explicitly found or implied reliance on hallucinated content, and therefore understates the true incidence. Cited in the preface as the public record of the failure mode this book’s research method is designed to avoid. https://www.damiencharlotin.com/hallucinations/

Chemical & Engineering News. “EPA Plans More Environmental Deregulation in 2026.” February 11, 2026. Reports on EPA Administrator Lee Zeldin’s plans to continue rolling back vehicle emission rules in 2026, including the Model Year 2027 light- and medium-duty vehicle standards and the 2022 Heavy-Duty Engine and Vehicle NOx rule. Source for the trucking chapter’s account of the EPA-path actions extending into 2026. https://cen.acs.org/environment/epa-deregulation-zeldin-climate-endangerment-vehicle-emission-rules/104/web/2026/01.

Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc., 467 U.S. 837 (1984). https://supreme.justia.com/cases/federal/us/467/837/.

The Chicago Manual of Style. 17th ed. Chicago: University of Chicago Press, 2017. The book’s Endnotes section is formatted according to this manual’s bibliographic conventions for periodicals, books, court cases, federal statutes, and online sources. In-text citations follow Chicago author-date style. https://www.chicagomanualofstyle.org/home.html. See Endnote Commentary.

City of Cleburne v. Cleburne Living Center, Inc., 473 U.S. 432 (1985). The Supreme Court decision applying rational-basis review with unusual rigor to strike down a municipal

zoning ordinance that excluded a group home for persons with intellectual disabilities. The Court found that the ordinance's only conceivable rationale was prejudice against an unpopular group, which is not a legitimate state interest. The case is the cleanest illustration of the "rational basis with bite" doctrine that occasionally allows an equal-protection challenge to economic or social classifications to succeed where animus is identifiable. Cited in the chapter on objections considered as the narrow escape route from Williamson v. Lee Optical, the conditions for which the regulatory wall in this book's argument does not satisfy.
https://supreme.justia.com/cases/federal/us/473/432/.

Clapper v. Amnesty International USA, 568 U.S. 398 (2013).
https://supreme.justia.com/cases/federal/us/568/398/.

Clash-of-Leges. "A Bilingual Dataset for Conflict Detection and Explanation in Statutory Law." *Expert Systems with Applications*, 2025. A dataset derived from rulings of the Constitutional Court of the Italian Republic, built to train models on three tasks — deciding whether two legal articles conflict, explaining the conflict, and retrieving the legal basis for it. Cited in the preface as the academic step toward the search this book runs, limited by construction to conflicts a court has already declared.
https://www.sciencedirect.com/science/article/pii/S0957417425037972.

Clean Air Act Amendments of 1970, Pub. L. No. 91-604, 84 Stat. 1676. Codified as amended at 42 U.S.C. §§ 7401 et seq.
https://www.law.cornell.edu/uscode/text/42/chapter-85.

Clean Air Act Amendments of 1977, Pub. L. No. 95-95, 91 Stat. 685. Codified as amendments to 42 U.S.C. §§ 7401 et seq. Established the Prevention of Significant Deterioration program and the New Source Review program.
https://www.law.cornell.edu/uscode/text/42/chapter-85.

Clean Air Act Amendments of 1990, Pub. L. No. 101-549, 104 Stat. 2399. Codified as amendments to 42 U.S.C. §§ 7401 et seq. Expanded the toxic-air-pollutants program, added

the acid-rain trading regime, and authorized the Title VI implementation of the Montreal Protocol.
https://www.law.cornell.edu/uscode/text/42/chapter-85.

Coast Guard, Vessel Inspection Authority. 46 U.S.C. Subtitle II, Part B (Inspection and Regulation of Vessels). Codifies the Coast Guard's authority to inspect, certify, and regulate U.S.-flagged commercial vessels.
https://www.law.cornell.edu/uscode/text/46/subtitle-II/part-B.

Cobell v. Salazar, No. 96-1285 (D.D.C. settled December 8, 2009; final court approval June 20, 2011). $3.4 billion class action settlement of trust accounting claims on behalf of approximately 500,000 individual Indian trust beneficiaries; named plaintiff Elouise Cobell of the Blackfeet Nation. Authorized by the Claims Resolution Act of 2010, Pub. L. No. 111-291.
https://narf.org/cases/cobell/.

Code of Federal Regulations. "21 C.F.R. § 801.415 — Maximum Acceptable Level of Ozone." Food and Drug Administration regulation classifying ozone as a toxic gas with no known useful medical application in specific, adjunctive, or preventive therapy. Electronic Code of Federal Regulations.
https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-801/subpart-F/section-801.415.

Code of Federal Regulations. "40 C.F.R. § 19.4 — Civil Monetary Penalty Inflation Adjustment, Table 1." U.S. Environmental Protection Agency. The current adjusted civil penalty maxima EPA uses for Clean Air Act violations. For tampering with emission-control devices on a highway motor vehicle under Clean Air Act § 203(a)(3) (42 U.S.C. § 7522(a)(3)), the schedule sets the per-violation maximum at $5,911 for any person other than a manufacturer or dealer and $59,114 for a manufacturer or dealer (as inflation-adjusted against the statutory caps of $2,500 and $25,000 in 42 U.S.C. § 7524(a)). Each

tampered vehicle or engine, and each defeat-device part, counts as a separate violation. Source for the per-violation civil-penalty figures cited in the trucking chapter. https://www.ecfr.gov/current/title-40/chapter-I/subchapter-A/part-19/section-19.4.

Code of Federal Regulations. “40 C.F.R. § 86.1848-10 — Compliance with Emission Standards for the Purpose of Certification.” U.S. Environmental Protection Agency. The rule requiring a manufacturer of new light-duty vehicles and light-duty trucks to obtain an EPA certificate of conformity covering the vehicles before selling, offering for sale, introducing into commerce, or importing them. Cited in the chapter on objections considered to establish that a faithful new copy of a patent-expired vehicle cannot be lawfully sold without a current-tier certificate of conformity — a prohibition on sale, not a penalty. https://www.law.cornell.edu/cfr/text/40/86.1848-10.

Code of Federal Regulations. “40 C.F.R. § 1068.120 — Requirements for Rebuilding Engines.” U.S. Environmental Protection Agency. Requires anyone rebuilding an engine to have a reasonable technical basis for knowing that its emission controls perform at least as well as in its certified configuration, and requires an engine that stays in the same equipment to be rebuilt to its original configuration or to a certified configuration of the same or a later model year. Cited in the Patent Clause prong chapter for the point that a rebuilt engine is held to the standards of its own model year, not the current ones. https://www.ecfr.gov/current/title-40/chapter-I/subchapter-U/part-1068/subpart-B/section-1068.120.

Code of Federal Regulations. “40 C.F.R. § 1068.240 — Exempting New Replacement Engines.” U.S. Environmental Protection Agency. Lets a certificate holder, and no one else, build a new engine to the configuration of an earlier model year with less stringent standards, solely to replace an engine already placed into service, in equipment no more than forty years old, and only after determining that no engine certified to current standards has the physical or performance characteristics to fit. Cited in the Patent Clause prong

chapter as a federal time-shift for engines, confined to a certificate holder and to machines already built.
https://www.ecfr.gov/current/title-40/chapter-I/subchapter-U/part-1068/subpart-C/section-1068.240. See Endnote Commentary.

Code of Federal Regulations. “40 C.F.R. Part 1068 — General Compliance Provisions for Highway, Stationary, and Nonroad Programs.” U.S. Environmental Protection Agency. The general-compliance regulatory framework that sets EPA’s requirements for rebuilding engines (§ 1068.120) and its exemption for new replacement engines (§ 1068.240), cited in the trucking chapter as a regulatory model for independent engineering certification of patent-expired heavy-duty truck platforms.
https://www.ecfr.gov/current/title-40/chapter-I/subchapter-U/part-1068.

Coin Laundry Association (CLA). Organizational homepage. Founded in 1960 and headquartered in Oakbrook Terrace, Illinois, the national not-for-profit trade body for the self-service laundry industry; members include laundry owners, prospective owners, equipment distributors, and manufacturers. The association rebranded to CLA on July 1, 2024 and retains its original domain. Identified in the plaintiff-categories chapter as the trade body for the laundromat operator, whose governing hook — 42 U.S.C. § 6295(o)(1) — names clothes washers as a covered product by name. Two figures should not be conflated: the association reports more than 300 manufacturer and distributor members and roughly 1,500 owner members, while the national self-service-laundry industry it serves runs to approximately 30,000 storefronts.
https://www.coinlaundry.org/. See Endnote Commentary.

Coke, Edward. *The Third Part of the Institutes of the Laws of England*. 2nd ed. London: M. Flesher for W. Lee and D. Pakeman, 1648. Chapter “Of Monopolies.” Coke defines a monopoly as an allowance by the King, “by his grant, commission, or otherwise,” to “any person or persons” for “the sole buying, selling, making, working, or using of any thing,” whereby others “are sought to be restrained of any freedom or liberty that they had before.” The test turns on the restraint placed on everyone else, not on how many hold the grant. The passage falls at page 181 of the 1669 edition. Cited in the Preface for the definition behind

the book's title, and in the chapter *Four Problems, One Remedy*, which answers the same objection by back-reference.
https://scholarship.law.wm.edu/british-colonial-antecedents/32/.

Collins, Andrew P. "Ford CEO Jim Farley's Right To Repair Comment Should Make Every Car Owner Uncomfortable." *The Drive*, June 11, 2026. Transcribes the exchange between Ford chief executive Jim Farley and a *Detroit Free Press* reporter at an event announcing a $5 million Ford and Bloomberg Philanthropies investment in the Detroit Public Schools Community District Foundation's auto technician program, days after the June 3, 2026 White House meeting between the administration and automobile executives on right-to-repair policy. Asked whether he wanted owners repairing their own vehicles, Farley drew the distinction the foreword turns on: he says he has no difficulty working on a 1973 Bronco, but that a current-model Bronco requires specialty tools and that owner repair of it would put lives at risk. The admission is that the older machine remains serviceable by its owner and the new one does not — a difference of regulatory and electronic architecture, not of mechanical necessity. The foreword by Gord Magill quotes the exchange from this transcription. *Road & Track* reported the same footage under the headline "Ford CEO Jim Farley Clarifies Blue Oval's Position on 'Right to Repair' Rules After Trump Comments" and renders Farley's opening sentence differently, so The Drive is the source of record for the wording used. The underlying video, published June 8, 2026, is at
https://www.freep.com/videos/money/cars/ford/2026/06/08/ford-ceo-jim-farley-right-to-repair/90465881007/. This footnote is also for the Foreword.
https://www.thedrive.com/news/ford-ceo-jim-farleys-right-to-repair-comment-should-make-every-car-owner-uncomfortable.

Compco Corp. v. Day-Brite Lighting, Inc., 376 U.S. 234 (1964). Companion case to Sears, Roebuck & Co. v. Stiffel Co., decided the same day. Together the two cases established that the federal patent system's policy of leaving the unpatented public domain free for all to copy is supreme over state unfair-competition laws that would prohibit such copying. Cited

in the chapter on objections considered as part of the Supreme Court doctrinal line that supports the book's structural Patent Clause argument.
https://supreme.justia.com/cases/federal/us/376/234/.

Congressional Record. 100th Cong., 1st sess., vol. 133, pt. 3 (February 17, 1987): 3371–3374. National Appliance Energy Conservation Act of 1987 (S. 83), Senate floor debate. Includes Senate Roll Call Vote No. 28 (89–6) and remarks by Senators Phil Gramm (R-TX), J. Bennett Johnston (D-LA), James A. McClure (R-ID), Robert Dole (R-KS), and Alan Cranston (D-CA). U.S. Government Publishing Office, GPO-CRECB-1987-pt3-6.pdf. https://www.govinfo.gov/app/details/GPO-CRECB-1987-pt3.

Congressional Research Service. *Diesel Fuel and Engines: An Analysis of EPA's New Regulations.* CRS Report RL30737. May 1, 2001. Neutral congressional analytical document reviewing the EPA heavy-duty diesel rule signed December 21, 2000 by Administrator Carol Browner and promulgated as final rule at 66 Fed. Reg. 5002 (January 18, 2001). The report explicitly characterizes industry reactions to the rule as "mixed," documenting concerns expressed by engine manufacturers and refiners about technological feasibility, cost, and the diesel-fuel sulfur-reduction component, all addressed by EPA through its formal Response to Comments document published with the final rule under the notice-and-comment rulemaking process of the Administrative Procedure Act, 5 U.S.C. § 553. Source for the trucking chapter's hedged characterization that the affected manufacturers are reported as having opposed the 2007 rule when it was proposed.
https://www.everycrsreport.com/reports/RL30737.html.

Connecticut Humanities. "Albert Augustus Pope, Transportation Pioneer." ConnecticutHistory.org. Pope began by importing English bicycles in the 1870s and contracting the Weed Sewing Machine Company of Hartford to build them under the Columbia name. By 1892 he was selling a safety bicycle that carried the Starley frame, tangential spoke lacing, the Renold chain, and pneumatic tires — British designs, every one, built in Connecticut. That same year he took majority control of the Hartford Rubber Works, then employing roughly one hundred seventy people and running around the clock, along with a steel company and a nickel-plating works. The strategy was vertical

integration, and it worked because the underlying designs cost him nothing. A firm that had to license the frame, the chain, and the tire could not have absorbed its suppliers; it would have been paying rent on the machine itself.
https://connecticuthistory.org/albert-augustus-pope-1843-1909/.

Conrad, Kylie, and John D. Graham. “The Benefits and Costs of Automotive Regulations for Low-Income Americans.” *Journal of Benefit-Cost Analysis* 12, no. 3 (Fall 2021): 518–549. Aggregates per-vehicle regulatory-cost data from agency Regulatory Impact Assessments compiled in the Office of Management and Budget’s annual reports to Congress, covering the period 1967–2015. Table 5 (p. 535) itemizes the per-vehicle cumulative cost addition in 2016 dollars: $2,708.23 for safety regulations, $1,049.19 for emissions regulations, and $2,439.14 for fuel-economy regulations, summing to $6,196.56. The authors describe the total burden as “in the vicinity of $6,000 to $7,000 per vehicle” and note that, against an average new-vehicle transaction price “approaching $40,000,” cumulative federal regulation accounts for 15 to 20 percent of the price the consumer pays. The cars-and-trucks chapter uses the Conrad and Graham (2021) figure as the post-1970 cost magnitude and Sperling et al. (2004) as the proportional-attribution evidence that essentially all of that cumulative burden was added after 1970.
https://doi.org/10.1017/bca.2021.12.

Consumer Product Safety Act of 1972, Pub. L. No. 92-573, 86 Stat. 1207. Codified as amended at 15 U.S.C. §§ 2051 et seq. Establishes the Consumer Product Safety Commission and authorizes the issuance of mandatory product-safety standards and recalls under section 15.
https://www.law.cornell.edu/uscode/text/15/chapter-47.

Consumer Product Safety Commission. “Requirements for Bicycles.” 16 C.F.R. Part 1512. First promulgated at 43 Fed. Reg. 60034 (December 22, 1978), with amendments in 1980, 1981, 1995, and 2003. Bicycles were among the first product categories the Commission regulated after Congress created it in 1972. The date carries the weight here. Every American bicycle discussed in this chapter’s historical section — the Columbia safeties of the 1890s, the Indianapolis chain that went to Kitty Hawk, the balloon-tire machines of the

1930s, the middleweights and lightweights of the 1950s — was designed, built, and sold before any federal design standard for bicycles existed. Secondary accounts disagree about whether the operative date is 1975, 1976, or 1978, and the Commission's own documents are not consistent; the Federal Register citation above is the one to rely on. https://www.federalregister.gov/documents/2011/05/13/2011-11742/requirements-for-bicycles.

Continental Paper Bag Co. v. Eastern Paper Bag Co., 210 U.S. 405 (1908). Decided June 1, 1908. Held that a patentee who has never used its invention may still enjoin others from using it: the right to exclude is "the very essence of the right conferred by the patent," and nonuse alone is no ground to deny an injunction. Justice McKenna wrote for the Court; Justice Harlan dissented. Cited in the Preface, and in the chapter *Four Problems, One Remedy*, as the third authority for the book's title: a monopoly lies in the power to exclude, which needs no production, and the patent's power to exclude is what outlives the term. https://supreme.justia.com/cases/federal/us/210/405/.

Corner Post, Inc. v. Board of Governors of the Federal Reserve System, 603 U.S. 799 (2024). Decided July 1, 2024, three days after Loper Bright and in the same term. Held, 6–3, that a claim under the Administrative Procedure Act accrues when the plaintiff is injured by an agency rule rather than when the rule was issued, so the six-year limitations period of 28 U.S.C. § 2401(a) runs from injury and not from promulgation. Justice Barrett wrote for the Court; Justice Kavanaugh concurred; Justice Jackson dissented, joined by Justices Sotomayor and Kagan. Cited in the filable-case chapter as the companion rule to the end of Chevron deference: the regulatory wall was built between 1970 and 1987, and under the older reading of the limitations period most of it was beyond challenge by anyone not already in court decades ago. https://supreme.justia.com/cases/federal/us/603/22-1008/.

Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027–2031. Final Rule, 89 Fed. Reg. 52396 (June 24, 2024). National Highway Traffic Safety Administration. https://www.federalregister.gov/documents/2024/06/24/2024-13243/corporate-average-fuel-economy-standards-for-passenger-cars-and-light-trucks-for-model-years.

Coudray, Nicolas, et al. “Classification and Mutation Prediction from Non-Small Cell Lung Cancer Histopathology Images Using Deep Learning.” *Nature Medicine* 24, no. 10 (2018): 1559-1567. Six of the ten most commonly mutated genes in lung adenocarcinoma predicted from stained tissue, with areas under the curve between 0.733 and 0.856.

Cummings, Mary L. (“Missy”). Quoted in “Driverless Big Rigs Are Coming to American Highways, and Soon,” distributed by the New York Times News Service and syndicated at Journal-News (Cox Media Group), April 6, 2026. Cummings is director of the Mason Autonomy and Robotics Center at George Mason University and a former senior safety advisor to the National Highway Traffic Safety Administration. The article surfaces her phantom-braking warning — “There is no identified solution on the horizon for the problem of phantom braking. And it will not be addressed soon, because nobody wants to admit that it’s happening” — cited in the autonomous-driving chapter. https://www.journal-news.com/driverless-big-rigs-are-coming-to-american-highways-and-soon/article_f2d5f6e0-bfd5-4917-8ba0-17828a7c9492.html.

Daily Signal. “A New Car Will Cost You at Least $3,800 Extra Because of Government Regulation.” Heritage Foundation analysis, March 28, 2016. Estimates the minimum federal-regulatory cost addition on a new American vehicle at $3,800 in 2016 dollars; the analysis flags the figure as understated and likely higher with subsequent inflation. The cars-and-trucks chapter uses this as a floor and sanity check on the more detailed pre-1971 versus post-1970 cost-attribution figures derived from the Journal of Benefit-Cost Analysis study. https://www.dailysignal.com/2016/03/28/a-new-car-will-cost-you-at-least-3800-extra-because-of-government-regulation/.

D’Angelo, James, and Brent Ranalli. “The Dark Side of Sunlight: How Transparency Helps Lobbyists and Hurts the Public.” *Foreign Affairs* 98, no. 3 (May/June 2019): 155–168. https://www.foreignaffairs.com/united-states/dark-side-sunlight ; https://www.jstor.org/stable/26798160.

DaSilva, Amber. “Human Drivers Avoid Crashes 99.999819% Of The Time, Self-Driving Cars Need To Be Even Safer.” Jalopnik, February 28, 2023. Derives the per-mile crash-free rate from 2021 NHTSA data: 2.903 trillion miles driven against 5.250 million crashes, yielding a 0.000181 percent crash rate per mile, i.e., 99.999819 percent crash-free. Cited in the autonomous-driving chapter as the safety bar an autonomous vehicle would have to clear merely to match the human driver. https://www.jalopnik.com/self-driving-car-vs-human-99-percent-safe-crash-data-1850170268/.

Davis, William D. “Saddle.” U.S. Patent No. 568,939. Filed March 25, 1896; issued October 6, 1896. Patent for the spring-seat saddle, claiming a saddle “different from the common saddle in having two girths, springs under the seat, and a different form of pommel.” Davis was a soldier in the Tenth U.S. Cavalry stationed at Fort Assiniboine, Montana. https://patents.google.com/patent/US568939A.

de Bono, Edward. *The Use of Lateral Thinking*. London: Jonathan Cape, 1967. https://www.amazon.com/Use-Lateral-Thinking-Edward-Bono/dp/0140137882.

de Moor, Oege. “Inside the Rise of Autonomous AI Hackers.” Talk at AI Ascent 2026, May 6, 2026. XBOW’s founder describes Bing image search as one of the best secured systems in the world, very well secured by the engineers at Microsoft, and hammered by thousands of hackers from all over the world trying to get in. Cited in the preface. https://www.youtube.com/watch?v=eHsr1Fl2jNA.

Declaratory Judgment Act of 1934, 28 U.S.C. §§ 2201–2202. Authorizes federal courts to declare the rights of any party in a case of actual controversy without requiring anyone to

first violate the law they seek to challenge. Originally enacted as Pub. L. No. 73-343, 48 Stat. 955 (June 14, 1934).
https://www.law.cornell.edu/uscode/text/28/2201.

Deere & Company. “S-Series Combines.” Manufacturer product page for the John Deere S-Series, including the S7 600 cited in the farm-equipment chapter as the U.S. counterpart to the India-market John Deere W70.
https://www.deere.com/en/harvesting/s-series-combines/.

Detroit Reman. “History.” Company history of the Detroit Diesel remanufacturing operation, including the 1966 “reliabilt” program that set standardized quality requirements and distributor licensing for remanufacturing. Cited in the objections chapter as evidence that the original engine maker entered the remanufacturing business itself.
https://archive.detroitreman.com/about-us/history.

Diesel Engine Consent Decrees, 1998. United States v. Caterpillar Inc., Cummins Engine Company, Detroit Diesel Corporation, Mack Trucks Inc., Navistar International Transportation Corporation, Renault Véhicules Industriels, and Volvo Truck Corporation. Settlements lodged October 22, 1998, in the U.S. District Court for the District of Columbia (Judge Henry H. Kennedy Jr.), entered as final judgments July 1, 1999. The decrees resolved Clean Air Act claims that the seven manufacturers had installed undisclosed defeat-device software (altering fuel injection timing) in heavy-duty diesel engines from the late 1980s through 1998 that allowed engines to pass EPA emissions tests while emitting up to two to three times the NOx standard during normal highway operation. The combined civil penalty was $83.4 million — the largest civil penalty in environmental enforcement history at the time — with twenty-five percent of each company’s share directed to the California Air Resources Board in parallel settlements. Including injunctive relief (accelerated compliance with stricter NOx-plus-NMHC standards by October 2002, new not-to-exceed and emission surface limits, supplemental testing, recalls and fixes for in-use engines, and offset projects), the total settlement cost the manufacturers well over one billion dollars. Source for the consent-decree reference in the trucking chapter and the

assertion that the 1998 settlements stripped heavy-duty engine manufacturers of credibility to resist tougher standards.
https://www.epa.gov/enforcement/heavy-duty-diesel-engine-settlements-1998. Companion primary source: U.S. Department of Justice Press Release No. 499 (October 22, 1998), https://www.justice.gov/archive/opa/pr/1998/October/499_enr.htm.

DieselNet. “United States: Heavy-Duty Onroad Engines.” Emission standards reference compiled by Ecopoint Inc. The authoritative public summary of U.S. EPA particulate matter, NOx, and hydrocarbon standards for heavy-duty highway diesel engines from 1988 through the present, cross-referenced against 40 C.F.R. Part 86 and the underlying Federal Register notices. Documents the PM standard progression: 0.60 g/bhp-hr (model years 1988–1990), 0.25 g/bhp-hr (1991–1993), 0.10 g/bhp-hr (1994–2006), and 0.01 g/bhp-hr (2007–onward); and the in-use unregulated baseline of approximately 1.0 g/bhp-hr for pre-1988 engines. Source for Table 16.1 in the trucking chapter and for the chapter’s reconciliation of the eighty-three-percent reduction from the 1988 federal baseline with the approximately ninety-percent reduction from the unregulated pre-1988 in-use level.
https://dieselnet.com/standards/us/hd.php.

DOE, “The History of the Light Bulb.” U.S. Department of Energy. Documents that the first spiral compact fluorescent lamp was created by engineer Edward E. Hammer at General Electric in 1976, in response to the 1973 oil crisis, and that early compact fluorescents reached the consumer market in the 1980s (the first successful integral product being the Philips Model SL, 1980). General Electric patented Hammer’s spiral work; those foundational compact-fluorescent patents, and the integral-lamp, ballast, and phosphor patents underlying the ordinary screw-base compact fluorescent, have long since expired — the latest clearly relevant foundational U.S. patent identified in the author’s review reaching expiration in 2015, and most expiring well before that; that expiration-date synthesis is this argument’s own, drawn from the underlying U.S. patent records rather than from the DOE history. The compact fluorescent was therefore public-domain technology when the April 2024 general service lamp rule raised the efficiency threshold beyond its reach. Cited in the

healthy-lighting chapter for the recurrence of the withdrawal pattern in the technology that replaced the incandescent.
https://www.energy.gov/articles/history-light-bulb

DOE Energy Conservation Standards for Consumer Products and Industrial Equipment, 10 C.F.R. Parts 430 and 431. Department of Energy regulations implementing the energy-efficiency provisions of the Energy Policy and Conservation Act of 1975 and its amendments, including the National Appliance Energy Conservation Act of 1987. Cover refrigerators, freezers, room and central air conditioners, water heaters, furnaces, boilers, dishwashers, ovens and ranges, and a long list of other consumer durables.
https://www.ecfr.gov/current/title-10/chapter-II/subchapter-D.

DOE General Service Lamps — Backstop Final Rule (2022). Energy Conservation Program: Energy Conservation Standards for General Service Lamps. 87 Fed. Reg. 27439 (May 9, 2022). U.S. Department of Energy (RIN: 1904-AF09). Final rule effective July 25, 2022; full enforcement on manufacture and import beginning January 2023 and on retail and distribution beginning July 2023. Codified the 45 lumens-per-watt minimum efficacy backstop standard for general service lamps under 42 U.S.C. § 6295(i)(6)(A)(v). This is the operative rule that pulled the ordinary incandescent light bulb from the United States residential market and is the principal subject of the healthy-lighting chapter.
https://www.federalregister.gov/documents/2022/05/09/2022-09477/energy-conservation-program-energy-conservation-standards-for-general-service-lamps.

DOE General Service Lamps — Definitions Final Rule (2022). Energy Conservation Program: Definitions for General Service Lamps. 87 Fed. Reg. 27461 (May 9, 2022). U.S. Department of Energy (RIN: 1904-AF22). Final rule effective July 8, 2022. Reinstated the Obama-era GSL and GSIL definitions DOE had withdrawn in September 2019, expanding the regulatory scope to include LED, OLED, decorative candle and globe lamps, and reflector lamps that had been exempt from federal coverage.
https://www.federalregister.gov/documents/2022/05/09/2022-09480/energy-conservation-program-definitions-for-general-service-lamps.

DOE General Service Lamps — 120 Lumens-per-Watt Final Rule (2024). Energy Conservation Program: Energy Conservation Standards for General Service Lamps. 89 Fed. Reg. 28859 (April 19, 2024). U.S. Department of Energy. Final rule effective July 3, 2024; compliance required on and after July 25, 2028. Raises the minimum efficacy threshold from 45 to more than 120 lumens-per-watt, effectively eliminating compact fluorescent lamps as compliant alternatives alongside the ordinary incandescent already eliminated by the 2022 rules.
https://www.federalregister.gov/documents/2024/04/19/2024-07831/energy-conservation-program-energy-conservation-standards-for-general-service-lamps.

DOE Lighting Market Characterization 2020. U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, Solid-State Lighting Program, published April 2024 (baseline year 2020). The most recent comprehensive federal characterization of the U.S. lighting market by lamp type, sector, technology, and installed-stock distribution. Documents approximately 819 million incandescent lamps installed in the U.S. residential sector at the 2020 baseline (approximately 12.6 percent of the residential installed lighting stock, comprising approximately 161 million A-shape general service incandescent lamps, 294 million decorative lamps, 334 million reflector lamps, and miscellaneous types). Companion to the 2015 Lighting Market Characterization report. Source for the healthy-lighting chapter's installed-stock figure documenting the public-right being exercised at the moment the 2022 backstop rule withdrew the manufacturing future of those lamps.
https://www.energy.gov/sites/default/files/2024-08/ssl-lmc2020_apr24.pdf.

DOE Process Rule Notice of Proposed Rulemaking (2026). U.S. Department of Energy, Office of Critical Minerals and Energy Innovation. Notice of Proposed Rulemaking issued July 2, 2026, announced by Secretary of Energy Chris Wright. The Department's public announcement described the action as moving to end home-appliance and equipment efficiency mandates; the proposed rule itself amends the Department's internal "Process Rule" governing how future energy-conservation standards are evaluated and does not repeal any existing standard, including the general service lamp standards. Trade-press and legal analysis noted that any subsequent attempt to repeal an existing standard would

implicate the anti-backsliding provision at 42 U.S.C. § 6295(o)(1) and would likely be challenged in litigation. Cited in the healthy-lighting chapter as the second test, after Executive Order 14154, of the executive path's closure.
https://www.energy.gov/documents/process-rule-nopr

Doenges, J. L. Autopsy series reporting spiral organisms in forty-three percent of 242 human stomach specimens, 1938. Cited in the chapter on outsiders and blind doctrine for the fact that the organisms had been counted in a published human series and connected to no disease. Known here through later reviews of *Helicobacter pylori*; the full citation has not been confirmed against the original.
See Endnote Commentary.

Dolan v. City of Tigard, 512 U.S. 374 (1994). The Supreme Court's holding that regulatory conditions imposed on the use of property must be "roughly proportional" to the impact of the proposed use, building on *Nollan v. California Coastal Commission* and supplying the second prong of the modern proportionality test reaffirmed in *Sheetz v. County of El Dorado* (2024).
https://supreme.justia.com/cases/federal/us/512/374/.

Drais, Karl von. The running machine Karl von Drais first rode in 1817 and protected by a ten-year Baden privilege in 1818. Baden had no patent law at the time, so what Drais received on 12 January 1818 was a *Großherzogliches Privileg* — a grand-ducal privilege running ten years and effective only within Baden, which the German Patent and Trade Mark Office describes as the equivalent of a patent today. His first public ride on the *Laufmaschine* was at Mannheim on 12 June 1817, and a French *brevet* followed the Baden grant. Cited in the chapters on pre-1900 American prosperity and on the sectors where the bargain still works for the proto-bicycle, or "dandy horse," that opens the historical lineage running forward through the safety bicycle of the 1880s and 1890s; the design was widely copied in the United States and Britain once the brief and territorially limited term ended.

Deutsches Patent- und Markenamt, “World Day of the Bicycle,” last updated 16 February 2026. https://www.dpma.de/english/our_office/publications/background/sports-leisure-ip/worlddayofthebicycle/index.html.

Drug Price Competition and Patent Term Restoration Act of 1984 (the Hatch-Waxman Act), Pub. L. No. 98-417, 98 Stat. 1585. Codified in part at 21 U.S.C. § 355(j). Created the Abbreviated New Drug Application pathway for generic drug manufacturers, requiring proof of bioequivalence rather than full safety and efficacy review, and restored some patent term consumed by the FDA pre-market approval process for originators. Co-sponsored by Senator Orrin Hatch (R-Utah) and Representative Henry Waxman (D-California). Cited in the policy-framework chapter as the American precedent for opening a blocked patent commons through statutory action; the chapter contrasts the statutory mechanism Hatch-Waxman required with the constitutional remedy this book argues SCOTUS can deliver without congressional action. https://www.law.cornell.edu/uscode/text/21/355.

Dun & Bradstreet. “Fuel Dealers in the United States — Business Directory.” D&B Business Directory. Source for the count of fuel-dealer firms in the United States, listing roughly sixteen thousand two hundred company profiles; cited in the trucker chapter’s fuel-hauler subsection as evidence that the trade is dominated by small, independent, frequently multigenerational family operators rather than national chains. https://www.dnb.com/business-directory/company-information.fuel_dealers.us.html.

Duniho, Fergus, Jean-Louis Cazaux, and Hans Bodlaender. “Janggi — 장기 — Korean Chess.” The Chess Variant Pages. The principal English-language scholarly reference page on Janggi maintained by the chess-variants research community since the late 1990s; current version mostly rewritten by Duniho, based on earlier text by Cazaux and Bodlaender, and ultimately on Chapter LXXIV (“Tjyang-Keui—Chess”) of Stewart Culin’s *Korean Games* (1895). The page acknowledges, on its face, that “certain parts of Cazaux’s text were taken from Roleigh Martin and from David B. Pritchard” — placing Martin’s 1995 web page alongside Pritchard’s *Encyclopedia of Chess Variants* (1994) as a canonical

English-language source for the rules of the game. The page further notes that “an earlier version had lifted large portions of text from Roleigh Martin’s page describing Korean Chess, and the rule in question was the interpretation of Martin.” The site’s persons index separately credits Martin as the author of “Korean Chess” (an extensive description) and “Chinese and Korean Chess Links,” with reference to his *ShareDebate International*. Cited in the preface as the independent third-party corroboration of the author’s claim to have published the first comprehensive English-language webpage on Korean Chess. https://www.chessvariants.com/oriental.dir/koreanchess.html.

Dunlop, John Boyd. Pneumatic tire. U.S. Patent 435,995, granted 1890. The specification describes an expansible rubber tube inside a protective cover reinforced with cloth or canvas. Dunlop had patented the design in Great Britain in December 1888, and British courts invalidated that patent in 1890 on Robert William Thomson’s prior art — Thomson having patented the same principle in France in 1846 and the United States in 1847. The American patent is the one that matters for this chapter’s argument. A Scottish veterinary surgeon living in Belfast, holding no American rights by virtue of his British filing, came to the United States Patent Office and accepted the same terms the Constitution offers every inventor. Seventeen years from grant, the design belonged to the American public. https://patents.google.com/patent/US435995A/en.

Eastern Railroad Presidents Conference v. Noerr Motor Freight, Inc., 365 U.S. 127 (1961), and *United Mine Workers v. Pennington*, 381 U.S. 657 (1965). Together the source of the Noerr-Pennington doctrine, under which concerted efforts to petition the government for favorable legislation or regulation are immune from antitrust liability, even where the petitioner’s purpose is to disadvantage a competitor. Cited in *Objections Considered* as the doctrine that would defeat an antitrust theory built on the lobbying activity described in the chapter, *Why Nobody Noticed*. https://supreme.justia.com/cases/federal/us/365/127/.

Easton, Thomas A. *Off the Main Sequence*. I. O. Evans Studies in the Philosophy and Criticism of Literature, no. 31. Rockville, MD: Wildside Press, 2006. Essays by the

science-fiction critic and author, including discussion of the early-1990s electronic magazine *ShareDebate International* (see Martin, Roleigh, ed., below). https://www.amazon.com/dp/080951205X.

EdisonReport. “About That Executive Order…” *EdisonReport*lighting trade-press analysis, January 27, 2025. Reports the National Electrical Manufacturers Association’s confirmation, in the week following Executive Order 14154 (Unleashing American Energy), that the executive order could not override the statutory anti-backsliding lock at 42 U.S.C. § 6295(o)(1) on the federal general service lamps efficiency standards. Source for the healthy-lighting chapter’s account of the executive path’s closure. https://edisonreport.com/2025/01/27/about-that-executive-order/.

Eldred v. Ashcroft, 537 U.S. 186 (2003). Seven-to-two decision upholding the 1998 Sonny Bono Copyright Term Extension Act against a “limited Times” challenge under Article I, Section 8, Clause 8. Justice Ginsburg’s majority opinion held that an extension is permissible as long as the resulting term remains “limited, not perpetual.” Justices Stevens and Breyer dissented; Breyer’s dissent argued that the extension produced a term worth 99.8 percent of the value of a perpetual copyright. Cited in the chapter on objections considered as the leading adverse precedent for the Patent Clause structural prong, and distinguished there as a copyright case in a calendar-adjustment posture rather than a patent case in a structural-ratchet posture. https://supreme.justia.com/cases/federal/us/537/186/.

Electronic Code of Federal Regulations. “49 CFR Part 562 — Lighting and Marking on Agricultural Equipment.” National Highway Traffic Safety Administration. Codifies the lighting and marking standards adopted by NHTSA in 2016 under the Moving Ahead for Progress in the 21st Century Act (MAP-21) and discussed in the farm-equipment chapter as the third of the three regulatory layers that captured the American tractor. https://www.ecfr.gov/current/title-49/part-562.

Electronic Stability Control Rule (FMVSS No. 126). Final Rule, “Federal Motor Vehicle Safety Standards; Electronic Stability Control Systems; Controls and Displays,” 72 Fed.

Reg. 17236 (April 6, 2007). National Highway Traffic Safety Administration. https://www.federalregister.gov/documents/2007/04/06/E7-5921/federal-motor-vehicle-safety-standards-electronic-stability-control-systems-controls-and-displays.

Elster, Jon, ed. *Secrecy and Publicity in Votes and Debates*. Cambridge: Cambridge University Press, 2015. Edited volume of essays by political theorists on the effects of voting transparency in deliberative bodies. Includes Daniela Giannetti's chapter documenting the use of *voto segreto* in the post-war Italian parliament as a strategic tool of legislators against external coercive pressures, the scholarly basis for the Hybrid Voting chapter's reference to the parliament's 119-year use of the practice. https://www.amazon.com/dp/1107084334. See Endnote Commentary.

Emissions standards, 2010 versus current. A vehicle certified around 2010 met the federal Tier 2 program, whose fleet-average standard was 0.07 grams per mile NOx (Bin 5); California's LEV II program reached full phase-in the same year, placing much of the fleet at ULEV levels. The current Tier 3 program sets a fleet-average NMOG-plus-NOx target of 0.030 grams per mile, reached in model year 2025, and lowered gasoline sulfur from 30 to 10 parts per million beginning in 2017, while extending required emission-control durability from 120,000 to 150,000 miles. A remanufactured vehicle built on a roughly 2010-era patent-expired platform is therefore about one regulatory generation behind a new vehicle certified to the current standard, and is no dirtier than the vehicles of that generation already lawful on the road. https://www.epa.gov/regulations-emissions-vehicles-and-engines/final-rule-control-air-pollution-motor-vehicles-tier-3.

Encyclopedia of Indianapolis. "Diamond Chain Company." Indianapolis: Indiana Historical Society and IU Indianapolis. Hans Renold's roller chain, developed in Manchester, made the chain-driven safety bicycle possible, and imported chain stayed beyond the reach of most buyers. On December 24, 1890, Arthur C. Newby, Edward C. Fletcher, and Glenn Howe invested five thousand dollars to begin manufacturing bicycle chain in Indianapolis, starting with four machines and four operators in rented rooms above a tinner's shop. The encyclopedia states the consequence plainly: manufacturing the drive chains in Indianapolis

made bicycles affordable to the masses in the 1890s. By 1904 the firm produced two-thirds of the bicycle chain used in America, and it grew to supply roughly sixty percent of all bicycles sold in the country. Wilbur and Orville Wright, running a bicycle shop in Dayton, were agents for the company; seven specially designed chains drove the propellers of their flying machine at Kitty Hawk in 1903. The company was later renamed Diamond Chain. No license was sought and none was needed.
https://indyencyclopedia.org/diamond-chain-company/.

Energy Independence and Security Act of 2007, Pub. L. No. 110-140, 121 Stat. 1492. Codified as amended at scattered sections of 42 U.S.C., including the lighting-efficiency provisions amending the Energy Policy and Conservation Act at 42 U.S.C. §§ 6291 et seq.
https://www.law.cornell.edu/uscode/text/42/chapter-77.

Energy Information Administration. “Electric Power Monthly, Table 5.6.A: Average Price of Electricity to Ultimate Customers by End-Use Sector.” U.S. Department of Energy. Monthly publication of average retail electricity prices to U.S. ultimate customers by end-use sector and state. Source for the healthy-lighting chapter’s residential-electricity-rate progression: annual averages of 15.04 cents per kilowatt-hour (2022), 16.00 cents per kilowatt-hour (2023), 16.48 cents per kilowatt-hour (2024), and 17.30 cents per kilowatt-hour (2025) — a fifteen-percent increase across the period in which DOE’s general-service-lamp rule savings were projected to accrue.
https://www.eia.gov/electricity/monthly/epm_table_grapher.php?t=epmt_5_6_a.

Energy Policy and Conservation Act of 1975, Pub. L. No. 94-163, 89 Stat. 871. Codified as amended at various sections of 42 U.S.C. and 49 U.S.C., including the Corporate Average Fuel Economy program at 49 U.S.C. § 32902.
https://www.law.cornell.edu/uscode/text/49/32902.

Environmental Protection Agency. “Heavy-Duty Engine and Vehicle Standards and Highway Diesel Fuel Sulfur Control Requirements: Regulatory Impact Analysis.” EPA420-R-00-026. December 2000. The Regulatory Impact Analysis accompanying the heavy-duty diesel rule signed December 21, 2000 by Administrator Carol Browner and promulgated as

final rule at 66 Fed. Reg. 5002 (January 18, 2001). Chapter VII (Benefit-Cost Analysis), Section D, Table VII-19 documents the primary estimate of 8,300 $PM_{2.5}$-related avoided premature deaths annually at full implementation in 2030 from the combined engine/vehicle standards and 15 ppm diesel fuel sulfur requirements. Premature mortality comprises approximately eighty-nine percent of the rule's approximately $70.4 billion in total monetized annual benefits. Alternative concentration-response functions in the RIA produce ranges from approximately 9,400 to 24,200; 8,300 is the central-case primary estimate. Source for the trucking chapter's reference to EPA's projected 8,300 premature deaths prevented annually.
https://nepis.epa.gov/Exe/ZyPDF.cgi/P100K576.PDF?Dockey=P100K576.pdf.

Environmental Protection Agency. "Civil Monetary Penalty Inflation Adjustment." Final Rule, 90 Fed. Reg. 1375 (January 8, 2025). The annual rulemaking that updates the table in 40 C.F.R. § 19.4 with the inflation-adjusted civil penalty maxima EPA uses across its statutes, including the Clean Air Act tampering provisions. The 2025 adjusted figures (the most recent currently in effect: under Office of Management and Budget Memorandum M-26-11, no further cost-of-living adjustment was issued for 2026) set the Clean Air Act § 203 tampering maximum at $5,911 per violation for an individual and $59,114 per violation for a manufacturer or dealer. Source for the current per-violation tampering penalty figures cited in the trucking chapter.
https://www.federalregister.gov/documents/2025/01/08/2025-00206/civil-monetary-penalty-inflation-adjustment.

Environmental Protection Agency. *Our Nation's Air: Trends Through 2024.* Annual report on national criteria-pollutant trends. U.S. Environmental Protection Agency, 2025.
https://gispub.epa.gov/air/trendsreport/.

Environmental Protection Agency. "Administrator Zeldin Takes Additional Measures to Address Diesel Exhaust Fluid (DEF) Issues for American Farmers and Truckers." Press release, U.S. Environmental Protection Agency, February 3, 2026. The follow-up to the August 2025 guidance reversing five-mile-per-hour DEF derates. Announces a Section 208(a) information-collection demand against the top fourteen on-road and nonroad diesel

manufacturers and previews rulemaking to permanently eliminate DEF derates from new and existing trucks. Source for the trucking chapter's account of the EPA-path actions through February 2026.
https://www.epa.gov/newsreleases/administrator-zeldin-takes-additional-measures-address-diesel-exhaust-fluid-def-issues.

Environmental Protection Agency. *The 2026 EPA Automotive Trends Report*. EPA-420-R-26-001. Washington, DC: U.S. Environmental Protection Agency, February 2026. Annual EPA report documenting trends in light-duty vehicle fuel economy, emissions, and technology adoption from model year 1975 forward. Source for the manual-transmission share trajectory cited in the surveillance chapter: 34.6 percent of new American light-duty vehicles in 1980; 22.2 percent in 1990; 9.7 percent in 2000; 3.8 percent in 2010; 1.1 percent in 2020; 0.9 percent in 2024.
https://www.epa.gov/system/files/documents/2026-02/420r26001.pdf.

Environmental Protection Agency. *Battery Collection Best Practices Report to Congress*. Washington, DC: U.S. Environmental Protection Agency, May 2026. The report Congress required under Section 70401(b) of the 2021 Infrastructure Investment and Jobs Act. It finds no widely accepted U.S. recycling rate for lithium-ion batteries, notes that the commonly cited five-percent figure is European and nearly a decade old, and states that used electric-vehicle batteries are only beginning to enter the waste stream with no collection system of their own yet in place — against a 99 percent industry-reported recycling rate for the lead-acid battery in a conventional vehicle. Cited in the environmental chapter for the point that the technology the regulatory wall favors carries lifecycle costs no regulation prices.
https://www.epa.gov/system/files/documents/2026-05/battery-collection-best-practices-report-to-congress_508.pdf.

Environmental Protection Agency. "Regulations for Emissions from Heavy Equipment with Compression-Ignition Engines." U.S. EPA portal for the nonroad compression-ignition (diesel) emissions program, codifying Tier 1 (1994), Tiers 2 and 3 (1998), and Tier 4 (2004) standards expressly covering farm tractors and other nonroad agricultural equipment.

Source for the second of the three regulatory layers discussed in the farm-equipment chapter.
https://www.epa.gov/regulations-emissions-vehicles-and-engines/regulations-emissions-heavy-equipment-compression.

Environmental Protection Agency. “Regulations for Emissions from Marine Compression-Ignition (Diesel) Engines.” Codified at 40 C.F.R. Part 1042. U.S. Environmental Protection Agency. The marine compression-ignition (diesel) emissions program governing engines installed in commercial vessels such as tugs, towboats, and barges; the program’s near-term Tier 3 and longer-term Tier 4 aftertreatment-based standards for engines under thirty liters per cylinder were finalized in the June 30, 2008 final rule (73 Fed. Reg. 37243). Cited in the plaintiff-categories chapter as the federal hook for the independent tug and barge operator, distinct from the Part 1039 nonroad rule that governs land-based heavy equipment.
https://www.ecfr.gov/current/title-40/chapter-I/subchapter-U/part-1042.

EPA Final Rule for Control of Emissions of Air Pollution From Highway Heavy-Duty Engines. 62 Fed. Reg. 54694 (October 21, 1997). U.S. Environmental Protection Agency. The 1997 final rule that established the 2.4 g/bhp-hr NMHC+NOx standard (or the alternative 2.5 g/bhp-hr with a 0.5 g NMHC cap) for heavy-duty diesel engines beginning with the 2004 model year — a fifty-percent reduction from the 1998 NOx standard of 4.0 g/bhp-hr. The October 1998 Consent Decrees pulled the 2004 standard forward by fifteen months to October 2002, the implementation point that pushed every signatory toward cooled exhaust gas recirculation. Source for the trucking chapter’s account of the first wave of post-1999 regulatory tightening that ended the mechanical-engine era of heavy-duty diesel and defines the 1999 model year as the dream-truck cutoff.
https://www.epa.gov/regulations-emissions-vehicles-and-engines/final-rule-control-emissions-air-pollution-highway-heavy.

EPA Heavy-Duty Highway Engine and Vehicle Standards and Highway Diesel Fuel Sulfur Control Requirements. Final Rule, 66 Fed. Reg. 5002 (January 18, 2001). U.S. Environmental Protection Agency. The 2001 final rule (commonly referenced as the 2007

rule for its principal compliance date) established the 0.01 grams per brake-horsepower-hour particulate-matter standard that made diesel particulate filters and the full aftertreatment stack effectively mandatory on every Class 8 truck sold after 2007. Source for the trucking chapter's account of the regulatory transition from engine-design cleanup to mandatory aftertreatment.
https://www.federalregister.gov/documents/2001/01/18/01-2/control-of-air-pollution-from-new-motor-vehicles-heavy-duty-engine-and-vehicle-standards-and-highway.

EPA Standards for Small Nonroad Spark-Ignition Engines and Equipment. Phase 1 final rule, 60 Fed. Reg. 34581 (July 3, 1995). Phase 3 final rule, 73 Fed. Reg. 59259 (October 8, 2008). Codified at 40 C.F.R. Part 1054. U.S. Environmental Protection Agency. Standards apply to engines used in lawn and garden equipment, snow blowers, and other small nonroad applications.
https://www.federalregister.gov/documents/2008/10/08/E8-21093/control-of-emissions-from-nonroad-spark-ignition-engines-and-equipment.

Epstein, Richard A. *Takings: Private Property and the Power of Eminent Domain*. Cambridge, MA: Harvard University Press, 1985. The leading libertarian-property-rights statement of regulatory-takings doctrine; counter-authority to Joseph Sax's narrower police-power analysis. Cited in the chapter on objections considered as part of the doctrinal landscape against which the Takings Clause prong of the book's argument is built.
https://www.hup.harvard.edu/books/9780674867307.

Equal Access to Justice Act, 28 U.S.C. § 2412; 5 U.S.C. § 504. The statute waiving federal sovereign immunity for attorney-fee awards to parties who prevail in non-tort civil litigation against the United States. Recoverable fees are capped at $125 per hour, subject to increase for cost of living or a special factor; recovery is limited to individuals with a net worth of $2 million or less and to businesses worth $7 million or less with no more than 500 employees, while organizations exempt under Internal Revenue Code § 501(c)(3) are not subject to those caps. No award is available if the government shows its litigating position was substantially justified, and fee shifting runs one way only. Cited in *Objections Considered* for the proposition that the fee structure, rather than the merits, explains why no

firm has brought the Patent Clause case since the doctrinal obstacle to it was removed in 2024.
https://www.congress.gov/crs-product/IF11246.

Erlich, Mark. *With Our Hands: The Story of Carpenters in Massachusetts*. Philadelphia: Temple University Press, 1986. Oral history of the Massachusetts building trades, built from interviews with carpenters whose working lives ran from the 1910s through the 1980s. Chapter 14, “New Tools, New Materials, New Methods,” is the source for the preface’s account of how the trade received the portable circular saw: that hand labor dominated on-site construction until after the Second World War; that the early models carried only marginal safety features and often proved dangerous in the hands of a first-time user; that one carpenter carried his Skilsaw to the roof of a building and threw it off, saying the saw was too fast and he would cut by hand; and that as early as 1932 the Springfield District Council proposed fining union members twenty-five dollars for furnishing electric hand-driven machinery to a contractor. Read free in the open-access edition published by Temple University Press and North Broad Press.
https://temple.manifoldapp.org/read/with-our-hands-the-story-of-carpenters-in-massachusetts/section/5a3bd679-a298-470e-8e0e-702802bd49e9. See Endnote Commentary.

Erman, Sam. *Almost Citizens: Puerto Rico, the U.S. Constitution, and Empire*. Cambridge University Press, 2018; Studies in Legal History; LCCN 2018035531; ISBN 9781108415491. The paperback of 2019 is widely cited as the first edition and is not. The nearest work in the literature to the argument that individuals without formal legal training make a difference at law, advanced there about the constitutional absorption of empire.

European Parliament and Council. Regulation (EU) 2018/858 of 30 May 2018 on the approval and market surveillance of motor vehicles and their trailers, and of systems, components and separate technical units intended for such vehicles. Official Journal of the European Union L 151, 14 June 2018, pp. 1–218. The Whole Vehicle Type Approval

framework that governs motor vehicle market access across the twenty-seven EU member states.
https://eur-lex.europa.eu/eli/reg/2018/858/oj/eng.

Evans, Oliver. U.S. Patent for the high-pressure steam engine, granted 1804 (term expired 1818 under the fourteen-year rule of the Patent Act of 1790). The Evans engine made high-pressure steam economical for the first time and is the steam-engine entry in the chapter's eight-product table. Background and patent history at the Linda Hall Library's history-of-science collection.
https://www.lindahall.org/about/news/scientist-of-the-day/oliver-evans.

Event Data Recorder Rule. 49 C.F.R. § 563 (Event Data Recorders). National Highway Traffic Safety Administration. Original final rule promulgated 2006; multiple amendments since. Required for vehicles voluntarily equipped with event data recorders, with prescribed data fields and survivability standards.
https://www.ecfr.gov/current/title-49/subtitle-B/chapter-V/part-563.

Executive Office of the President. "Memorandum on Delivering Emergency Price Relief for American Families and Defeating the Cost-of-Living Crisis." Presidential Memorandum of January 20, 2025. 90 Fed. Reg. 8245 (January 28, 2025) (Federal Register Doc. 2025-01904). Directs the heads of all executive departments and agencies to pursue actions reducing housing costs, eliminating administrative expenses and rent-seeking practices that raise health-care costs, and removing "counterproductive requirements that raise the costs of home appliances." This document is a Presidential Memorandum, not Executive Order 14148; the chapter prose mis-cites it as an Executive Order and should be corrected at edit time.
https://www.federalregister.gov/documents/2025/01/28/2025-01904/delivering-emergency-price-relief-for-american-families-and-defeating-the-cost-of-living-crisis.

Executive Order 14154 of January 20, 2025. "Unleashing American Energy." 90 Fed. Reg. 8353 (January 29, 2025) (Federal Register Doc. 2025-01956). Directs federal agencies to review and, where appropriate, withdraw, rescind, or postpone regulations and orders that

impede the development of domestic energy resources, including consumer-appliance and equipment energy-efficiency mandates. The Trump administration's February 14, 2025 Department of Energy announcement postponing seven Biden-administration appliance-efficiency mandates was issued under the authority of this Executive Order and the related Presidential Memorandum on Delivering Emergency Price Relief. https://www.federalregister.gov/documents/2025/01/29/2025-01956/unleashing-american-energy.

Farm Action. Organizational homepage. Farmer-led watchdog organization focused on corporate consolidation and antitrust enforcement in agriculture; policy work has been developed into twenty pieces of congressional legislation. Identified in the farm-equipment chapter as one of five candidate plaintiff organizations under associational standing established in Hunt v. Washington State Apple Advertising Commission (1977). https://farmaction.us/.

Farm and Ranch Freedom Alliance (FARFA). Organizational homepage. Coalition-building organization founded in 2006 representing independent small-scale farmers, ranchers, and value-added producers; works with fifty or more farm organizations on regulatory and antitrust questions. Identified in the farm-equipment chapter as one of five candidate plaintiff organizations under associational standing. https://farmandranchfreedom.org/.

Farrand, Max, ed. *The Records of the Federal Convention of 1787*. Vol. 2. New Haven: Yale University Press, 1911. Entries for August 18, 1787, recording Madison's proposal to secure to literary authors their copyrights "for a limited time" and the separate proposal offered the same day to grant patents for useful inventions, and for September 5, 1787, recording the Committee of Eleven's report of the merged clause securing "for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries." Cited in the chapter on objections considered for the drafting history of the plural "Times." https://memory.loc.gov/ammem/amlaw/lwfr.html.

Fawzi, Alhussein, et al. “Discovering Faster Matrix Multiplication Algorithms with Reinforcement Learning.” *Nature* 610 (2022): 47-53. The forty-seven multiplication result for four-by-four matrices holds in arithmetic modulo two and not over the real numbers, where two-level Strassen’s forty-nine still stands.
See Endnote Commentary.

Federal Aviation Administration, Aircraft Type Certification. 49 U.S.C. § 44704; 14 C.F.R. Parts 21, 23, 25, 27, and 29. Statutory and regulatory framework requiring type certificates for new aircraft designs, production certificates for manufacturing facilities, and airworthiness certificates for individual aircraft.
https://www.ecfr.gov/current/title-14/chapter-I/subchapter-C/part-21.

Federal Aviation Administration. Type Certificate Data Sheet 3A12 (Cessna 172 series), issued November 4, 1955, on an application of July 11, 1955, and amended since; and Advisory Circular 21.101-1B, Establishing the Certification Basis of Changed Aeronautical Products. The data sheet shows that the 172R (amendment of June 21, 1996) and 172S (amendment of May 1, 1998) were added to the 1955 certificate rather than certified afresh, and that the certificate passed to Textron Aviation on July 29, 2015. The advisory circular, applying 14 C.F.R. § 21.101, states that a change to a type-certificated product is evaluated against its existing certification basis and that earlier regulations are acceptable where the latest amendments are not appropriate. Cited in The Patent Clause Prong as the clearest instance of a federal power securing a patent-expired original design to one private party without an end date; and in How a Remake Is Built and Certified for the point that a design’s own certification history can still govern (newer Part 23 amendments apply to the changed and affected areas of a design change, while earlier requirements are retained where the latest amendments are not appropriate) and as the working precedent for platform-year certification, together with the General Aviation Revitalization Act of 1994, Public Law 103-298, which restored Cessna’s production through liability reform and waived no airworthiness standard. The continuity runs with the certificate, which is transferable property; a third party gets it only by acquiring the certificate or licensing it

from the holder under 14 C.F.R. § 21.55, and the 172S carries a model-specific basis at several Part 23 amendment levels rather than the 1955 rules alone. https://drs.faa.gov/browse/excelExternalWindow/DRSDOCID1538016001202506271647 31.0001; https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_21.101-1B.pdf; https://www.law.cornell.edu/cfr/text/14/part-21/subpart-B

Federal Aviation Administration. Supplemental Type Certificate framework, 14 C.F.R. Part 21 Subpart E. The FAA's standing regulatory architecture under which modifications to a previously certified aircraft type design are approved through an independent engineering certification process distinct from the original type certificate held by the airframe manufacturer. Cited in the trucking chapter as the regulatory model for independent engineering certification of authentic remanufactured patent-expired heavy-duty truck platforms. https://www.ecfr.gov/current/title-14/chapter-I/subchapter-C/part-21/subpart-E.

Federal Motor Carrier Safety Administration. “Electronic Logging Devices and Hours of Service Supporting Documents.” Final Rule, 80 Fed. Reg. 78292 (December 16, 2015). Codified at 49 C.F.R. Parts 385, 386, 390, and 395. Requires most commercial motor vehicles engaged in interstate commerce to use a federally certified electronic logging device to record drivers’ hours of service, location, engine status, and motion. Compliance was required by December 18, 2017 for carriers previously using paper logs. Cited in the trucking-section discussion of federally mandated continuous driver surveillance in the surveillance chapter. This footnote is also for the Foreword. https://www.federalregister.gov/documents/2015/12/16/2015-31336/electronic-logging-devices-and-hours-of-service-supporting-documents.

Federal Motor Vehicle Safety Standards: date-of-manufacture applicability. NHTSA, “Importation and Certification FAQs.” Each vehicle’s certification label states that it conforms to all applicable FMVSS in effect on its date of manufacture, under 49 C.F.R. Part 567; the standards bind by manufacture date, and a vehicle lawfully built to the

standards then in force is not required to retrofit to later standards for continued operation or resale.
https://www.nhtsa.gov/importing-vehicle/importation-and-certification-faqs-8.

Federal Register. “Lighting and Marking on Agricultural Equipment.” 81 Fed. Reg. 40540 (June 22, 2016). National Highway Traffic Safety Administration final rulemaking implementing the lighting and marking provisions Congress mandated through the Moving Ahead for Progress in the 21st Century Act (MAP-21). The preamble contains NHTSA’s acknowledgment, quoted in the farm-equipment chapter, that the agency had not historically regulated most agricultural equipment because such equipment was not designed primarily for public road use.
https://www.federalregister.gov/documents/2016/06/22/2016-14571/lighting-and-marking-on-agricultural-equipment.

Federal Reserve Economic Data (FRED). Federal Reserve Bank of St. Louis. Database hosting more than 800,000 economic time series, including BEA Industry Economic Accounts series used in the cumulative wealth-transfer calculation discussed in Chapter 8.
https://fred.stlouisfed.org/.

Federal Rules of Civil Procedure (especially Rule 10(a), caption and names of parties); *Federal Rules of Appellate Procedure* (especially Rule 3, notice of appeal, and Rule 12, docketing the appeal); *Rules of the Supreme Court of the United States* (especially Rule 12, review on writ of certiorari). The three federal rule sets that establish the party-label conventions used at each level of the federal courts — plaintiff and defendant at the trial court, appellant and appellee at the United States Court of Appeals, petitioner and respondent at the Supreme Court. The Federal Rules of Civil Procedure and the Federal Rules of Appellate Procedure are promulgated by the Supreme Court under the Rules Enabling Act, 28 U.S.C. § 2072; the Rules of the Supreme Court are promulgated by the Court itself. Cited in the Endnote Commentary as the authority for the procedural-

terminology shifts a reader encounters when following the same litigant from the district court to the Supreme Court.
https://www.law.cornell.edu/rules/frcp; https://www.law.cornell.edu/rules/frap; https://www.supremecourt.gov/filingandrules/2023RulesOfTheCourt.pdf. See Endnote Commentary.

Federal Trade Commission. “FTC Takes Action Against General Motors for Sharing Drivers’ Precise Location and Driving Behavior Data Without Consent.” Press release. January 16, 2025. Proposed order banning General Motors and OnStar from disclosing consumers’ precise geolocation and driver behavior data to consumer reporting agencies for a period of five years; order finalized January 2026. Source for the FTC enforcement action documented in the surveillance chapter.
https://www.ftc.gov/news-events/news/press-releases/2025/01/ftc-takes-action-against-general-motors-sharing-drivers-precise-location-driving-behavior-data.

First English Evangelical Lutheran Church of Glendale v. County of Los Angeles, 482 U.S. 304 (1987).
https://supreme.justia.com/cases/federal/us/482/304/.

Fisher, Robert S., Jay N. Acharya, Fiona M. Baumer, et al. “Visually sensitive seizures: An updated review by the Epilepsy Foundation.” *Epilepsia* 63, no. 4 (2022): 739–768. The current peer-reviewed Epilepsy Foundation review of visually triggered seizures. Documents that approximately three percent of people with epilepsy experience photoparoxysmal responses to flashing lights or patterns; that the most provocative flicker frequency range is approximately three to sixty hertz (peak around fifteen to twenty hertz) at intensities above approximately twenty candelas per square meter, occupying ten to twenty-five percent of the visual field; and that modern LED light sources with elevated luminance can trigger photosensitive seizures in susceptible individuals. Source for the healthy-lighting chapter’s clinical anchor on LED flicker and photosensitive epilepsy.
https://doi.org/10.1111/epi.17175.

Florida Prepaid Postsecondary Education Expense Board v. College Savings Bank, 527 U.S. 627 (1999).
https://supreme.justia.com/cases/federal/us/527/627/.

Ford Motor Company. “Advertisement for the 1925 Ford Model T, ‘Within the Means of Millions.’” The Henry Ford Museum, Artifact #355535. Created by Ford Motor Company in 1925. The advertisement, held in The Henry Ford Museum’s digital collection, depicts factory workers leaving a plant in their personal Ford Model T cars; the five-word tagline “Within the / means of / millions” is inscribed in the lower-left corner, and a secondary safety slogan “MAKE SAFETY YOUR RESPONSIBILITY” appears at the bottom. The cars-and-trucks chapter quotes the tagline once as a one-sentence framing of the affordability the 1925 patent-expired commons in personal transportation actually delivered.
https://www.thehenryford.org/collections-and-research/digital-collections/artifact/355535/.

Ford Motor Company. Model T Runabout production and pricing records, 1925, as compiled in Bruce W. McCalley, *Model T Ford: The Car That Changed the World* (Iola, WI: Krause Publications, 1994), the standard scholarly reference work on Model T production and pricing, drawing on Ford Archives data. Source for the 1925 Model T Runabout MSRP of $260 nominal (the full 1925 Model T lineup, as documented in March 1925 archival records: Touring $290, Runabout $260, Tudor $580, Fordor $660, Coupe $520, Chassis $225), and for the 1925 Runabout production figure of 298,642 units (264,436 with electric starter and demountable rims, 34,206 base models, plus 33,795 separately accounted pickup-body Runabouts). The Model T Ford Club of America’s 1925 encyclopedia entry reproduces the McCalley pricing and production data as a freely accessible secondary anchor.
https://www.mtfca.com/model_t_encyclopedia/doc25/.

Ford Motor Company. “1969 Ford Falcon Specifications and Configuration.” Ford Heritage Vault and contemporaneous dealer literature. Source for the 1969 Ford Falcon as the

chapter’s intermediate datapoint between the 1925 Model T Runabout and the 2025 Ford Escape.
https://apps.abacus.ai/chatllm/share/conversation/15e770a9d4:13f78811d8 ;
https://falconclub.com/falcons-pages/the-ford-falcon/1969-ford-falcon/.

Ford Motor Company. “2025 Ford Escape Active.” Ford Motor Company configurator and model-year pricing. Source for the 2025 Ford Escape base MSRP of $28,150 cited in the cars-and-trucks chapter.
https://www.ford.com/suvs/escape/.

Ford Motor Company. “2025 Ford F-150 Lariat 4×4 SuperCrew 5.5′ Box.” Ford Motor Company configurator and 2025 model-year pricing schedule. Primary source for the chapter title figure of $72,495 representing the out-the-door total — base MSRP plus destination charge plus state and local sales tax plus title and license fees plus dealer documentation fee — for the Lariat 4×4 SuperCrew 5.5′ Box trim with the standard 3.5-liter V6 EcoBoost engine. Reconstruction of the line-item components, including the trim spread across the 2025 F-150 lineup, appears in the corresponding Endnote Commentary entry.
https://www.ford.com/trucks/f150/2025/. See Endnote Commentary.

Foster, Derek (Derek Redmond), ed. “Jeeps in India: Mahindra & Mahindra.” *CJ3B.info* (Willys CJ3B historical archive). Accessed May 9, 2026.
https://cj3b.info/World/India2.html.

Freedberg, A. Stone, and Louis E. Baron. Report of spiral organisms in the human stomach, Beth Israel Hospital, 1940. The finding was not reproduced and no negative result was published. Known here through later reviews; the full citation has not been confirmed against the original.

Friedman, Milton, and Rose D. Friedman. *Free to Choose: A Personal Statement.* New York: Harcourt Brace Jovanovich, 1980.
https://www.amazon.com/Free-Choose-Personal-Statement-Friedman/dp/0156334607.

Garner-Bunch, Linda. “EPA Will ‘Reconsider’ Emissions Standards for Heavy-Duty Trucks, Says Zeldin.” *The Trucker*, March 20, 2025. Trade-press account of EPA Administrator Lee Zeldin’s March 12, 2025 announcement reopening the Biden-administration greenhouse gas standards for model year 2027 and later heavy-duty trucks, alongside parallel reconsiderations of the light- and medium-duty and NOx rules. Source for the trucking chapter’s account of the EPA-path activity beginning in early 2025. https://www.thetrucker.com/trucking-news/equipment-tech/epa-will-reconsider-emissions-standards-for-heavy-duty-trucks-says-zeldin.

Gautret, Philippe, Jean-Christophe Lagier, Philippe Parola, et al. “Hydroxychloroquine and Azithromycin as a Treatment of COVID-19: Results of an Open-Label Non-Randomized Clinical Trial.” International Journal of Antimicrobial Agents 56, no. 1 (July 2020): 105949. The IHU Méditerranée-Infection study from Marseille. Retracted by the journal in January 2025, four and a half years after publication; the retraction notice appears at International Journal of Antimicrobial Agents 65, no. 1 (January 2025): 107416. The paper remains available in full from the authors’ own institution and is archived: https://www.mediterranee-infection.com/wp-content/uploads/2020/03/Hydroxychloroquine_final_DOI_IJAA.pdf (archived at https://web.archive.org/web/20260308204950/https://www.mediterranee-infection.com/wp-content/uploads/2020/03/Hydroxychloroquine_final_DOI_IJAA.pdf).
See Endnote Commentary.

Geier v. American Honda Motor Co., 529 U.S. 861 (2000). https://supreme.justia.com/cases/federal/us/529/861/.

General Aviation Revitalization Act of 1994, Pub. L. No. 103-298, 108 Stat. 1552 (Aug. 17, 1994). Created an eighteen-year statute of repose for product-liability suits against manufacturers of general-aviation aircraft and their parts. Cited in How a Remake Is Built and Certified for the point that the Act was liability reform and waived no airworthiness standard: Cessna resumed single-engine production in 1996 under its original type

certificate, and the certificate's existing certification basis survived on its own through the FAA's changed-product rule, before and after the Act.
https://www.govinfo.gov/content/pkg/STATUTE-108/pdf/STATUTE-108-Pg1552.pdf

General Motors. "GM Enhances Privacy Controls and Discontinues OnStar Smart Driver." News release. April 24, 2024. Formal discontinuation of the OnStar Smart Driver program and unenrollment of drivers previously enrolled; follow-on action to the March 22, 2024 announcement that GM had stopped sharing OnStar Smart Driver customer data with LexisNexis Risk Solutions and Verisk. Source for the surveillance chapter's reconstruction of the GM data-sharing timeline.
https://news.gm.com/home.detail.html/Pages/news/us/en/2024/apr/0424-driver.html.

Gilbert, Ruth, Georgia Salanti, et al. "Infant Sleeping Position and the Sudden Infant Death Syndrome: Systematic Review of Observational Studies and Historical Review of Recommendations from 1940 to 2002." *International Journal of Epidemiology* 34 (2005). Pooled odds ratio for front against back sleeping reached 2.93 (95 percent confidence interval 1.15 to 7.47) by 1970, and 3.00 (1.69 to 5.31) across five studies by 1986. The authors estimate that a systematic review conducted in 1970 might have prevented more than ten thousand infant deaths in the United Kingdom. See also their reply to Guntheroth and Spiers, same volume, page 1166, on the greater cost of introducing the advice as against the delay in withdrawing it.

Gilder Lehrman Institute of American History. "Statistics of the American Economy, 1920s." Historical statistics reference materials. Source for the 1925 American working man's median income figure of $1,200 nominal ($22,043 in 2025 dollars after CPI conversion) cited in the cars-and-trucks chapter.
https://www.gilderlehrman.org/history-resources/teacher-resources/historical-statistics-united-states.

Global NCAP. "Mahindra Thar (2020)." Crash test report, Global New Car Assessment Programme, November 2020. Documents the four-star ratings for adult and child occupant

protection awarded to the second-generation Mahindra Thar under the GNCAP testing protocol.
https://www.globalncap.org/indiaresults.

Gideon v. Wainwright, 372 U.S. 335 (1963). The Supreme Court's unanimous holding, written by Justice Black, that the Sixth Amendment right to counsel binds the States through the Fourteenth Amendment. Clarence Earl Gideon wrote his own petition in pencil; the Court appointed Abe Fortas to argue the case, and the winning argument was his.
https://supreme.justia.com/cases/federal/us/372/335/.

Goldberger, Joseph. "The Etiology of Pellagra: The Significance of Certain Epidemiological Observations with Respect Thereto." *Public Health Reports* 29, no. 26 (June 26, 1914): 1683-1686.

Golan v. Holder, 565 U.S. 302 (2012). Six-to-two decision, Justice Kagan taking no part, upholding Section 514 of the Uruguay Round Agreements Act, which restored United States copyright to foreign works that had already passed into the American public domain. Justice Ginsburg's majority opinion rejected the premise that the Constitution renders the public domain largely untouchable by Congress and read the "limited Times" language as no bar to protecting works already in it; the opinion refers to the provision as the Copyright and Patent Clause. Justice Breyer, joined by Justice Alito, dissented. Cited in the chapter on objections considered, alongside Eldred v. Ashcroft, as adverse authority on the reach of the public domain, and distinguished there as a copyright decision reviewing an act of Congress rather than a Patent Clause challenge to an agency-built certification barrier.
https://supreme.justia.com/cases/federal/us/565/302/.

Golden v. United States, 955 F.3d 981 (Fed. Cir. 2020). The Federal Circuit's reaffirmation of *Celgene Corp. v. Peter*, rejecting a takings claim arising from inter partes review; the opinion repeats that *Oil States Energy Services v. Greene's Energy Group* did not suggest that patents are not property for purposes of the Due Process or Takings Clauses, and declines to decide the broader status of patents under the Takings Clause because Celgene controlled the result. Cited in the chapter on objections considered, with Celgene, as the

Federal Circuit's application of the question Oil States reserved; neither decision reaches the public's right to build a patent-expired design. https://law.justia.com/cases/federal/appellate-courts/cafc/19-2134/19-2134-2020-04-10.html.

Goldman Sachs. "Robotaxis Are Forecast to Become a $400 Billion Market in 2035." Goldman Sachs Research, 2026. Reports the research team's projection that the all-in operating cost of an autonomous Class 8 truck, before corporate overhead, was approximately $8.56 per mile in 2025, falling toward $2.03 per mile by 2035; that autonomous and human-driven trucking costs are projected to cross around 2028; that the autonomous sensor and computing stack adds $125,000 to $150,000 per truck; that the comparable per-mile cost of a human-driven truck is projected to rise from $2.55 to $2.84 over the same period on expected driver-wage increases; that the autonomous stack is projected to fall to $35,000 to $40,000 per truck by 2035 as hardware improves and production scales; that vehicle depreciation falls from roughly $0.35 per mile in 2025 to about $0.14 in 2035 as the ratio of vehicles per remote supervisor improves from 6-to-1 today to 26-to-1 in 2035; that autonomous trucks are not subject to the 11-hour daily driving limit and mandatory breaks that bind human drivers, and so can cover more miles; and that the per-mile cost of owning a personal vehicle ($0.77 to $1.00) remains below the more-than-$2.00 cost of the average ride-hail trip. Cited throughout the autonomous-vehicle chapter. https://www.goldmansachs.com/insights/articles/robotaxis-to-become-a-400-billion-dollar-market-in-2035.

Google. Big Sleep discovery of CVE-2025-6965 in SQLite, July 15, 2025, as reported in "Google AI 'Big Sleep' Stops Exploitation of Critical SQLite Vulnerability Before Hackers Act," *The Hacker News*, July 16, 2025. Big Sleep, an AI agent built by Google DeepMind and Project Zero, identified a memory-corruption flaw in the SQLite database engine after Google's threat-intelligence group saw indications that an attacker was staging an exploit but could not isolate the vulnerability; Kent Walker of Google called it the first time an AI agent had been used to directly foil an exploit in the wild. Cited in the preface as one of

three cases in which a machine found, by reading a system's rules, a flaw its expert defenders had not.
https://thehackernews.com/2025/07/google-ai-big-sleep-stops-exploitation.html.

Google LLC v. Oracle America, Inc., 593 U.S. 1 (2021). Six-to-two decision, Justice Barrett taking no part, holding that Google's reimplementation of the declaring code of the Java SE application programming interface, to let programmers use familiar calls in a new platform, was fair use. The Court assumed copyrightability without deciding it. Cited for the room the fair-use doctrine leaves a rewriter who needs an interface to work with existing code.
https://supreme.justia.com/cases/federal/us/593/18-956/.

Gosker, Robert, and J. F. Degenhart. "Pack-Saddle." U.S. Patent No. 39,285. Issued July 21, 1863. Pack-saddle improvement designed for long-distance, mixed-cargo overland travel in the western territories.
https://patents.google.com/patent/US39285A.

Graham v. John Deere Co., 383 U.S. 1 (1966).
https://supreme.justia.com/cases/federal/us/383/1/.

Griffin, G. Edward. *World Without Cancer: The Story of Vitamin B17*. Westlake Village, CA: American Media, 1974. Multiple revised editions through 2019.
https://www.amazon.com/World-Without-Cancer-Story-Vitamin/dp/0912986190.

Haartsen, Jacobus Cornelis. "Contemporaneous Connectivity to Multiple Piconets." U.S. Patent 6,026,297, filed September 17, 1997, issued February 15, 2000. Assigned to Telefonaktiebolaget LM Ericsson AB. Foundational Bluetooth patent covering simultaneous connectivity to multiple piconets; expired 2017–2018. Cited in the surveillance chapter as one of the foundational expired patents from which a remanufactured vehicle's Bluetooth radio can be built.
https://patents.google.com/patent/US6026297A/en.

Haartsen, Jacobus Cornelis. "Frequency Hopping Piconets in an Uncoordinated Wireless Multi-User System." U.S. Patent 6,590,928, filed September 17, 1997, issued July 8, 2003.

Assigned to Telefonaktiebolaget LM Ericsson AB. Foundational Bluetooth patent on the frequency-hopping piconet architecture; expired August 2018. https://patents.google.com/patent/US6590928B1/en.

Haley, Daniel. *Politics in Healing: The Suppression and Manipulation of American Medicine*. Washington, DC: Potomac Valley Press, 2000. https://www.amazon.com/Politics-Healing-Suppression-Manipulation-American/dp/0972751416.

Hamblin, Michael R. “Mechanisms and applications of the anti-inflammatory effects of photobiomodulation.” *AIMS Biophysics* 4, no. 3 (2017): 337–361. Peer-reviewed review of the photobiomodulation literature. Documents that red and near-infrared light — wavelengths abundantly emitted by incandescent sources and substantially reduced or absent from blue-shifted LED output — are absorbed by cytochrome c oxidase in mitochondria, dissociating inhibitory nitric oxide, increasing ATP production, reducing oxidative stress, and producing measurable anti-inflammatory effects at the cellular level. Situates the work of Glen Jeffery and others in a substantial peer-reviewed body of research on the systemic and ocular consequences of long-wavelength light exposure. https://pmc.ncbi.nlm.nih.gov/articles/PMC5523874/.

Hamil, Robert C. “Improvement in Saddles.” U.S. Patent No. 139,668. Issued June 10, 1873. Riding-saddle improvement specifying a design with two girths passing round the body, addressing the slip problem of the Stickney single-girth design on long ascents. https://patents.google.com/patent/US139668A.

Handmer, Casey. “Reading Ancient Scrolls.” Personal blog, 5 August 2023. https://caseyhandmer.wordpress.com/2023/08/05/reading-ancient-scrolls/. The first substantial evidence of ink inside an unopened Herculaneum scroll, found by examining publicly released X-ray scan segments and identifying a cracked-mud surface texture in the shape of Greek letters. The Vesuvius Challenge awarded the First Ink Prize of ten thousand dollars on 12 October 2023.

Hardin, Garrett. “The Tragedy of the Commons.” *Science* 162, no. 3859 (December 13, 1968): 1243–1248. https://www.science.org/doi/10.1126/science.162.3859.1243.

Heckler v. Chaney, 470 U.S. 821 (1985). An agency’s decision not to undertake an enforcement action is presumptively committed to the agency’s discretion and unreviewable under the Administrative Procedure Act. Cited in the anti-backsliding chapter to distinguish enforcement discretion, the power not to prosecute a violation, from the power to repeal a standard, which the executive does not possess. https://supreme.justia.com/cases/federal/us/470/821/.

Hill, Kashmir. “Automakers Are Sharing Consumers’ Driving Behavior with Insurance Companies.” *New York Times*, March 11, 2024. First article in Hill’s 2024 investigation documenting that General Motors, Honda, Hyundai, Kia, Subaru, and Mitsubishi were transmitting driving-behavior data from their connected vehicles to LexisNexis Risk Solutions and Verisk, which compiled the data into individual driver-risk reports and sold them to insurance companies. https://www.nytimes.com/2024/03/11/technology/carmakers-driver-tracking-insurance.html.

Hill, Kashmir. “Florida Man Sues G.M. and LexisNexis Over Sale of His Cadillac Data.” *New York Times*, March 14, 2024. Reports the first private civil action filed against General Motors and LexisNexis arising from the data-sharing arrangement Hill’s reporting documented. https://www.nytimes.com/2024/03/14/technology/gm-lexis-nexis-driving-data.html.

Hill, Kashmir. “General Motors Quits Sharing Driving Behavior with Data Brokers.” *New York Times*, March 22, 2024. Reports GM’s announcement that it would discontinue the data-sharing arrangements with LexisNexis Risk Solutions and Verisk; precursor to the April 24, 2024 formal discontinuation of the OnStar Smart Driver program. https://www.nytimes.com/2024/03/22/technology/gm-onstar-driver-data.html.

Hill, Kashmir. “How G.M. Tricked Millions of Drivers into Being Spied On (Including Me).” *New York Times*, April 23, 2024. First-person investigative account of how the OnStar Smart Driver enrollment consent flow was structured to capture consent at vehicle delivery; documents the granular data fields transmitted, including hard-braking events, rapid accelerations, and speeding events.
https://www.nytimes.com/2024/04/23/technology/general-motors-spying-driver-data-consent.html.

Hill, Kashmir. “Automakers Sold Driver Data for Pennies, Senators Say.” *New York Times*, July 26, 2024. Reports on the Wyden–Markey letter to the Federal Trade Commission and the underlying oversight responses obtained directly from the automakers, including per-vehicle broker payments on the order of twenty-six cents from Honda and sixty-one cents from Hyundai.
https://www.nytimes.com/2024/07/26/technology/driver-data-sold-for-pennies.html.

HM Revenue and Customs. “CIRD123000 — R&D Tax Reliefs: Reformed Reliefs: ERIS: R&D Intensity Condition.” Corporate Intangibles Research and Development Manual. HMRC guidance establishing enhanced R&D-intensive support: an additional deduction of 86 percent of qualifying expenditure, giving a total deduction of 186 percent, together with a payable tax credit at 14.5 percent, for accounting periods beginning on or after April 1, 2024.
https://www.gov.uk/hmrc-internal-manuals/corporate-intangibles-research-and-development-manual/cird123000.

Honda Motor Co., Ltd. “2025 Civic Sedan: Trims and Pricing.” Official Honda Civic vehicle pages, accessed May 2026. Base LX trim MSRP $24,250; wireless Apple CarPlay and Android Auto included as standard equipment on all trims. Cited in the surveillance chapter as one of the three 2025-MY consumer-sedan reference points showing that the licensed-software stack has been absorbed into baseline pricing rather than remaining a discrete trim premium.
https://automobiles.honda.com/civic.

Hope, Justus R. *Ivermectin for the World*. Hope Pressworks International, 2021. https://www.amazon.com/Ivermectin-World-Justus-R-Hope/dp/B09JJG4HQ7.

Hopkins, Samuel. U.S. Patent for an "Improvement in the Making of Pot ash and Pearl ash by a New Apparatus and Process," signed by President Washington on July 31, 1790. The first patent issued under the Patent Act of 1790, with a fourteen-year term that expired in 1804. The potash entry in the chapter's eight-product table. https://www.uspto.gov/learning-and-resources/journeys-innovation/historical-stories/first-us-patent-issued-today-1790.

Horne v. Department of Agriculture, 576 U.S. 351 (2015). https://supreme.justia.com/cases/federal/us/576/350/.

H.R. 10 (1st Cong., 1789). "A bill to promote the progress of science and useful arts, by securing to authors and inventors the exclusive right to their respective writings and discoveries." 3 Documentary History of the First Federal Congress 94 (DePauw, Bickford, and Hauptman eds. 1977), as quoted in Eldred v. Ashcroft, 537 U.S. 186 (2003) (Stevens, J., dissenting). The First Congress's first implementing bill; debated but not enacted, a similar bill becoming the Patent Act of 1790. https://en.wikisource.org/wiki/Eldred_v._Ashcroft/Dissent_Stevens

H.R. 3341 (119th Cong., 2025) — Liberating Incandescent Technology Act of 2025 (House Companion). Introduced May 13, 2025 by Representative Craig A. Goldman (R-TX-12), with cosponsors Representatives Michael A. Rulli (R-OH-6) and Robert F. Onder (R-MO-3). The identical House companion to S. 1568 (LIT Act of 2025) introduced May 1, 2025 by Senator Mike Lee (R-UT). Referred to the House Committee on Energy and Commerce on the date of introduction. No hearings, no markup, and no floor action through the period covered by the healthy-lighting chapter. No Democratic representative has cosponsored, supported, or opposed the bill in the Congressional Record. https://www.congress.gov/bill/119th-congress/house-bill/3341.

Huberman, Andrew. "Red Light to Improve Metabolism and Harmful Effects of LED Light Exposure | Dr. Glen Jeffery." *Huberman Lab* podcast, Stanford School of Medicine and

Scicomm Media. Podcast episode discussing the peer-reviewed research of Dr. Glen Jeffery at University College London on the mitochondrial and retinal benefits of long-wavelength incandescent and natural-sunlight exposure, and the corresponding harms from blue-shifted LED exposure. Cited in the healthy-lighting chapter as one of the credentialed scientific voices the plaintiff coalition's outreach naturally reaches. https://www.hubermanlab.com/episode/red-light-to-improve-metabolism-and-harmful-effects-of-led-glen-jeffery.

Hunt v. Washington State Apple Advertising Commission, 432 U.S. 333 (1977). https://supreme.justia.com/cases/federal/us/432/333/. The associational-standing decision establishing the three-prong test under which a trade or membership association may sue on behalf of its members: (1) at least one member must have standing to sue individually; (2) the interests the lawsuit protects must be germane to the association's purpose; and (3) neither the claim nor the relief may require individual member participation. Cited in the farm-equipment chapter as the doctrinal basis for adding farm associations as a fifth category of plaintiff with standing to challenge the regulatory ratchet.

Hurst, G. Samuel. "Electrographic Sensor for Determining Planar Coordinates." U.S. Patent 3,798,370, filed April 17, 1972, issued March 19, 1974. Assigned to Elographics, Inc. Foundational electrographic-touchscreen patent in the corpus underlying modern capacitive touchscreens; expired in the early 1990s. Cited in the surveillance chapter as one of the foundational expired patents underlying a remanufactured vehicle's in-dash display. https://patents.google.com/patent/US3798370.

Hyundai Motor America. "2025 Sonata: Pricing." Hyundai USA model pages, accessed May 2026. Base SE trim MSRP approximately $26,900; 12.3-inch touchscreen with wireless Apple CarPlay and Android Auto included as standard equipment on all trims. Cited in the surveillance chapter as one of the three 2025-MY consumer-sedan reference points. https://www.hyundaiusa.com/us/en/vehicles/2025-sonata.

IAMERS (International Association of Medical Equipment Remarketers and Servicers). Industry association representing independent medical-equipment service organizations. Position papers on third-party servicing access.
https://www.iamers.org/.

IBISWorld. “Number of Businesses statistics: Waste Collection Services; Ready-Mix Concrete Manufacturing; and Septic, Drain & Sewer Cleaning Services in the US.” IBISWorld Industry Reports. Source for the firm counts cited in the trucker chapter’s specialized-trades subsections: approximately twenty-one thousand waste-collection businesses (2024–2025), approximately three thousand seven hundred ready-mix concrete manufacturers (2024–2025), and approximately seven thousand three hundred septic, drain, and sewer cleaning businesses (2024). Each industry is characterized by low market-share concentration and a preponderance of small, independent regional operators.
https://www.ibisworld.com/united-states/number-of-businesses/waste-collection-services/1506/.

IDTechEx. “Why Driving Will Be Outlawed by 2050.” IDTechEx research article. A technology-analyst forecast arguing that, as autonomous vehicles surpass human drivers in safety, manual driving will eventually be prohibited on public roads and confined to closed tracks as a sport. Cited in the autonomous-vehicle chapter on the documented movement to outlaw human driving.
https://www.idtechex.com/en/research-article/why-driving-will-be-outlawed-by-2050/24622.

iFixit. “Right to Repair: Medical Devices.” Repair-advocacy resource maintained by iFixit, the consumer- and equipment-repair documentation platform.
https://www.ifixit.com/Right-to-Repair.

Imported Vehicle Safety Compliance Act of 1988; the twenty-five-year exemption. Codified in the importation framework at 49 C.F.R. Part 591 and implemented through 49 C.F.R. Part 571, the rule exempts motor vehicles twenty-five years old or older from the

FMVSS that would otherwise bar importation, allowing an older design to lawfully enter and remain in U.S. commerce without meeting the current year's standards. https://www.ecfr.gov/current/title-49/subtitle-B/chapter-V/part-571.

In re Trade-Mark Cases, 100 U.S. 82 (1879). https://supreme.justia.com/cases/federal/us/100/82/.

India. Finance Act 2016, amending Section 35(2AB) of the Income-tax Act, 1961. Reduced the weighted deduction for approved in-house scientific research from 200 percent of qualifying expenditure to 150 percent effective April 1, 2017, and to 100 percent thereafter. Section 35 applied different weighted rates to different categories of research expenditure, making it the clearest documented instance of a single statute tuning the multiplier by category. Documented in KPMG India, "Department of Scientific and Industrial Research Amends Guidelines for Approval of In-House R&D Centres." https://www.in.kpmg.com/taxflashnews/KPMG-Flash-News-Department-of-Scientific-and-Industrial-Research-3.pdf.

India. Ministry of Road Transport and Highways. "AIS 137 – Part 6 – Administrative Procedure for Type Approval and Conformity of Production for BSVI Norms." Automotive Industry Standard issued under Central Motor Vehicles Rules 115, 116, and 126. Records the notified migration to Bharat Stage VI emission norms for two-, three-, and four-wheeled vehicles effective 1 April 2020, pursuant to Gazette Notification G.S.R. 889(E) of 16 September 2016. Cited in the international-comparison chapter for the proposition that Indian compliance obligations attach to the date a vehicle is manufactured. No legacy-platform, grandfathering, or platform-year provision was located in the retrieved text of this standard or of the Central Motor Vehicles Rules; that absence is the argument's own finding and is identified as such in the chapter. https://morth.gov.in/ais-137-%E2%80%93-part-6-%E2%80%93-administrative-procedure-type-approval-and-conformity-production-bsvi-norms.

Infrastructure Investment and Jobs Act of 2021 (also styled the Bipartisan Infrastructure Law), Pub. L. No. 117-58, 135 Stat. 429. Section 24208 directs the Department of

Transportation to issue performance standards for crash avoidance technology, including automatic emergency braking and forward collision warning, on all passenger motor vehicles. Codified in part at 49 U.S.C. §§ 30129–30130. https://www.law.cornell.edu/uscode/text/49/30129.

Inland Revenue Authority of Singapore. “Enterprise Innovation Scheme (EIS).” Establishes a total deduction of 400 percent on the first S$400,000 of qualifying research and development expenditure, comprising the 100 percent base deduction under Section 14C of the Income Tax Act 1947, the additional 150 percent deduction under Section 14D(1), and a further 150 percent enhanced deduction under the scheme, for Years of Assessment 2024 through 2028. The scheme sets a separate expenditure cap for each qualifying activity. https://www.iras.gov.sg/schemes/disbursement-schemes/enterprise-innovation-scheme-(eis).

Inland Revenue Board of Malaysia. Public Ruling No. 10/2021: Tax Treatment of Research and Development Expenditure, Part II. Sets out the double deduction available under Sections 34A and 34B of the Income Tax Act 1967, with worked examples showing qualifying expenditure multiplied by two. https://phl.hasil.gov.my/pdf/pdfam/PR_10_2021.pdf.

Inside.lighting. “Watt’s Going On? No, Incandescents Aren’t Back.” *Inside.lighting* analysis, January 2025. Lighting trade-press confirmation that Executive Order 14154 of January 20, 2025 (Unleashing American Energy) does not and cannot override the federal efficiency standards governing light bulbs, because the standards rest on statutory authority under the Energy Policy and Conservation Act and its anti-backsliding provision at 42 U.S.C. § 6295(o)(1). Companion source to EdisonReport in the healthy-lighting chapter’s account of the executive path. https://inside.lighting/news/25-01/watts-going-no-incandescents-arent-back.

Institute for Justice. Pro bono constitutional litigation organization founded in 1991, based in Arlington, Virginia. Has litigated more than thirteen Supreme Court cases, winning eleven, with a current docket of more than one hundred cases in thirty-seven states focused

on property rights, economic liberty, and constitutional limits on government. Cited in the trucking chapter as one of three public-interest legal foundations that carry constitutional cases of the kind the Patent Clause argument requires at no cost to the plaintiff. https://ij.org/.

Insurance Institute for Highway Safety. “Fatality Facts 2023: Bicyclists.” IIHS-HLDI, posted July 2025. Annual summary of bicyclist fatality data compiled from the U.S. Department of Transportation’s Fatality Analysis Reporting System (FARS). Source for the bicycles section figures on bicyclist deaths under age 20 (786 in 1975 falling to approximately 93 in 2023, an 88 percent decline), adult bicyclist deaths (212 in 1975 rising to 1,041 in 2023, a near-fivefold increase), and total bicyclist deaths (1,155 in 2023, the highest annual figure ever recorded). The same IIHS page documents the crossover year (1989) when adult bicyclist deaths first exceeded child bicyclist deaths in the United States. https://www.iihs.org/topics/fatality-statistics/detail/bicyclists.

Internal Revenue Code § 170(e)(3). Added by the Tax Reform Act of 1976 for contributions made after October 4, 1976. Allows a corporation contributing inventory for the care of the ill, the needy, or infants to deduct the lesser of basis plus one-half of the appreciation or twice the basis of the contributed property. The twice-basis ceiling is the longest-standing above-cost deduction in the United States Code. https://irc.bloombergtax.com/public/uscode/doc/irc/section_170.

International Agency for Research on Cancer. *IARC Monographs on the Identification of Carcinogenic Hazards to Humans, Volume 105: Diesel and Gasoline Engine Exhausts and Some Nitroarenes*. Lyon, France: World Health Organization, 2014. The 2012 IARC working-group reclassification of diesel engine exhaust from Group 2A (probably carcinogenic to humans) to Group 1 (carcinogenic to humans), based on sufficient evidence of an association with lung cancer. Source for the Group 1 classification cited in the trucking chapter's discussion of the genuine public-health basis for diesel-particulate regulation. https://publications.iarc.fr/Book-And-Report-Series/Iarc-Monographs-On-The-Identification-Of-Carcinogenic-Hazards-To-Humans/Diesel-And-Gasoline-Engine-

Exhausts-And-Some-Nitroarenes-2014.

International Association of Certified Home Inspectors (InterNACHI). “Standard Estimated Life Expectancy Chart for Homes.” InterNACHI member resource. The de facto industry-standard life-expectancy chart for residential systems and appliances, widely used by home inspectors and consumers in the absence of a comparable American Society of Home Inspectors (ASHI) publication. The chart provides estimated service-life ranges assuming regular maintenance and normal usage patterns. Cited in the cars-and-trucks chapter for the household appliance replacement-cycle figures alongside the National Association of Home Builders (NAHB) 2007 Study of Life Expectancy of Home Components.
https://www.nachi.org/life-expectancy.htm.

International Council on Clean Transportation (ICCT). Heavy-duty vehicle emissions-compliance cost analysis, May 2020. Estimates the manufacturing cost of the post-2007 heavy-duty diesel aftertreatment stack — diesel particulate filter, selective catalytic reduction catalyst, and diesel-exhaust-fluid system — at approximately $4,400 in 2019 dollars for a thirteen-liter engine, a figure that corroborates the Manufacturers of Emission Controls Association component-share estimate cited in the trucking chapter, and places the diesel-particulate-filter replacement market range at approximately $2,000 to $8,000. Source for the trucking chapter’s aftertreatment hardware-cost and DPF-replacement figures.
https://theicct.org/wp-content/uploads/2021/06/HDV-emissions-compliance-cost-may2020.pdf.

International Maritime Organization Conventions: SOLAS, MARPOL, COLREG. International Convention for the Safety of Life at Sea (SOLAS, 1974, as amended); International Convention for the Prevention of Pollution from Ships (MARPOL, 1973/1978, as amended); Convention on the International Regulations for Preventing Collisions at Sea (COLREG, 1972, as amended). Implemented in U.S. law primarily through the Coast Guard’s vessel inspection regime under 46 U.S.C.
https://www.imo.org/en/About/Conventions/Pages/Default.aspx.

IQVIA Institute for Human Data Science. *The Use of Medicines in the U.S.*. Annual report on American prescription-drug use, spending, and the generic-versus-brand-name share of total prescriptions dispensed. Source for the figure that generic drugs rose from approximately nineteen percent of American prescriptions before the Drug Price Competition and Patent Term Restoration Act of 1984 (the Hatch-Waxman Act) to roughly ninety percent today. Cited in the policy-framework chapter for the post-Hatch-Waxman market-share trajectory the vehicle remake industry is expected to follow once the constitutional remedy lands.
https://www.iqvia.com/insights/the-iqvia-institute/reports-and-publications/reports/the-use-of-medicines-in-the-us-2024.

Iran Khodro Industrial Group (IKCO). “Soren Plus.” Iran Khodro product page. Source for the Iranian-market retail price of the Soren Plus sedan, a modernized vehicle built around the patent-expired Peugeot 405 platform that IKCO has manufactured under license since 1991. The cars-and-trucks chapter cites the price at approximately $7,500–$9,900 USD-equivalent in 2025 at Iran’s free-market exchange rate (the rate most economists and journalists use for purchasing-power comparisons against Iran’s sanctioned and currency-controlled economy). Earlier drafts of this book cited the price as approximately $13,100, which is what the Soren Plus converts to at Iran’s official exchange rate; the free-market rate is the cited figure in the current text. The price of the Soren Plus inside Iran has been documented by Living in Tehran, an Iran-focused market reporting site, with rial-Toman pricing supplied from August 2025 (765 million to 1.010 billion Toman across variants).
https://ikcoglobal.com/vehicle/overview?soren-plus. USD-equivalent figures derived from the author’s published rupee-to-dollar conversion analysis at
https://x.com/i/grok/share/dDvRNvFNqsfdA59JhT54zOGCO. Note: at the time of publication, Iran’s national internet was offline due to wartime conditions, rendering *ikcoglobal.com* intermittently unreachable; archived snapshots of the cited page are available through the Internet Archive Wayback Machine at
https://web.archive.org/web/*/ikcoglobal.com/vehicle/overview?soren-plus.

Irons, Peter. *Justice at War: The Story of the Japanese American Internment Cases*. Oxford University Press, 1983.
See Endnote Commentary.

Irons, Peter. *A People's History of the Supreme Court*. Viking, 1999. Noted in the chapter on outsiders and blind doctrine as adjacent work by a researcher who also appears in it as a subject.

Isaacs, Davida H. "Not All Property Is Created Equal: Why Modern Courts Resist Applying the Takings Clause to Patents, and Why They Are Right to Do So." *George Mason Law Review* 15, no. 1 (2007): 1. Argues that a patent's status as property is a relevant but incomplete answer to the constitutional question, because the Supreme Court has held that some interests — federal benefits in particular — receive Due Process Clause protection without receiving Takings Clause protection, and that patents are federal entitlements of that kind. Cited in the objections chapter as the strongest scholarly statement of the objection to the Takings Clause prong.
https://digitalcommons.law.umaryland.edu/fac_pubs/879/ ;
https://ssrn.com/abstract=968417.

Jaffe, Richard A. *Galileo's Lawyer: Courtroom Battles in Alternative Health, Complementary Medicine, and Experimental Treatments*. Houston: Thumbs Up Press, 2008.
https://www.amazon.com/Galileos-Lawyer-Courtroom-Battles-Complementary/dp/0977476626.

James v. Campbell, 104 U.S. 356 (1882).
https://supreme.justia.com/cases/federal/us/104/356/.

Janis, Irving L. *Victims of Groupthink: A Psychological Study of Foreign-Policy Decisions and Fiascoes*. Boston: Houghton Mifflin, 1972.
https://www.amazon.com/Victims-groupthink-psychological-foreign-policy-decisions/dp/0395140021.

Jefferson, Thomas. “Thomas Jefferson to Benjamin Vaughan, 27 June 1790.” Founders Online, National Archives. Written from New York while Jefferson, as Secretary of State, sat on the statutory patent board created by the Patent Act of 1790; he reports the Act had given a spring to invention beyond his conception. https://founders.archives.gov/documents/Jefferson/01-16-02-0342.

Jefferson, Thomas. “Letter to Isaac McPherson, August 13, 1813.” Founders Online, National Archives. https://founders.archives.gov/documents/Jefferson/03-06-02-0322. See Endnote Commentary.

Jeffery, Glen, et al. Peer-reviewed clinical study on long-wavelength red light exposure and mitochondrial function in human visual decline. *Scientific Reports* (Nature), 2021. University College London research documenting measurable improvements in retinal function following brief exposure to deep-red (670 nm) light, with implications for the harms imposed by blue-shifted LED-only residential lighting environments. Anchored in the healthy-lighting chapter through the corresponding *Huberman Lab* podcast treatment. https://www.nature.com/articles/s41598-021-02311-1.

Jeffery, Glen, et al. Peer-reviewed follow-up research on long-wavelength light therapy and cellular function. *Scientific Reports* (Nature), 2025. University College London research extending the 2021 findings on the systemic and ocular benefits of deep-red light exposure and the corresponding harms from blue-shifted LED-only lighting environments. https://www.nature.com/articles/s41598-025-09785-3.

John Deere W70. India product listing on IndiaMART. Source for the $21,000–$24,700 retail price range and the 2.5-acres-per-hour capacity figure for the 100-horsepower, 14-foot, self-propelled wheeled combine harvester cited in the farm-equipment chapter as the India-market counterpart to the U.S. John Deere S7 600. https://dir.indiamart.com/items/john-deere-w70-14-feet-100-hp-combine-harvester-4-straw-walker-s22473.html.

Journal of Benefit-Cost Analysis. “The Benefits and Costs of Automotive Regulations for Low-Income Americans.” Cambridge University Press. Provides the pre-1971 versus post-1970 federal-regulatory cost split per vehicle cited in the cars-and-trucks chapter: $250–$400 in 2025 dollars for the pre-1971 layer (Motor Vehicle Safety Act of 1966 and Air Quality Act of 1967) against $6,600–$8,600 in 2025 dollars for the post-1970 layer (Clean Air Act Amendments of 1970 and Energy Policy and Conservation Act of 1975). Article author and exact publication date pending verification at edit time. https://www.cambridge.org/core/journals/journal-of-benefit-cost-analysis.

J. W. Hampton, Jr., & Co. v. United States, 276 U.S. 394 (1928). The Supreme Court upheld a delegation of rulemaking authority to the executive, holding that Congress may delegate such authority so long as it lays down an intelligible principle to guide the body exercising it. Cited in the chapter on the Patent Clause prong as the source of the modern non-delegation rule, and for the limit that rule carries: an agency may exercise only authority Congress possessed and could lawfully delegate, and Congress possessed no authority to cancel the public’s ownership of an expired invention. https://supreme.justia.com/cases/federal/us/276/394/.

Kaplan, Sonia, Malika Nurullaeva, and John Paul Helveston. “Comparing the Costs of Autonomous and Conventional Taxi Services.” Transport Policy 159 (December 2024): 108–119. DOI: 10.1016/j.tranpol.2024.10.010. Component-level cost model grounded in deployed U.S. robotaxi operations: an autonomous taxi at about $1.99 per mile against $2.74 for a conventional taxi, with labor still $1.02 per mile because remote support, cleaning, and depot work remain human. Assumes 65,000 miles per vehicle-year at 50 percent utilization. Cited in The Last Human Driver as the peer-reviewed cost work that follows the Massachusetts Institute of Technology study without overturning it.

Kauers, Manuel, and Jakob Moosbauer. “Flip Graphs for Matrix Multiplication.” In *Proceedings of the 2023 International Symposium on Symbolic and Algebraic Computation* (ISSAC ’23), Tromsø, Norway, 381–388. Association for Computing Machinery, 2023. doi:10.1145/3597066.3597120. Preprint arXiv:2212.01175. The authors state that in their quick response to Fawzi and colleagues they announced further four-by-four schemes at

forty-seven multiplications in characteristic two and a reduction of the five-by-five case to ninety-five, against AlphaTensor's ninety-six; they also reduced a rectangular format from sixty-three to sixty-two. The flip-graph search method is their own, ran on a small computer in seconds, and used AlphaTensor's output as a starting point. Cited in the chapter on outsiders and blind doctrine for the proposition that the machine result dislodged a settled belief that the question was closed, after which mathematicians who had held the capability all along went further than the machine had.

Kelley Blue Book. "2025 Ford F-150 SuperCrew Cab Lariat." Kelley Blue Book 2025 model-year pricing reference. Tertiary verification source for the $72,495 chapter title figure, confirming the Lariat 4×4 SuperCrew 5.5′ Box configuration as the typical family-trim price point in the F-150 lineup.
https://www.kbb.com/ford/f-150/2025/.

Kelley Blue Book / Cox Automotive. "New-Vehicle Transaction Prices in June Post Smallest Annual Gain in Nearly 4 Years." Cox Automotive Market Insights, July 11, 2023. Source for the U.S. average new-vehicle transaction price of $48,808 in June 2023 cited in the cars-and-trucks chapter.
https://www.coxautoinc.com/market-insights/kbb-atp-june-2023/.

Kahle v. Gonzales, 487 F.3d 697 (9th Cir. 2007). The Ninth Circuit's affirmance of the dismissal of a challenge to the shift from an opt-in to an opt-out copyright system, holding that the plaintiffs made essentially the argument the Supreme Court had rejected in *Eldred v. Ashcroft*. The opinion nonetheless states that *Eldred* did not purport to directly address whether the phrase "limited Times" places substantive limits on duration, citing Eldred's own footnote that the outer-boundary question was not before the Court. Judge Farris wrote for a unanimous panel.
https://law.justia.com/cases/federal/appellate-courts/F3/487/697/622521/. See Endnote Commentary.

Kellogg Co. v. National Biscuit Co., 305 U.S. 111 (1938). Justice Brandeis, for the Court, held that on expiration of the Perky shredded-wheat patents there passed to the public "not

only the right to make the article as it was made during the patent period, but also the right to apply thereto the name by which it had become known," and that upon expiration "the form, as well as the name, was dedicated to the public." The only obligation resting on the newcomer was to identify its own product so that buyers were not deceived as to source. Cited in the chapter on the Patent Clause prong for the Court's plainest statement that the form of the article, and not merely its underlying idea, enters the public domain when the patent expires, and in the chapter on objections considered for the source-identification duty that answers the trademark objection to remanufacturing. The suit arose in diversity as a common-law unfair-competition claim; the case is cited for what it holds about post-expiration rights, not for a constitutional proposition. It completes the line running from *Singer Manufacturing Co. v. June Manufacturing Co.* (1896) through *Scott Paper Co. v. Marcalus Manufacturing Co.* (1945) to *Sears, Roebuck & Co. v. Stiffel Co.* and *Compco Corp. v. Day-Brite Lighting, Inc.* (1964).
https://supreme.justia.com/cases/federal/us/305/111/.

Kempe, C. Henry, Frederic N. Silverman, Brandt F. Steele, William Droegemueller, and Henry K. Silver. "The Battered-Child Syndrome." *Journal of the American Medical Association* 181, no. 1 (1962): 17-24. The first state mandatory reporting statutes followed in 1963; all fifty states had enacted one by 1967.

Kennedy, Robert F., Jr. *The Real Anthony Fauci: Bill Gates, Big Pharma, and the Global War on Democracy and Public Health*. New York: Skyhorse, 2021.
https://www.skyhorsepublishing.com/9781510766808/the-real-anthony-fauci/.

Kenworth Truck Company. Model T680 and T800 pricing. Verified through dealer-network listings, including Kenworth Sales Co. published manufacturer's suggested retail pricing for the 2025 and 2026 model years. Source for the trucking chapter's American-truck price points: 2026 T680 day cab approximately $161,850, 2025 T680 sleeper with Cummins X15 approximately $172,900, and 2026 T800 heavy-spec approximately $284,900.
https://www.kenworth.com/.

Kelo v. City of New London, 545 U.S. 469 (2005). The Supreme Court's holding that transferring private property to a private developer for economic development satisfies the public-use requirement of the Takings Clause. Justices O'Connor and Thomas dissented. The development was never built and the anchor tenant left New London in 2009; the decision produced eminent-domain reform legislation in a majority of States. https://supreme.justia.com/cases/federal/us/545/469/.

Kewanee Oil Co. v. Bicron Corp., 416 U.S. 470 (1974). Six-to-two decision, Justice Powell taking no part, holding that Ohio trade-secret law is not preempted by the federal patent laws. Chief Justice Burger, for the Court, reasoned that trade-secret protection is weaker than a patent, withdraws nothing from the public domain, and poses no real risk of deterring inventors from seeking patents. Justice Marshall concurred in the result; Justice Douglas, joined by Justice Brennan, dissented. Cited in the chapter on objections considered as adverse authority for the proposition that federal patent policy does not displace every other legal regime bearing on an invention, and distinguished there on the ground that trade-secret law leaves the public free to reverse-engineer and copy what it learns, while the certification barrier forbids the lawful sale of the copy itself. https://supreme.justia.com/cases/federal/us/416/470/.

Kimble v. Marvel Entertainment, LLC, 576 U.S. 446 (2015). https://supreme.justia.com/cases/federal/us/576/446/.

King, Martin Luther, Jr. "Letter from Birmingham Jail." April 16, 1963. The Martin Luther King, Jr. Research and Education Institute, Stanford University. https://kinginstitute.stanford.edu/letter-birmingham-jail.

Knoll, Shaun, et al. U.S. Patent 5,615,404, filed October 31, 1994, issued March 25, 1997. Assigned to Intel Corporation. Foundational Intel-assigned patent in the corpus establishing the Universal Serial Bus protocol; expired October 31, 2014. Cited in the surveillance chapter as one of the foundational expired USB patents from which a remanufactured vehicle's USB port can be built. https://patents.google.com/patent/US5615404A/en.

Knoll, Shaun, et al. U.S. Patent 5,623,610, filed October 31, 1994, issued April 22, 1997. Assigned to Intel Corporation. One of the foundational Intel-assigned USB patents; expired October 31, 2014.
https://patents.google.com/patent/US5623610A/en.

Knoll, Shaun, et al. U.S. Patent 5,742,847, filed October 31, 1994, issued April 21, 1998. Assigned to Intel Corporation. One of the foundational Intel-assigned USB patents; expired October 31, 2014.
https://patents.google.com/patent/US5742847A/en.

Knorozov, Yuri V. “Drevnyaya pis’mennost’ Tsentral’noy Ameriki” [Ancient Writing of Central America]. *Sovetskaya Etnografiya*, 1952. The paper that read the de Landa signs as a syllabary rather than an alphabet.
See Endnote Commentary.

Kory, Pierre, Gianfranco Umberto Meduri, Joseph Varon, Jose Iglesias, and Paul E. Marik. “Review of the Emerging Evidence Demonstrating the Efficacy of Ivermectin in the Prophylaxis and Treatment of COVID-19.” American Journal of Therapeutics 28, no. 3 (May/June 2021): e299–e318. Meta-analysis of eighteen randomized controlled trials. The journal issued an Expression of Concern in 2022 (American Journal of Therapeutics 29, no. 2: e231) and an erratum in 2021 (28, no. 6: e813); the Expression of Concern states that it does not imply the methodology was incorrect.
https://doi.org/10.1097/MJT.0000000000001377. See Endnote Commentary.

Kory, Pierre, with Jenna McCarthy. *The War on Ivermectin: The Medicine That Saved Millions and Could Have Ended the Pandemic*. New York: ICAN Press / Skyhorse, 2023.
https://www.skyhorsepublishing.com/9781510774070/the-war-on-ivermectin/.

Korematsu v. United States, 323 U.S. 214 (1944). The Supreme Court’s decision upholding the wartime exclusion of Japanese Americans from the West Coast, with Justices Roberts, Murphy, and Jackson dissenting. The claim of military necessity was made while contrary

evidence sat in the government's files; a federal court vacated Fred Korematsu's conviction on coram nobis in 1983 after the record surfaced. https://supreme.justia.com/cases/federal/us/323/214/.

Kuhn, Thomas S. *The Structure of Scientific Revolutions*. Chicago: University of Chicago Press, 1962. https://press.uchicago.edu/ucp/books/book/chicago/S/bo13179781.html. Cited in the chapter on outsiders and blind doctrine, and already invoked in the preface: normal science operates inside a settled paradigm that determines which questions count, and that framework is what renders some fully observable facts invisible.

Kuran, Timur. *Private Truths, Public Lies: The Social Consequences of Preference Falsification*. Cambridge, MA: Harvard University Press, 1995. https://www.hup.harvard.edu/books/9780674707580.

Lakshman, Nandini. "Mahindra and Mahindra: A Journey Flagged Off by the Willys Jeep." *Mint* (Livemint), September 17, 2010. https://www.livemint.com/Companies/fOMKVnNwjxhkIlJqChnVfO/Mahindra-and-Mahindra-Group-A-journey-flagged-off-by-the-Wi.html.

Landa, Diego de. *Relación de las cosas de Yucatán*, written circa 1566. First published by Charles Étienne Brasseur de Bourbourg as *Relation des choses de Yucatan de Diego de Landa*, Paris, 1864. The material Knorozov used, in print for eighty-eight years before he used it.

Lanham Act of 1946 (Trademark Act of 1946), Pub. L. No. 79-489, 60 Stat. 427. Codified as amended at 15 U.S.C. §§ 1051 et seq. https://www.law.cornell.edu/uscode/text/15/chapter-22.

Largey, Matt. "The 'Bad Grade' That Changed The U.S. Constitution." *All Things Considered*, National Public Radio, May 5, 2017. Originally produced for *Pop-Up Magazine* in collaboration with KUT-FM Austin, Texas. The principal popular source for Chapter 31's account of Gregory Watson's 1982–1992 campaign to ratify the Twenty-Seventh Amendment, featuring interviews with Watson and with his University of Texas

professor Sharon Waite, whose 2017 reconsideration of the original C grade led to the formal university grade change from C to A+ on March 1, 2017. Largey's reporting also documents the chronology of state ratifications during the 1982–1992 campaign. The 1792 Kentucky ratification, rediscovered after Michigan's May 7, 1992 vote — and which makes Alabama (May 5, 1992) the technical thirty-eighth ratifying state — is documented across multiple state-archive sources collected in the U.S. National Archives Twenty-Seventh Amendment file.
https://www.npr.org/2017/05/05/526900818/the-bad-grade-that-changed-the-u-s-constitution.

Larsson, Fredrik, Petra Andersson, Per Blomqvist, and Bengt-Erik Mellander. "Toxic Fluoride Gas Emissions from Lithium-Ion Battery Fires." *Scientific Reports* 7 (2017): 10018. Fire tests on seven types of commercial lithium-ion batteries measured hydrogen fluoride emissions of 20 to 200 milligrams per watt-hour of nominal battery capacity, with phosphoryl fluoride detected in some tests; the authors extrapolate a release of 400 to 1,200 grams of hydrogen fluoride from a 10 kWh automotive pack in a fire. Cited in the environmental chapter for the point that the technology the regulatory wall favors carries lifecycle costs no regulation prices.
https://www.nature.com/articles/s41598-017-09784-z.

Last Emperor, The. Directed by Bernardo Bertolucci. Screenplay by Mark Peploe and Bernardo Bertolucci. Columbia Pictures, 1987. Winner of nine Academy Awards (60th Academy Awards, 1988), including Best Picture and Best Director. The Hybrid Voting chapter cites the tutor-scene dialogue in which Reginald Johnston (Peter O'Toole) tells the fifteen-year-old Pu Yi (Wu Tao), in response to the Emperor's question why precise words matter: "If you cannot say what you mean, Your Majesty, you will never mean what you say. And a gentleman should always mean what he says." The exchange arises from Johnston's correction that Scotsmen wear kilts, not skirts — "a matter of words, perhaps, but words are important." The line is original to the Peploe-Bertolucci screenplay and does not appear in Pu Yi's memoir *From Emperor to Citizen* (1964) or in Reginald Johnston's

memoir *Twilight in the Forbidden City* (1934). Quote attribution verified against the film's audio and against the Internet Movie Database's quotes page.
https://www.imdb.com/title/tt0093389/quotes/.

Lawrence v. Texas, 539 U.S. 558 (2003). The Supreme Court's holding that a Texas statute criminalizing same-sex intimate conduct violates the Due Process Clause, overruling *Bowers v. Hardwick*. Justice O'Connor concurred in the judgment on equal-protection grounds. The corrected historical record came from a brief filed by historians outside the legal profession.
https://supreme.justia.com/cases/federal/us/539/558/.

Lead Phase-Out, Regulation of Fuels and Fuel Additives, 40 C.F.R. Part 80, Subpart B (lead phase-down rules first promulgated in 1973; complete prohibition of leaded gasoline for on-road motor vehicles effective January 1, 1996, codified at 40 C.F.R. § 80.22).
https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-80.

Leake, John, and Peter A. McCullough. *The Courage to Face COVID-19: Preventing Hospitalization and Death While Battling the Bio-Pharmaceutical Complex*. New York: Skyhorse, 2022.
https://www.skyhorsepublishing.com/9781510776807/the-courage-to-face-covid-19/.

Levy, Karen. *Data Driven: Truckers, Technology, and the New Workplace Surveillance*. Princeton: Princeton University Press, 2022. Sociologist's ethnographic study of the electronic-logging-device mandate and its downstream consequences for American long-haul truck drivers, documenting how federal regulation supplied the legal floor while commercial vendors built the surveillance stack above it, including the Omnitracs Driver Retention Model that analyzes hours-of-service data to predict which drivers are at risk of voluntary termination. Cited in the trucking-section of the surveillance chapter for the proposition that federal mandate, vendor product, employer surveillance, and behavioral prediction move in one direction only.
https://press.princeton.edu/books/hardcover/9780691175300/data-driven.

Li, David H. *The Genealogy of Chess*. Bethesda, MD: Premier Publishing Co., 1998. The minority-view book in the chess-origin debate, advancing the thesis that an early form of chess was invented in China around 203–204 BC by General Han Xin during the Chu-Han Contention, and that the Indian *chaturanga* tradition derives from a westward transmission of the Chinese game rather than the reverse. Named Book of the Year 1998 by the editors of *GAMES Magazine*. Cited in the preface as one side of the historians' debate over whether chess migrated from India to China or from China to India.
https://www.amazon.com/genealogy-chess-David-H-Li/dp/0963785222.

Lieberman, Ben. "Home Appliance Regulations Are Out of Control." *City Journal*, March 13, 2025. Senior fellow Lieberman (Competitive Enterprise Institute) surveys the Department of Energy efficiency-standards regime applied to washing machines, refrigerators, furnaces, dishwashers, central air conditioners, light bulbs, and water heaters; documents DOE-estimated cost increases such as the $953 added cost on the latest electric water heater standard; and reports the post-standards performance degradation that has produced two-hour dishwasher cycles. The article does not contain the Kansas-family dishwasher anecdote attributed to it in some derivative writing; the chapter prose attribution should be corrected at edit time.
https://www.city-journal.org/article/home-appliance-regulations-department-of-energy-efficiency-standards.

Liu, Wenzhe. *The Chinese School of Chess*. Translated by Yasser Seirawan and Ken Smith. London: Batsford, 2002. Liu, a leading Chinese international master and theoretician, takes a pro-China view of the origins of the chess family and traces, from inside the modern game, the distinctively Chinese contribution to opening theory and strategic outlook. Cited in the preface together with Li (1998) and Murray (1913) as a representative voice in the historians' debate over whether chess migrated from India to China or from China to India.
https://www.chess.com/article/view/quotchinese-school-of-chessquot-by-liu-wenzhe.

Loper Bright Enterprises v. Raimondo, 603 U.S. 369 (2024). This footnote is also for the Foreword.
https://supreme.justia.com/cases/federal/us/603/22-451/.

Loretto v. Teleprompter Manhattan CATV Corp., 458 U.S. 419 (1982). https://supreme.justia.com/cases/federal/us/458/419/.

Loudon, Irvine. “Ignaz Phillip Semmelweis’ Studies of Death in Childbirth.” *Journal of the Royal Society of Medicine* 106, no. 11 (November 2013): 461–463. doi:10.1177/0141076813507844. Source for the maternal mortality rates of 98.4 per thousand births in the first clinic against 36.2 in the second across 1840 to 1846, and for the fact that women were admitted to the two clinics on alternate days with no clinical selection. https://pmc.ncbi.nlm.nih.gov/articles/PMC3807776.

Low-Emission Vehicle Program (LEV I). California Air Resources Board, adopted 1990; standards applicable to model years 1994–2003. 13 C.C.R. § 1960.1. California Air Resources Board. https://ww2.arb.ca.gov/our-work/programs/low-emission-vehicle-program.

Low-Emission Vehicle Program (LEV II). California Air Resources Board, adopted 1999; standards phased in from model year 2004. 13 C.C.R. § 1961. California Air Resources Board. https://ww2.arb.ca.gov/our-work/programs/low-emission-vehicle-program.

Low-Emission Vehicle Program (LEV III). California Air Resources Board, adopted January 2012; combined NMOG+NOx standards phased in over model years 2015–2025. 13 C.C.R. § 1961.2. California Air Resources Board. https://ww2.arb.ca.gov/our-work/programs/low-emission-vehicle-program.

Low Volume Motor Vehicle Manufacturers Act of 2015. Enacted as Section 24405 of the Fixing America’s Surface Transportation Act (FAST Act), Pub. L. No. 114-94, 129 Stat. 1312 (December 4, 2015). Codified at 49 U.S.C. § 30114(b). Authorizes specialty manufacturers to produce replica cars of designs at least twenty-five years old without full current-year federal motor vehicle safety standards compliance, subject to a ceiling of three hundred twenty-five replica vehicles per manufacturer per year across the entire United States and the requirement that the replicas meet current-year EPA emissions standards. The

Act's amendment to Clean Air Act § 206 deems a replica emissions-compliant only if it is equipped with an engine, including all emissions-control equipment, already covered by an EPA certificate of conformity for a current model year — so the statute that reopens the old body to production requires, in its own text, that the original powertrain be replaced with a modern certified one. Cited in the chapter on objections considered as the empirical test in which Congress waived the safety standards for old designs yet left the emissions wall standing.
https://www.law.cornell.edu/uscode/text/49/30114.

Lucas v. South Carolina Coastal Council, 505 U.S. 1003 (1992).
https://supreme.justia.com/cases/federal/us/505/1003/.

Lujan v. Defenders of Wildlife, 504 U.S. 555 (1992).
https://supreme.justia.com/cases/federal/us/504/555/.

Madison, James. "The Federalist No. 43: The Same Subject Continued: The Powers Conferred by the Constitution Further Considered." January 23, 1788. In *The Federalist Papers*. Avalon Project, Yale Law School.
https://avalon.law.yale.edu/18th_century/fed43.asp.

Magill, Gord. Interview by Tucker Carlson. *The Tucker Carlson Show*, January 24, 2024. Approximately forty minutes; covers the imprisoned "Coutts Four" and the 2022 Freedom Convoy protests against the Canadian government. The episode is available at https://podcasts.apple.com/us/podcast/gord-magill/id1719657632?i=1000642825841, with a transcript at https://podcasts.happyscribe.com/the-tucker-carlson-podcast/gord-magill. See Endnote Commentary.

Magill, Gord. Interview on *Bannon's WarRoom*, April 15, 2025. Discussion of foreign labor in the trucking workforce and the "driver shortage" framing. Show-notes page at https://warroom.org/select-bannons-warroom-show-clip-roundup-15-april-25-pm/; no transcript published. See Endnote Commentary.

Magill, Gord. *End of the Road: Inside the War on Truckers*. Nashville, TN: Creed & Culture, March 24, 2026. A third-generation trucker's firsthand account of how the once-

iconic American occupation has been ground down by surveillance technology, immigration scams that put untrained drivers behind the wheel, and the regulatory and corporate apparatus that treats drivers as inputs rather than people. Cited in the trucking chapter as the firsthand companion to the constitutional argument: Magill describes the war from inside the cab; the chapter describes the constitutional brief that explains why the war was lawful to wage. Magill's continuing Substack work, *Autonomous Truck(er)s*, is available at
https://substack.com/@autonomoustruckers/p-201861755. This footnote is also for the Foreword. https://creedandculture.com/books/end-of-the-road-inside-the-war-on-truckers/. The autonomous-freight chapter, “Truckzilla: Invasion of the RoboTrucks,” is the firsthand companion to this book’s chapter on autonomous vehicles. Magill also discussed the autonomous-trucking timeline and the working driver’s preference for pre-2000 trucks in a telephone conversation with the author, May 3, 2026 (thirteen minutes); as a personal communication it carries no URL.

Magill, Gord. “Self-driving trucks are about controlling the roads — not making them safer.” *Blaze Media* (Align section), May 28, 2026. Magill’s column-length argument that the autonomous-trucking program rests on a safety case the technology has not earned, and that the human-driven alternative is being engineered off the road by parallel regulatory and labor pipelines. Source for the Waymo Philippines and Kodiak remote-operations claims in the autonomous-driving chapter, and the aggregator surfacing the New York Times Cummings phantom-braking quote (see Cummings entry above).
https://www.theblaze.com/align/self-driving-trucks-are-about-controlling-the-roads-not-making-them-safer.

Magill, Gord. Interview by Tucker Carlson. *The Tucker Carlson Show*, June 1, 2026. Approximately sixty-five minutes; covers the war-on-truckers thesis of Magill’s *End of the Road* (entry above), the foreign-ownership and remote-operations dynamics in U.S.

trucking, and the legislative landscape as of mid-2026. Cited as a primary-source record of the foreword author's positions on the subject matter of the autonomous-driving chapter. https://tuckercarlson.com/tucker-show-gord-magil-060126. A full transcript is available at https://singjupost.com/tucker-carlson-show-w-ex-trucker-gord-magill-june-1-2026-transcript/. See Endnote Commentary.

Mahindra & Mahindra Ltd. "Mahindra Sets New Safety Benchmarks as Thar ROXX, XUV 3XO, and XUV400 Earn 5-Star Bharat-NCAP Rating." Press release, November 2024. Documents the milestone of the Thar ROXX becoming the first body-on-frame SUV to achieve a five-star Bharat NCAP rating. https://auto.mahindra.com/press-release/mahindra-sets-new-safety-benchmarks-as-thar-roxx-xuv-3xo-and-xuv400-earn-5-star-bharat-ncap-rating.html.

Mahindra & Mahindra Ltd. "About Us." Corporate page. Source for Mahindra & Mahindra Limited's $19.4 billion multinational scale cited in the farm-equipment chapter. Mahindra is the parent corporation of Mahindra Tractor and the Mahindra Auto division whose product pages are cited separately in this apparatus. https://www.mahindra.com/about-us.

Mahindra & Mahindra Ltd. "Global Presence." Mahindra Auto division corporate page documenting Mahindra's operations in over 100 countries. Cited in the farm-equipment chapter alongside ZX Auto's sales-network maps as one of two manufacturer pages from which forty-three countries can be individually extracted, against a claimed total of ninety-plus countries. https://auto.mahindra.com/global.html.

Mahindra & Mahindra Ltd. "Mahindra Bolero." Mahindra product page. Source for the Indian-market retail price of the Mahindra Bolero, a Jeep CJ-lineage vehicle fitted with a modern Mahindra diesel, at $8,400–$10,400 USD-equivalent in 2025 (₹8.00 lakh base B4 to ₹9.90 lakh top B8, ex-showroom, October 2025 launch update). https://auto.mahindra.com/suv/bolero/BOL.html. USD-equivalent figures derived from the author's published rupee-to-dollar conversion analysis at

https://x.com/i/grok/share/dDvRNvFNqsfdA59JhT54zOGCO.

Mahindra & Mahindra Ltd. “Mahindra Thar.” Mahindra product page. Source for the Indian-market retail price range of the Mahindra Thar 3-door, a modernized successor to the Jeep CJ-5 and CJ-7, at $10,500–$18,500 USD-equivalent in 2025 (₹9.99 lakh base AXT RWD Diesel MT to ₹17.63 lakh top LXT 4WD Diesel AT, ex-showroom, 2025 facelift).
https://auto.mahindra.com/suv/thar/THRN.html. USD-equivalent figures derived from the author’s published rupee-to-dollar conversion analysis at
https://x.com/i/grok/share/dDvRNvFNqsfdA59JhT54zOGCO.

Mahindra Tractor. “Mahindra 475 DI XP Plus.” Mahindra Tractor product page for the 44-horsepower 475 DI XP Plus tractor sold in the Indian market for approximately $7,300 at current exchange rates. Source for the India-market tractor pricing comparison anchored in the opening of the farm-equipment chapter.
https://www.mahindratractor.com/tractors/mahindra-475-di-xp-plus.

Mahindra USA. “Mahindra 4540 Configurator.” Mahindra USA product configurator page for the 41-horsepower Mahindra 4540 tractor sold in the United States at approximately $27,992 MSRP. Source for the U.S.-market tractor pricing comparison anchored in the opening of the farm-equipment chapter.
https://mahindrausa.com/build-your-own/add-ons/?byo_id=331407.

Mankowitz, Daniel J., et al. “Faster Sorting Algorithms Discovered Using Deep Reinforcement Learning.” *Nature* 618 (2023): 257-263. The resulting fixed-sort routines were merged into the LLVM standard C++ library.

Manufacturers of Emission Controls Association (MECA). Trade association of the emission-controls component-manufacturers serving the automotive, heavy-duty truck, off-road, and stationary engine markets. Membership includes Corning Inc. (the principal manufacturer of ceramic substrates used in diesel particulate filters), Johnson Matthey, BASF Catalysts, Umicore, and other component suppliers. Cited in the trucking chapter as the trade-association beneficiary of the EPA 2007 heavy-duty diesel rule, which

transformed the heavy-duty truck emission-controls market from a segment that did not exist at commercial scale before the rule into a multi-billion-dollar segment after it. https://www.meca.org/.

Manufacturers of Emission Controls Association (MECA). “Technology Feasibility for Heavy-Duty Diesel Trucks in Achieving 90% Lower NOx Standards in 2027.” White paper, 2020. Documents that the MECA-member aftertreatment-hardware share (diesel oxidation catalyst, diesel particulate filter substrate, selective catalytic reduction catalyst) on a current Class 8 line-haul tractor with a twelve- to thirteen-liter engine, certified to the EPA 2010 heavy-duty standards, runs approximately $3,500 to $4,600 in 2019 dollars, based on a MECA member survey. Source for the trucking chapter’s figure that every Class 8 truck sold after 2007 carries roughly three to five thousand dollars in MECA-member components. The total aftertreatment-system cost paid by the buyer (including original-equipment-manufacturer integration, sensors, tanks, dosing equipment, on-board diagnostics, and dealer markup) is materially higher; the MECA-member component share isolated by this white paper is the figure the chapter cites. https://www.meca.org/wp-content/uploads/resources/MECA_2027_Low_NOx_White_Paper_FINAL.pdf.

Marik, Paul E. *Cancer Care: The Role of Repurposed Drugs and Metabolic Interventions in Treating Cancer*. 2nd ed. FLCCC Alliance / Independent Medical Alliance, 2024. https://imahealth.org/research/cancer-care/.

Markopolos, Harry. *No One Would Listen: A True Financial Thriller*. Wiley, 2010. See Endnote Commentary.

Martin, Roleigh, ed. *ShareDebate International*. Bloomington and Edina, MN: Applied Foresight, Inc., 1990–1995. ISSN 1054-0695. A diskette-distributed (later optical-disk) freeware electronic magazine, published twice yearly and circulated to between 550 and 1,450 bulletin-board systems and shareware dealers across twenty-five countries via the Shareware Distribution Network, PC-SIG CD-ROMs, BIX, GEnie, and anonymous FTP from wuarchive.wustl.edu. The author’s first publishing venture on the constitutional and

economic questions developed in this book; Issue 1 (1990) opened Debate Topic #4, “Japanese VS American Patent Law Debate,” framing the fairness and economic impact of U.S. patent law as a question for public deliberation thirty-five years before the present book reviewed patent law in a deeper, more constitutional form.
https://portal.issn.org/resource/ISSN/1054-0695. See Endnote Commentary.

Martin, Roleigh. “Privatizing Scientific Medical Research Funding with a Multi-Value Tax Deduction.” Originally published as “Privatizing Welfare By Funding It With a Double-Value Tax Deduction” in ShareDebate International, June 26, 1994; reprinted on X, April 29, 2024. The author’s original proposal of the multi-value tax deduction — a deduction of two or three dollars for each dollar spent on a favored activity, cheaper to the Treasury than a tax credit and more inducing than an ordinary deduction. Cited in the environmental chapter as the market mechanism by which the newest patent-expired emissions-control technology can be made the economical default in remanufactured products without a direct subsidy.
https://x.com/RoleighMartin/status/1784804443508138171.

Martin, Roleigh. “Korean Chess.” Applied Foresight, Inc., circa 1996. Original page hosted at timeless-eye.tzo.com:8448/afi/koreanch.htm; the author’s introduction to Janggi (Korean Chess), published as the first English-language web page on the game. Archived March 10, 2005 via the Internet Archive Wayback Machine. Cited in the preface for the rule-bound habit of mind a chess player develops, applied by the author to the reading of the Patent Clause.
https://web.archive.org/web/20050310062250/http://timeless-eye.tzo.com:8448/afi/koreanch.htm.

Martin, Roleigh. “How Another Legislature Avoided Coercion for 74 Years.” Published May 12, 2024. The author’s treatment of the Italian parliament’s *voto segreto* practice from 1848 to 1922 under the *Statuto Albertino*, and the 1922 Mussolini suspension. Source for the 74-year pre-Fascist span within those 119 years, and for the structural argument that secret voting protected the parliament against external coercion. X article:
https://x.com/RoleighMartin/status/1789698216511844465.

Martin, Roleigh. “The History of Secret Legislative Voting in the US Congress.” Published June 16, 2024. The author’s survey of the 180-year history of unrecorded and secret voting in the U.S. Congress prior to the Legislative Reorganization Act of 1970, including the use of the Committee of the Whole as an opacity venue, voice voting on amendments, and the routine application of Article I, Section 5, Clause 3’s exception clause. X article: https://x.com/RoleighMartin/status/1802209703352508636.

Martin, Roleigh. “What is Hybrid Legislative Voting – Why D.O.G.E. needs it to help reduce the national debt.” Published January 13, 2025. The author’s original article proposing Hybrid Voting as the structural reform to restore Congress’s pre-1970 ability to reduce the national debt; the chapter develops and re-sources the article’s central material. X article: https://x.com/RoleighMartin/status/1878818484303417711. Substack: https://postpandemicchanges.substack.com/p/what-is-hybrid-legislative-voting?r=2pw9k.

Martin, Roleigh. “If DOGE audits Congress, scan its history to reveal why it can’t cut debt.” Published February 23, 2025. The author’s original presentation of the central empirical claim of this chapter: that fiscal year 1957 is the last fiscal year in which the total outstanding national debt of the United States fell below the level recorded the previous fiscal year, sourced from the U.S. Treasury Historical Debt Outstanding dataset. X article: https://x.com/RoleighMartin/status/1893529780290756633.

Martin, Roleigh. “The Congressional Research Institute Findings on how Congress became controllable in 1971.” Published April 5, 2025. The author’s summary of the political-economy work of James D’Angelo and the Congressional Research Institute documenting the post-1970 explosion of lobbying activity and its causal relationship to the recorded-voting reforms of the Legislative Reorganization Act of 1970. Includes the 175-to-2,445 firm growth figure (1971–1982) cited in this chapter. X article: https://x.com/RoleighMartin/status/1908691753432940663. Substack: https://postpandemicchanges.substack.com/p/the-congressional-research-institute?r=2pw9k.

Martin, Roleigh. “Truck vs Calculator Prices: The Hidden War on Expired Auto Patents.” Published June 21, 2025. X article: https://x.com/RoleighMartin/status/1936617506203947297. Substack: https://postpandemicchanges.substack.com/p/truck-vs-calculator-prices-the-hidden.

Martin, Roleigh. “The Vanishing Promise of Expired Patents — Lessons from History & Abroad for MALA.” Published October 26, 2025. X article: https://x.com/RoleighMartin/status/1982462873310773675. Substack: https://postpandemicchanges.substack.com/p/the-vanishing-promise-of-expired.

Martin, Roleigh. “Saddles Unshackled the Frontier — Why Expired Patent Bans Prevent Cheap Wheels Today!” Published November 1, 2025. X article: https://x.com/RoleighMartin/status/1984810182367482096. Substack: https://postpandemicchanges.substack.com/p/saddles-unshackled-the-frontierwhy.

Martin, Roleigh. “Countries with newly remade patent-expired cars and trucks for sale.” Published November 23, 2025, on X and on Substack. The article and its embedded primary-source URLs constitute the source apparatus for the international comparison the cars-and-trucks chapter relies on, including the Mahindra Thar 3-door and Thar ROXX, the Mahindra Bolero, the Maruti Suzuki Alto K10, the Iran Khodro Soren Plus and Arisun 2 Pickup, the ZX Auto Grand Tiger and Terralord, the Force Motors Force Gurkha, and Mobius Motors Kenya. The article’s embedded Grok analysis at https://x.com/i/grok/share/dDvRNvFNqsfdA59JhT54zOGCO is the source of the rupee-to-dollar and other USD-equivalent conversions used in the chapter and in the corresponding manufacturer-page endnotes. X article: https://x.com/RoleighMartin/status/1992685422736982434. Substack: https://postpandemicchanges.substack.com/p/countries-with-newly-remade-patent.

Martin, Roleigh. “The $22 Trillion Constitutional Theft—How Energy & Safety Laws Created Perpetual Patents.” Published December 6, 2025. X article: https://x.com/RoleighMartin/status/1997437445906792931. Substack: https://postpandemicchanges.substack.com/p/the-22-trillion-constitutional-theft.

Martin, Roleigh. “Can Federal Courts Strike Down Regulations That Turned Expired Patents into Permanent Monopolies?” Working paper, February 2, 2026. SSRN Abstract ID 6166206. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6166206.

Martin, Roleigh. “The Constitutional Violation No One Reports: Who Profits from Blocking Expired-Patent Competition?” Published February 15, 2026. X article: https://x.com/RoleighMartin/status/2023078933416071558. Substack: https://postpandemicchanges.substack.com/p/the-constitutional-violation-no-one.

Martin, Roleigh. “Farm Associations Open a Fifth Front Against the Lawfare War Against Expired Patents.” Published March 28, 2026. The source article for the farm-equipment chapter, opening the fifth plaintiff category (farm associations) to complement the four prior categories developed across the campaign (county sheriffs, governors and state attorneys general, tribal nations, and public-interest amicus organizations). Anchors the Mahindra and John Deere price comparisons, the regulatory-ratchet diagnostic for tractors, the USDA Food Dollar Series analysis, and the candidate-plaintiff treatment of the National Farmers Union, Farm Action, the Farm and Ranch Freedom Alliance, the National Sustainable Agriculture Coalition, and the National Family Farm Coalition. X article: https://x.com/RoleighMartin/status/2038039210737676595. Substack: https://postpandemicchanges.substack.com/p/farm-associations-open-a-fifth-front.

Martin, Roleigh. “Semi-Truck Owner-Operators Pay $5,000 for a Filter the Constitution Says Should Not Be Required.” Post-Pandemic Changes (Substack), April 1, 2026. The author's long-form treatment of the diesel-particulate-filter mandate as a Patent Clause violation, expanded in this chapter. The chapter develops and independently re-sources the article’s central material — the five-thousand-dollar filter as the worked example, the eighty-three percent pre-2007 particulate-matter reduction achieved through engine design alone, and the international price comparison against the Sinotruk HOWO. https://postpandemicchanges.substack.com/.

Martin, Roleigh. “The Forgotten Half of the Patent Clause: What was Promised When Patents Expired.” Published April 10, 2026. The published companion methodology article for the three-tier breakdown of the expired-patent economy. Its section “A note on the methodology beneath the two charts” sets out the derivation: the major categories of expired-patent commerce were weighted by their share of U.S. manufacturing output and household consumer spending, drawing on the Bureau of Labor Statistics Consumer Expenditure Survey, the Bureau of Economic Analysis Industry Economic Accounts, and the Census Bureau Annual Survey of Manufactures. Each sub-category was then assigned to a tier by degree of regulatory interference: blocked where post-1970 federal regulation makes domestic manufacture of expired-patent designs effectively impossible, hindered where FDA and medical-licensing capture limits but does not fully block use of expired patents, and flowing where the public domain is at least partially honored. A second chart splits the flowing tier between imports and domestic production using Census Bureau International Trade data. The article states plainly that these are informed estimates, order-of-magnitude accurate but not rigorously calculated, because the counterfactual they represent — what American commerce would look like if regulation did not block use of the expired-patent public domain — is inherently difficult to specify with precision. Readers wanting harder numbers for any single sub-category are directed there to the Census Bureau Annual Survey of Manufactures and the International Trade Administration’s trade data, which together supply the authoritative federal baseline. X article: https://x.com/RoleighMartin/status/2042726307931287738. Substack: https://postpandemicchanges.substack.com/p/the-forgotten-half-of-the-patent-e04.

Martin, Roleigh. Interview by Gord Magill. *Autonomous Truck(er)s* (Substack), released June 13, 2026. “The Unconstitutional Highway — on the “Forgotten Half of the Patent Clause” with Roleigh Martin.” A roughly ninety-minute conversation, recorded in late May 2026 and released after editing on June 13, in which the author and the foreword writer discuss the book’s constitutional argument; Magill states that he has been asked to write the foreword and that his podcast listeners and Substack readers will be among the first to

know when the book finds a publisher. Cited in the foreword placeholder as the source of Magill's public commitment to write the foreword. Available at https://substack.com/@autonomoustruckers/posts; also on Spotify at https://open.spotify.com/episode/4K75yoUnFD2G0y2YA5t3wK. #Martin-Podcast

Martin, Roleigh. Interview by Timothy Dooner. *The Dooner Show* (YouTube), September 23, 2026. "The Legal Loophole That Could Bring Back Old-School Trucks." A conversation in which the author explains the book's constitutional argument to the trucking audience of Dooner's program. Available at https://www.youtube.com/watch?v=zjM_nb8W23M&t=100s. #Martin-Podcast

Maruti Suzuki India Limited. "Alto K10." Maruti Suzuki product page. Source for the Indian-market retail price range of the Maruti Suzuki Alto K10, a compact city car descended from the 1979 Suzuki Alto kei car and built on Suzuki's HEARTECT platform with patent-expired mechanical underpinnings, at $3,900–$5,700 USD-equivalent in 2025 (₹3.69 lakh base STD to ₹5.45 lakh top VXi Plus AGS, ex-showroom, post-GST revision of September 22, 2025). Earlier drafts of this book incorrectly described the Alto K10 as built on the Fiat 126 design lineage; the actual lineage is the Japanese Suzuki Alto kei car (model SS30V, 1979 onward), with no platform, engineering, corporate, or design relationship to the Fiat 126. https://www.marutisuzuki.com/arena/alto-k10. USD-equivalent figures derived from the author's published rupee-to-dollar conversion analysis at https://x.com/i/grok/share/dDvRNvFNqsfdA59JhT54zOGCO.

Massachusetts v. Environmental Protection Agency, 549 U.S. 497 (2007). https://supreme.justia.com/cases/federal/us/549/497/.

McCormick Harvesting Machine Co. v. Aultman, 169 U.S. 606 (1898). https://supreme.justia.com/cases/federal/us/169/606/.

McKinsey & Company. McKinsey Center for Future Mobility, autonomous-mobility research. As reported in trade coverage of the firm's findings, McKinsey senior partner Philipp Kampshoff stated that, as robotaxi operating costs fall and the cost of owning a

personal vehicle rises, owning a car would no longer make economic sense for roughly half the residents of larger cities. Cited in the autonomous-vehicle chapter on the projected displacement of private car ownership.
https://www.wardsauto.com/autonomous-adas/robotaxi-adas-growth-fueled-by-falling-tech-costs.

MedImmune, Inc. v. Genentech, Inc., 549 U.S. 118 (2007).
https://supreme.justia.com/cases/federal/us/549/118/.

MEMA, The Vehicle Suppliers Association. “The History of Remanufacturing.” Industry timeline. Source for the twentieth-century U.S. remanufacturing record cited in the objections chapter: Ford’s 1934 dealer-level engine-exchange program establishing core-exchange logistics, the rise of production engine rebuilders, and the 1966 decision by Detroit Diesel and Cummins to assign monetary value to used cores and stand up factory remanufacturing programs.
https://www.mema.org/reman-history/timeline.

Menominee Tribe of Indians v. United States, 391 U.S. 404 (1968).
https://supreme.justia.com/cases/federal/us/391/404/.

Metzl, Jamie. “Why I Co-Authored a Book with ChatGPT.” *The Free Press*, April 21, 2026. Metzl’s defense of naming GPT-5 as named co-author of his book *The AI Ten Commandments: A New Moral Code for Humanity* (Transcend Books, 2026) — the first major work of nonfiction to do so. Metzl describes a workflow in which the human serves as sole conceptualizer, framer, and editorial controller while the AI serves as drafting and synthesis partner across thousands of iterative cycles. Cited in this book’s note on AI-assisted authorship as a contemporary publishing precedent that arrived alongside this book; the methodologies closely parallel each other, the principal difference being that Metzl’s book presents an AI synthesis at its core while this book presents a human-discovered constitutional argument that the AI helped execute.
https://www.thefp.com/p/why-i-co-authored-a-book-with-chatgpt

Michelman, Frank I. “Property, Utility, and Fairness: Comments on the Ethical Foundations of ‘Just Compensation’ Law.” *Harvard Law Review* 80 (1967): 1165–1258. Companion analytical foundation to Joseph Sax’s 1964 Yale article, addressing the line between compensable and noncompensable regulatory burdens through reference to settled expectations and fairness norms. Cited in the chapter on objections considered alongside the Sax articles as the doctrinal terrain the Takings Clause prong of the book’s argument must occupy.
https://www.jstor.org/stable/1339276.

Microsoft Security Response Center. CVE-2026-32194 and CVE-2026-32191, “Microsoft Bing Images Remote Code Execution Vulnerability.” Released March 19, 2026. Two Critical advisories, CVSS 3.1 base score 9.8, for command-injection flaws in Microsoft Bing Images, each acknowledging XBOW. XBOW’s technical account, “A Shell Is Worth a Thousand Images: Bing Images RCEs” (July 23, 2026, https://xbow.com/blog/bing-images-rce-vulnerabilities), maps CVE-2026-32194 to the public “search by image” upload path. Cited in the preface as one of three cases in which a machine found, by reading a system’s rules, a flaw its expert defenders had not.
https://msrc.microsoft.com/update-guide/vulnerability/CVE-2026-32194 and
https://msrc.microsoft.com/update-guide/vulnerability/CVE-2026-32191.

Million, Matthieu, Jean-Christophe Lagier, Philippe Gautret, et al. “Early Treatment of COVID-19 Patients with Hydroxychloroquine and Azithromycin: A Retrospective Analysis of 1,061 Cases in Marseille, France.” Travel Medicine and Infectious Disease 35 (May/June 2020): 101738. The larger Marseille cohort. Not retracted.
https://doi.org/10.1016/j.tmaid.2020.101738.

Ministry of Finance and State Taxation Administration of the People’s Republic of China. Announcement on Further Improving the Pre-tax Additional Deduction Policy for Research and Development Expenses, Announcement No. 7 of 2023. Issued March 26, 2023, effective retroactively from January 1, 2023, with no expiry date. Permits an additional deduction of 100 percent of qualifying research expenditure, for a total pre-tax deduction of 200 percent. The rate was raised to 100 percent for manufacturing enterprises in 2021

before being extended to all eligible enterprises, an instance of a multiplier tuned by sector before being generalized. Documented in Dezan Shira & Associates, “China R&D Expenses Pre-Tax Super Deduction.” https://www.china-briefing.com/news/china-rd-expenses-pre-tax-super-deduction-explainer/.

Minnesota v. Rettmann, Scott County District Court, Minnesota (1998). State enforcement action under deceptive-trade-practices and consumer-fraud law against the operator of a Rife frequency device, brought by Minnesota Attorney General Hubert Humphrey III. Procedural summary at Quackwatch. https://quackwatch.org/consumer-education/news/rife/.

Montreal Protocol on Substances that Deplete the Ozone Layer, 1522 U.N.T.S. 3 (September 16, 1987, as amended). Multilateral environmental agreement implemented in U.S. law primarily through Title VI of the Clean Air Act Amendments of 1990 (42 U.S.C. §§ 7671 et seq.). Phases down ozone-depleting substances including CFCs (R-12 phaseout, 1995) and HCFCs (R-22 phaseout, 2010–2020). https://ozone.unep.org/treaties/montreal-protocol.

Müller, Julian F., and Jan Gogoll. “Should Manual Driving Be (Eventually) Outlawed?” *Science and Engineering Ethics* 26, no. 3 (June 2020): 1549–1567. A peer-reviewed paper in applied ethics arguing that, in a future in which autonomous cars are significantly safer than human drivers, manually driven cars should for moral reasons be banned from ordinary traffic. Cited in the autonomous-vehicle chapter on the documented movement to outlaw human driving. https://doi.org/10.1007/s11948-020-00190-9.

Morriss, Jeff Charles, et al. U.S. Patent 5,694,555, filed March 19, 1996 (claiming priority to a parent application filed October 31, 1994), issued December 2, 1997. Assigned to Intel Corporation. One of the foundational Intel-assigned USB patents; expired. https://patents.google.com/patent/US5694555A/en.

Morriss, Jeff Charles, et al. U.S. Patent 5,909,556, filed December 1, 1997 (claiming priority to a parent application filed October 31, 1994), issued June 1, 1999. Assigned to Intel Corporation. One of the foundational Intel-assigned USB patents; expired April 2018. https://patents.google.com/patent/US5909556A/en.

Moss, Ralph W. *The Cancer Industry: The Classic Exposé on the Cancer Establishment*. Updated edition. Brooklyn: Equinox Press, 1996. https://www.amazon.com/Cancer-Industry-Classic-Establishment-Equinox/dp/1881025098.

Moss, Ralph W. *Doctored Results: The Suppression of Laetrile at Sloan-Kettering Memorial Cancer Center*. Brooklyn: Equinox Press, 2014. https://www.amazon.com/Doctored-Results-Suppression-Sloan-Kettering-Memorial/dp/1881025519.

Mossoff, Adam. "Who Cares What Thomas Jefferson Thought About Patents? Reevaluating the Patent 'Privilege' in Historical Context." *Cornell Law Review* 92, no. 5 (2007): 953–1012. https://scholarship.law.cornell.edu/clr/vol92/iss5/2.

Motion Picture Patents Co. v. Universal Film Manufacturing Co., 243 U.S. 502 (1917). Held (6—3) that a patentee may not, by notice attached to a patented machine, extend the monopoly to unpatented supplies used with it; an unconditional sale exhausts the patent right. In the course of that holding the Court stated, as a rule settled since *Pennock v. Dialogue* (1829), that the primary purpose of the patent laws is not the creation of private fortunes for patent owners but the promotion of the progress of science and the useful arts. Cited in the Patent Clause prong chapter as the decision elevating that reasoning toward settled rule. https://supreme.justia.com/cases/federal/us/243/502/.

Muller v. Oregon, 208 U.S. 412 (1908). Justice Brewer, for a unanimous Court, upheld an Oregon statute limiting women in factories and laundries to ten hours a day. The decision is

cited in this book for the brief rather than the holding; see the entry under Brandeis and Goldmark for the filing that gave the Brandeis brief its name.
https://supreme.justia.com/cases/federal/us/208/412/.

Murray, H. J. R. *A History of Chess*. Oxford: Clarendon Press, 1913. The classical and exhaustive scholarly reference on the history of the chess family, supporting the long-standing mainstream view that the game originated as Indian *chaturanga* around the sixth century CE, spread westward via Persian *shatranj*, and that Chinese Xiangqi and Korean Janggi developed as eastward derivatives or parallels of the same ancestor. Cited in the preface as the principal representative of the mainstream Indian-origin position against which Li (1998) argues. Public-domain digital edition via the Internet Archive.
https://archive.org/details/historyofchess0000hjrm.

Natanson, Hannah, Jeff Stein, Dan Diamond, and Rachel Siegel. “DOGE Builds AI Tool to Cut 50 Percent of Federal Regulations.” *The Washington Post*, July 26, 2025. Reports, from a July 1 PowerPoint and four officials, that the “DOGE AI Deregulation Decision Tool” was built to analyze roughly 200,000 federal regulations for those no longer required by law, had made decisions on 1,083 regulatory sections at the Department of Housing and Urban Development, and had written the Consumer Financial Protection Bureau’s deregulations; HUD employees said it misread the language of the law on several occasions. Cited in the preface as the nearest existing use of AI on the regulatory state — asking whether a statute still requires a rule, not whether the statute and rule together do what the Constitution forbade.
https://www.washingtonpost.com/business/2025/07/26/doge-ai-tool-cut-regulations-trump/.

National Aeronautics and Space Administration. “Artificial Intelligence, NASA Data Used to Discover Eighth Planet Circling Distant Star.” News release, December 14, 2017. Reports the discovery of Kepler-90i by a neural network built by Christopher Shallue, a software engineer at Google, trained on fifteen thousand previously vetted Kepler signals and applied to 670 star systems already examined by professional astronomers. Shallue originated the approach and was not an astronomer; the astronomy was supplied by Andrew Vanderburg, then a NASA Sagan postdoctoral fellow at the University of Texas at Austin,

who worked the problem with him and co-authored the resulting paper. The finding tied Kepler-90 with the solar system for the most planets known to orbit a single star. Cited in the preface as an instance in which an outsider supplied the method and specialists inside the field supplied the domain judgment.
https://www.nasa.gov/news-release/artificial-intelligence-nasa-data-used-to-discover-eighth-planet-circling-distant-star/

National Appliance Energy Conservation Act of 1987, Pub. L. No. 100-12, 101 Stat. 103 (March 17, 1987). Codified primarily at 42 U.S.C. §§ 6291–6309, with the anti-backsliding provision at § 6295(o)(1). Senate Roll Call 28, February 17, 1987 (89–6); House voice vote, March 3, 1987.
https://www.law.cornell.edu/uscode/text/42/chapter-77.

National Association of Home Builders (NAHB). *Study of Life Expectancy of Home Components*. Washington, D.C.: NAHB and Bank of America Home Equity, February 2007. Industry-standard reference for residential appliance and component lifespans, compiled from a 2006 survey of manufacturers, trade associations, and researchers. The figures represent industry-wide averages assuming normal usage and adherence to maintenance schedules; the study notes that actual service life depends substantially on installation quality, usage frequency, and climate conditions, and that appliances are often replaced for reasons of technological advancement or changing consumer preferences well before they cease to function. Cited in the cars-and-trucks chapter for the appliance replacement-cycle figures, including the refrigerator (13 years), dishwasher (9 years), clothes dryer (13 years), microwave oven (9 years), gas range (15 years), and electric range (13 years). The chapter's per-appliance figures are conservative midpoints calibrated against the NAHB single-point estimates and the InterNACHI ranges.
https://hcmuddox.com/wp-content/uploads/2022/10/nahb20study20of20life20expectancy20of20home20components.pdf.

National Association of Landscape Professionals (NALP). Organizational homepage. The national trade association for the landscape, lawn care, irrigation, and tree care trades,

headquartered in Fairfax, Virginia. Formed January 1, 2005 by the merger of the Associated Landscape Contractors of America (founded 1961) and the Professional Lawn Care Association of America as the Professional Landcare Network, and rebranded to its present name April 1, 2015. Identified in the plaintiff-categories chapter as the trade body for the landscaping and grounds-maintenance operator, whose governing hook is EPA Part 1054, the standard for small spark-ignition engines. The association has actively opposed restrictive small-engine emissions rules and state-level gasoline-equipment bans. Membership figures appear in two forms and should not be conflated: roughly 100,000 landscape industry professionals in membership, within an industry the association describes as approaching one million professionals.
https://www.landscapeprofessionals.org/.

National Automobile Chamber of Commerce. *Automobile Facts and Figures*, 1926 edition. Industry trade-association annual compilation of U.S. motor-vehicle production, sales, and registration data. Source for the 1925 U.S. motor-vehicle production figures cited in the cars-and-trucks chapter: 3,735,171 passenger cars and 4,265,830 total motor vehicles (passenger cars plus 530,659 trucks). The NACC was the primary industry body collecting and publishing automotive production statistics in the 1920s; comprehensive retail-sales data was not compiled and made public until after World War II, so the chapter's reference to four million new cars sold in 1925 represents factory production, which the industry of the period used interchangeably with factory sales. Archived by the Federal Reserve Bank of St. Louis, FRASER digital library, alongside related commercial-car registration series.
https://fred.stlouisfed.org/series/M0146DUSM543NNBR.

National Electrical Manufacturers Association v. United States Department of Energy, No. 17-1341 (4th Cir.). Petition for review filed by NEMA in 2017 challenging the U.S. Department of Energy's January 2017 expanded definitions of general service lamp and general service incandescent lamp. Settled by agreement on July 11, 2017; the settlement led directly to the Trump-administration September 2019 rollback of the expanded definitions (84 Fed. Reg. 46661, September 5, 2019). Cited in the healthy-lighting chapter's bootleggers-and-Baptists section as documentary evidence that NEMA's posture across the

2007–2024 arc included active litigation against DOE on the scope of the rule, complicating any straight-line account of industry support for the regulatory architecture. https://www.law.nyu.edu/sites/default/files/settlement-agreement.pdf.

National Family Farm Coalition (NFFC). Organizational homepage. Coalition representing family farmers across dozens of state-level farmer organizations; multi-state structure provides the geographic and circuit diversity that strengthens a constitutional filing. Identified in the farm-equipment chapter as one of five candidate plaintiff organizations under associational standing.
https://nffc.net/.

National Farmers Union (NFU). Organizational homepage. Founded 1902; represents more than 200,000 family farms and ranches across 33 state divisions through a grassroots policy process. President Rob Larew, Vice President Jeff Kippley. The NFU 124th Anniversary Convention (March 7–9, 2026, New Orleans) highlighted the 46 percent year-over-year increase in farm bankruptcy filings reported by the American Farm Bureau Federation. Identified in the farm-equipment chapter as the strongest single candidate plaintiff among the five farm associations developed under associational standing.
https://nfu.org/.

National Highway Traffic Safety Administration. “Bicycle Safety.” U.S. Department of Transportation. Official federal portal for bicycle safety statistics, educational materials, and crash data drawn from the agency’s Fatality Analysis Reporting System (FARS). Cited in the bicycles section of the modern-bargain chapter as the federal-data anchor for cyclist fatality trends and as the agency-level confirmation that no federal product safety regulation operates on the bicycle itself.
https://www.nhtsa.gov/road-safety/bicycle-safety.

National Institute of Mental Health. “Seasonal Affective Disorder.” NIMH Health Publications, U.S. Department of Health and Human Services. The current authoritative federal mental-health publication on Seasonal Affective Disorder (SAD), a recurring depressive disorder linked to seasonal variation in light exposure. NIMH states that

millions of Americans experience SAD, with prevalence rising at higher latitudes and higher rates in women. Treated with controlled-spectrum light therapy that requires careful filtering against the blue-shifted output of standard residential LED lighting. Cited in the healthy-lighting chapter's coalition section as the federal-source anchor for the SAD-affected population.
https://www.nimh.nih.gov/health/publications/seasonal-affective-disorder.

National Psoriasis Foundation. "Phototherapy." NPF patient-resource page. Authoritative patient-organization summary of phototherapy as an approved treatment modality for psoriasis. Documents that ultraviolet B (UVB) light, present in natural sunlight and delivered via therapeutic lamps, is an effective treatment for psoriasis because it penetrates the skin and slows the excessive growth of affected skin cells. Psoriasis affects approximately 7.5 to 8 million American adults per the NPF's prevalence data, of whom a non-trivial fraction depends on UVB phototherapy delivered through specialized incandescent and halogen lamp architectures. Cited in the healthy-lighting chapter's coalition section.
https://www.psoriasis.org/phototherapy/.

National Sustainable Agriculture Coalition (NSAC). Organizational homepage. Coalition of grassroots organizations advocating for sustainable and organic farmers; published analyses document that four firms control 45 percent of global farm equipment sales. Identified in the farm-equipment chapter as one of five candidate plaintiff organizations under associational standing.
https://sustainableagriculture.net/.

National Technology Transfer and Advancement Act of 1995, Pub. L. No. 104-113, 110 Stat. 775. Directs federal agencies to use technical standards developed by private-sector voluntary consensus standards bodies (such as SAE, ASTM, UL, and ANSI) where practicable, in lieu of developing federal standards from scratch. Implemented through OMB Circular A-119.
https://www.law.cornell.edu/uscode/text/15/272.

Neri, Cassio. “Shorter and Faster than Sort3AlphaDev.” arXiv:2307.14503 (2023). Supplies counterexamples to the claim that no shorter three-element sorting program exists.

New Civil Liberties Alliance (NCLA). Pro bono litigation organization founded in 2017, headquartered in Washington, DC. Focused on constitutional challenges to administrative-state overreach. NCLA was lead counsel on *Relentless, Inc. v. Department of Commerce*, which the Supreme Court consolidated with *Loper Bright Enterprises v. Raimondo* — the 2024 decision that ended Chevron deference. Cited in the trucking chapter as one of three public-interest legal foundations carrying constitutional cases of the kind the Patent Clause argument requires at no cost to the plaintiff.
https://nclalegal.org/.

Newbold, Charles. U.S. Patent for the cast-iron plow, granted 1797 (term expired 1811 under the fourteen-year rule of the Patent Act of 1790). The single-piece cast-iron moldboard plow that, after expiration, became the dominant American farming implement of the early-to-mid nineteenth century. The cast-iron plow entry in the chapter’s eight-product table.
https://en.wikipedia.org/wiki/Charles_Newbold.

Nichia Corporation and the LED-to-laser lighting succession. The efficient blue gallium-nitride light-emitting diode was developed by Shuji Nakamura at Nichia Corporation of Japan in the early 1990s; Nichia paired a blue chip with a yellow phosphor to produce the first commercial white LED in 1996, the architecture in most household LED bulbs today. Nakamura shared the 2014 Nobel Prize in Physics for the blue LED. Nichia held the core white-LED patents, earned a reported fortune from them, and litigated to defend them; those patents, filed in the 1990s, have only recently begun to expire. As the LED portfolio lapses, the same inventor and the same company have moved to laser-based white light: Nakamura’s post-LED work through SoraaLaser (SLD Laser) and Nichia’s GigaWhite blue-laser-plus-phosphor module, developed for automotive, specialty, and industrial lighting, with Nichia in 2016 publicly prioritizing laser diodes as the next lighting source. Cited in the healthy-lighting chapter for the observation that the regulatory withdrawal

pattern recurs one technology at a time, each expired-patent design displaced as its patented successor arrives.
https://led-ld.nichia.co.jp/en/product/ld_gigawhite.html. Nobel Prize background:
https://en.wikipedia.org/wiki/Shuji_Nakamura

Nixon v. United States, 506 U.S. 224 (1993). The Supreme Court's leading modern application of the political-question doctrine to internal Congressional procedure: the Senate's procedure for trying an impeachment was held nonjusticiable because the Constitution textually commits the matter to the Senate and the courts lack judicially manageable standards for second-guessing the chamber's procedural choices. Cited in the Hybrid Voting chapter as the doctrinal anchor for the proposition that federal courts have shown strong reluctance to police the chamber's internal voting mechanics, including the interpretive scope of the Yeas-and-Nays override in Article I, Section 5, Clause 3.
https://supreme.justia.com/cases/federal/us/506/224/.

Nollan v. California Coastal Commission, 483 U.S. 825 (1987). The Supreme Court's holding that regulatory conditions imposed on the use of property must bear an "essential nexus" to the harm the regulation claims to address, supplying the first prong of the proportionality test later completed in *Dolan v. City of Tigard* (1994) and reaffirmed in *Sheetz v. County of El Dorado* (2024).
https://supreme.justia.com/cases/federal/us/483/825/.

Nordlinger v. Hahn, 505 U.S. 1 (1992). The Supreme Court's decision upholding California's acquisition-value property-tax scheme under Proposition 13, which assessed newly purchased property at its purchase price while capping annual increases on long-held property at two percent. Justice Blackmun wrote for the Court, which found the classification rationally related to legitimate state interests and confined *Allegheny Pittsburgh Coal Co. v. County Commission of Webster County* (1989) to the rare case in which the facts precluded any plausible inference that an acquisition-value policy explained the disparity. Justice Thomas concurred in part while stating that *Allegheny Pittsburgh* could not be distinguished; Justice Stevens dissented. The decision is the strongest

authority against pleading this book's equality argument as a standalone equal-protection claim, and it is taken up in that posture in the objections chapter.
https://supreme.justia.com/cases/federal/us/505/1/.

Northeast States for Coordinated Air Use Management (NESCAUM). "Heavy-Duty Omnibus Low-NOx Rule: Frequently Asked Questions." September 2024. The current authoritative public summary of state adoption of the California Air Resources Board Heavy-Duty Omnibus (Low-NOx) regulation under Clean Air Act § 177. Documents the nine states that have specifically adopted the CARB Heavy-Duty Omnibus standards as of 2024–2025: Colorado (model year 2027), Massachusetts (MY 2025), New Jersey (MY 2027), New Mexico (MY 2027), New York (MY 2026), Oregon (MY 2025), Rhode Island (MY 2027), Vermont (MY 2026), and Washington (MY 2026). The broader seventeen-state Section 177 group that adopted CARB light-duty (LEV/ZEV/ACC II) or general vehicle standards is distinct from the nine-state Heavy-Duty Omnibus group. Source for the trucking chapter's reference to the nine states whose Heavy-Duty Omnibus adoption layered the federal mandate with a parallel state regime.
https://www.nescaum.org/documents/HD-Low-NOx-Omnibus-FAQ_website-version_clean_09-23-24.pdf.

Novak Conversions. *The Novak Guide to the Spicer Model 18 Transfer Case*. Reference history of the Jeep transfer case. The civilian Jeep's core drivetrain was supplied and patented by independent component makers rather than by Willys–Overland or its successors: the Borg-Warner T-90 three-speed transmission and the Spicer (Dana) Model 18 transfer case and Dana axles. The underlying utility patents on these mid-century components expired under the seventeen-year-from-grant term then in force, placing the CJ drivetrain in the public domain by the late 1980s.
https://www.novak-adapt.com/knowledge/transfer-cases/model-18/.

Nunes, Ashley, and Kristen D. Hernandez. "Autonomous Vehicles and Public Health: High Cost or High Opportunity Cost?" Massachusetts Institute of Technology, 2019. A full-cost study finding that a conventional human-driven vehicle costs about $0.72 per mile to operate, while a robotaxi—once licensing, insurance, cleaning, remote oversight, and the

low utilization of the taxi fleet are counted—costs between $1.58 and $6.01 per mile. Cited in the autonomous-vehicle chapter on robotaxi economics. https://www.researchgate.net/publication/332743767_Autonomous_Vehicles_and_Public_Health_High_Cost_or_High_Opportunity_Cost.

Occupational Safety and Health Act of 1970, Pub. L. No. 91-596, 84 Stat. 1590. Codified as amended at 29 U.S.C. §§ 651 et seq. Establishes federal workplace safety standards through the Occupational Safety and Health Administration; the General Duty Clause is at 29 U.S.C. § 654. https://www.law.cornell.edu/uscode/text/29/chapter-15.

Occupational Safety and Health Administration. “29 CFR 1928.51 — Roll-Over Protective Structures for Agricultural Tractors.” U.S. Department of Labor. Source for the first of the three regulatory layers discussed in the farm-equipment chapter: the 1976 OSHA rollover rule requiring rollover protective structures and seatbelts on tractors manufactured after October 25, 1976, when operated by employees. https://www.osha.gov/laws-regs/regulations/standardnumber/1928/1928.51.

Oil States Energy Services, LLC v. Greene’s Energy Group, LLC, 584 U.S. 325 (2018). Seven-to-two decision holding that inter partes review, the Patent Office proceeding for reconsidering the validity of an issued patent, violates neither Article III nor the Seventh Amendment. Justice Thomas’s majority opinion characterized a patent as a public franchise and patent validity as a matter of public rights that Congress may assign to an executive agency; it described the holding as narrow, declined to decide whether inter partes review may be applied to patents issued before the Leahy-Smith America Invents Act, and stated that the decision should not be read as holding that patents are not property for purposes of the Due Process Clause or the Takings Clause. In the same passage the majority repeated, from its own older cases, that the grant of a patent is a matter between the public, who are the grantors, and the patentee, and that in issuing a patent the Patent Office takes from the public rights of immense value and bestows them upon the patentee. Justice Breyer, joined by Justices Ginsburg and Sotomayor, concurred. Justice Gorsuch, joined by Chief Justice Roberts, dissented on the ground that an issued patent is private property revocable only by

an Article III court. Cited in the chapter on objections considered as adverse authority on the government's power over the patent grant, and distinguished there as a decision about the grant side of the Clause that says nothing about the invention after its patent expires. https://supreme.justia.com/cases/federal/us/584/16-712/case.pdf.

Old Tacoma Marine. "Fairbanks-Morse." Marine engine history page. States that after the expiration of Rudolf Diesel's American licence in 1912, Fairbanks-Morse entered the large-engine business, its Model Y semi-diesel becoming a standard workhorse in sugar, rice, timber, and mine mills. The page's 1912 date refers to Diesel's American licence; the underlying patent, U.S. Patent No. 608,845, ran to 1915 (see that entry). Cited in the trucking chapter for Fairbanks-Morse as an American firm that entered the diesel-engine business after Diesel's American patent rights ran out. https://www.oldtacomamarine.com/engines/fairbanks/index.html.

Olson, Mancur. *The Logic of Collective Action: Public Goods and the Theory of Groups*. Cambridge, MA: Harvard University Press, 1965. https://www.hup.harvard.edu/books/9780674537514.

Ostrom, Elinor. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge: Cambridge University Press, 1990. https://doi.org/10.1017/CBO9780511807763. See Endnote Commentary.

Owner-Operator Independent Drivers Ass'n, Inc. v. United States Department of Transportation, No. 15-3756 (7th Cir. Oct. 31, 2016), cert. denied, 137 S. Ct. 2246 (2017). Seventh Circuit decision rejecting OOIDA's Fourth Amendment challenge to the FMCSA Electronic Logging Device rule. The court held that the trucking industry's status as a "pervasively regulated industry" under *New York v. Burger*, 482 U.S. 691 (1987), permits the warrantless inspections the ELD rule effectively requires. Cited in the trucking-section of the surveillance chapter as the doctrinal anchor for the proposition that the Fourth Amendment frame has lost on the question of continuous federally-mandated driver

surveillance — and as the open door for the Patent Clause and Takings Clause prongs this book develops.
https://caselaw.findlaw.com/court/us-7th-circuit/1752477.html.

Owner-Operator Independent Drivers Association (OOIDA). “2024 Freight Rate Survey: Final Report.” OOIDA Foundation, January 2025. The current authoritative source for OOIDA’s self-described membership: approximately 150,000 members in all fifty states, collectively owning and operating more than 240,000 individual heavy-duty trucks. Source for the trucking chapter’s 150,000-member figure used in the standing section.
https://www.ooida.com/wp-content/uploads/2025/01/2024-Freight-Rate-Survey-Final-Report.pdf.

Pacific Legal Foundation. Pro bono constitutional litigation organization founded in 1973, based in Sacramento, California. Has won eighteen of its last twenty Supreme Court cases, including *Sheetz v. County of El Dorado* (2024), cited above. Focused on property rights, economic liberty, and constitutional limits on regulatory action. Cited in the trucking chapter as one of three public-interest legal foundations carrying constitutional cases of the kind the Patent Clause argument requires at no cost to the plaintiff.
https://pacificlegal.org/.

Palmer, E. D. Histological study of gastric biopsies from a thousand subjects concluding that spiral organisms were contaminants introduced from the mouth, 1954. Cited in the chapter on outsiders and blind doctrine as the study that closed the question for thirty years. Known here through later reviews; the full citation has not been confirmed against the original.

Pardee, Joseph T. “Unusual Currents in Glacial Lake Missoula, Montana.” *Geological Society of America Bulletin* 53, no. 11 (1942): 1569-1600. Identified the water source that resolved the Scablands dispute in Bretz’s favour.

Patent Act of 1790, ch. 7, 1 Stat. 109 (April 10, 1790). Federal Reserve Archival System for Economic Research (FRASER), Federal Reserve Bank of St. Louis. The First Congress did not merely permit an exclusive right; it deployed national authority to secure one. Section 1

directed that letters patent issue “in the name of the United States,” signed by the President. Section 4 armed the right with a federal cause of action — damages assessed by a jury and forfeiture of the infringing articles to the person aggrieved, recoverable in the circuit courts of the United States. This is the affirmative, coercive sense of the Constitution’s verb “securing.” Note one correction often made in secondary summaries: the familiar treble-damages remedy (three times the price the patentee usually charged) is not in the 1790 Act; it belongs to the Patent Act of 1793, § 5, 1 Stat. 318.
https://fraser.stlouisfed.org/title/patent-act-1790-5734/fulltext.

Patent Act of 1861, ch. 88, 12 Stat. 246 (March 2, 1861). Extended the U.S. utility patent term from fourteen years to seventeen years from the date of grant. The seventeen-year rule remained in force until 1995, when the Uruguay Round Agreements Act harmonized U.S. patent terms with TRIPS at twenty years from the date of filing.
https://en.wikipedia.org/wiki/Term_of_patent_in_the_United_States.

Patent Clause Letter Campaign. Hub page for the campaign’s free, browser-based letter and brochure generators, which produce mailing packages addressed to county sheriffs, governors and state attorneys general, and tribal nations; readers can consult the hub to see whether a generator has been added for a given plaintiff category.
https://marti124.github.io/Patent_Letters/.

Patterson v. Kentucky, 97 U.S. 501 (1878). Foundational Supreme Court decision holding that the existence of a federal patent does not exempt the tangible product embodying the invention from state police-power regulation aimed at the safety of the public. The Kentucky statute at issue banned the sale of a patented “improved burning oil” that the state inspector had condemned as unsafe; the patentee’s argument that his federal patent placed the oil beyond state regulatory reach was rejected unanimously. Cited in the chapter on objections considered as the principal authority for the proposition (which the book accepts) that patents do not confer immunity from regulation.
https://supreme.justia.com/cases/federal/us/97/501/.

Peace Corps, Establishment of. Executive Order 10924, "Establishment and Administration of the Peace Corps in the Department of State," March 1, 1961, signed by President John F. Kennedy; authorized by Congress in the Peace Corps Act, Pub. L. No. 87-293, 75 Stat. 612, approved September 22, 1961. Cited in the remedy chapter as the verified historical instance of a President standing up a federal institution by executive order, later ratified by statute. National Archives, "Executive Order 10924: Establishment of the Peace Corps." https://www.archives.gov/milestone-documents/executive-order-10924.

Penn Central Transportation Co. v. City of New York, 438 U.S. 104 (1978). https://supreme.justia.com/cases/federal/us/438/104/.

Penn State Extension. "A Summary of Laws and Regulations Related to Agricultural Safety and Machinery." Synthesizing source for the federal regulatory landscape surrounding agricultural equipment manufacture and operation, drawn upon in the farm-equipment chapter's diagnostic of the three regulatory layers (OSHA, EPA, NHTSA). https://extension.psu.edu/a-summary-of-laws-and-regulations-related-to-agricultural-safety-and-machinery.

Pennock v. Dialogue, 27 U.S. (2 Pet.) 1 (1829). Held that an inventor who permits public use or sale of his invention before applying for a patent forfeits the right to one (an ancestor of the on-sale bar). In reasoning the Court has treated as settled ever since, Justice Story explained that the main object of the patent system is to promote the progress of science and useful arts, best served by giving the public at large the right to make, construct, use, and vend the invention at as early a period as possible, with due regard to the rights of the inventor. The opinion construes the Patent Act of 1793. Cited in the Patent Clause prong chapter for the public's half of the bargain, stated early and by a Madison appointee. https://supreme.justia.com/cases/federal/us/27/1/.

Pennsylvania Coal Co. v. Mahon, 260 U.S. 393 (1922). https://supreme.justia.com/cases/federal/us/260/393/.

Perkins, Jacob. U.S. Patent for a nail-making machine, granted 1795 (term expired 1809 under the fourteen-year rule of the Patent Act of 1790). The Perkins machine and its post-

expiration competitors collapsed the American price of nails by roughly seventy percent across the nineteenth century. The nail-making entry in the chapter's eight-product table. https://en.wikipedia.org/wiki/Jacob_Perkins.

Pipeline and Hazardous Materials Safety Administration. "Specification for DOT-406 Cargo Tank Motor Vehicles and Hazardous Materials Transportation Requirements." Codified at 49 C.F.R. Parts 171–180. U.S. Department of Transportation. Source for the distinct federal hazardous-materials regime governing fuel-tanker operators: bulk packaging above 119 gallons triggers hazmat and tanker-endorsement requirements, the cargo tank itself must meet the DOT-406 construction specification, and the driver must hold a hazardous-materials endorsement and clear a Transportation Security Administration background check. Cited in the trucker chapter's fuel-hauler subsection. https://www.ecfr.gov/current/title-49/subtitle-B/chapter-I/subchapter-C.

Poplin, Ryan, Avinash V. Varadarajan, et al. "Prediction of Cardiovascular Risk Factors from Retinal Fundus Photographs via Deep Learning." *Nature Biomedical Engineering* 2 (2018): 158-164. Age, sex, smoking status, systolic blood pressure and major adverse cardiac events predicted from the photograph alone. See Endnote Commentary.

Printz v. United States, 521 U.S. 898 (1997). https://supreme.justia.com/cases/federal/us/521/898/.

Railway Labor Executives' Association v. Gibbons, 455 U.S. 457 (1982). The Supreme Court's holding that Congress may not use the Commerce Clause to enact a non-uniform bankruptcy law, because permitting it would eradicate from the Constitution the uniformity limitation the Bankruptcy Clause places on the bankruptcy power. The specific-governs-the-general anchor for this book's argument that a limitation written into one clause of Article I, Section 8 is not escaped by legislating under another clause of the same section. *Gibbons* concerned the Bankruptcy Clause; carrying its reasoning to the Patent Clause's "limited Times" limitation is this book's step and not the Court's. https://supreme.justia.com/cases/federal/us/455/457/.

Ranalli, Brent, James D'Angelo, and David King. *The Sunshine Reforms and the Transformation of Congressional Lobbying*. Congressional Research Institute, 2018. https://congressionalresearch.org/Ranalli2018SunshineReformsAndTransformationOfLobbying.html. See Endnote Commentary.

Resource Conservation and Recovery Act of 1976, Pub. L. No. 94-580, 90 Stat. 2795. Codified as amended at 42 U.S.C. §§ 6901 et seq. Establishes the federal regulatory framework for the generation, transportation, treatment, storage, and disposal of solid and hazardous waste. Subtitle C governs hazardous waste; Subtitle D governs nonhazardous solid waste. https://www.law.cornell.edu/uscode/text/42/chapter-82.

Risch, Harvey A. "Early Outpatient Treatment of Symptomatic, High-Risk COVID-19 Patients That Should Be Ramped Up Immediately as Key to the Pandemic Crisis." American Journal of Epidemiology 189, no. 11 (November 2020): 1218–26. Yale School of Public Health. Not retracted; the journal published letters in opposition and the author's reply in the same issue. https://doi.org/10.1093/aje/kwaa093.

Risner, Genevieve. "Parts of a Western Saddle and the Variations." Buffalo Bill Center of the West (Cody, WY), August 20, 2021, updated February 10, 2025. A museum-curated walkthrough of the principal American saddle types — Vaquero, stock, McClellan, and Pony Express — with photographs from the Center's collection. The Buffalo Bill Center of the West is a Smithsonian Affiliate. https://centerofthewest.org/2021/08/20/parts-of-a-western-saddle-and-the-variations/. See Endnote Commentary.

Romer v. Evans, 517 U.S. 620 (1996). https://supreme.justia.com/cases/federal/us/517/620/.

Rosenberg, Jeremy. "The Birth of Sprawl: How Ending the Great Depression Meant Inventing the Suburbs." PBS SoCal, "Laws That Shaped LA" series, October 1, 2012. Documents the role of the Federal Housing Administration's mortgage insurance and

lending standards in shaping the design of post-war American suburbs, including the FHA's preference for wide road rights-of-way, cul-de-sacs, and developments that discouraged through traffic. Cited in the bicycles section of the modern-bargain chapter as a source for the FHA's role in producing the modern American residential road environment. https://www.pbssocal.org/history-society/the-birth-of-sprawl-how-ending-the-great-depression-meant-inventing-the-suburbs.

Rubin, Vera C., and W. Kent Ford Jr. "Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions." *Astrophysical Journal* 159 (1970): 379.

Ruckelshaus v. Monsanto Co., 467 U.S. 986 (1984). https://supreme.justia.com/cases/federal/us/467/986/.

S. 1568 (119th Cong., 2025) — Liberating Incandescent Technology (LIT) Act of 2025. Introduced May 1, 2025 by Senator Mike Lee (R-UT), Chairman of the Senate Committee on Energy and Natural Resources, with five original cosponsors — Senators John Curtis (R-UT), Jim Justice (R-WV), Tom Cotton (R-AR), Bill Hagerty (R-TN), and Eric Schmitt (R-MO) — joined after introduction by Senator Bernie Moreno (R-OH), bringing the total to six; House companion (H.R. 3341) introduced May 13, 2025 by Representative Craig Goldman (R-TX), with cosponsors Representatives Michael Rulli (R-OH) and Robert Onder (R-MO). The bill repeals Section 325(i) of the Energy Policy and Conservation Act, repeals the two May 2022 DOE general service lamps final rules (87 Fed. Reg. 27439 and 87 Fed. Reg. 27461), and repeals the April 2024 DOE final rule (89 Fed. Reg. 28859). Referred to the Senate Committee on Energy and Natural Resources and the House Committee on Energy and Commerce; no hearing held, no committee mark-up, no floor action through the period covered by the healthy-lighting chapter. https://www.energy.senate.gov/2025/5/chairman-lee-introduces-bill-to-end-biden-era-lightbulb-ban.

S. 3007 (119th Cong., 2025) — Diesel Truck Liberation Act. Introduced October 2025 by Senator Cynthia Lummis (R-Wyoming). Companion bill introduced in the House in March 2026. Removes Environmental Protection Agency authority over emission-control devices

on heavy-duty diesel trucks. Status as of this writing: in committee. Source for the legislative-parallel reference in the trucking chapter.
https://www.congress.gov/bill/119th-congress/senate-bill/3007.

Sackett v. Environmental Protection Agency, 598 U.S. 651 (2023). A landowning couple challenged the reach of a federal Clean Water Act rule over their own property and prevailed at the Supreme Court without first incurring a penalty. Cited in the plaintiff chapter as an example of a directly regulated party reaching the Court through a pre-enforcement challenge — the party the rule injured, suing in its own name, without first breaking the rule.
https://supreme.justia.com/cases/federal/us/598/651/.

SAVADeck Bicycles. "V3 Carbon Road Bike 18S US." Product page, SAVADeck Bicycles, model-year 2026 listing. Pricing and specification reference for the SAVA Warwind V3 ($1,099 retail; Toray T800 carbon-fiber frame and fork; Shimano Sora eighteen-speed groupset; Tektro V-brakes; UCI-certified for competitive racing) cited in the bicycles section of the modern-bargain chapter as the post-expiration counterpart to the 2005 Trek Madone 5.2.
https://savadeck-bike.com/products/v3-carbon-road-bike-18s.

Sax, Joseph L. "Takings and the Police Power." *Yale Law Journal* 74 (1964): 36–76. The leading mid-twentieth-century analytical framework for distinguishing compensable takings from non-compensable police-power regulations. Cited in the chapter on objections considered as part of the doctrinal terrain the Takings Clause prong of the book's argument must occupy.
https://openyls.law.yale.edu/handle/20.500.13051/14962.

Sax, Joseph L. "Takings, Private Property and Public Rights." *Yale Law Journal* 81 (1971): 149–186. Follow-on article reframing the takings question in terms of the conflict between private property rights and public rights in resources. Cited in the chapter on objections

considered for its narrower police-power analysis, against which Richard Epstein's libertarian Takings (1985) operates as counter-authority. https://openyls.law.yale.edu/handle/20.500.13051/15453.

Schelling, Thomas C. *Micromotives and Macrobehavior*. New York: W. W. Norton, 1978. https://wwnorton.com/books/Micromotives-and-Macrobehavior/.

Scott Paper Co. v. Marcalus Manufacturing Co., 326 U.S. 249 (1945). https://supreme.justia.com/cases/federal/us/326/249/.

Sears, Roebuck & Co. v. Stiffel Co., 376 U.S. 225 (1964). https://supreme.justia.com/cases/federal/us/376/225/.

Self-certification and certification by simulation. The United States does not pre-approve motor vehicles before sale; the National Traffic and Motor Vehicle Safety Act establishes a self-certification process under which manufacturers certify that their vehicles comply with all applicable Federal Motor Vehicle Safety Standards. NHTSA has confirmed that a manufacturer exercising reasonable care may certify compliance through simulation or engineering analysis, not only through the agency's physical test procedures. Independent engineering and homologation firms perform such certification work today, at both the whole-vehicle and component level. NHTSA, Notice Regarding the Applicability of NHTSA FMVSS Test Procedures to Certifying Manufacturers, 85 Fed. Reg. 83,502 (Dec. 21, 2020). https://www.federalregister.gov/documents/2020/12/21/2020-28107/notice-regarding-the-applicability-of-nhtsa-fmvss-test-procedures-to-certifying-manufacturers.

Semmelweis, Ignaz. *Die Ätiologie, der Begriff und die Prophylaxis des Kindbettfiebers*. Pest, Vienna and Leipzig: C. A. Hartleben, 1861. The dates for the division of the Vienna General Hospital's obstetric clinics and for the introduction of chlorine washing are taken from *Epidemiology and Infection* 136, no. 1 (7 June 2007): 1–9, doi 10.1017/S0950268807008746, whose figure legend gives them directly; the authors of that

article have not been confirmed. Sources differ on whether the clinics divided by gender in 1839 or on 19 October 1840.
See Endnote Commentary.

Shallue, Christopher J., and Andrew Vanderburg. “Identifying Exoplanets with Deep Learning: A Five-Planet Resonant Chain around Kepler-80 and an Eighth Planet around Kepler-90.” *Astronomical Journal* 155, no. 2 (February 2018): 94. doi:10.3847/1538-3881/aa9e09. A neural network trained on fifteen thousand previously vetted Kepler signals, classifying a held-out test set correctly ninety-six percent of the time, then directed at 670 star systems already known to hold multiple planets. Kepler-90i is listed in the NASA Exoplanet Archive.
See Endnote Commentary.

Sheetz v. County of El Dorado, 601 U.S. 267 (2024). https://supreme.justia.com/cases/federal/us/601/22-1074/.

Shoshone Tribe of Indians v. United States, 299 U.S. 476 (1937). https://supreme.justia.com/cases/federal/us/299/476/.

Shoshone Tribe v. United States (1937; covered in the Takings Clause prong chapter). The federal government had settled another tribe on land guaranteed to the Shoshone by treaty, diminishing their interest. The Supreme Court held the loss compensable. Justice Cardozo’s opinion supplied the phrase this book relies on — “Spoliation is not management” — and the principle it carries: a collective, non-individually-titled interest is property the government must pay to take. The book cites the case for that principle, not for any holding about the patent commons.

Sierra Saddlery School (Las Cruces, New Mexico). “Some History of the Western Saddle.” Posted October 18, 2012. A capable narrative summary of the trans-Atlantic genealogy of the American Western saddle — from Moorish horsemen of the Dark Ages, to the Spanish War Saddle brought to the New World, to the post-conquest evolution of the saddle into a working tool for the Mexican vaquero, to the Texican-and-Californio split that produced

today’s swell-fork (square-skirted, double-rigged) and slick-fork (round-skirted, center-fire) Western styles.
https://saddleschool.wordpress.com/2012/10/18/some-history-of-the-western-saddle/. See Endnote Commentary.

Signify N.V. Annual Report 2025. Signify N.V. (formerly Philips Lighting), Eindhoven, Netherlands. The current annual report of the lighting company that emerged from the May 2016 separation of Philips’ Lighting division. Documents that Signify holds more than twenty thousand patents (with the cumulative portfolio reflecting decades of Philips Lighting investment in light-emitting-diode and earlier lighting technologies; the foundational compact-fluorescent patents dating to the 1980 Philips SL lamp have long since expired, while the in-force portfolio is concentrated in solid-state lighting); that annual research and development spending is reported at approximately 250 to 271 million euros, the great majority of it on solid-state lighting and adjacent technologies. Cited in the healthy-lighting chapter’s bootleggers-and-Baptists section as the primary-source anchor for the patent-protected business model that benefits structurally from the displacement of the unpatentable incandescent.
https://www.signify.com/static/2025/signify-annual-report-2025.pdf.

Sinclair, Upton. *The Jungle*. New York: Doubleday, Page & Co., 1906. Cited in the chapter on why nobody noticed as the first of four Progressive-Era cases in which public awareness alone forced reform against a profitable concealed practice. Sinclair’s account of conditions in the Chicago meatpacking plants prompted the Neill-Reynolds investigation and the passage, within the same year, of the Pure Food and Drug Act (Pub. L. No. 59-384, 34 Stat. 768, June 30, 1906) and the Federal Meat Inspection Act (34 Stat. 1260, June 30, 1906).
https://www.gutenberg.org/ebooks/140.

Singer Manufacturing Co. v. June Manufacturing Co., 163 U.S. 169 (1896).
https://supreme.justia.com/cases/federal/us/163/169/.

Sinotruk (Hong Kong) Limited. Corporate page and HOWO 6×4 tractor product listings. The Sinotruk corporate page reports product sales in more than ninety countries; the

HOWO 6×4 tractor is the company's current-production heavy semi-truck platform, with verified export pricing from suppliers and international trade-listing sites in the range of $28,000 to $35,000 FOB China for standard 371–420 horsepower configurations. The price comparison is cited in the trucking chapter as evidence of what the global market delivers for a no-aftertreatment Class 8 platform built to Euro II emission standards — not as a recommendation to import.
https://www.sinotrukgroup.com/.

Skidmore v. Swift & Co., 323 U.S. 134 (1944).
https://supreme.justia.com/cases/federal/us/323/134/.

Skilsaw and the portable circular saw, 1924. Edmond Michel, a French immigrant living in New Orleans, built the first electric handsaw in 1923 by mounting a six-inch blade on a wooden frame and driving it with a motor taken from a malted-milk mixer. Joseph W. Sullivan read a newspaper account of the invention, sought Michel out, and the two founded the Michel Electric Handsaw Company in Chicago in 1924; they named the saw the Skilsaw. The firm was renamed Skilsaw Incorporated in 1926 after Michel left it, and the die-cast aluminum Model E followed in 1928. Cited in the preface for the date the portable circular saw reached the market, which the trade did not adopt in quantity until after the Second World War. Skil, “History” (corporate timeline).
https://www.skileurope.com/uk/history. Companion trade-press account: Industrial Equipment News, “IEN 75th Anniversary Perspective & Timeline: Skil Power Tools,” December 2, 2008, https://www.ien.com/operations/news/21098125/ien-75th-anniversary-perspective-timeline-skil-power-tools (metered access).

Somers, Suzanne. *Knockout: Interviews with Doctors Who Are Curing Cancer—And How to Prevent Getting It in the First Place*. New York: Crown, 2009.
https://www.penguinrandomhouse.com/books/24617/knockout-by-suzanne-somers/.

Sonalika Group. Corporate homepage. India-based tractor and farm-equipment manufacturer. Source for the production volume (more than 120,000 units per year) and export reach (more than 150 nations, approximately one-third of India’s total tractor

exports) cited in the farm-equipment chapter alongside Mahindra and TAFE as the principal Indian manufacturers serving global farm-equipment markets at price points unreachable in the United States.
https://www.sonalika.com/.

Southworth, Michael, and Eran Ben-Joseph. “Street Standards and the Shaping of Suburbia.” *Journal of the American Planning Association* 61, no. 1 (1995): 65–81. DOI: 10.1080/01944369508975620. The foundational scholarly paper on the historical evolution of American suburban residential street standards. Documents the role of the Federal Housing Administration — described by the authors as “the most ambitious suburbanization plan in United States history” — in establishing the 50-foot minimum right-of-way for federally-backed developments, the preference for cul-de-sacs, and the standards that discouraged through traffic. Hosted on Ben-Joseph’s MIT Department of Urban Studies and Planning page. Cited in the bicycles section of the modern-bargain chapter for the FHA’s role in producing the modern American residential road environment.
https://web.mit.edu/ebj/www/doc/JAPAv61n1.pdf.

Sperling, Daniel, Ethan C. Abeles, David S. Bunch, Andrew Burke, Belinda S. Chen, Kenneth S. Kurani, and Thomas S. Turrentine. “The Price of Regulation.” *ACCESS Magazine*, no. 25 (Fall 2004): 9–18. University of California Transportation Center, summarizing research prepared for the California Air Resources Board under contract 02-310. The article estimates total federal regulatory cost (safety, emissions, and fuel-economy compliance) on a 2001 passenger vehicle at $2,500–$4,020 in then-current dollars, against approximately $11 for the same regulatory categories on a 1967 vehicle. The proportional finding — that more than 99 percent of the cumulative federal regulatory cost on a new vehicle by 2001 was added after 1970 — is the source the cars-and-trucks chapter relies on for the post-1970 attribution. The authors’ preferred estimate is the lower figure ($2,500, of which approximately $1,000 is emissions compliance and less than $1,500 is safety compliance); the higher figure ($4,020) is the BEA/BLS estimate compiled in Ward’s Automotive Yearbook.
https://www.accessmagazine.org/fall-2004/price-regulation/.

Standard Oil Co. of New Jersey v. United States, 221 U.S. 1 (1911). The antitrust decision dissolving the Standard Oil trust, cited in the chapter on why nobody noticed as the legal endpoint of the public pressure that Ida Tarbell's *McClure's* series had built. https://supreme.justia.com/cases/federal/us/221/1/.

Star Athletica, L.L.C. v. Varsity Brands, Inc., 580 U.S. 405 (2017). Establishes the governing separability test under 17 U.S.C. § 101: a feature incorporated into the design of a useful article is eligible for copyright protection only if it can be perceived as a two- or three-dimensional work of art separate from the useful article and would qualify as a protectable pictorial, graphic, or sculptural work on its own or in another medium if imagined separately from the article. The Court noted that the test does not render the underlying useful article itself eligible for copyright. Cited in the chapter *Objections Considered* for the limit it places on copyright claims over vehicles: a separable badge or graphic may qualify, the vehicle may not. https://supreme.justia.com/cases/federal/us/580/15-866/.

State-of-the-art defense, products liability. The "state-of-the-art" defense varies by jurisdiction: some states codify it as a full defense to a design-defect claim (e.g., Iowa Code § 668.12), some as a rebuttable presumption that a conforming product is not defective (e.g., Colo. Rev. Stat. § 13-21-403), some as one factor for the jury to weigh (e.g., S.D. Codified Laws § 20-9-10.1), and some (e.g., Illinois) recognize no affirmative state-of-the-art defense, admitting era-relative evidence only within the risk-utility analysis. Federal motor-vehicle safety standards generally do not preempt state tort law; see 49 U.S.C. § 30103(e) and Williamson v. Mazda Motor of America, Inc., 562 U.S. 323 (2011). Cited in the chapter on how a remake is built and certified. https://www.bloomberglaw.com/external/document/X8S1SDG000000/litigation-overview-state-of-the-art-tort-defense.

Statute of Monopolies, 21 Jac. I, c. 3 (1624). The National Archives (UK), legislation.gov.uk. https://www.legislation.gov.uk/aep/Ja1/21/3.

Statuto Albertino. 1848. Constitution granted by King Charles Albert of Sardinia, March 4, 1848; extended to the unified Kingdom of Italy upon its proclamation on March 17, 1861; officially abolished when the Constitution of the Italian Republic took effect on January 1, 1948. Article 63 provided, in English translation: *Votes shall be taken by rising, by division, and by secret ballot. The latter method, however, shall always be employed for the final vote on a law and in all cases of a personal nature.* Article 63 is the constitutional foundation for the Italian parliament's *voto segreto* practice from 1848 to 1922 under direct constitutional mandate — the first span within the 119 years the Hybrid Voting chapter refers to. The post-war Republican Constitution's Article 64 transferred chamber voting procedure to each chamber's own rules of procedure, under which *voto segreto* continued in practice from 1948 until the October 1988 chamber-rules reform under the De Mita government.
https://en.wikisource.org/wiki/Statuto_Albertino. See Endnote Commentary.

Steffens, Lincoln. *The Shame of the Cities*. New York: McClure, Phillips & Co., 1904. The collected *McClure's* exposé of municipal political corruption, cited in the chapter on why nobody noticed as the third of the four Progressive-Era cases in which exposure alone drove reform.
https://www.gutenberg.org/ebooks/4904.

Stickney, Thomas. "Improvement in Saddles." U.S. Patent No. X-187. Issued 1797. The first American saddle patent under the Patent Act of 1790, claiming a saddle "different from the common saddle in having but one girth, and that passing between the legs instead of round the body." The single-girth design dropped the saddle's weight from twenty-five or thirty pounds to roughly fifteen.
https://apps.abacus.ai/chatllm/share/conversation/18d6a5d7:13f78811d8 ;
https://datamp.org/patents/advance.php?pn=187X&id=36942&set=3

Stigler, George J. "The Theory of Economic Regulation." *Bell Journal of Economics and Management Science* 2, no. 1 (Spring 1971): 3–21.
https://www.jstor.org/stable/3003160.

Susan B. Anthony List v. Driehaus, 573 U.S. 149 (2014). https://supreme.justia.com/cases/federal/us/573/149/.

Table Saw Advance Notice of Proposed Rulemaking. Consumer Product Safety Commission, "Safety Standard Addressing Blade-Contact Injuries on Table Saws," 76 Fed. Reg. 62678 (October 11, 2011). Issued under the Consumer Product Safety Act, 15 U.S.C. §§ 2051 et seq. https://www.federalregister.gov/documents/2011/10/11/2011-26183/safety-standard-addressing-blade-contact-injuries-on-table-saws.

Table Saw Notice of Proposed Rulemaking. Consumer Product Safety Commission, "Safety Standard Addressing Blade-Contact Injuries on Table Saws," 82 Fed. Reg. 22300 (May 12, 2017). Proposed performance standard requiring automatic blade-contact-detection technology on table saws. Not finalized as of this writing. https://www.federalregister.gov/documents/2017/05/12/2017-09657/safety-standard-addressing-blade-contact-injuries-on-table-saws.

Tarbell, Ida M. *The History of the Standard Oil Company*. New York: McClure, Phillips & Co., 1904. Originally serialized in *McClure's Magazine* from 1902 to 1904. Cited in the chapter on why nobody noticed as the second of the four Progressive-Era cases; the dissolution of the trust followed in *Standard Oil Co. of New Jersey v. United States* (1911). https://archive.org/details/historyofstandar01tarb.

Terry, Eli. U.S. Patent for the wooden-movement shelf clock, granted 1816 (term expired 1830 under the fourteen-year rule of the Patent Act of 1790). The shelf clock entry in the chapter's eight-product table; the Terry-pattern wooden shelf clock fell from a during-patent price of about one hundred and fifty dollars to thirty dollars or less by the 1890s, a sixty-four percent real-terms drop across roughly seventy-five years. https://en.wikipedia.org/wiki/Eli_Terry.

Textron Aviation. "Textron Aviation Expands Global Training Reach with Order of Ten Cessna Skyhawks from Egyptian Aviation Academy." Press release, Farnborough International Airshow, July 21, 2026. Confirms that the Cessna 172 Skyhawk, which first

flew in 1955, remains in production, with more than 45,000 delivered. Cited in How a Remake Is Built and Certified for the Cessna 172's continued production under Textron Aviation.
https://media.txtav.com/260118-textron-aviation-expands-global-training-reach-with-order-of-ten-cessna-skyhawks-from-egyptian-aviation-academy/.

Tier 1 Motor Vehicle Emission Standards. Final Rule, "Standards for Emissions from New Gasoline-Fueled and Diesel-Fueled Light-Duty Vehicles and Light-Duty Trucks," 56 Fed. Reg. 25724 (June 5, 1991); phased in over model years 1994–1997. U.S. Environmental Protection Agency.
https://www.federalregister.gov/citation/56-FR-25724.

Tier 2 Motor Vehicle Emission Standards. Final Rule, "Control of Air Pollution from New Motor Vehicles: Tier 2 Motor Vehicle Emission Standards and Gasoline Sulfur Control Requirements," 64 Fed. Reg. 26004 (May 13, 1999). U.S. Environmental Protection Agency.
https://www.epa.gov/regulations-emissions-vehicles-and-engines/final-rule-control-air-pollution-new-motor-vehicles-tier.

Tier 3 Motor Vehicle Emission and Fuel Standards. Final Rule, "Control of Air Pollution from Motor Vehicles: Tier 3 Motor Vehicle Emission and Fuel Standards," 79 Fed. Reg. 23414 (April 28, 2014). U.S. Environmental Protection Agency.
https://www.federalregister.gov/documents/2014/04/28/2014-06954/control-of-air-pollution-from-motor-vehicles-tier-3-motor-vehicle-emission-and-fuel-standards.

Tire Pressure Monitoring System Rule (FMVSS No. 138). Final Rule, "Federal Motor Vehicle Safety Standards; Tire Pressure Monitoring Systems; Controls and Displays," 70 Fed. Reg. 18136 (April 8, 2005). National Highway Traffic Safety Administration.
https://www.federalregister.gov/documents/2005/04/08/05-6741/federal-motor-vehicle-safety-standards-tire-pressure-monitoring-systems-controls-and-displays.

Tobacco Master Settlement Agreement (1998). Agreement entered between the attorneys general of forty-six states and the major United States tobacco manufacturers. Cited in the

chapter on why nobody noticed as the fourth Progressive-Era-type case: a profitable concealment of known harm — the industry's internal knowledge of addiction and disease — that collapsed into regulation once leaked documents and whistleblower testimony made the private knowledge public. National Association of Attorneys General. https://www.naag.org/our-work/naag-center-for-tobacco-and-public-health/the-master-settlement-agreement/.

Tompsett, Michael Francis. "Charge Transfer Imaging Devices." U.S. Patent 4,085,456, filed August 30, 1972, issued April 18, 1978. Assigned to Bell Telephone Laboratories, Inc. Foundational charge-coupled-device imaging patent in the corpus underlying automotive backup-camera image sensors; expired in the 1990s. https://patents.google.com/patent/US4085456.

Torchinsky, Jason. "This Is Why Mahindra Can Build Tiny Jeeps." *Jalopnik*, March 15, 2018. https://www.jalopnik.com/this-is-why-mahindra-can-build-tiny-jeeps-1823472625.

Townsend, Harry Pond. Driving and braking mechanism for cycles. U.S. Patent 850,077, filed October 10, 1898 (Serial No. 693,117), granted April 9, 1907; assigned to the New Departure Manufacturing Company of Bristol, Connecticut. Expired 1924. Contemporary trade accounts state that the Townsend patent contained the broad and controlling claims for the coaster brake. Attribution here is contested and worth stating plainly. Daniel Stover and William Hance of Freeport, Illinois are widely credited with an 1889 coaster brake, and they did receive an 1889 patent — but the specification describes a lever pivoted above the bottom bracket bearing on the rim of the wheel, which is a back-pedal spoon brake, not a coaster brake. It has no freewheel, it sits outside the hub, and it was obsolete once pneumatic tires arrived, since a friction pad rubbing a pneumatic tire destroys it. The coaster brake proper — drive, coast, and brake combined inside the hub — arrives with the 1898 filings. Townsend, James S. Copeland of Pope Manufacturing, and William Robinson of Brooklyn all filed within roughly a year of one another, and Alexander P. Morrow of Elmira, New York followed with U.S. Patent 773,333, filed August 1, 1902 and granted October 25, 1904. New Departure lacked the machinery to build Townsend's design and

contracted P. & F. Corbin of New Britain for the first five thousand brakes; within a decade the coaster brake was standard equipment on American-made bicycles. https://patents.google.com/patent/US850077A/en.

Toyota Motor Sales, USA, Inc. “2025 Camry Product Information.” e-Brochure, accessed May 2026. Base LE trim MSRP $28,400; wireless Apple CarPlay and Android Auto included as standard equipment on all trims. Cited in the surveillance chapter as one of the three 2025-MY consumer-sedan reference points showing that the licensed-software stack has been absorbed into baseline pricing rather than remaining a discrete trim premium. https://www.toyota.com/content/dam/toyota/brochures/pdf/2025/my25-camry-ebrochure.pdf.

TractorGyan. “Mahindra Tractors.” India-market tractor pricing aggregator. Cross-reference for the India-market Mahindra tractor pricing cited in the farm-equipment chapter against the manufacturer’s own product pages. https://tractorgyan.com/tractor/Mahindra.

TractorGyan. “Tractor Implements in India: Combine Harvester.” India-market combine-harvester pricing aggregator. Cross-reference for the India-market combine pricing cited in the farm-equipment chapter against the IndiaMART John Deere W70 listing. https://tractorgyan.com/tractor-implements-in-india/combine-harvester.

TractorHouse. “John Deere 5050E Listings.” U.S. tractor listing aggregator. Cross-reference for U.S.-market Deere tractor pricing cited in the farm-equipment chapter against India-market Mahindra and Deere pricing. https://www.tractorhouse.com/listings/for-sale/john-deere/5050e/40-hp-to-99-hp-tractors/1110.

Tractors and Farm Equipment Limited (TAFE). Corporate homepage. India-based agricultural equipment manufacturer, licensee of the Massey Ferguson tractor brand. Source for TAFE’s export reach to more than 100 countries cited in the farm-equipment chapter

alongside Mahindra and Sonalika as the principal Indian manufacturers serving global farm-equipment markets.
https://www.tafe.com/.

TrafFix Devices, Inc. v. Marketing Displays, Inc., 532 U.S. 23 (2001). Holds that a functional product feature cannot receive trade dress protection under the Lanham Act, and that an expired utility patent is strong evidence that the features it claimed are functional. Where the expired patent claimed the feature in question, the party seeking trade dress protection carries a heavy burden of showing that the feature is merely ornamental, incidental, or arbitrary. Cited in the chapter *Objections Considered* as the doctrinal bar that prevents trade dress from being used to recapture a functional design after the patent on it has expired — the trademark-side counterpart to the patent-side rule of *Bonito Boats, Inc. v. Thunder Craft Boats, Inc.* and *Sears, Roebuck & Co. v. Stiffel Co.*.
https://supreme.justia.com/cases/federal/us/532/23/.

TrueCar. “2025 Ford F-150 Lariat 4×4 SuperCrew 5.5′ Box Listings.” TrueCar dealer-inventory price aggregator. Secondary verification source for the chapter title figure of $72,495: real-world Lariat 4×4 SuperCrew PowerBoost Hybrid configurations are listed at MSRP-with-destination totals of $72,480, $72,580, and $72,690 — converging on the $72,495 figure from a second direction (base Lariat out-the-door total against hybrid Lariat MSRP-with-destination).
https://www.truecar.com/prices-new/ford/f-150-pricing/.

Tucker Act, 28 U.S.C. § 1491; limitations period at 28 U.S.C. § 2501. Confers jurisdiction on the United States Court of Federal Claims over non-tort monetary claims against the United States founded on the Constitution, including claims for just compensation under the Fifth Amendment, subject to a six-year limitations period running from the accrual of each claim. Because the Administrative Procedure Act’s waiver of sovereign immunity at 5 U.S.C. § 702 extends only to relief other than money damages, a plaintiff seeking compensation rather than declaratory or injunctive relief must proceed here instead. Cited

in *Objections Considered* for the six-year ceiling on any damages theory built on the regulatory wall.
https://www.law.cornell.edu/uscode/text/28/1491.

Trump v. Hawaii, 585 U.S. 667 (2018). The Supreme Court's decision upholding the entry restrictions of Presidential Proclamation 9645. Cited here for one sentence: the Court stated that *Korematsu* had been gravely wrong the day it was decided and has no place in law under the Constitution — seventy-four years after the fact.
https://supreme.justia.com/cases/federal/us/585/17-965/.

Tyler v. Hennepin County, Minnesota, 598 U.S. 631 (2023).
https://supreme.justia.com/cases/federal/us/598/22-166/.

Tyson, Brian, George Fareed, and Mathew Crawford. *Overcoming the COVID Darkness: How Two Doctors Successfully Treated 7000 Patients*. Self-published, 2022.
https://www.amazon.com/Overcoming-COVID-Darkness-Successfully-Patients/dp/B09NRJSCFB.

UL 987 — Stationary and Fixed Electric Tools. Voluntary safety standard published by Underwriters Laboratories; widely adopted by retailers as a purchasing requirement for stationary and fixed power tools sold in the United States. Incorporated by reference in some federal and state regulatory schemes via the National Technology Transfer and Advancement Act of 1995.
https://www.shopulstandards.com/ProductDetail.aspx?productId=UL987.

UL 62841-3-1 — Electric Motor-Operated Hand-Held Tools, Transportable Tools, and Lawn and Garden Machinery — Safety — Part 3-1: Particular Requirements for Transportable Table Saws. Voluntary safety standard published by Underwriters Laboratories; widely adopted by retailers as a purchasing requirement for transportable table saws sold in the United States.
https://www.shopulstandards.com/ProductDetail.aspx?productId=UL62841-3-1.

United Food and Commercial Workers Union Local 751 v. Brown Group, Inc., 517 U.S. 544 (1996). Construing the third prong of the associational-standing test of *Hunt v.*

Washington State Apple Advertising Commission. The Court explained that individual participation by members is not normally necessary when an association seeks prospective or injunctive relief on their behalf, but would be required in an action for damages, so that an association's damages action running solely to its members is barred for want of standing. Cited in the citizen's-playbook chapter and in *Objections Considered* for the proposition that a membership association is available as plaintiff for the remedy this book seeks and unavailable for a damages claim.
https://www.law.cornell.edu/supct/html/95-340.ZO.html.

United States Code. "17 U.S.C. § 101 — Definitions ('useful article' and 'pictorial, graphic, and sculptural works')." Copyright Act of 1976, as amended. Defines a useful article as one having an intrinsic utilitarian function that is not merely to portray the appearance of the article or to convey information, and provides that the design of a useful article is protected only to the extent it incorporates features that can be identified separately from, and are capable of existing independently of, the utilitarian aspects of the article. Cited in the chapter *Objections Considered*.
https://www.law.cornell.edu/uscode/text/17/101.

United States Code. "17 U.S.C. § 302(c) — Duration of Copyright: Anonymous Works, Pseudonymous Works, and Works Made for Hire." Provides that copyright in a work made for hire endures for ninety-five years from the year of first publication or one hundred twenty years from the year of creation, whichever expires first. Cited in the chapter *Objections Considered* as the source of the ninety-five-year figure a correspondent attributed to whole vehicles; the term is correctly stated but does not attach to a useful article.
https://www.law.cornell.edu/uscode/text/17/302.

United States Code. "42 U.S.C. § 6295(o)(1) — Energy Conservation Program for Consumer Products: Anti-Backsliding Provision." Cornell University Legal Information Institute.
https://www.law.cornell.edu/uscode/text/42/6295.

United States Code. “42 U.S.C. § 7521(a)(1) — Emission Standards for New Motor Vehicles or New Motor Vehicle Engines.” Cornell University Legal Information Institute. The Clean Air Act provision granting EPA authority to prescribe emission standards “applicable to the emission of any air pollutant from any class or classes of new motor vehicles or new motor vehicle engines.” Cited in the trucking chapter as the statutory grandfathering principle: EPA’s authority reaches engines at the time of manufacture, fixing each model year’s emission certification as the perpetual standard for the life of that engine. A 1999 model-year heavy-duty diesel certified to the 1998 standards (4.0 g/bhp-hr NOx, 0.10 g/bhp-hr PM) remains subject to those standards, and only those standards, for its operational life.
https://www.law.cornell.edu/uscode/text/42/7521.

United States Code. “49 U.S.C. § 30112(a) — Prohibitions on Manufacturing, Selling, and Importing Noncomplying Motor Vehicles” and “49 U.S.C. § 30115 — Certification of Compliance.” Cornell University Legal Information Institute. Section 30112(a) forbids any person to manufacture for sale, sell, offer for sale, introduce or deliver for introduction into interstate commerce, or import a motor vehicle manufactured on or after the date an applicable Federal Motor Vehicle Safety Standard takes effect unless the vehicle complies with the standard and is covered by a § 30115 certification. Cited in the chapter on objections considered to establish that a faithful new copy of a patent-expired vehicle, built to its original platform-year design, cannot be lawfully manufactured for sale or sold today — the standards binding by date of manufacture, so a vehicle built now must meet the standards in force now.
https://www.law.cornell.edu/uscode/text/49/30112 and
https://www.law.cornell.edu/uscode/text/49/30115.

United States Code. “49 U.S.C. § 30112(b)(9) — Nonapplication of Federal Motor Vehicle Safety Standards to Vehicles at Least Twenty-Five Years Old.” Cornell University Legal Information Institute. The provision exempting a motor vehicle that is at least twenty-five years old from the Federal Motor Vehicle Safety Standards whose noncompliance would otherwise bar its manufacture, sale, or importation. Cited in the policy-framework chapter

as standing proof that federal law already grandfathers older vehicle designs to the standards of an earlier era; the chapter's grandfather-exemption leg extends that established principle, replacing the statute's rolling twenty-five-year age threshold with the fixed platform year the remanufactured design is certified to.
https://www.law.cornell.edu/uscode/text/49/30112.

United States Copyright Office. "Useful Articles." Registration guidance, Visual Arts. States that copyright in a work portraying a useful article extends only to the artistic expression of the author of the pictorial, graphic, or sculptural work and does not extend to the design of the article portrayed; a drawing or photograph of an automobile may be copyrighted, but that does not give the artist or photographer the exclusive right to make automobiles of the same design. Cited in the chapter *Objections Considered*.
https://www.copyright.gov/register/va-useful.html.

United States Department of Agriculture v. Moreno, 413 U.S. 528 (1973).
https://supreme.justia.com/cases/federal/us/413/528/.

United States International Trade Commission. *Remanufactured Goods: An Overview of the U.S. and Global Industries, Markets, and Trade*. Investigation No. 332-525, USITC Publication 4356. Washington, DC: USITC, October 2012. Source for the figure cited in the chapter on the wall that the United States is the world's largest producer of remanufactured goods, valued at no less than $43 billion in annual production over 2009–2011 and supporting some 180,000 full-time jobs, with motor-vehicle parts among the three largest sectors. The chapter cites the report to distinguish the thriving trade in remanufactured *components* below the certification gate from the assembled patent-expired *products* the wall blocks at the gate.
https://www.usitc.gov/publications/332/pub4356.pdf.

United States Patent and Trademark Office. "2701 Patent Term [R-10.2019]." *Manual of Patent Examining Procedure* (MPEP). The authoritative reference for the U.S. patent term: seventeen years from grant for utility patents filed before June 8, 1995, with the post-1995 transitional rule giving the greater of seventeen years from grant or twenty years from

filing. Under either rule, all utility patents on the saddles, plows, gins, engines, and clocks discussed in this chapter expired no later than the late 1980s. Accessed May 9, 2026. https://www.uspto.gov/web/offices/pac/mpep/s2701.html.

United States v. 50 Acres of Land, 469 U.S. 24 (1984). https://supreme.justia.com/cases/federal/us/469/24/.

United States v. Ballin, 144 U.S. 1 (1892). The closest Supreme Court case to address the Journal Clause of Article I, Section 5, Clause 3. In *Ballin* the Court held the House Journal to be conclusive evidence of what occurred during a contested quorum vote, but did not interpret whether the Yeas-and-Nays override ("at the Desire of one fifth of those Present, be entered on the Journal") constitutionally requires individual named records or only an aggregate tally. The case is cited in the Hybrid Voting chapter as the doctrinal anchor for the proposition that no Supreme Court holding has squarely decided the interpretive scope of the Yeas-and-Nays override, leaving the question one of first impression. https://supreme.justia.com/cases/federal/us/144/1/.

United States v. Lopez, 514 U.S. 549 (1995). Five-to-four decision holding the Gun-Free School Zones Act of 1990 beyond the commerce power. Cited for the shape of an enumerated-powers question: the Court asked whether the regulated activity fell within the power, not whether Congress could rationally have thought the statute wise. https://supreme.justia.com/cases/federal/us/514/549/.

United States v. Moghadam, 175 F.3d 1269 (11th Cir. 1999), cert. denied, 529 U.S. 1036 (2000). The Eleventh Circuit's decision upholding the federal anti-bootlegging statute under the Commerce Clause while assuming that the Copyright Clause could not sustain it, live performances being unfixed and so outside the term "Writings." The court held that the commerce power may not be used where the legislation is fundamentally inconsistent with the limitations of Article I, Section 8, Clause 8, and its footnote identified the statute's lack of any express time limit as the likely site of that inconsistency. The court did not decide the point; the appellant had not preserved it. https://law.justia.com/cases/federal/appellate-courts/F3/175/1269/637322/.

United States v. Sioux Nation of Indians, 448 U.S. 371 (1980). https://supreme.justia.com/cases/federal/us/448/371/.

U.S. Department of Agriculture, Economic Research Service. “Food Dollar Series.” Source for the farm share (15.9 cents) and marketing share (84.1 cents) of every retail dollar Americans spend on food, 2023 data, cited in the farm-equipment chapter’s grocery cost projection. https://www.ers.usda.gov/data-products/food-dollar-series. Companion data: U.S. Department of Agriculture, Economic Research Service. “Food Prices and Spending” (Ag and Food Statistics: Charting the Essentials). Source for the $2.58 trillion in total U.S. food spending in 2024 and the 10.4 percent of disposable personal income that figure represents. https://www.ers.usda.gov/data-products/ag-and-food-statistics-charting-the-essentials/food-prices-and-spending. Methodological credit: the author’s analysis of the USDA Food Dollar Series data underlying the grocery cost projection in the source article was developed in collaboration with the Claude session at https://claude.ai/share/1d635ab2-1915-442c-88ca-ce33e6ad92bc.

U.S. Department of Agriculture, Economic Research Service. “Fewer Farms, More Milk: The Changing Structure and Costs of U.S. Dairy Farming.” Amber Waves. Source for U.S. milk production of roughly two hundred twenty-six billion pounds in 2024, nearly all of it moved off the farm by truck every day or every other day; cited in the trucker chapter’s milk-hauler subsection. https://www.ers.usda.gov/amber-waves/2026/february/fewer-farms-more-milk-the-changing-structure-and-costs-of-us-dairy-farming.

U.S. Department of Energy. “Energy Department Acts to Lower Prices and Increase Consumer Choice with Household Appliances.” Press release, February 14, 2025. Announces the postponement of implementation for seven Biden-administration appliance-efficiency rules covering central air conditioners, clothes washers and dryers, general service lamps, walk-in coolers and freezers, gas instantaneous water heaters, commercial refrigeration equipment, and air compressors, together with the creation of a new energy-efficiency category exempting natural gas tankless water heaters. The action was taken

under the authority of Executive Order 14154 and the related Presidential Memorandum on Delivering Emergency Price Relief.
https://www.energy.gov/articles/energy-department-acts-lower-prices-and-increase-consumer-choice-household-appliances.

U.S. Department of Justice, Environment and Natural Resources Division. “Tribal Trust Case Settlements.” Last updated June 21, 2023. Official Justice Department account of the wave of breach-of-trust litigation tribes brought against the United States beginning in January 2001. Reports that by January 2007 there were over one hundred such cases, brought by about one hundred twenty tribes, and that the government later reached settlement agreements with more than one hundred ten tribes for a total of about $3.7 billion. Cited in the plaintiff chapter as a measure of the frequency with which tribal nations bring suit against the federal government, not as a measure of merits outcomes; the matters resolved by negotiated settlement rather than judgment.
https://www.justice.gov/enrd/natural-resources-section/tribal-trust-case-settlements.

U.S. Department of the Treasury. “Historical Debt Outstanding.” Treasury Fiscal Data. Official Treasury dataset reporting the total outstanding U.S. national debt at the end of every fiscal year since 1790. Primary source for the chapter’s 1957 datum (the last fiscal year in which total outstanding national debt fell below the previous year), for the Coolidge-era 1923–1929 debt reduction figures, and for the FY 1998–2001 surplus-years data demonstrating that a budget surplus is not always accompanied by a reduction in total outstanding debt.
https://fiscaldata.treasury.gov/datasets/historical-debt-outstanding/.

U.S. Food and Drug Administration. “Sanitary Transportation of Human and Animal Food.” Final Rule under the FDA Food Safety Modernization Act, codified at 21 C.F.R. Part 1, Subpart O. Source for the distinct federal food-safety regime governing bulk milk haulers, layered with state bulk-milk permits requiring truck inspection, driver training and

testing on cleaning and sample collection, and scheduled tank sanitation. Cited in the trucker chapter's milk-hauler subsection.
https://www.fda.gov/food/food-safety-modernization-act-fsma/fsma-final-rule-sanitary-transportation-human-and-animal-food.

U.S. Food and Drug Administration. "Regenerative Medicine Framework." Final guidance documents issued November 2017 (four documents bundled), establishing the agency's policy on human cell, tissue, and cellular and tissue-based products considered "more than minimally manipulated."
https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/regenerative-medicine-advanced-therapy-designation.

U.S. Food and Drug Administration. *FDA Report on the Quality, Safety, and Effectiveness of Servicing of Medical Devices*. May 2018. Concludes that independent medical-device servicers are critical to U.S. healthcare and that adverse events from third-party servicing constitute under one percent of reported incidents.
https://www.fda.gov/medical-devices/medical-device-servicing-and-remanufacturing/quality-safety-and-effectiveness-medical-device-servicing.

U.S. Food and Drug Administration. "Why You Should Not Use Ivermectin to Treat or Prevent COVID-19." Consumer Update. Most recently updated 2024.
https://www.fda.gov/consumers/consumer-updates/why-you-should-not-use-ivermectin-treat-or-prevent-covid-19.

U.S. International Trade Commission. "Remanufactured Goods: An Overview of the U.S. and Global Industries, Markets, and Trade." Investigation No. 332-525, Publication 4356 (2012). Government source corroborating the scale and history of the U.S. remanufacturing industry cited in the objections chapter.
https://www.usitc.gov/publications/332/pub4356.pdf.

U.S. Patent No. 39,285 (Gosker and Degenhart). "Improvement in Pack-Saddles." Issued 1863. Google Patents.
https://patents.google.com/patent/US39285A/en.

U.S. Patent No. 139,668 (Hamil). “Improvement in Saddles.” Issued 1873. Google Patents. https://patents.google.com/patent/US139668A/en.

U.S. Patent No. 568,939 (Davis). “Saddle.” Issued 1896. Google Patents. https://patents.google.com/patent/US568939A/en.

U.S. Patent No. 608,845 (Diesel). “Internal Combustion Engine.” Issued August 9, 1898. Rudolf Diesel’s basic U.S. patent on the compression-ignition engine; under the seventeen-year term then in force it expired in 1915. Cited in the trucking chapter for the date Diesel’s American patent ran out. Google Patents. https://patents.google.com/patent/US608845A/en.

U.S. Patent No. 5,357,751 (Orzel). “Air/Fuel Control System Providing Catalytic Monitoring.” Filed April 8, 1993; issued October 25, 1994. Assigned to Ford Motor Company. Google Patents. https://patents.google.com/patent/US5357751A/en.

U.S. Patent No. 5,899,062 (Jerger et al.). “Catalyst Monitor Using Arc Length Ratio of Pre- and Post-Catalyst Sensor Signals.” Filed January 22, 1997; issued May 4, 1999. Assigned to Ford Global Technologies. Google Patents. https://patents.google.com/patent/US5899062A/en.

U.S. Patent No. 6,195,986 (Davey et al.). “Method and System for Monitoring a Catalytic Converter.” Filed June 21, 1999; issued March 6, 2001. Assigned to Ford Global Technologies. Google Patents. https://patents.google.com/patent/US6195986B1/en.

U.S. Patent No. 6,397,583 (Davey et al.). “Method for Catalyst Monitoring Using Flex Fuel.” Filed March 8, 2001; issued June 4, 2002. Assigned to Ford Global Technologies. Google Patents. https://patents.google.com/patent/US6397583B1/en.

U.S. Patent No. 6,532,734 (Nader et al.). “On-Board Diagnostic Catalyst Monitoring System.” Filed February 1, 2002; issued March 18, 2003. Assigned to Ford Global

Technologies. Google Patents.
https://patents.google.com/patent/US6532734B1/en.

U.S. Patent No. 6,550,237 (Adamczyk et al.). “Method and System for Monitoring a Catalytic Converter.” Filed September 25, 2002; issued April 22, 2003. Assigned to Ford Global Technologies. Google Patents.
https://patents.google.com/patent/US6550237B1/en.

U.S. Patent No. 10,626,779 (Bagnasco et al.). “Method and System for Improving Diagnosis of a Catalyst.” Filed July 19, 2018; issued April 21, 2020; lapsed for non-payment of maintenance fees effective April 21, 2024. Assigned to Ford Global Technologies, LLC. Google Patents.
https://patents.google.com/patent/US10626779B2/en.

U.S. Patent No. 10,947,910 (Bagnasco et al.). “Method and System for Catalyst Feedback Control.” Filed May 7, 2019; issued March 16, 2021. Assigned to Ford Global Technologies, LLC. Google Patents.
https://patents.google.com/patent/US10947910B2/en.

U.S. Patent No. 12,196,640 (Martinez Leyva et al.). “System and Method for Diagnosing an Exhaust Gas Sensor.” Filed June 27, 2022; issued January 14, 2025. Assigned to Ford Global Technologies, LLC. Google Patents.
https://patents.google.com/patent/US12196640B2/en.

U.S. Public Interest Research Group (PIRG). “Right to Repair.” Consumer-advocacy resource on the right to repair across consumer goods and medical equipment.
https://pirg.org/right-to-repair/.

U.S. Senate Committee on Agriculture, Nutrition, and Forestry. “Revisiting Farm Production Expenses.” Minority Blog. Source for the figure of approximately $500 billion in U.S. farm production expenditures in 2023 and the 28 percent two-year increase (the

sharpest on record) cited in the farm-equipment chapter's treatment of how equipment cost flows through the food supply chain.
https://www.agriculture.senate.gov/newsroom/minority-blog/revisiting-farm-production-expenses.

Vargo, Jason, Benjamin G. Gerhardstein, Geoffrey P. Whitfield, and Arthur Wendel. "Bicyclist Deaths Associated with Motor Vehicle Traffic — United States, 1975–2012." *Morbidity and Mortality Weekly Report* 64, no. 31 (August 14, 2015): 837. Centers for Disease Control and Prevention. Analyzes FARS bicycle fatality data for the period 1975–2012 and reports cyclist mortality on a per-population rate basis (a methodology that addresses denominator concerns inherent in absolute-count comparisons). Cyclist mortality rate fell from 0.41 to 0.23 deaths per 100,000 across the study period — a 44 percent decline. The steepest decline occurred among children aged under 15; age-specific mortality rates for adults aged 35–74 increased over the same period. The bicycles section of the modern-bargain chapter cites the Vargo et al. rate figure as the population-adjusted counterpart to the chapter's absolute-count IIHS figure, making possible an apples-to-apples comparison against the per-vehicle-mile fatality rate the chapter cites for the automobile.
https://www.cdc.gov/mmwr/preview/mmwrhtml/mm6431a1.htm.

Vaughn, Mark. "FCA Wins Legal Battle Against Mahindra and Roxor." *Autoweek*, June 12, 2020.
https://www.autoweek.com/news/industry-news/a32851849/fca-wins-legal-battle-against-mahindra-and-roxor/. The dispute, before the U.S. International Trade Commission, turned on Jeep's trade dress—the vehicle's ornamental appearance, including its boxy body, flat vertical sides, and round headlights—and not on the vehicle's mechanical design or on Mahindra's underlying license. The Commission found the 2018–2019 Roxor infringed that trade dress while violating none of FCA's registered trademarks. Mahindra restyled the front end, and in December 2020 the Commission cleared the redesigned vehicle, which resumed sale in the United States mechanically unchanged.

Verge, The. Trade-press report on General Motors CEO Mary Barra's October 2025 confirmation that the company will phase out support for Apple CarPlay and Android Auto across its entire vehicle lineup — internal-combustion and electric — in favor of a proprietary infotainment system built on Google's Android Automotive operating system, with full transition expected by the 2028 model year. October 22, 2025. https://www.theverge.com/transportation/804562/gm-apple-carplay-android-auto-gas-cars-mary-barra.

Vesuvius Challenge. Project site and results archive. Documents the open competition to read the carbonized Herculaneum papyri without unrolling them. In August 2023 Casey Handmer, a former Jet Propulsion Laboratory physicist, identified a faint "crackle" texture in the scan data as ink — the first ink found inside an unopened scroll — and took the First Ink Prize for it. Building on Handmer's discovery, Luke Farritor, then a twenty-one-year-old undergraduate in computer science with no training in papyrology or classical Greek, trained a model on the crackle shapes and in October 2023 recovered the Greek word for purple from inside an unopened scroll, winning the First Letters Prize; papyrologists confirmed the reading. Farritor subsequently shared the competition's $700,000 Grand Prize with Youssef Nader and Julian Schilliger for recovering roughly fifteen columns of a previously unread philosophical text. Cited in the preface as an instance in which an outsider supplied the method and specialists inside the field supplied the domain judgment. https://scrollprize.org/grandprize See Endnote Commentary.

Vesuvius Challenge. "First Word Discovered in Unopened Herculaneum Scroll by 21yo Computer Science Student." 12 October 2023. https://scrollprize.org/firstletters. Records the First Letters Prize of forty thousand dollars to Luke Farritor for more than ten characters recovered from roughly four square centimetres of scroll, the word πορφύρας, the second-place award to Youssef Nader, and the First Ink Prize to Casey Handmer whose crackle pattern supplied Farritor's training data. The seven-hundred-thousand-dollar grand prize followed in December 2023 to Farritor, Nader and Julian Schilliger for about two thousand characters. Scans produced by Brent Seales and EduceLab at the University of Kentucky; readings confirmed by papyrologists.

von Neumann, John, and Oskar Morgenstern. *Theory of Games and Economic Behavior*. Princeton: Princeton University Press, 1944. https://press.princeton.edu/books/paperback/9780691130613/theory-of-games-and-economic-behavior.

Walk-Behind Power Lawn Mower Safety Standard. Consumer Product Safety Commission, 16 C.F.R. Part 1205. Final Rule, 47 Fed. Reg. 1880 (January 14, 1982). Mandatory product-safety standard issued under the Consumer Product Safety Act, 15 U.S.C. §§ 2051 et seq. https://www.ecfr.gov/current/title-16/chapter-II/subchapter-B/part-1205.

Walterscheid, Edward C. *The Nature of the Intellectual Property Clause: A Study in Historical Perspective*. Buffalo, NY: William S. Hein, 2002. https://www.amazon.com/Nature-Intellectual-Property-Clause-Perspective/dp/1575887096.

Washington Examiner. "Industry, Not Environmentalists, Killed Incandescent Bulbs." Washington Examiner opinion analysis. Documents the role of patent-protected lighting incumbents — through the National Electrical Manufacturers Association, representing GE Lighting, Signify (formerly Philips), Osram, Cree, and Acuity — in lobbying for the federal efficiency standards that displaced their unpatentable incandescent competitor. Cited in the bootleggers-and-Baptists section of the healthy-lighting chapter. https://www.washingtonexaminer.com/opinion/2372297/industry-not-environmentalists-killed-incandescent-bulbs/.

Webber v. Virginia, 103 U.S. 344 (1880). Supreme Court decision reaffirming the Patterson v. Kentucky principle that letters patent granted by the United States do not exclude from state tax or license law the tangible property in which the invention or discovery is embodied. Cited in the chapter on objections considered as part of the Supreme Court line establishing that patents and regulation coexist. https://supreme.justia.com/cases/federal/us/103/344/.

West Virginia v. Environmental Protection Agency, 597 U.S. 697 (2022). https://supreme.justia.com/cases/federal/us/597/20-1530/.

Westerman, Wayne, and John G. Elias. “Method and Apparatus for Integrating Manual Input.” U.S. Patent 6,323,846, filed January 25, 1999, issued November 27, 2001. Assigned to the University of Delaware. Foundational multi-touch capacitive touchscreen patent; expired 2019. Cited in the surveillance chapter as the foundational expired multi-touch patent from which a remanufactured vehicle’s in-dash capacitive touchscreen can be built. https://patents.google.com/patent/US6323846B1.

Wheaton v. Peters, 33 U.S. (8 Pet.) 591 (1834). Foundational Supreme Court decision holding that there is no common-law copyright after publication; intellectual-property rights exist only as Congress grants them under the Patent and Copyright Clause. Cited in the chapter on objections considered as the holding that establishes the Patent Clause as the sole constitutional source of intellectual-property rights, foreclosing the defense that a patent-equivalent commercial protection without an expiration date is something other than a patent right. https://supreme.justia.com/cases/federal/us/33/591/.

Whitney, Eli. U.S. Patent for the cotton gin, granted 1794 (term expired 1808 under the fourteen-year rule of the Patent Act of 1790). The cotton gin entry in the chapter’s eight-product table. Whitney’s during-patent commercial model was service-based at a rate equivalent to roughly nineteen hundred dollars per gin per year; after expiration, mass-produced gins were selling for about twelve hundred dollars by the 1830s and roughly five hundred dollars by the 1880s, a fifty-six percent real-terms drop. https://en.wikipedia.org/wiki/Cotton_gin.

Whittemore, Amos. U.S. Patent for the cotton carding machine, granted 1797 (term expired 1811 under the fourteen-year rule of the Patent Act of 1790). The cotton carding entry in the chapter’s eight-product table. The Whittemore design fell from a during-patent price of about thirty cents per pound of carded cotton to ten cents by the 1880s, a fifty percent real-terms drop. https://en.wikipedia.org/wiki/Amos_Whittemore.

Wickard v. Filburn, 317 U.S. 111 (1942). The Supreme Court held that Congress may, under the Commerce Clause, regulate even wheat a farmer grew for consumption on his own farm, on the ground that the aggregate effect of such activity on interstate commerce brought it within the commerce power. The decision marks the modern outer reach of that power. Cited in the chapter on the Patent Clause prong as the authority the regulatory wall's defenders rely on for the breadth of federal regulatory jurisdiction, and as the case the prong answers: the commerce power, however broad, does not extend to nullifying what the Patent Clause commands at the end of the patent term.
https://supreme.justia.com/cases/federal/us/317/111/.

Wilkinson, David. U.S. Patent for a screw-cutting machine, granted 1798 (term expired 1812 under the fourteen-year rule of the Patent Act of 1790). The screw-cutting entry in the chapter's eight-product table. The Wilkinson design fell from a during-patent price of about four dollars per pound of cut screws to roughly sixty cents per pound by the 1880s, a sixty-four percent real-terms drop, as automated threading entered competing American shops.
https://en.wikipedia.org/wiki/David_Wilkinson_(mechanic).

Williamson v. Lee Optical of Oklahoma, Inc., 348 U.S. 483 (1955). The unanimous Supreme Court decision establishing the modern rational-basis floor for federal economic classifications: a court asks only whether the classification is rationally related to some conceivable legitimate government interest, not whether the stated rationale is the real one. The decision upheld an Oklahoma statute requiring eyeglass replacement lenses to be made only from prescriptions written by licensed optometrists or ophthalmologists, against the challenge that the statute was economic protectionism dressed as health regulation. Cited in the chapter on objections considered as the doctrinal wall that almost every Fifth Amendment equal-protection challenge to federal economic regulation runs into and breaks on, and the principal reason equality under the law cannot serve as a standalone constitutional prong of the book's argument.
https://supreme.justia.com/cases/federal/us/348/483/.

World Trade Organization. Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS). Annex 1C of the Marrakesh Agreement Establishing the World Trade

Organization, signed at Marrakesh on 15 April 1994. Article 33 establishes the minimum patent term as twenty years from the filing date. The Agreement binds all 166 WTO member nations.
https://www.wto.org/english/docs_e/legal_e/27-trips_04c_e.htm.

Wyden, Ron, and Edward J. Markey. Letter to FTC Chair Lina M. Khan. July 26, 2024. United States Senate. Joint senatorial inquiry to the Federal Trade Commission urging investigation into automaker disclosures of connected-vehicle driving-behavior data to consumer-reporting brokers; draws on oversight responses obtained directly from General Motors, Honda, and Hyundai documenting per-vehicle broker payments on the order of twenty-six cents (Honda) and sixty-one cents (Hyundai). Source for the surveillance chapter's documentation of the broker-payment magnitudes.
https://www.wyden.senate.gov/imo/media/doc/wyden-markey_auto_privacy_letter_to_ftc.pdf.

XBOW. First non-human to top HackerOne's United States leaderboard, June 2025, as reported in "How an AI-Based 'Pen Tester' Became a Top Bug Hunter on HackerOne," *Dark Reading*, August 2025; HackerOne co-founder Michiel Prins confirmed the ranking to Bloomberg. Roughly 1,060 vulnerability reports submitted over a few months, including a previously unknown flaw in Palo Alto Networks' GlobalProtect affecting more than 2,000 hosts. Cited in the preface.
https://www.darkreading.com/vulnerabilities-threats/ai-based-pen-tester-top-bug-hunter-hackerone.

Yandle, Bruce. "Bootleggers and Baptists: The Education of a Regulatory Economist." *Regulation* 7, no. 3 (May/June 1983): 12–16.
https://www.cato.org/sites/cato.org/files/serials/files/regulation/1983/5/v7n3-3.pdf.

Yao, Xiaoxi, et al. "Artificial Intelligence-Enabled Electrocardiograms for Identification of Patients with Low Ejection Fraction: A Pragmatic, Randomized Clinical Trial." *Nature Medicine* 27, no. 5 (2021): 815-819. One hundred twenty primary care teams, 22,641 adults; new diagnoses rose from 1.6 to 2.1 percent, an odds ratio of 1.32.

Zoltek Corp. v. United States, 442 F.3d 1345 (Fed. Cir. 2006) (per curiam), reh'g en banc denied, 464 F.3d 1335 (Fed. Cir. 2006), cert. denied, vacated in part, 672 F.3d 1309 (Fed. Cir. 2012) (en banc). The panel held that a patent holder whose invention the United States uses may not bring a Fifth Amendment takings claim, and that the statutory remedy Congress supplied is the only one available. Judge Plager dissented from the panel opinion; Judge Newman dissented from the denial of rehearing en banc. In 2012 the court sitting en banc vacated the 2006 opinion, giving as its grounds the panel's construction of that statute. Commentators divide on what the vacatur did to the constitutional holding: some treat the takings analysis as vacated with the rest, others report that the reasoning was never repudiated and that later filings continue to cite the case for it. A Federal Circuit decision, not a Supreme Court decision. Cited in the objections chapter.
https://law.justia.com/cases/federal/appellate-courts/F3/442/1345/642048/.

Zwicky, Fritz. "Die Rotverschiebung von extragalaktischen Nebeln." *Helvetica Physica Acta* 6 (1933): 110-127. Applied the virial theorem to the Coma cluster and named the discrepancy *dunkle Materie*.
See Endnote Commentary.

ZX Auto (Hebei Zhongxing Automobile Co., Ltd.). "Corporate Introduction." ZX Auto Global Official Site. Source for ZX Auto's claim of operations in more than 90 countries and regions, cited in the farm-equipment chapter alongside Mahindra's Global Presence page as one of two manufacturer pages from which 43 countries can be individually extracted by name against the manufacturers' own claimed totals.
https://en.zxauto.com.cn/CORPORATE.html.

ZX Auto (Hebei Zhongxing Automobile Co., Ltd.). "TERRALORD." ZX Auto Global Official Site. Manufacturer product page for the ZX Terralord pickup truck. Specifications include 4WD diesel powertrain, 6AT/6MT transmission, double cab, three trim levels (Standard, Luxury, Supreme Luxury), four-wheel disc brakes, and 265/65R17 tires. The chapter on patent-expired international vehicles cites the Terralord alongside the Grand Tiger as a working pickup truck derived from late-1990s Toyota Tacoma platform

technology, exported by ZX Auto to more than ninety countries including Kenya, Ghana, Nigeria, the Philippines, Bolivia, and across Latin America. https://en.zxauto.com.cn/TERRALORD.html.

ZX Auto (Hebei Zhongxing Automobile Co., Ltd.). “GRANDTIGER.” ZX Auto Global Official Site. Manufacturer product page for the ZX Grand Tiger pickup truck. Specifications include 2.4L gasoline and 2.8T diesel engine options, 5-speed manual transmission, 2WD/4WD, double cab, and a 5310×1812×1800mm body. The chapter on patent-expired international vehicles cites the Grand Tiger as a working pickup truck derived from the 1997-to-2004 Toyota Hilux platform whose core mechanical patents have been in the public domain for more than a decade. Export pricing runs roughly $13,000 to $18,000 in the buyer markets the chapter discusses. ZX Auto, founded in 1949, is described on its own page as China’s first independent pickup brand. https://en.zxauto.com.cn/GRANDTIGER.html.

Endnote Commentary

(Ordered to match the Endnotes section, keyed to the corresponding entry above. Substantive notes only; not every Endnote has a Commentary entry.)

***The Chicago Manual of Style*. 17th ed.** The book follows the 17th edition (2017) because it is the most widely used scholarly standard for U.S. nonfiction trade and academic publishing, and because most U.S. literary agents and academic-trade publishers expect endnote apparatus formatted according to its conventions. The 18th edition, published in 2024, refines a number of digital-citation conventions but does not alter the core bibliographic forms the book’s apparatus uses. Where this book’s conventions diverge from strict Chicago style, the divergence is intentional and noted. The principal divergence is that every Endnote entry concludes with a working URL pointing to the source itself — the publisher’s canonical page for periodicals, books, and reports; Justia for U.S. Supreme Court cases; the Cornell University Legal Information Institute for federal statutes; Founders Online for Founding-era documents — rather than to a citation database or

aggregator. The aim is to put the reader in front of the authoritative location of each source. Some publisher canonical pages sit behind paywalls; the book does not provide workarounds for those paywalls, but it does name the canonical location so that any reader with institutional access, or any reader who chooses to purchase, can verify the citation directly at the source. Where the publisher's canonical URL is unavailable, the URL hierarchy falls back through DOI or scholarly database, with Amazon as a last resort for older works without stable publisher URLs. Court cases are linked to Justia. Federal statutes are linked to the Cornell University Legal Information Institute. Founding-era documents are linked to Founders Online at the National Archives.
See Endnote for full citation.

Boulware, David R., et al. "A Randomized Trial of Hydroxychloroquine as Postexposure Prophylaxis for Covid-19." The Minnesota trial is the most-cited randomized evidence against hydroxychloroquine, and the chapter offers it as an exhibit rather than as a result. The critique that follows is this book's own and is not the journal's: that the dose administered exceeded the range the early-treatment protocols specify, that zinc was not co-administered although the protocols treat it as necessary, and that participants were enrolled past the window in which early treatment is said to work. On the last point the published record supplies a mechanism: the drug was shipped to participants, and the shipping interval meant a portion of the treatment arm did not begin dosing inside the four-day post-exposure window the trial reports. Readers should weigh the trial and the critique together. The point the chapter draws is narrower than either — that when a molecule is off patent, no sponsor has a commercial interest in designing the trial that would vindicate it.
See Endnote for full citation.

Class 8 tractor, remanufactured: the $80,000 should-cost figure. The book prices a faithfully remanufactured pre-2000 Class 8 highway tractor — a Peterbilt 379 or Kenworth W900 class chassis around a Caterpillar 3406E, first-year C-15, Cummins N14, or Detroit Series 60 certified to 1999 standards — at $80,000 in 2025 dollars, against $172,900 for a 2025 Kenworth T680 sleeper (see the Kenworth Truck Company endnote). The figure is a projection, and the chapters that use it say so; no such market exists while the wall stands.

It is built from three components. The floor is the bare platform: a new Sinotruk HOWO 6×4 tractor of comparable horsepower sells for $28,000 to $35,000 FOB China with no aftertreatment (see the Sinotruk endnote), which is what a public-domain heavy-truck platform costs to build where no patent royalty, no diesel particulate filter, no selective catalytic reduction system, and no post-2000 electronics are carried. The remainder is American: domestic labor and materials, the independent engineering certification the remedy requires, and a domestic margin. The result sits at less than half the new-truck price rather than at one-fifth, because the book prices an American-built truck at American wages, not an import. The chapters that carry the figure — the opening chapter, the environmental chapter, the trucking chapter, the autonomous-vehicle chapter, the plaintiff chapter, the plaintiffs' recovery chapter (which compares against the $161,850 T680 day cab), and the Epilogue — all use $80,000, and the per-mile arithmetic in the autonomous-vehicle chapter (about eleven cents of capital cost over 750,000 miles) is derived from it. The methodology chapter states the book's base year: unless a year is given, prices are in 2025 dollars.

Code of Federal Regulations. "40 C.F.R. § 1068.240 — Exempting New Replacement Engines." The replacement-engine exemption is the nearest thing outside aviation to the time-shift this book proposes, and its limits are the point. Under it a newly manufactured engine may be built to the standards of an earlier model year. Four conditions confine it. Only a certificate holder may build the engine. The engine may only replace one already placed into service. The equipment may be no more than forty years old. And the maker must first determine that no engine certified to current standards will fit; where an engine of an intermediate tier fits, that one must be used. Section 1068.120 does the parallel work for the rebuilder, who must hold a rebuilt engine to the certified configuration of its own model year or a cleaner one, not to today's standards. So the EPA already accepts that an engine of an earlier year's design may lawfully be made new and run. What it does not accept is that engine in a newly built machine, or a new engine of that kind made by anyone but a certificate holder. Compare the Cessna record (see the Federal Aviation Administration, Type Certificate Data Sheet 3A12 endnote): there the time-shift reached the whole airplane,

but again only the certificate's holder and those it licensed. In neither field does the time-shift reach the public. The comparison is this book's own; no court has drawn it.
See Endnote for full citation.

Coin Laundry Association (CLA). The laundromat is the cleanest statutory hook in the American economy, and the reason is that no gap exists between the regulated equipment and the business model. A federal efficiency standard naming clothes washers as a covered product governs the only capital asset the business owns, so damages trace to the regulation at essentially one hundred percent — a posture no other category in the plaintiff chapter can claim. The arithmetic is unglamorous and that is its strength. A commercial washer-extractor runs from roughly three thousand dollars for a mid-size unit to well above twenty thousand for a high-capacity machine, and a working store is a floor of twenty, thirty, or more such units, so a full equipment retool is a six-figure capital event spent entirely on machines a federal standard governs. The patent-expiration anchor is equally uncontroversial: the foundational washing-machine patents date to the early twentieth century and expired generations ago. One caution on the trade body. It is a smaller organization than the restaurant or hotel associations and carries less Washington presence, and laundromat ownership is a small-business field, frequently immigrant-owned and pragmatically cost-driven, which makes it genuine but resource-limited as a litigation partner.
See Endnote for full citation.

Competitive-discount floor of fifteen percent. Two different fifteen-percent figures appear in the book, and they describe two different things. In the methodology chapter, the retrospective calculation of the $15.73 trillion floor applied a competitive-pressure discount in a range of eight to fifteen percent — component two of that calculation, the price reduction felt by the buyers who still purchase a new item rather than a remanufactured one. That range is a minor adjustment riding atop the dominant component, the direct savings of roughly forty to sixty percent realized by the buyers who purchase remanufactured goods, conservatively drawn from an observed international spread of fifty to eighty-eight percent. The fifteen-percent figure used in the recovery chapters is a

forward-looking estimate, not a restatement of that historical range: it is the conservative floor of the competitive discount a reopened commons would deliver after a favorable ruling. Three independent reasons make fifteen percent conservative rather than optimistic. First, in a post-ruling economy no durable good sits atop a fully patented bill of materials; somewhere in every supply chain remanufacturing introduces discounting, so the effect compounds through the chain rather than capping at the final assembly. Second, the roughly ninety countries where patent-expired remanufacturing is already lawful show real-dollar prices well below a fifteen-percent discount off the American equivalent — often a half or a third of the U.S. price — so fifteen percent understates what those markets actually achieve. Third, the methodology's "flowing" tier deliberately excludes a large mass of goods that would in fact be remanufactured once the fifty-eight percent blocked tier is freed, so the conservative base itself omits real savings. A wholly independent verification path points the same way: applying a twenty-five to forty percent savings rate (a thirty-four percent midpoint) to an $81.1 trillion present-value base of consumer durable spending yields roughly $28 trillion, comfortably above the $15.73 trillion floor and confirming the floor is conservative from two directions. See the methodology chapter for the full calculation and the recovery chapters for the per-household figures.

Gautret, Philippe, et al. "Hydroxychloroquine and Azithromycin as a Treatment of COVID-19." The retraction is itself part of the record this chapter maps. The paper was published in March 2020, cited thousands of times, criticized on methodological grounds within months — the journal printed Rosendaal's review and the authors' replies in the same volume — and then retracted in January 2025, four and a half years later. A retraction that arrives half a decade after publication, in a journal that had already printed the criticism and the reply, is a different act from a retraction that corrects a fresh error. The chapter cites the paper for what it reported and records the retraction so that a reader can weigh both. The URL given in the Endnote is the authors' own institutional copy, with an archived snapshot, because the publisher's copy now carries the paper only behind the retraction notice.

See Endnote for full citation.

Doenges, J. L. The chapter compresses a ninety-year sequence into a paragraph, and the sequence deserves its dates. Bizzozero (1893) described spiral organisms living in the glands of the stomach — in dogs, a qualification the popular retelling drops. Doenges (1938) examined 242 human stomachs after death, found the same organisms in forty-three percent of them, and connected them to no disease. Freedberg and Baron (1940) found them in living patients; the result was never repeated, and nobody published a failure to repeat it either. Then Palmer (1954) studied stomach tissue from a thousand people and concluded the organisms were contamination carried in from the mouth. Medicine already believed the stomach was sterile, because stomach acid kills bacteria. Palmer's study told the field what it already believed, on a sample too large to argue with, and the question closed for thirty years. That is why the case is labeled mixed rather than filed as a clean unasked question. The slides were an unasked question for ninety years. The answer, once Warren and Marshall reopened the question in 1979 and 1981, came from the better tool: the bacterium had to be grown, the classical test for proving that a microbe causes a disease had to be met, and clinical trials had to follow. Both halves are true and the chapter needs both. The sharper point survives the labeling. The organisms were not merely present and unremarked. They had been counted, in a published series, in nearly half a large sample — and then explained away.
See Endnote for full citation.

Elster, Jon, ed. *Secrecy and Publicity in Votes and Debates*. The Italian parliament's *voto segreto* practice operated under two distinct legal regimes across its 119-year history, a feature of the record the chapter's single-sentence reference does not set out. The complete timeline:

Period	Authority for voto segreto
1848 – 1922	Constitutional mandate under Article 63 of the Statuto Albertino: secret balloting was the constitutionally required method for every final vote on a law and for all votes of a personal nature. Seventy-four years on direct constitutional authority.
1922 – 1943	Fascist suspension. Mussolini's March on Rome (October 1922) did not formally abolish the Statuto, but parliamentary practice was suspended in substance over the twenty-one years of Fascist rule.
1943 – 1948	Transition. Mussolini fell July 1943; the September 1943 armistice opened a five-year transitional period; a June 1946 referendum abolished the monarchy; a Constituent Assembly drafted a new Constitution.
1948 – 1988	Chamber rule under the *Regolamento*. The 1948 Republican Constitution formally abolished the Statuto Albertino in its entirety. Its Article 64 transferred voting procedure from constitutional authority to each chamber's own rules, where *voto segreto* continued for the next forty years on chamber-rule authority alone.
1988 – present	Reformed. On October 13, 1988, the De Mita government repealed *voto segreto* as the default method for final votes through amendments to the Camera and Senato *Regolamenti* — no constitutional amendment required, because the practice no longer rested on constitutional ground. *Voto segreto* survives today for "conscience" votes (life-and-death, family, individual moral status) and personal-nature votes (individual members or ministers) only.

Article 63 of the Statuto Albertino, in English translation: *Votes shall be taken by rising, by division, and by secret ballot. The latter method, however, shall always be employed for the final vote on a law and in all cases of a personal nature.* That is a direct constitutional mandate, not a chamber preference. The 1848 framers of the Statuto thought secret balloting on final votes was important enough to put into the constitutional text itself, not merely into chamber rules.

How the practice lost constitutional rank is a story in itself, and it has no American analogue. The Statuto was a *flexible* constitution. It set out no amendment procedure, and Article 65 gave the interpretation of laws to Parliament rather than to any court, so it carried no judicial review. Everything in it, Article 63 included, could be altered by ordinary legislation. Nothing was entrenched in the sense an American reader assumes.

The demotion therefore did not happen by amendment. It happened at a refounding. The 1948 Republican Constitution replaced the Statuto entirely, and the Constituent Assembly had to decide, provision by provision, what to carry forward. On 14 October 1947 the Christian Democrat leader Aldo Moro moved to strike secret voting from draft Article 69 of the new text, arguing that rules governing the legislative process belong in each chamber's own rules of procedure and not in a constitution. The Assembly approved the change the next day. A Communist amendment to keep the provision was rejected 135 to 160 — by secret ballot. The motives were strategic on both sides. The Christian Democrats, having learned from the June 1946 election that they could expect to govern, wanted open voting to enforce party discipline; the Communists, expecting permanent exclusion from government, wanted secret voting precisely because it would let them exploit divisions in future majorities. Both read the future correctly.

That single drafting decision is why October 1988 required no constitutional amendment. Because Article 64 of the 1948 Constitution left voting procedure to each chamber, the repeal needed only an amendment to the Regolamento, carried by absolute majority. It passed 323 in favour, 58 against, with 222 abstaining — seven votes above the 316 required. The constitutional objection was raised and nearly prevailed: three motions asked whether Parliament had the right under the Constitution to reform secret voting at all, and the floor resolved the question by secret ballot, 258 holding that reform violated no constitutional principle against 231 holding that it did.

Two points of precision. The Fascist interruption is dated here from 1922 because parliamentary practice was hollowed out from that year, but the formal abolition came later: Law No. 129 of 19 January 1939 created the Camera dei Fasci e delle Corporazioni and provided at Article 15 that voting is always open, the chamber rules of December 1938

having already said so. And the 119-year figure used in the chapter is the 140-year span from 1848 to 1988, which is the figure Giannetti uses, less the 21 years of the Fascist interruption. Both numbers describe the same record.

One further point belongs on the record, because it is the strongest argument against the reform this chapter proposes, and it comes from the Italian precedent itself. Italy abolished *voto segreto* because the secret ballot had become the instrument of the *franchi tiratori* — the snipers, factions inside governing parties who anonymously killed their own government's bills. Prime Minister Cossiga won an open confidence vote 329 in September 1980 and lost the secret vote on the same decree minutes later, 297 to 298, and resigned. Craxi won confidence 338 in June 1986 and lost the secret vote 266, a twenty-one percent desertion, and resigned. Governments fell not despite winning in the open but immediately after winning in the open.

The objection is real, and the answer is sequence. The Italian secret ballot came *second*, and destroyed what the open vote had just approved. Hybrid Voting reverses the order: the open vote comes first, and the secret ballot is reached only on a bill that has already failed. It cannot be used to sink a measure that passed in the open, because a measure that passed in the open never reaches it. The two systems deploy the same pair of instruments in opposite order, and the order is what determines whether the secret ballot destabilises a government or rescues a bill from the lobby.

The Hybrid Voting chapter's argument that the United States could adopt the procedure by simple chamber-rule majority under Article I § 5 cl. 2 finds its closest historical analogue not in the Statuto-era mandate but in the post-1948 chamber-rule regime: forty years of working secret-ballot practice, in a major Western parliamentary democracy, on chamber-rule authority alone, terminated when the chamber chose to terminate it. The American reform proposed in the Hybrid Voting chapter requires neither a constitutional amendment nor a chamber-rule mandate at the level the Statuto provided — only the simple-majority chamber-rule adoption the post-1948 Italian parliament already operated under for forty years.

See Endnote for full citation.

Erlich, Mark. *With Our Hands: The Story of Carpenters in Massachusetts*. Three things in Erlich's record deserve a longer note than the preface's use of it can carry. The first is that the carpenters' resistance was not simple Luddism. Oscar Pratt, one of Erlich's interview subjects, acknowledged that the men could not refuse to use the new tools even while they could see what those tools would do to the trade — which is to say the doubters misjudged how fast the tool would improve and read its effect on their craft correctly. The analogy in the preface is drawn to the first error and not to the second. The second is that the 1932 Springfield item was a proposal rather than an imposed penalty. Erlich documents a fine that did land: in 1964 a member of Local 685, Philador Lemay, was found to have violated union rules and fined twenty-five dollars after it emerged that his greater speed rested on supplying his own truck, four-foot level, and electric saber saw for his employer's use. By then union by-laws placed the duty to furnish power tools on the employer, which was the point of the 1932 proposal in the first place. The third is a textual variance worth recording. The United Brotherhood of Carpenters and Joiners of America retells the roof story on its own history pages, citing Erlich at page 146 and attributing it to a Massachusetts carpenter who joined the union in 1946; the union's version renders the carpenter's line as a fear of cutting his hand, where Erlich's text has him declaring he will cut by hand. This book follows Erlich, whose reading is the primary one and the one that parses with the man's refusal. United Brotherhood of Carpenters and Joiners of America, "Tool Revolutions." https://www.carpenters.org/tool-revolutions/. See Endnote for full citation.

Expired Patent Commons. This is the central term of art of this book, and the reason the book carries the title it does. When a patent reaches the end of its limited term, the invention it covered is meant to pass to the public — to be made, copied, improved, and remanufactured by anyone. The Supreme Court's cases call that destination the public domain. This book calls it the Expired Patent Commons, and the choice of the word "commons" is deliberate. A commons is not an empty space or the mere absence of a right; it is a shared resource the public can actually reach and actually use, as villagers once walked onto common land and grazed it. An invention has entered the Expired Patent Commons only when an American may in fact pick it up and build it — the term is defined

by use, not by paper. A patent that has lapsed on paper, but that federal regulation forbids anyone to manufacture, has not entered the commons; it is merely a patent that has expired. The Expired Patent Commons is the public's half of Jefferson's bargain — the forgotten half of the Patent Clause — and the book's argument is that the post-1970 regulatory wall does not raid this commons but prevents it from ever forming. The term is defined in full in the chapter developing the Patent Clause prong.

Fawzi, Alhussein, et al. "Discovering Faster Matrix Multiplication Algorithms with Reinforcement Learning." AlphaTensor is often folded into the same story as the attention results in the chapter's third movement, and it should not be. The system found a way to multiply certain small tables of numbers using fewer multiplications than the best method known since 1969. One qualifier is not optional: the headline four-by-four result holds in arithmetic modulo two, not in ordinary arithmetic, where Strassen's method still stands. Kauers and Moosbauer (2023) answered within about a week, and they did not merely match the machine. They found further four-by-four schemes at the same length, then went past it, cutting the five-by-five case from AlphaTensor's ninety-six multiplications to ninety-five and a rectangular format from sixty-three to sixty-two. Their method, which they call flip graphs, is their own. Their computer was small and the search took seconds. And they said plainly that they had used the machine's output as their starting point. The sorting case ran the same way by a different route. AlphaDev (Mankowitz 2023) found faster routines for sorting short lists at the level of individual machine instructions, and they were adopted into a standard library that a great deal of software uses. The paper also claimed that for the three-element case no shorter program existed. Neri (2023) produced counterexamples to the impossibility claim, and a plain implementation that ran faster besides. Two machine results, then, and two human responses. The machine results are the better tool: a machine built something that had not existed before, and that indicts nobody. The human responses are the unasked question. AlphaDev asserted an impossibility and was wrong, and being wrong in public is what provoked somebody to check. AlphaTensor asserted nothing of the kind. It reported an improved upper bound, which is all anyone reports in that field, and in doing so showed that ground the discipline had stopped digging in, sixty years after Strassen, was not exhausted. Within days, mathematicians who had held

the capability the whole time went further than the machine had. What the machines dislodged was not a technical barrier. In one case it was a false claim. In the other it was a settled belief that a question was closed.
See Endnote for full citation.

Federal court procedure: party labels at each level. A reader who has followed the book through its chapters on the filable case and on plaintiff categories will at some point notice an apparent inconsistency in how parties to the Supreme Court's cases are named. The chapters on filability and on who can sue use *plaintiff* and *defendant* throughout. The cases themselves — *Loper Bright*, *Cedar Point*, *Sheetz*, *Massachusetts v. EPA*, *Horne* — are captioned with the constitutional challenger sometimes first and sometimes second, and the Court's opinions speak of *petitioner* and *respondent* rather than of plaintiff and defendant. The inconsistency is not editorial. It is the way the federal court system labels the parties at each layer, and the labels shift as the case moves upward.

The terminology runs as follows. In federal district court, where every constitutional challenge to a federal regulation begins, the party who files the complaint — the grandfather-exemption advocate, in the case this book describes — is the *plaintiff*; the federal agency named as defendant (the EPA, NHTSA, or DOE, represented in court by the Department of Justice) is the *defendant*. The chapters of this book that describe who can file the case and what the complaint will ask for use those labels because that is where the case begins. At the United States Court of Appeals, whichever side lost in the district court and appealed becomes the *appellant*; the other side is the *appellee*. At the Supreme Court, the party who sought review by filing the petition for a writ of certiorari is the *petitioner*; the other side is the *respondent*. The case caption at the Supreme Court level always lists petitioner first.

Whether the grandfather-exemption advocate stands at the Supreme Court as petitioner or as respondent depends entirely on who won at the Court of Appeals. If the advocate loses at the circuit court and petitions the Supreme Court for review, the advocate is the *petitioner*. That is the procedural posture of every modern Supreme Court win this book builds on — *Loper Bright Enterprises v. Raimondo* (fishermen petitioned; won), *Cedar Point Nursery v.*

Hassid (growers petitioned; won), *Sheetz v. County of El Dorado* (landowner petitioned; won), *Massachusetts v. EPA* (states petitioned; won), *Horne v. Department of Agriculture* (raisin growers petitioned; won). The constitutional challenger as petitioner is the modal posture in the cases the book relies on. If the advocate instead wins at the circuit court and the federal government petitions the Supreme Court, the advocate is the *respondent*. *Ruckelshaus v. Monsanto Co.* is the canonical example on the Takings Clause side: Monsanto had won below; the EPA Administrator petitioned; Monsanto stood as respondent and won again.

The plaintiff and defendant labels do not disappear when the case reaches the Supreme Court. They remain attached to the case's history, and the briefs and the Court's opinion will refer back to "the plaintiff below" or "the district-court defendant" when recounting the procedural posture. But they are not the active labels at the Supreme Court. The advocates speak as petitioner and respondent. The caption reads "Petitioner v. Respondent."

One usage note for the reader who reads journalism or general-interest writing on Supreme Court cases. Both routinely use *plaintiff* loosely to mean "the original challenger" all the way through the litigation, including at the Supreme Court. The usage is imprecise but common, and a reader who carries the imprecise usage into the Court's own opinions will misread some of the citations. In a book written as a brief in waiting for an actual Supreme Court case, the precise terminology — plaintiff at the trial court, appellant or appellee at the circuit, petitioner or respondent at the Supreme Court — will read more credibly to the lawyer who eventually picks up the brief and works from it.
See Endnote for full citation.

Ford Motor Company. "2025 Ford F-150 Lariat 4×4 SuperCrew 5.5′ Box." The chapter's title figure of $72,495 is the out-the-door total a typical American family pays to drive a new F-150 home in 2025. It reflects the Lariat 4×4 SuperCrew 5.5′ Box — Ford's mid-tier family trim, positioned above the work-truck XL (base $37,250 MSRP) and the volume-trim XLT (approximately $47,590 MSRP) but below the luxury-tier Platinum ($74,905 base MSRP) and Raptor ($79,005 base MSRP). The Lariat is what Ford positions

as the family truck: the trim that combines hauling capacity with the interior appointments a household financing a new vehicle expects.

The components of the $72,495 total reconstruct as follows. The Lariat 4×4 SuperCrew 5.5′ Box base MSRP for model year 2025 was $65,255 at announcement, revised marginally during the production year as Ford adjusted incentives and option-package pricing. The destination and delivery charge added $1,995, producing a manufacturer's invoice subtotal of $67,250. State and local sales tax, computed at approximately 7 percent on the manufacturer's invoice subtotal — the U.S. weighted average across state and local jurisdictions in 2024–2025, per the Tax Foundation — added $4,708. Title, license, and registration fees added approximately $200, varying by state. The dealer documentation fee, which varies by state and dealership but commonly runs $300 to $700, added approximately $337 at the median. The total at the dealer's cashier — the amount the buyer's lender finances or the buyer writes in a check — is $72,490, within rounding of the $72,495 the chapter uses.

The figure converges from a second verification path. TrueCar dealer-inventory listings show real-world 2025 Lariat 4×4 SuperCrew 5.5′ Box PowerBoost Hybrid configurations at MSRP-with-destination totals of $72,480, $72,580, and $72,690 — before tax and fees. The hybrid drivetrain adds approximately $5,000 to the base Lariat's MSRP, but the verification is about the figure, not the engine: the base Lariat's out-the-door total and the hybrid Lariat's bare MSRP-with-destination converge at the same number, $72,495, from two different directions. The convergence is what gives the chapter title figure its specificity.

The $25,000 should-cost baseline against which the $72,495 is compared is the book-canonical figure established in the wealth-transfer chapter. The $47,495 per-vehicle wall premium is the arithmetic difference between what the family pays and what the family would pay for an equivalent-utility patent-expired vehicle absent the post-1970 federal regulatory architecture this book has been tracing. The reader who wants to reconstruct the out-the-door figure independently can do so from Ford's published 2025 MSRP schedule on shop.ford.com, the destination charge listed on every Ford window sticker, the reader's

own state and local sales tax rate (or the Tax Foundation's national average), and a representative dealer documentation fee for the reader's state.
See Endnote for full citation.

Irons, Peter. *Justice at War*. Aiko Herzig-Yoshinaga had no law degree, no university post, and no institutional standing of any kind. As a teenager she had been imprisoned at Manzanar, one of the camps where the United States held Japanese Americans during the Second World War. In the late 1970s she was working in the National Archives in Washington, and she found what forty years of litigation had not: the government's own original copy of General DeWitt's Final Report on the removal, containing passages that a later, revised version had taken out. The deleted material showed the government had known there was no time to hold individual loyalty hearings, and that the claim of military necessity made to the Supreme Court in 1944 had been made while contrary evidence sat in the file. Irons, a lawyer, carried the finding into court. Fred Korematsu had been convicted for refusing to go, and the Supreme Court had upheld the conviction in Korematsu v. United States (1944). In 1983 a federal court in San Francisco vacated it on a writ of coram nobis — an old and rarely used procedure that lets a court reopen a case when it learns the original proceeding was tainted. In Trump v. Hawaii (2018) the Supreme Court said Korematsu had been gravely wrong the day it was decided. The chapter calls this the most damning unasked question in its set, and the reason is the location of the evidence. It was not merely available somewhere. It was in the government's own files, put there by the government, kept by the government, for forty years. The finding and the argument sat in two people, which is the pattern the chapter's second movement is about.
See Endnote for full citation.

Jefferson, Thomas. "Letter to Isaac McPherson, August 13, 1813." Chapter 2 quotes a single line from this letter — the line in which ideas spread "like fire, expansible over all space, without lessening their density in any point." That line is one of three poetic images Jefferson stacks in a single passage, and it is not the most famous of the three. The full passage runs as follows:

> *If nature has made any one thing less susceptible than all others of exclusive property, it is the action of the thinking power called an idea, which an individual may exclusively possess as long as he keeps it to himself; but the moment it is divulged, it forces itself into the possession of every one, and the receiver cannot dispossess himself of it. Its peculiar character, too, is that no one possesses the less, because every other possesses the whole of it. He who receives an idea from me, receives instruction himself without lessening mine; as he who lights his taper at mine, receives light without darkening me. That ideas should freely spread from one to another over the globe, for the moral and mutual instruction of man, and improvement of his condition, seems to have been peculiarly and benevolently designed by nature, when she made them, like fire, expansible over all space, without lessening their density in any point, and like the air in which we breathe, move, and have our physical being, incapable of confinement or exclusive appropriation.*

Three images stand out. The first is the thinking-power image: an idea, once divulged, cannot be reclaimed. The second is the taper image — he who lights his taper at mine receives light without darkening me — which is the line most often quoted in patent scholarship and in economic writing on non-rival goods. The third is the fire-and-air image, which is the line Chapter 2 quotes. Each of the three has at one time or another been treated as the heart of the passage. The reader who wants Jefferson's full argument — the part the chapter pivots to — will find the entire letter at the URL listed in the corresponding Endnote.

See Endnote for full citation.

***Kahle v. Gonzales*.** Brewster Kahle, the engineer and archivist who built the Internet Archive, made a structural observation of exactly the kind this book makes. When Congress removed the requirement that copyright owners renew their copyrights, it turned an opt-in system into an opt-out one. On paper, no individual term changed. In practice, works stopped entering the public domain by default, because nobody now had to do anything to keep them out of it. That is an observation about how a system behaves, not about what a statute says. He lost. The Ninth Circuit told him he was making the argument the Supreme

Court had already rejected in Eldred v. Ashcroft (2003), wearing different clothes. Two things in the opinion matter to this book, and they cut opposite ways. The court admitted an opening. Eldred, it said, had not directly decided whether the words “limited Times” put a real ceiling on how long a copyright can last; the Supreme Court had expressly left that question open. Then the Ninth Circuit answered it, and answered it badly for anyone following. It held that the ceiling is found by weighing the encouragement longer terms give authors against the benefit shorter terms give the public, and that the weighing belongs to Congress, subject only to a test courts almost never fail anyone. One sentence keeps the door from shutting: future extensions may or may not survive review, but the current term is constitutional. The distinction this book rests on is that Kahle was about extending a term. The argument here alleges no extension of anything. The patent term ran and expired on schedule; the wall stands in the regulation that outlived it. Whether that distinction holds is for a court to say. It is stated here as this argument’s own, not as something Kahle decided.

See Endnote for full citation.

Knorozov, Yuri V. This is the purest case in the chapter, and the correction is what makes it so. In the sixteenth century the Spanish friar Diego de Landa wrote down what he took to be the Maya alphabet. His manuscript was published in Paris in 1864, and photographic copies of the surviving Maya books were in print by the end of that century. Everything needed was public. Scholars used it and failed, and not for want of trying. De Landa’s list is inconsistent, some of its signs never appear in real inscriptions, and — above all — everyone read it as an alphabet. In 1952 Knorozov, working in Leningrad, read it as a syllabary instead. When de Landa asked his Maya informant for the letter B, the informant gave him the syllable be. Knorozov added a rule about how syllables pair, and the writing began to open. The popular version says nobody asked. Many asked. What Knorozov brought was not evidence and not an instrument. It was a different idea about what kind of writing it was. That is why this case, of all the cases in the chapter, is the one that matches the Patent Clause. The Clause has been read continuously since 1787. Reading a text

wrongly is not the same as never reading it, and that is why forgetting is the word the book uses for what happened to the Clause's second half.

See Endnote for full citation.

Magill, Gord. The foreword author's media reach, referenced in the Preface. Gord Magill is a third-generation trucker with more than twenty-five years behind the wheel. He has written for *Newsweek*, *The American Conservative*, and *American Affairs*; publishes the *Autonomous Truck(er)s* Substack; and hosts the *Voice of GO(r)D* podcast. He has carried the trucking story onto the two largest platforms of the populist right. He appeared on *The Tucker Carlson Show* twice — on January 24, 2024, on the imprisoned "Coutts Four" and the 2022 Freedom Convoy, and on June 1, 2026, in a roughly sixty-five-minute conversation built around *End of the Road* and the long deregulation of trucking. The reach figures are drawn from contemporaneous reporting. CNN reported in June 2025 that Carlson's program averaged roughly 651,000 views an episode on YouTube in May 2025 and roughly 1.06 million in June 2025, with individual interviews drawing several million and a single teaser clip on X reaching 39.9 million. Magill also appeared on Steve Bannon's *WarRoom* on April 15, 2025, on foreign labor in the trucking workforce; *WarRoom* streams several times each weekday and has ranked for years among the most popular political podcasts in the United States, described by NBC News as routinely near the top of Apple's charts.

See Endnote for full citation.

Markopolos, Harry. *No One Would Listen*. Markopolos's own account gives the timing the chapter compresses. His firm asked him to copy a competitor's investment returns. The competitor was Bernard Madoff. Markopolos looked at the numbers and could not copy them, because they were not possible; he has said it took him about five minutes to see that, and about four hours to build the model that proved it. Those figures are his own, from the time, not a story improved in the retelling. He told the Securities and Exchange Commission in 2000. He told them again in 2001. In 2005 he sent the agency a document titled "The World's Largest Hedge Fund is a Fraud." The agency did not act, and the scheme collapsed in December 2008, taking roughly sixty-five billion dollars of client

money with it. The chapter opens its first movement with this case because it has the same shape as the book's argument. Nothing was hidden. Markopolos did not find a document. He read numbers the agency already had, that anyone could have read, and he was not an insider at the agency in any sense — no standing, no channel, no way to make anyone look. The parallel does not need labor. The rules in this book are published in the Federal Register. The patent expirations are docketed. Nobody has to find anything.
See Endnote for full citation.

Martin, Roleigh. "The Forgotten Half of the Patent Clause: What was Promised When Patents Expired." The $30 to $45 trillion range this book gives as the plausible true total rests on seven categories of cost the consumer-expenditure floor calculation excludes. Each is named here so that a reader can see what the floor leaves out rather than take the range on trust.

First, compounding economic effects: reduced business formation, lost innovation, and diminished entrepreneurship in sectors where the cost of regulatory compliance puts new entrants out of the market before they can begin. Second, indirect costs through the rest of the economy: higher housing costs caused by expensive construction equipment, higher food costs caused by expensive farm equipment, and higher transportation costs that ripple into the price of everything moved by truck. Third, the time value of money — the floor calculation performs no present-value adjustment, and one would raise the figure substantially before any other exclusion is added. Fourth, specialized industrial-equipment categories: mining, oil and gas, semiconductor manufacturing, pharmaceutical production, textile machinery, packaging equipment, and a long tail of others the floor did not attempt to price. Fifth, commercial vehicles beyond those counted: delivery vans, buses, and specialized trucks outside the categorical breakdown. Sixth, building materials and systems: windows, insulation, and roofing affected by post-1970 energy codes. Seventh, the innovation tax — products never developed because regulatory barriers made market entry impossible. See Endnote for full citation.

Martin, Roleigh. A note from the author, on the path that led to this book. The recognition that the post-1970 federal regulatory wall around expired-patent commerce constitutes a

constitutional violation, rather than merely an unfortunate policy outcome, did not come to me from a law degree or from a career in patent litigation. It came from two professional habits of mind I have spent a working life inside.

The first is game theory. My academic background trained me to look for the structure underneath an outcome — to ask, of any policy outcome whose stated rationale and observed effect do not match, what set of player incentives, what set of payoff matrices, and what set of moves the rules permit and forbid, would produce the observed effect from rational actors. Applying that lens to the post-1970 expansion of federal energy, safety, and emissions regulation produced, after some years of reading and re-reading the Patent Clause against the regulatory record, the recognition the book develops in detail: that the wall converts the patent bargain from a contract with two halves to a contract with one, and that the constitutional text already names the missing half. The half is in the words *limited Times.*

The second is the working habit of a man who spent thirty-four years in software — first as a programmer, then as a database programmer, inside the technical organization of a Fortune 25 company. The first joke I ever heard about my occupation, the one I have kept all the way through, is that *the worst programmer is the one who does exactly what you ask them, and nothing more.* The joke is funny because it is precisely right. A compiler does not interpret intent. It reads what you wrote, applies the language's rules, and returns either an executable result or a compilation error. It does not consult the spirit of the document; it consults the syntax of the document. A working life shaped by that machine produces a habit of reading any structured text as if it were code — including the Constitution.

Article I, Section 8, Clause 8, read by the compiler-reading mind, behaves like any other line of structured text. The phrase that arrests the reading is *limited Times.* It has a type signature. It is a duration. Durations have a start and an end. For a database programmer in particular, the same clause reads as a foreign-key constraint: the patent points at a defined term, the federal regulations that the rest of this book examines point at the patent-expired product, and the two tables join on the patent. When the patent's term expires, the parent row reaches end of life. Any regulation that continues to reference the patent-protected

status as if it were live is holding a foreign key to a row that no longer exists in the form the constraint requires. A database with referential integrity enabled refuses the operation. The federal government has been operating with referential integrity disabled since approximately 1970. The book is, in one of its readings, a fifty-year audit log of constitutional referential-integrity violations.

The two habits reinforce each other. Game theory asks what incentive structure would produce the wall as the equilibrium outcome of rational actors. Compiler-reading asks what the constitutional text, read as written, requires of the actors regardless of their incentives. The first habit explains why no one stopped the wall from being built. The second habit explains why the Constitution did not authorize what was built. The book is the long form of both observations.

The same compiler-reading habit, applied to the constitutional clause this book's Hybrid Voting chapter spends most of its constitutional argument on, returns a result the constitutional consensus has not. Article I, Section 5, Clause 3 reads in part: "the Yeas and Nays of the Members of either House on any question shall, at the Desire of one fifth of those Present, be entered on the Journal." Two phrases in that sentence admit ambiguities that two centuries of practice have papered over.

The first is *the Yeas and Nays of the Members.* Structurally a noun phrase, it could resolve to "the count of yeas and the count of nays" — a numeric tally — or to "the named position of each member" — a per-member roster. The text does not specify which. A compiler asked to resolve a noun phrase that admits two materially different parsings emits an error and demands clarification from a higher authority. The second is *on any question.* Structurally adverbial, it could resolve to "on every vote a chamber takes on the question" or to "on the question, once" — meaning a chamber that votes a bill twice could satisfy the clause by recording the named yeas and nays on either round. Again, the text does not specify. The constitutional consensus on both phrases is supplied by two centuries of legislative practice; legislative practice is not the type of authority a compiler accepts. A compiler accepts the text, the rules of the language, and binding constructions from a higher level of authority. Above the chamber, two such layers exist: the constitutional text itself,

which on these two questions is ambiguous, and the Supreme Court's construction of the text, which on these two questions does not exist. With both higher layers silent, the compiler accepts the next available authority — and the Constitution itself names what that authority is. Article I, Section 5, Clause 2 grants each chamber the power to *determine the Rules of its Proceedings,* and the rules of each chamber are, in practice, set by the procedural majority the chamber's elected leadership commands. The Speaker of the House and the Majority Leader of the Senate, each elected by the chamber that operates under the Yeas-and-Nays clause, hold that procedural majority. The compiler-reading version of the political-question doctrine, then, is not an evasion the courts performed; it is a delegation the Constitution wrote, with two ambiguous phrases handed downward through the layers of authority until they come to rest with the leadership the chamber elected. The identity of that leadership is therefore not procedural housekeeping. It is the runtime variable on which the meaning of two constitutional ambiguities is set, and the chamber that elects its leader is, in the act of that election, choosing which resolution the Yeas-and-Nays clause will receive.

The Hybrid Voting chapter develops both of these textual ambiguities at greater length and closes with a line from Bernardo Bertolucci's *The Last Emperor* — Reginald Johnston, played by Peter O'Toole, instructing the young Emperor Pu Yi: "If you cannot say what you mean, Your Majesty, you will never mean what you say. And a gentleman should always mean what he says." The Patent Clause meant what it said, on both halves of the bargain it described. The Yeas-and-Nays clause, on these two questions, did not.

See Endnotes for full citations to the author's articles and working paper.

Kory, Pierre, et al. "Review of the Emerging Evidence Demonstrating the Efficacy of Ivermectin." The Expression of Concern the journal attached to this paper repays reading closely, because of what it declines to say. It states that it "does not imply that the methodology used by Dr. Pierre Kory and his collaborators was incorrect," and that "the use of summary data published by others is a generally accepted approach in biomedical metanalytic research." A journal that concedes the method was sound, and attaches a notice anyway, has recorded something about the pressure on the literature rather than about the

paper. Kory is a critical-care physician, a founder of the Front-Line COVID-19 Critical Care Alliance, and a co-author of the curated cancer-care standard this chapter cites elsewhere. The chapter takes no position on ivermectin's clinical efficacy; it records that the hindrance pattern it describes reaches the published record and not only the pharmacy counter.

See Endnote for full citation.

Martin, Roleigh, ed. *ShareDebate International*. *ShareDebate International* was a small-press project at the leading edge of pre-Web electronic publishing. The magazine was a freeware diskette publication — which in 1990 meant a 1.44-megabyte 3.5-inch disk written in IBM PC format, copied by thousands of bulletin-board system operators, and read by their users one at a time on personal computers running DOS. After Issue 6 the publication moved from shareware to copyrighted freeware, eliminating the suggested subscription fee while preserving the distribution structure. Contributors across the magazine's run included Milton Friedman, Murray Rothbard, George Gilder, Doug Bandow, Eamonn Butler, Jerry Pournelle, Leonard Peikoff, Ben Bova, J. Neil Schulman, Victor Koman, Joseph Sobran, George Benston, George Kaufman, Lt. Gen. Daniel O. Graham, Paul Levinson, Douglas Ostrom of the Japan Economic Institute, Masanobu Taniguchi, the Libertarian Party, the Republican Liberty Caucus, and the League for Programming Freedom (with Richard Stallman). Ben Bova praised the magazine as "a bold, exciting venture" deserving of broad attention; J. Neil Schulman called it the arrival of "grassroots mass-market publishing."

Two threads in the magazine bear directly on the present book. The first is Debate Topic #4, the "Japanese VS American Patent Law Debate," opened by the editor in Issue 1 (1990) with the framing: *Is the Japanese Patent Law more just and more conducive to economic growth than America's Patent Laws? A side aspect to this topic is the fairness and economic impact of America's patent laws.* That side aspect — the fairness and economic impact of America's patent laws — is a variant of the question this book takes up thirty-five years later in a deeper constitutional form. Issue 3 carried a four-page engagement letter from Nobel Laureate Milton Friedman, dated July 26, 1990, on Hoover Institution

letterhead, praising the project's premise, raising three substantive critiques, and wishing the editor well. Friedman returned to the magazine across multiple subsequent issues. The second relevant thread is Debate Topic #6, "Government's Impact on Health Costs," which prefigures the broader cost-of-living argument the author's MALA (Making American Living Affordable) project has taken up in the years since. Third-party commentary on the magazine appears in Easton (2006), cited separately above. *See Endnote for full citation.*

Ostrom, Elinor. *Governing the Commons*. Elinor Ostrom was awarded the 2009 Nobel Prize in Economics for her work on the management of common-pool resources — the first woman to receive the prize, and one of the few political scientists ever to receive it. The award recognized a body of work spanning thirty years and hundreds of empirical cases: fisheries, irrigation systems, forests, grazing lands, groundwater basins. Her 1990 book *Governing the Commons* is the synthesis. Its central claim is that human communities, given the right institutional design, have repeatedly and successfully governed shared resources without resort to either state ownership or full privatization — the two solutions Garrett Hardin's "Tragedy of the Commons" had told mid-twentieth-century policy thinkers were the only options. Ostrom showed Hardin had been wrong about the empirical record. Real commons regimes, when they worked, depended on a small number of recurring design principles: clearly defined boundaries, rules matched to local conditions, collective-choice arrangements for those affected, monitoring, graduated sanctions, low-cost dispute resolution, and recognition by higher-level authorities. The reader who wants to see how those principles map onto the post-expiration patent commons described in this chapter — and why the post-1970 federal regulatory state has destroyed the higher-level recognition the commons depends on — will find the framework developed across the diagnostic and remedy chapters of the book.
See Endnote for full citation.

Poplin, Ryan, Avinash V. Varadarajan, et al. "Prediction of Cardiovascular Risk Factors from Retinal Fundus Photographs via Deep Learning." Doctors have been photographing the back of the eye for about a hundred years. The picture shows the retina and its blood vessels, and it is used to look for eye disease. Poplin and his colleagues

trained a system on retinal photographs from 284,335 patients and tested it on two separate groups. From the photograph alone, the system estimated a patient's age to within about three and a quarter years and read blood pressure to within about eleven points. It predicted whether the patient smoked, and whether the patient would go on to have a major cardiac event. It also identified the patient's sex, almost perfectly. No ophthalmologist can do that, and none suspected it could be done; the authors say plainly that these signals were not previously thought to be present in retinal images at all. The chapter files this as an unasked question without qualification, and the reason bears stating. The instrument was new. The information was not. It had been in every such photograph taken for a century.
See Endnote for full citation.

Ranalli, Brent, James D'Angelo, and David King. *The Sunshine Reforms and the Transformation of Congressional Lobbying*. Brent Ranalli is a historian; James D'Angelo is a political scientist; David King is a senior lecturer at the Harvard Kennedy School. Their joint work at the Congressional Research Institute is the most rigorous body of empirical research available on the structural transformation of the U.S. Congress after the 1970 Legislative Reorganization Act. The 2019 *Foreign Affairs* essay "The Dark Side of Sunlight," co-authored by D'Angelo and Ranalli, is the landmark synthesis; access to that essay sits behind a paywall. The 2018 Congressional Research Institute paper *The Sunshine Reforms and the Transformation of Congressional Lobbying* develops the same empirical case at greater length and is freely available on the Congressional Research Institute website (
https://congressionalresearch.org). The empirical record this book uses on lobbyist growth, on the explosion of legislation since 1970, and on the relationship between the "sunshine" transparency reforms and the rise of lobbyist surveillance derives from their joint research at that institute. Three findings are central. First, the registered-lobbyist count in Washington grew from a few hundred in 1970 to over thirteen thousand by 2024. Second, the volume of federal legislation tripled in the decade after the 1970 Act. Third, the federal government has not passed a budget that reduced the national debt in a single fiscal year since 1957. The reader who wants to understand why the modern American Congress produces the legislative outputs it does should begin with the Congressional Research

Institute's published work. American constitutional self-government depends on citizens understanding what was changed in October of 1970 and what the change has cost. See Endnote for full citation.

Risner, Genevieve. "Parts of a Western Saddle and the Variations." The Buffalo Bill Center of the West, located in Cody, Wyoming, is a Smithsonian Affiliate that operates five museums under one roof — the Buffalo Bill Museum, the Plains Indian Museum, the Whitney Western Art Museum, the Cody Firearms Museum, and the Draper Natural History Museum. Risner's 2021 article on the parts and types of the Western saddle is the chapter's recommended one-stop online reference for visual identification of the saddle types this chapter touches. The article identifies and photographs four distinct American saddle types from the Center's own collection: the Vaquero saddle, the stock saddle, the McClellan, and the Pony Express saddle. Each is presented with photographs from multiple angles and a short narrative on the working role it served. The article's evolution timeline runs from the eighth-century Moorish saddle, through the sixteenth-century Mexican vaquero adaptations that added the roping horn, through the 1850s adoption of the McClellan by the U.S. Military, through the 1860–1861 Pony Express. Although the article does not address the patent-expiration economics this chapter foregrounds, it is the standard accessible reference for the working forms the post-1797 commons produced.
See Endnote for full citation.

Semmelweis, Ignaz. The chapter states the two numbers and the admission rule; the rest of the case belongs here. The Vienna General Hospital ran two maternity clinics. Doctors and medical students staffed the first; midwives staffed the second. From 1840 through 1846 the death rate from childbed fever in the doctors' clinic was 98.4 per thousand births. In the midwives' clinic it was 36.2. The figures were in the hospital's own ledgers and printed in its annual returns. Women were admitted to the two clinics on alternate days, with no medical selection of any kind, so the obvious objection — that the doctors were taking sicker patients — is not available. The only systematic difference between the two groups of women was who delivered them. The hospital was running a controlled experiment without meaning to, publishing the results every year, and reading them as nothing.

Semmelweis saw it in 1847, after his colleague Jakob Kolletschka cut himself during an autopsy and died of what Semmelweis recognized as the same disease that was killing the mothers. The doctors came to deliveries straight from the dissecting room. The midwives did not. He ordered chlorine handwashing that May, and deaths fell. The correction the chapter insists on is the one the popular version omits. Semmelweis was not an outsider. He was a credentialed physician, an assistant in the very clinic whose figures he read. If a theory of blindness requires an outsider to break it, that theory deletes the best case medicine has. The blindness here was not caused by a shortage of credentials. It was caused by a settled sense of what those mortality figures were about — the mechanism Kuhn named, and the one this book says operated on the Patent Clause.

See Endnote for full citation.

Shallue, Christopher J., and Andrew Vanderburg. “Identifying Exoplanets with Deep Learning.” NASA’s Kepler telescope watched roughly 150,000 stars at once, looking for the minute dimming that occurs when a planet crosses in front of one. It produced an archive that was public, enormous, and worked over for years by professional astronomers who had extracted thousands of planets from it. Shallue was a software engineer at Google, not an astronomer. The professionals had concentrated on strong signals, for a defensible reason: a weak dip in a star’s brightness is usually noise, and chasing weak signals wastes time on false positives. Shallue’s idea was that a neural network might sort the weak ones — that the constraint had been human attention rather than the data. He trained a network on fifteen thousand Kepler signals that human scientists had already labelled as real planets or false alarms; on a held-out test set it classified them correctly ninety-six percent of the time. Then he and Vanderburg, the astronomer on the paper, pointed it at 670 star systems already known to hold multiple planets, on the reasoning that a system with several planets was a likely place to find another. It found Kepler-90i: about thirty percent larger than Earth, orbiting its star every 14.4 days, with a surface temperature above eight hundred degrees Fahrenheit. Kepler-90 had been identified in 2013 as the first seven-planet system Kepler found; the eighth planet’s signal was too weak for the methods that found the other seven, and its discovery made Kepler-90 the first system known to match our own for planet count. The same run turned up Kepler-80g. Vanderburg supplied what Shallue could

not: the judgment about where to look and which candidates survived scrutiny. The chapter labels the case a hybrid and separates the halves. The network was a new tool, and Shallue built it. But the tool is not what had been missing. The data was public and had been examined. What had not happened was anyone asking whether the signals the professionals had reasonably skipped were worth a second look. The question was the contribution. The network was how he could afford to ask it.

See Endnote for full citation.

Sierra Saddlery School. “Some History of the Western Saddle.” The chapter does not lean on this source for the dates and patent figures it cites — those come from the patent record itself and from the standard scholarly references. The Sierra Saddlery School blog post earns its place in the apparatus for a different reason: it is a competent and accessible narrative of how the working saddle of the American West came to look the way it does. The trans-Atlantic line runs from the Moorish horsemen of the Dark Ages, who carried into the Iberian peninsula a high-cantle saddle designed for armored war, to the Spaniards who adapted it into the Spanish War Saddle and brought it to the New World, to the post-conquest evolution of the saddle into a working tool of the Mexican vaquero. The vaqueros, in turn, were the saddle-makers who taught the trade to the working cowboy of Texas, California, and the Great Basin, and the Texican-and-Californio split among working cowboys produced the two saddle silhouettes the modern reader still recognizes — the swell-fork, square-skirted, double-rigged Texican saddle suited to the heavy brush and “hard-and-fast” roping of the Texas range, and the slick-fork, round-skirted, center-fire-rigged Californio saddle suited to the milder climate, the dally style of roping, and the time the vaquero had for tooled-leather artistry. The reader who wants the longer saddle-history story than the chapter has room to tell will find a serviceable summary at the URL given in the corresponding Endnote, with the caveat that the post is a saddle-school blog rather than a museum or scholarly source. The Buffalo Bill Center of the West (Risner 2021) is the better one-stop reference for visual identification of the saddle types the chapter discusses.

See Endnote for full citation.

Statuto Albertino. The full timeline of the Italian parliament's *voto segreto* practice under the Statuto and under the post-1948 chamber-rule regime, together with Article 63 in English translation, is set out in the Elster commentary entry above.
See Endnote for full citation.

Vesuvius Challenge. Two people acted on the same scans, and the chapter's third movement separates what each of them did, because the sequence is the point. In AD 79 Vesuvius buried Herculaneum and with it a villa library of roughly 1,800 papyrus scrolls. The eruption carbonized them, which is why they survived and why they were unreadable: the ink is carbon, the papyrus is carbon, and unrolling one destroys it. In 2019 Brent Seales and his laboratory at the University of Kentucky put two scrolls through a particle accelerator and produced X-ray scans at resolutions down to a few micrometres. In 2023 the scans were released publicly as an open competition with a grand prize of seven hundred thousand dollars, and hundreds of people downloaded them. Casey Handmer, a physicist formerly at the Jet Propulsion Laboratory with no papyrology and no Greek, downloaded them too, and built nothing. He opened the flattened scan segments and looked at them for hours, and he noticed a faint cracked-mud texture appearing in shapes that looked like Greek letters. He called it crackle and published it on his blog in August 2023. The carbon ink was invisible because it has almost the same X-ray density as the carbon papyrus. What Handmer saw was the disturbance the ink had left in the surface it dried on — the mark of a thing rather than the thing. It was the first substantial evidence of ink inside an unopened scroll, and the competition awarded him ten thousand dollars for it. Handmer has described his own part as modest. It was the hinge on which everything after it turned. Luke Farritor, a twenty-one-year-old computer science undergraduate at the University of Nebraska–Lincoln, read Handmer's post and did what Handmer could not. He took the crackle pattern as training data, built a machine-learning model to find more of it, fed the letters it found back in as further training, and iterated. The model returned ink strokes, then characters, then, in a patch about four square centimetres across, more than ten of them in sequence: πορφύρας, purple. Farritor won forty thousand dollars in October 2023; Youssef Nader reached the same word independently and took ten thousand. By December, Farritor, Nader and Julian Schilliger had recovered about two thousand

characters and shared the grand prize. Then papyrologists at Naples, at Pisa, and elsewhere read the Greek, confirmed the readings, and worked out what the text was. Four steps, and the labels matter. An institution supplied the capability. An outsider supplied the observation — the unasked question in its purest modern form, since no instrument was missing and the scans were in hundreds of hands. A second outsider supplied the machine — the better tool. Insiders supplied the judgment about whether any of it was true. That is the arrangement behind this book, and the chapter says so rather than leaving a reader to find it out. The middle step is the only one a machine performs. Somebody has to notice, and somebody else has to check.

See Endnote for full citation.

Zwicky, Fritz. Zwicky measured how fast galaxies were moving inside the Coma cluster and compared that with how much matter he could see. The galaxies were moving far too fast. At those speeds the cluster should have flown apart long ago; something invisible was holding it together, and there was a great deal of it. He applied the virial theorem to published measurements and named the missing material dunkle Materie. The calculation used standard physics, and any competent astronomer of 1933 could have run it. It was largely neglected for about forty years, and the chapter labels the case mixed for a reason. The 1933 result was an unasked question: the measurements were in print and the arithmetic was ordinary. The resolution took better tools. Acceptance came when Rubin and Ford (1970) measured how fast stars orbit inside individual galaxies, starting with Andromeda, and when radio astronomers reached further out than optical telescopes could. Both halves belong in the label, and the chapter's rule is to say which part is which rather than round the case toward the answer it wants.

See Endnote for full citation.

Appendix A: SCOTUS Case Master Table

The Second Monopoly

Eighty-nine rulings surveyed, 1829–2024 · cross-reference table, sorted by decision year

Scope. Every Supreme Court ruling this book surveys appears here: eighty-nine rulings spanning one hundred ninety-five years, from Pennock v. Dialogue (1829) to Loper Bright Enterprises v. Raimondo, Sheetz v. County of El Dorado, and Corner Post, Inc. v. Board of Governors (2024). Inclusion is not endorsement, and survey is not citation. Some of these rulings are discussed in the running text. Others are here because a defender of the wall would raise them, because the open-book examination puts them in play, or because they bear on the argument at one remove. A row earns its place by belonging to the field the argument must cross, not by appearing in a chapter. The table lists every case the argument relies on, every case cited to concede a point, and every case a defender of the wall would cite in response — each described for what it actually holds. Several cut against parts of the argument, and the "Why Case is Relevant" note says so plainly. Each row also records how the Court divided, in the Score field explained in the key that follows the table. Cases decided on state-law preemption or copyright grounds are marked as such; they are cited for the principle they state, not for holdings they do not contain. Each row is one Supreme Court ruling. Nine of those rulings decided more than one separately docketed case in a single opinion — Shoshone Tribe of Indians v. United States (two), Brown v. Board of Education (four), Graham v. John Deere Co. (three), Railway Labor Executives' Association v. Gibbons (two), Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc. (three), Allegheny Pittsburgh Coal Co. v. County Commission of Webster County (two), Printz v. United States (two), West Virginia v. Environmental Protection Agency (four), and Loper Bright Enterprises v. Raimondo (two) — so the eighty-nine rulings listed here resolved one hundred one cases. Bolling v. Sharpe is listed on its own line because it was decided independently of Brown, on Fifth Amendment grounds, and was not among the four cases Brown consolidated. A companion case argued and decided the same day but carrying its own opinion is not part of the ruling it accompanies: Compco Corp. v. Day-Brite Lighting is a companion to Sears, Roebuck & Co. v. Stiffel Co., not part of it. Not one of the eighty-nine holds that a federal agency may prohibit the remanufacture of a patent-expired design. The Court has never been asked.

#	SCOTUS Case	Description
1	Pennock v. Dialogue (1829) 27 U.S. (2 Pet.) 1 (1829) **Score:** 6x0*8	**General Description:** An inventor who allowed the public to use and buy his hose invention for years before applying forfeited the patent. Justice Story, writing for a unanimous Court, held the patent void.
	In the reasoning the Court has treated as settled ever since, Story explained that the "main object" of the patent system is to promote the progress of science and useful arts, best served by giving the public at large the right to make, construct, use, and vend the invention at as early a period as possible, with due regard to the rights of the inventor. The opinion construes the Patent Act of 1793. The holding itself enforces the public's side of the bargain against a tardy inventor; holding and rationale point the same way. **Why Case is Relevant:** *The Patent Clause Prong* — The earliest judicial statement of the public's half of the bargain, from a Justice Madison appointed while framers still lived. Cited for the principle that the public's eventual right to make and vend the invention is the system's main object. Presented as the Court's reasoning, not as its holding.	
2	Wheaton v. Peters (1834) 33 U.S. (8 Pet.) 591 (1834) **Score:** —*10	**General Description:** Henry Wheaton, the Supreme Court's third Reporter of Decisions, sued his successor Richard Peters for republishing his volumes in condensed form. Wheaton claimed a perpetual common-law copyright in the reports.
	Justice McLean, for the Court, rejected a federal common-law copyright in published works: post-publication protection exists only by statute, and the statutory formalities must be strictly met. The Court added, unanimously on this point, that no reporter can hold any copyright in the written opinions of the Court, and that the Justices cannot confer such a right. Justices Thompson and Baldwin dissented from the judgment. The decision is a copyright case, not a patent case; it is cited for the constitutional structure the two clauses share — that the exclusive right is a statutory grant on the public's terms, not a natural right that precedes it. **Why Case is Relevant:** The Patent Clause Prong — Cited for the proposition that exclusive rights under the Patent and Copyright Clause are creatures of statute granted on the public's terms, not pre-existing natural rights. Copyright authority, offered for the shared clause structure; it is not a patent holding and is not presented as one.	
3	Patterson v. Kentucky (1878) 97 U.S. 501 (1878) **Score:** 8x0 **In Exam**	**General Description:** The State of Kentucky banned the sale of a patented "improved burning oil" that the state inspector had condemned as unsafe. The patentee argued that his federal patent placed the product beyond state regulatory reach.
	The Supreme Court rejected the argument unanimously. The patent secured the inventor's exclusive right to manufacture and sell against private competitors; it did not exempt the tangible product from state police-power regulation aimed at the safety of the public. The case is the foundational holding that a patent confers no immunity from generally applicable regulation — a holding the book accepts and builds on. The remanufacturer of a patent-expired design asks for the same regulation, not exemption from it. **Why Case is Relevant:** *Objections Considered* — The book's lead authority for the objection that a patent already carries no exemption from regulation — the remanufacturer asks for the same regulation everyone follows, not exemption from it. *Reason the Constitutional Argument Yourself* — Supplied among the Supreme Court holdings the reader reasons through: a patent does not exempt the patented article from generally applicable state safety regulation.	

#	SCOTUS Case	Description
4	Baker v. Selden (1879) 101 U.S. 99 (1879) **Score:** 9x0*20	**General Description:** Charles Selden copyrighted a book explaining a system of bookkeeping, with ruled forms for putting it into practice. Baker published a book using a similar system with differently arranged forms. Selden's estate sued for copyright infringement, contending that the copyright in the book carried with it the exclusive right to the system the book described.
	Justice Bradley, for the Court, held that the copyright secured the author's explanation, not the art it explained. A method or system may be secured, if at all, only by patent; describing it in a copyrighted book gives no exclusive right to use it, and the forms necessary to practice it are open to the public. The case is the root of the rule that copyright reaches expression and not function. **Why Case is Relevant:** *Objections Considered — The first of three reasons the copyright in engine-control firmware is narrower than its ninety-five-year term suggests: the maker's copyright covers its lines of code, not the job those lines do. A remanufacturer who writes its own program to run the same engine copies nothing copyright reaches.*	
5	Webber v. Virginia (1880) 103 U.S. 344 (1880) **Score:** —*7 **In Exam**	**General Description:** Two years after Patterson v. Kentucky, the Supreme Court reaffirmed the Patterson principle in the tax setting.
	Virginia imposed a license tax on the sale of sewing machines, and a patentee argued that his federal patent exempted the patented machines from the tax. The Court rejected the argument, holding that letters patent granted by the United States do not exclude the tangible property in which the invention is embodied from state tax or license law. The case is part of the Patterson line establishing that the existence of a patent does not place the patented product beyond ordinary state and federal regulation. **Why Case is Relevant:** *Objections Considered* — Cited as reaffirming the Patterson principle in the tax setting — a patent does not place the article beyond general state law. *Reason the Constitutional Argument Yourself* — Exam holding: a patent confers the right to make and sell, but does not place the article beyond the reach of general law.	
6	James v. Campbell (1882) 104 U.S. 356 (1882) **Score:** 6x1 **In Exam**	**General Description:** An inventor held patents on improvements used in machinery the federal government employed to print postage stamps and currency, and claimed the government's use infringed them.
	The Supreme Court held that the government's use of a patented invention is a taking of private property for which the inventor must be compensated — establishing that a patent is property in the full constitutional sense, not a privilege the government may draw on at will. **Why Case is Relevant:** *The Patent Clause Prong* — Establishes that a patent is property and that the government's own use of a patented invention is a taking — anchoring the property premise of the first prong. *The Takings Clause Prong* — Cited for the same property point on the takings side: when the federal government uses a patented invention without leave, it must pay. *Reason the Constitutional Argument Yourself* — Exam holding: a patent is private property; the government's use of a patented invention is a taking requiring compensation.	

#	SCOTUS Case	Description
7	United States v. Ballin (1892) 144 U.S. 1 (1892) **Score:** 8x0	**General Description:** The closest Supreme Court case to address the Journal Clause of Article I, Section 5, Clause 3. In Ballin the Court held the House Journal to be conclusive evidence of what occurred during
	a contested quorum vote, but did not interpret whether the Yeas-and-Nays override ("at the Desire of one fifth of those Present, be entered on the Journal") constitutionally requires individual named records or only an aggregate tally. The case is cited in the Hybrid Voting chapter as the doctrinal anchor for the proposition that no Supreme Court holding has squarely decided the interpretive scope of the Yeas-and-Nays override, leaving the question one of first impression. **Why Case is Relevant:** *Hybrid Legislative Voting* — The closest authority on the Journal Clause: the House Journal is conclusive evidence of what occurred, without resolving whether names or only totals must be recorded — leaving room for the chapter's hybrid-voting proposal.	
8	Singer Manufacturing Co. v. June Manufacturing Co. (1896) 163 U.S. 169 (1896) **Score:** —*7 **In Exam**	**General Description:** The Singer sewing machine had been built under patents that had long since expired. A competitor began making machines of the same form and selling them as "Singer" machines. Singer sued, claiming an exclusive right to both the design and the name.
	The Supreme Court refused. Once the patents expired, the Court held, "the right to make the patented article and to use the generic name passed to the public with the dedication resulting from the expiration of the patent"; and the name "Singer," having come to denote the type of machine rather than its single maker, passed to the public as a generic term, subject only to the competitor's duty to identify its own source of manufacture so buyers were not deceived. The case is cited for the Court's description of the post-expiration interest as property that becomes the public's, while the maker's trademark survives — the holding concerns patent expiration and trademark, not the Takings Clause. **Why Case is Relevant:** *The Takings Clause Prong* — Cited for the holding that at expiration "the right to make the thing formerly covered by the patent becomes public property" — the vested public right the prong protects. *Objections Considered* — Answers the "monopoly will be destroyed" objection: Singer kept its market lead for generations on its sales network, financing, and name — everything except the patent monopoly. *Reason the Constitutional Argument Yourself* — Exam holding: when the patents on a machine expire, its form becomes public property, free for anyone to make.	
9	McCormick Harvesting Machine Co. v. Aultman (1898) 169 U.S. 606 (1898) **Score:** —*7 **In Exam**	**General Description:** After a patent had been issued, its owner returned it to the Patent Office to have certain claims reissued; the Office instead treated the original patent as cancelled. The Supreme Court held the Patent Office had no such power.
	Once a patent issues, it becomes the property of the patentee, and only a court — not the executive agency that granted it — may set it aside. The case fixes the separation-of-powers line this chapter relies on. **Why Case is Relevant:** *The Patent Clause Prong* — Draws the line the chapter needs: once a patent issues it is the patentee's property, and only a court — not the granting agency — may set it aside. *Reason the Constitutional Argument Yourself* — Exam holding: once a patent issues it is the property of the patentee; only a court, not the agency, may set it aside.	

#	SCOTUS Case	Description
10	Muller v. Oregon (1908) 208 U.S. 412 (1908) **Score:** 9x0	**General Description:** Oregon limited women in factories and laundries to ten hours a day, and an employer challenged the law as an interference with freedom of contract. Justice Brewer, for a unanimous Court, upheld it.
	The decision is remembered less for its holding, which rested on reasoning about women that the law has long since abandoned, than for what the State filed. Of the one hundred thirteen pages of the brief prepared by Louis Brandeis and Josephine Goldmark, roughly ninety-eight consisted of factory-inspection reports, medical testimony, and foreign labor statutes. The Court took formal notice of the material, and the Brandeis brief has been the ancestor of every filing since that brings a court a body of evidence rather than a chain of citations. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The founding example of an outsider supplying the material a court needed. Goldmark had no law degree and did not argue the case; she assembled the record that changed how the case was decided.	
11	Continental Paper Bag Co. v. Eastern Paper Bag Co. (1908) 210 U.S. 405 (1908) **Score:** 8x1*22	**General Description:** Eastern Paper Bag held a patent on a self-opening bag machine it had never put into use, and sued Continental for infringing it. The Court held that a patentee who never uses its invention may still enjoin others from using it.
	Justice McKenna, for the Court, held that the right to exclude others is not lost by the patentee's nonuse: exclusion is "the very essence of the right conferred by the patent," and nonuse alone is no ground to deny an injunction. Justice Harlan dissented, arguing that public policy should bar relief. **Why Case is Relevant:** Preface — The third authority behind the book's title. A monopoly lies in the power to exclude, which needs no production, and that power is what outlives the patent: it passes to one federal rulebook that still decides who may build the expired design. *Four Problems, One Remedy* — The same point, by back-reference to the Preface.	
12	Standard Oil Co. of New Jersey v. United States (1911) 221 U.S. 1 (1911) **Score:** 9x0*1	**General Description:** The antitrust decision dissolving the Standard Oil trust, cited in the chapter on why nobody noticed as the legal endpoint of the public pressure that Ida Tarbell's McClure's series had
	built. **Why Case is Relevant:** *Why Nobody Noticed* — Cited as the historical trust that Ida Tarbell's reporting helped break up — an illustration of how public attention is mobilized.	

#	SCOTUS Case	Description
13	Motion Picture Patents Co. v. Universal Film Mfg. Co. (1917) 243 U.S. 502 (1917) **Score:** 6x3*9	**General Description:** A patent owner attached a notice restricting its patented projector to films it approved. The Court held the restriction void: an unconditional sale exhausts the patent, and the patentee may not extend the monopoly to unpatented supplies.
	Writing for the Court, Justice Clarke stated that since Pennock v. Dialogue (1829) the Court has consistently held that the primary purpose of the patent laws is not the creation of private fortunes for patent owners but the promotion of the progress of science and the useful arts. Holmes dissented, joined by McKenna and Van Devanter; McReynolds concurred in the result only. **Why Case is Relevant:** *The Patent Clause Prong* — The decision that elevated Pennock's public-interest reasoning toward settled rule, letting the argument say the Court has treated the public purpose as dominant "consistently" across a century. Cited for the ceiling on how far a patent's commercial benefit may be pushed, not for its exhaustion holding.	
14	Pennsylvania Coal Co. v. Mahon (1922) 260 U.S. 393 (1922) **Score:** 7x1 **In Exam**	**General Description:** A Pennsylvania statute forbade underground coal mining that would cause the surface above to collapse, even where the coal company had sold the surface rights while expressly keeping the right to mine the coal below.
	Justice Holmes, writing for the Court, held that the statute went too far and effected a taking. The decision is the origin of regulatory-takings law: it established that although property may be regulated, a regulation that goes too far will be recognized as a taking requiring compensation. **Why Case is Relevant:** *The Environmental Question* — Listed among the takings doctrines (regulatory takings) that the Patent Clause remedy does not disturb. *The Takings Clause Prong* — The origin of regulatory-takings doctrine — a regulation that goes "too far" is a taking — the doctrinal root the prong builds on. *Reason the Constitutional Argument Yourself* — Exam holding: a regulation that diminishes property value too far may constitute a taking.	
15	J. W. Hampton, Jr., & Co. v. United States (1928) 276 U.S. 394 (1928) **Score:** 9x0	**General Description:** An importer challenged a tariff rate the President had raised under the flexible-tariff provision of the Tariff Act of 1922, arguing that Congress had unconstitutionally delegated its legislative power. Chief Justice Taft, for a unanimous Court, upheld the delegation.
	The opinion supplies the intelligible-principle test: Congress may authorize the Executive to act where it lays down by legislative act a principle intelligible enough to guide the exercise. The rule is a limit on how a granted power may be handed to an agency; it says nothing about whether the power existed to grant. **Why Case is Relevant:** *The Patent Clause Prong* — Cited for the principle that an agency cannot hold a power its creating Congress did not have. A delegation may be intelligible and still be void if what was delegated lay outside Congress's own authority.	

#	SCOTUS Case	Description
16	Shoshone Tribe of Indians v. United States (1937) 299 U.S. 476 (1937) **Score:** 8x0	**General Description:** The federal government had settled another tribe on land guaranteed to the Shoshone by treaty, diminishing their interest. The Supreme Court held the loss compensable.
	Justice Cardozo's opinion supplied the phrase this book relies on — "Spoliation is not management" — and the principle it carries: a collective, non-individually-titled interest is property the government must pay to take. The book cites the case for that principle, not for any holding about the patent commons. **Why Case is Relevant:** *The Takings Clause Prong* — Cardozo's "Spoliation is not management" — cited for the principle that a collective, non-individually-titled interest is still protected property, fitting the public's post-expiry right.	
17	Aetna Life Insurance Co. v. Haworth (1937) 300 U.S. 227 (1937) **Score:** 9x0	**General Description:** Aetna had issued disability-income policies to Edwin Haworth and had been paying benefits for several years when it concluded that Haworth was no longer disabled within the meaning of the contracts;
	Haworth disagreed but had not yet sued for past or future benefits. Aetna sought a federal-court declaration that the policies were void and no further benefits were owed. The Supreme Court held that the federal Declaratory Judgment Act of 1934 allowed Aetna to file: where a controversy is real and immediate, a federal court may declare the parties' legal rights without waiting for one of them to be sued or prosecuted. The case is the foundation of the constitutional-litigation path this book describes — the path on which the patent-bargain case will be filed before any plaintiff has been arrested, fined, or enjoined. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The leading early declaratory-judgment case: upheld the Declaratory Judgment Act against an Article III challenge, confirming a real controversy can be adjudicated before enforcement. *The Filable Case* — Cited for the same point in the filability chapter — a pre-enforcement dispute can be a genuine case or controversy.	

#	SCOTUS Case	Description
18	Kellogg Co. v. National Biscuit Co. (1938) 305 U.S. 111 (1938) **Score:** 6x2	**General Description:** National Biscuit claimed the exclusive right to the name "shredded wheat" and to the pillow shape of the biscuit. Both had been made under the Perky patents, which expired in 1912. Kellogg began making the same product in the same shape in 1922.
	Justice Brandeis, for the Court, held that on expiration of the patents there passed to the public "not only the right to make the article as it was made during the patent period, but also the right to apply thereto the name by which it had become known." The pillow shape went with it: upon expiration of the patents "the form, as well as the name, was dedicated to the public." The sole surviving obligation on the newcomer was to identify its own product so that buyers were not deceived as to source. The suit arose in diversity as a common-law unfair-competition claim, and the case is cited for what it holds about post-expiration rights, not for a constitutional proposition. Justices McReynolds and Butler would have affirmed the court below, stating that Kellogg was fraudulently seeking to appropriate a goodwill built up at great cost by the respondent. Eight Justices sat; Cardozo had died in July 1938 and Frankfurter was not yet confirmed. At 120 the Court rested the rule on *Singer*, at 185: "on the termination of the patent there passes to the public the right to make the machine in the form in which it was constructed during the patent." The Court then gave a second and independent reason the shape had to remain available. At 122 it found the pillow shape functional: the evidence was persuasive that the cost of the biscuit would be increased and its high quality lessened if some other form were substituted. Requiring a different form would therefore have imposed a penalty on the newcomer — an observation this argument draws from the Court's reasoning, not a holding the Court announced. **Why Case is Relevant:** *The Patent Clause Prong* — The Court's plainest statement that the form of the article, and not merely its underlying idea, enters the public domain when the patent expires. The faithful remake is the right the Court described, and Kellogg completes the line that runs from Singer (1896) through Scott Paper (1945) to Sears and Compco (1964). *Objections Considered* — The source-identification duty is the answer to the trademark objection: the remanufacturer must mark its own goods as its own, and that is all it must do. The functionality finding is the closer analogue still: the Court would not force a competitor into a different form when the original form was the one that worked.	
19	Wickard v. Filburn (1942) 317 U.S. 111 (1942) **Score:** 9x0	**General Description:** Roscoe Filburn grew wheat on his Ohio farm in excess of his federal allotment, intending the surplus for his own use, not for sale.
	The Supreme Court upheld the penalty against him, reasoning that even purely local activity may be regulated under the Commerce Clause when its aggregate effect on interstate commerce is substantial. The case is the high-water mark of the modern commerce power and the authority a defender of the regulatory wall will cite for the proposition that Congress may regulate the manufacture of vehicles in commerce. The book does not dispute that power. The Patent Clause prong answers it on a different ground: a power broad enough to reach the wheat on Filburn's farm is still not a power to cancel the public's ownership of an expired invention, because no grant of federal power authorizes Congress to nullify what another clause of the Constitution commands. **Why Case is Relevant:** *The Patent Clause Prong* — Cited to concede the breadth of the Commerce Clause power — the book does not dispute Congress's authority to regulate; its argument is about the Patent Clause limitation.	

#	SCOTUS Case	Description
20	Betts v. Brady (1942) 316 U.S. 455 (1942) **Score:** 6x3	**General Description:** An indigent defendant tried for robbery in Maryland asked the court to appoint counsel and was refused. Justice Roberts, for a six-Justice majority, held that due process did not require a State to furnish counsel in a noncapital case. Justice Black dissented, joined by Justices Douglas and Murphy.
	The case stood for twenty-one years and was overruled in Gideon v. Wainwright (1963), where Justice Black wrote for a unanimous Court. The book cites it not for either holding but for the interval between them, and for the fact that the Court had been looking for a vehicle in which to reconsider Betts before Gideon supplied one. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — Cited to show that the case a court needs and the theory that decides it rarely come from the same person. Gideon supplied the case; Abe Fortas supplied the argument.	
21	Skidmore v. Swift & Co. (1944) 323 U.S. 134 (1944) **Score:** 9x0	**General Description:** Held that an agency interpretation is entitled to judicial respect proportional to its power to persuade — the "Skidmore deference" standard.
	Why Case is Relevant: *You No Longer Need to Break a Law to Challenge One* — Cited for "Skidmore weight" — after Loper Bright, agency interpretations earn respect only to the extent persuasive, not deference — which matters to the patent-bargain case. *The Filable Case* — Same point in the filability chapter: the agency's view earns respect only if persuasive; "the thumb came off the scale."	
22	Korematsu v. United States (1944) 323 U.S. 214 (1944) **Score:** 6x3	**General Description:** The Court upheld the wartime exclusion of Japanese Americans from the West Coast. Justice Black wrote for the majority; Justices Roberts, Murphy, and Jackson each dissented.
	The claim of military necessity was made to the Court while contrary evidence sat in the government's own files. In 1983 a federal court in San Francisco vacated Fred Korematsu's conviction on a writ of coram nobis after that record surfaced, and in Trump v. Hawaii (2018) the Court stated that Korematsu had been gravely wrong the day it was decided. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The most damning instance of a question nobody asked. The evidence was not merely available somewhere; it was in the government's files, put there by the government, for forty years.	

#	SCOTUS Case	Description
23	Scott Paper Co. v. Marcalus Manufacturing Co. (1945) 326 U.S. 249 (1945) **Score:** 6x2 **In Exam**	**General Description:** Marcalus had assigned a patent to Scott Paper, then later built a product using a design covered by a different patent that had already expired.
	Scott Paper invoked a contract doctrine — assignor estoppel — to stop its former assignor from competing. The Supreme Court refused. The public's right to use an invention whose patent has expired, the Court held, is a matter of patent policy that a private estoppel rule cannot override. The patent laws, the Court held, "dedicate to public use the invention of an expired patent," and do not contemplate that anyone may "withhold from the public the use of the invention of an expired patent." **Why Case is Relevant:** *Four Problems, One Remedy* — Listed among the cases affirming that what a patent covered belongs to the public at expiration. *The Patent Clause Prong* — Authority a plaintiff sets before the court: a patent holder cannot use a private contract to keep control after the patent expires. *The Takings Clause Prong* — Cited for describing the expired invention as dedicated to public use — the vested right the prong protects. *Reason the Constitutional Argument Yourself* — Referenced in the examination's synthesis of the public-domain cases (with Graham and Singer).	
24	Brown v. Board of Education (1954) 347 U.S. 483 (1954) **Score:** 9x0	**General Description:** Chief Justice Warren, for a unanimous Court, held that separate educational facilities are inherently unequal and that state-imposed school segregation violates the Equal Protection Clause of the Fourteenth Amendment.
	The work of Kenneth and Mamie Clark appears in footnote eleven of the opinion, in a list of social-science authorities. The book cites the case for what that footnote does and does not show: the Clarks were psychologists, neither was a lawyer, and their studies did not decide the case so much as supply a sentence the Court was prepared to write. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — Cited against the popular version of its own story. Outsiders furnished evidence; lawyers and Justices carried it to a holding.	
25	Bolling v. Sharpe (1954) 347 U.S. 497 (1954) **Score:** 9x0 **In Exam**	**General Description:** The federal companion to Brown v. Board of Education, decided the same day. Brown held that state-mandated school segregation violated the Fourteenth Amendment's Equal Protection Clause.
	Bolling faced the parallel question for the District of Columbia, which is not a state and to which the Fourteenth Amendment does not apply. The Court held that the Fifth Amendment's Due Process Clause incorporates an equal-protection guarantee against the federal government. Federal action classifying similarly situated persons unequally is subject to constitutional review under the Fifth Amendment even where the Fourteenth Amendment has no purchase. The case is what makes any federal-equal-protection framing of the regulatory wall possible at all. **Why Case is Relevant:** *The Patent Clause Prong* — Cited for the equal-protection obligation the Fifth Amendment imposes on the federal government (the wall is federal), though economic classifications get only rational-basis review. *The Takings Clause Prong* — Supplies the bridge applying equal protection to the federal government, reaffirmed in Adarand. *Reason the Constitutional Argument Yourself* — Exam holding: the Fifth Amendment's Due Process Clause imposes a federal equal-protection obligation.	

#	SCOTUS Case	Description
26	Williamson v. Lee Optical of Oklahoma, Inc. (1955) 348 U.S. 483 (1955) **Score:** 8x0	**General Description:** Oklahoma forbade opticians to fit or duplicate lenses without a prescription. Justice Douglas, for the Court, upheld the statute; Justice Harlan took no part.
	The opinion is the canonical statement of rational-basis review for economic regulation: the law may be needless and wasteful, and it is for the legislature rather than the courts to balance advantage against disadvantage. A court need only be able to conceive of a state of facts that would justify the classification. **Why Case is Relevant:** *Objections Considered* — The reason equality cannot be pleaded as a standalone third prong. An equal-protection claim against the regulatory wall would meet rational-basis review and lose, which is why equality operates inside each of the two prongs rather than beside them.	
27	Aro Manufacturing Co. v. Convertible Top Replacement Co. (1961) 365 U.S. 336 (1961) **Score:** 6x3*2	**General Description:** The Supreme Court's repair-and-reconstruction decision, arising from replacement tops for Ford and General Motors convertibles.
	The Court held that replacing a worn, unpatented element of a patented combination is permissible repair, not infringing reconstruction, and that no unpatented element of a combination is entitled to the patent monopoly, however essential to the combination it may be. Cited for the principle that a combination patent reaches only the totality of its claimed elements. **Why Case is Relevant:** *Objections Considered* — Cited on combination patents: no single unpatented element carries the monopoly — answering an objection about component-level rights.	
28	Eastern Railroad Presidents Conference v. Noerr Motor Freight, Inc. (1961) 365 U.S. 127 (1961) **Score:** 9x0	**General Description:** Twenty-four eastern railroads ran a publicity campaign against the trucking industry, and truckers sued under the Sherman Act. Justice Black, for a unanimous Court, held that the campaign was not an antitrust violation even though its purpose was to injure competitors.
	With United Mine Workers v. Pennington (1965) the case established the Noerr-Pennington doctrine, which immunizes petitioning the government from antitrust liability. The doctrine is why the regulatory wall cannot be attacked as a conspiracy: the lobbying that raised it is protected activity. **Why Case is Relevant:** *Objections Considered* — Cited to close an alternative theory rather than to open one. A plaintiff who reached for an antitrust account of how the wall was built would meet Noerr-Pennington, which is one reason the case must be constitutional.	

#	SCOTUS Case	Description
29	Gideon v. Wainwright (1963) 372 U.S. 335 (1963) **Score:** 9x0	**General Description:** Clarence Earl Gideon, tried in Florida without counsel, wrote his own petition to the Supreme Court in pencil. Justice Black wrote for a unanimous Court, overruling Betts v. Brady and holding that the Sixth Amendment right to counsel binds the States through the Fourteenth.
	Gideon is the archetype everyone reaches for when an outsider is said to have changed constitutional law, and the book cites it to show that the archetype does not fit. The Court appointed Abe Fortas to argue the case, and the winning argument was his. Gideon supplied the vehicle, not the theory. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The case that most nearly answers the objection that an outsider never wins, examined and found not to answer it. The two jobs — finding the injury and arguing the law — sat in different people.	
30	Sears, Roebuck & Co. v. Stiffel Co. (1964) 376 U.S. 225 (1964) **Score:** 9x0*11	**General Description:** Stiffel's pole-lamp patents were held invalid for want of invention. Sears copied the lamp and sold it cheaply. Stiffel won below on an Illinois unfair-competition theory.
	Justice Black, for a unanimous Court, reversed. An article on which the patent has expired, like an unpatentable article, is in the public domain and may be made and sold by whoever chooses to do so. A State may not use unfair-competition law to give what amounts to a patent monopoly on an unpatented article, though it may require labeling to prevent deception as to source. The holding runs against a State, not against a federal agency. The same opinion, at 230, restates what expiration does: when the patent expires the monopoly expires with it, and the right to make the article — "including the right to make it in precisely the shape it carried when patented" — passes to the public, citing *Kellogg* and *Singer*. That passage is the Court's reasoning, not its holding, and is cited as such. **Why Case is Relevant:** The Patent Clause Prong — The Court's plainest statement that the expired or unpatentable article belongs to the public and may be made by anyone. Cited for that principle. The holding is a state-law preemption holding: it bars a State from re-erecting patent-like protection. The Court has never applied the principle against a federal regulator, and this argument does not claim it has.	
31	Brulotte v. Thys Co. (1964) 379 U.S. 29 (1964) **Score:** 8x1	**General Description:** A hop-picking machine was sold under a license requiring royalties that ran past the expiration of every patent embodied in it. Justice Douglas, for the Court, held the post-expiration royalty provisions unenforceable. Justice Harlan dissented.
	The patents in the case expired on schedule and no term was extended by a day. The Court struck the arrangement down anyway, because the practical consequence of expiration — the public's unrestricted freedom in the released invention — had been withheld by another instrument. Reaffirmed on stare decisis grounds in Kimble v. Marvel Entertainment (2015). **Why Case is Relevant:** *Objections Considered* — First rung of the ladder answering the enumerated-powers objection. Had "limited Times" been a scheduling instruction, Brulotte would have been decided the other way in a paragraph.	

#	SCOTUS Case	Description
32	Graham v. John Deere Co. (1966) 383 U.S. 1 (1966) **Score:** 8x0 **In Exam**	**General Description:** A dispute over the validity of a patent on a clamp for plow shanks. The Supreme Court used the case to set out how courts decide whether an invention is "non-obvious" enough to deserve a
	patent — and, more important for this book, to describe the constitutional nature of the Patent Clause itself. The Court called the clause both a grant of power to Congress and a limitation on that power. In the Court's own words, the clause "is both a grant of power and a limitation," and "Congress in the exercise of the patent power may not overreach the restraints imposed by the stated constitutional purpose." The Court continued, at 5–6, that Congress "may not authorize the issuance of patents whose effects are to remove existent knowledge from the public domain, or to restrict free access to materials already available." **Why Case is Relevant:** *Four Problems, One Remedy* — Listed among the cases affirming that an expired patent's subject belongs to the public. *The Patent Clause Prong* — The chapter's keystone authority: Graham holds the Patent Clause is "both a grant of power and a limitation upon the exercise of that power." *Reason the Constitutional Argument Yourself* — Exam holding: the Patent Clause is both a grant of power and a limitation on it; Congress may not overreach in exercising the patent power.	
33	Menominee Tribe of Indians v. United States (1968) 391 U.S. 404 (1968) **Score:** 6x2	**General Description:** The Menominee Tribe sued the United States in the Court of Claims for compensation after federal legislation terminated the tribe's recognized status.
	The Supreme Court held that treaty-guaranteed hunting and fishing rights survived the termination, because Congress had not clearly abrogated them. In this book the case is one of three showing a settled pattern: a tribe may bring a compensation claim against the federal government through ordinary litigation, breaking no law to do so. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — Example of the older plaintiff pattern — the Tribe filed in the Court of Claims for compensation — illustrating that a plaintiff need not break a law to sue.	
34	United States Department of Agriculture v. Moreno (1973) 413 U.S. 528 (1973) **Score:** 7x2 **In Exam**	**General Description:** Held that a food-stamp restriction aimed at excluding "hippie" households failed rational-basis review; bare desire to harm a group is not a legitimate interest.
	Why Case is Relevant: *Objections Considered* — Paired with Romer as authority that a classification resting only on a bare desire to harm a disfavored group fails rational-basis review. *Reason the Constitutional Argument Yourself* — Exam holding: a classification fails rational-basis review where its only rationale is a bare desire to harm a disfavored group.	

#	SCOTUS Case	Description
35	Kewanee Oil Co. v. Bicron Corp. (1974) 416 U.S. 470 (1974) **Score:** 6x2*13	**General Description:** Ohio trade-secret law protected the process Harshaw Chemical used to grow synthetic crystals. The Sixth Circuit held that federal patent law preempted the state protection. The Supreme Court reversed.
	Chief Justice Burger, for the Court, held that state trade-secret law survives federal patent preemption. The reasoning: trade-secret protection is weaker than a patent, it withdraws nothing from the public domain, and it poses no real risk of deterring inventors from seeking patents, so partial preemption as to patentable subject matter would burden the States without serving federal patent policy. Justice Marshall concurred in the result; Justice Douglas, joined by Justice Brennan, dissented on the view that a State may not confer protection on what Congress left free. The holding is about a parallel state regime. It does not address what a federal agency may do to a design whose patent has already expired; that distinction is this argument's, not the Court's. **Why Case is Relevant:** *Objections Considered* — Adverse authority — cited for the proposition that federal patent policy does not displace every other legal regime touching an invention, and distinguished on the ground that trade-secret law leaves the public free to reverse-engineer and copy what it learns, while the certification barrier forbids the lawful sale of the copy itself.	
36	Hunt v. Washington State Apple Advertising Commission (1977) 432 U.S. 333 (1977) **Score:** 8x0	**General Description:** North Carolina passed a statute requiring closed shipping containers of apples to display the U.S. grade rather than any state grade.
	The Washington State Apple Advertising Commission, representing Washington apple growers, sued, claiming the statute burdened interstate commerce. The Supreme Court used the case to set out the three-prong test for when a trade or membership association may sue on behalf of its members: at least one member must have standing in his own right, the interests at stake must be germane to the association's purpose, and neither the claim nor the relief may require the participation of individual members. The case is the doctrinal basis for associational standing — the theory under which trucker associations, farm associations, and remanufacturer associations could sue in the patent-bargain case. **Why Case is Relevant:** *Direct and Associational Plaintiff Categories* — The governing associational-standing rule: an association may sue on its members' behalf when they would have standing to sue individually. *Why You Want to Be the Plaintiff* — Cited for the same associational-standing rule in the chapter on choosing the plaintiff.	

#	SCOTUS Case	Description
37	Penn Central Transportation Co. v. City of New York (1978) 438 U.S. 104 (1978) **Score:** 6x3 **In Exam**	**General Description:** New York City's landmarks law blocked the owner of Grand Central Terminal from constructing an office tower above the station. The owner claimed the restriction took its property.
	The Supreme Court ruled for the city and, in doing so, set out the framework that still governs most regulatory-takings claims: courts weigh the economic impact of the regulation on the owner, its interference with reasonable investment-backed expectations, and the character of the government action. Penn Central is the balancing test for regulations that diminish a property's value without destroying it. **Why Case is Relevant:** *The Environmental Question* — The investment-backed-expectations test, cited to show the strategic builder who enters after a regulation has weak takings footing. *The Takings Clause Prong* — The multi-factor regulatory-takings balancing test — one of the two roads the prong analyzes. *Objections Considered* — Cited (with Lucas) to show a latecomer loses the takings analysis for lack of investment-backed expectations. *Reason the Constitutional Argument Yourself* — Exam holding: whether a regulation is a taking is judged by weighing economic impact, interference with expectations, and the character of the government action.	
38	Aronson v. Quick Point Pencil Co. (1979) 440 U.S. 257 (1979) **Score:** 9x0 **In Exam**	**General Description:** An inventor licensed an unpatented keyholder design to a manufacturer in exchange for a royalty, with a contractual provision reducing the royalty if no patent ever issued. No patent did.
	The manufacturer eventually argued that the royalty contract was unenforceable as inconsistent with federal patent policy, since the keyholder's design was free for all to copy. The Supreme Court rejected the argument. Federal patent law leaves the unpatented public domain free for everyone, but it does not bar private parties from contracting on their own terms for the use of a design, and a state contract law that lets them do so is not preempted. The case is the modern restatement of the Patterson v. Kentucky principle that patent law and ordinary regulation coexist. **Why Case is Relevant:** *Objections Considered* — Cited as the modern restatement that federal patent law does not displace ordinary state law governing a product. *Reason the Constitutional Argument Yourself* — Exam holding: federal patent law does not displace ordinary state regulation of a product.	

#	SCOTUS Case	Description
39	Andrus v. Allard (1979) 444 U.S. 51 (1979) **Score:** 9x0 **In Exam**	**General Description:** The federal Eagle Protection Act, joined by the Migratory Bird Treaty Act, made it unlawful to sell artifacts incorporating eagle parts even where the parts had been lawfully obtained before the statute.
	The owners of pre-existing inventories of Indian artifacts containing eagle feathers sued, claiming the sale ban was a Fifth Amendment taking of their property. The Supreme Court unanimously rejected the claim. A regulation that destroys one of the strands in the bundle of property rights — here, the right to sell — is not necessarily a taking, the Court reasoned, because possession, transport, and noncommercial transfer of the artifacts remained lawful. The case is the leading adverse Takings precedent the patent-bargain case will have to distinguish; the chapter on objections considered develops the distinction. **Why Case is Relevant:** *Objections Considered* — Opens the adverse-precedent discussion: a federal ban on commercial sale is not necessarily a taking where other uses remain. *Reason the Constitutional Argument Yourself* — Exam holding: a ban on the commercial sale of lawfully held property is not necessarily a taking where other uses remain.	
40	United States v. Sioux Nation of Indians (1980) 448 U.S. 371 (1980) **Score:** 8x1	**General Description:** Congress seized the Black Hills of South Dakota from the Sioux by an 1877 Act, in violation of an earlier treaty.
	The Supreme Court held the seizure a taking that required just compensation, though the land was held by the Sioux collectively rather than by any individual owner. The book uses the case two ways: as proof that a taking claim can be litigated without anyone breaking a law, and as authority that common, non-individually-titled property falls within the Just Compensation Clause. The book does not claim the case reached the Expired Patent Commons; it concerned Indian treaty lands. It claims only that the case settles the governing principle — property need not be privately and individually titled to be compensable — and that extending that principle to the commons is a short, logical step the chapter asks the court to take. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The most directly relevant plaintiff-pattern case — the Tribe sued for compensation for the Black Hills — no law needed to be broken. *The Takings Clause Prong* — Cited for the holding that Congress's 1877 seizure of the Black Hills was a taking requiring just compensation, though the land belonged to the tribe collectively.	

#	SCOTUS Case	Description
41	Loretto v. Teleprompter Manhattan CATV Corp. (1982) 458 U.S. 419 (1982) **Score:** 6x3 **In Exam**	**General Description:** A New York statute required landlords to permit cable-television operators to install a small cable box and a few feet of cable on the exterior of apartment buildings, in exchange for a one-time payment of one dollar set by a state commission.
	Jean Loretto, a New York City landlord, challenged the mandate as a taking. The Supreme Court held that any permanent physical occupation of property authorized by government, however minor, is a per se taking — categorically requiring compensation regardless of the economic impact on the owner or the public interest in the occupation. The case is the per se physical-occupation line that runs through Horne v. Department of Agriculture (2015) and Cedar Point Nursery v. Hassid (2021). **Why Case is Relevant:** *The Environmental Question* — Listed among the takings categories (physical occupation) the Patent Clause remedy does not disturb. *The Takings Clause Prong* — The per se physical-occupation rule that predates Lucas: "a permanent physical occupation of property is a taking," however small. *Reason the Constitutional Argument Yourself* — Exam holding: a permanent physical occupation of property is a per se taking.	
42	Alfred L. Snapp & Son, Inc. v. Puerto Rico (1982) 458 U.S. 592 (1982) **Score:** 8x0	**General Description:** Puerto Rico sued Virginia apple growers alleging discrimination against Puerto Rican farmworkers in favor of temporary foreign labor. Justice White, for the Court, held that Puerto Rico had parens patriae standing; Justice Powell took no part.
	The opinion is the modern statement of the doctrine: a sovereign may sue on a quasi-sovereign interest, which includes the health and well-being — physical and economic — of a substantial segment of its population, provided the sovereign articulates an interest apart from those of particular private parties. **Why Case is Relevant:** *Sovereign and Public-Fiduciary Plaintiffs* — The controlling authority for a State attorney general suing on behalf of households injured by the wall. Supplies the quasi-sovereign interest the chapter's sovereign plaintiff categories rest on.	
43	Railway Labor Executives' Association v. Gibbons (1982) 455 U.S. 457 (1982) **Score:** 9x0*15	**General Description:** Congress enacted a statute governing the liquidation of a single bankrupt railroad and the claims of its employees. Justice Rehnquist, for the Court, held that the statute was a bankruptcy law, that it was not uniform, and that the Commerce Clause could not be used to enact it.
	To hold that Congress may enact non-uniform bankruptcy laws under the commerce power, the Court reasoned, would eradicate from the Constitution the limitation the Bankruptcy Clause imposes. The case is the Court's clearest application of the specific-governs-the-general canon to two clauses standing side by side in Article I, Section 8. **Why Case is Relevant:** *The Patent Clause Prong* — The anchor for the argument that a limitation written into one clause of Article I, Section 8 cannot be escaped by legislating under another. *Objections Considered* — Answers the enumerated-powers objection directly. Carrying the reasoning from the Bankruptcy Clause to "limited Times" is the book's step and not the Court's, and the entry says so.	

#	SCOTUS Case	Description
44	Ruckelshaus v. Monsanto Co. (1984) 467 U.S. 986 (1984) **Score:** 7x1*3 **In Exam**	**General Description:** Monsanto challenged a federal pesticide statute that allowed the EPA to use and disclose the confidential test data that companies submitted for registration.
	The Supreme Court held that a trade secret is property protected by the Takings Clause, even though it has no physical form, and that a regulation appropriating the value of that confidential data can be a taking — though whether it is depends on whether the company held a reasonable investment-backed expectation that the data would stay confidential. The case establishes that intangible rights of defined economic value are property for Fifth Amendment purposes. **Why Case is Relevant:** *The Takings Clause Prong* — The case the prong leans on hardest — trade secrets are protected property under the Takings Clause. *Reason the Constitutional Argument Yourself* — Exam holding: intangible property, such as a trade secret, is protected under the Takings Clause.	
45	United States v. 50 Acres of Land (1984) **Score:** 9x0	**General Description:** Held that a public condemnee is entitled to fair market value, not the cost of a substitute facility, as just compensation.
	Why Case is Relevant: *The Environmental Question* — Compensation is fair market value at the time of the taking — the remanufacturer who tools a factory to attract a taking gets no windfall.	
46	Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc. (1984) 467 U.S. 837 (1984) **Score:** 6x0	**General Description:** Justice Stevens, for the Court, upheld the EPA's plantwide definition of a stationary source under the Clean Air Act and set out the two-step framework that governed judicial review of agency statutory interpretation for forty years. Justices Marshall, Rehnquist, and O'Connor took no part.
	Chevron deference is the reason the regulatory wall stood unexamined for so long: a court asked whether an agency had exceeded its statute deferred to the agency's own answer wherever the statute was ambiguous. Overruled by Loper Bright Enterprises v. Raimondo (2024), which is case 84 in this table. **Why Case is Relevant:** *Why Nobody Noticed: How a Constitutional Violation Hides in Plain Sight* — The doctrinal reason the wall was never tested. *The Filable Case: Why 2026 and Not Before* — Its overruling is the first of the four conditions that make the case filable now and not before.	
47	Heckler v. Chaney (1985) 470 U.S. 821 (1985) **Score:** 9x0	**General Description:** An agency's decision not to undertake an enforcement action is presumptively committed to the agency's discretion and unreviewable under the Administrative Procedure Act.
	Why Case is Relevant: *The Anti-Backsliding Provision* — Cited for the holding that an agency's decision not to enforce is presumptively unreviewable — relevant to how the anti-backsliding regime operates.	

#	SCOTUS Case	Description
48	City of Cleburne v. Cleburne Living Center, Inc. (1985) 473 U.S. 432 (1985) **Score:** 9x0*16	**General Description:** A Texas city denied a special-use permit for a group home. Justice White, for a six-Justice majority, declined to treat the intellectually disabled as a quasi-suspect class but struck the ordinance down as applied under rational-basis review. Justices Marshall, Brennan, and Blackmun concurred in the judgment in part and dissented in part.
	The case is the leading example of rational basis with teeth: a classification resting on irrational prejudice fails even the deferential standard. Justice Marshall's separate opinion objected that the Court's review was heightened scrutiny under another name. **Why Case is Relevant:** *Objections Considered* — The escape route from Williamson v. Lee Optical, and the reason the escape is not taken. Cleburne shows what a successful equality claim needs; the regulatory wall has stated rationales that Cleburne's ordinance lacked.	
49	Bowers v. Hardwick (1986) 478 U.S. 186 (1986) **Score:** 5x4	**General Description:** Justice White, for a five-Justice majority, upheld a Georgia sodomy statute, stating in the course of the opinion that proscriptions against such conduct had ancient roots. Justices Blackmun and Stevens each dissented.
	Seventeen years later a brief filed by historians in Lawrence v. Texas took the historical claim apart using sources that had been in print the whole time. The record did not say what Bowers said it said, and the category of person the opinion described did not exist in the period it pointed to. The Court overruled Bowers in 2003. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The cleanest instance of the unasked question. Nothing was discovered; a court asserted a fact, everyone believed it, and for seventeen years nobody checked it against scholarship that already said otherwise.	
50	First English Evangelical Lutheran Church of Glendale v. County of Los Angeles (1987) 482 U.S. 304 (1987) **Score:** 6x3 **In Exam**	**General Description:** A county ordinance temporarily barred a church from rebuilding on flood-prone land it owned. The church argued the ordinance had taken its property for the period the ban was in force.
	The Supreme Court used the case to settle what the Takings Clause requires: it does not forbid the government from taking property, but it requires the payment of just compensation when the government does. A finding that a regulation effects a taking establishes that compensation is owed; it is not, by itself, an order striking the regulation down. **Why Case is Relevant:** *The Takings Clause Prong* — Cited for the remedy point: the Takings Clause does not forbid the taking; it requires compensation. *Reason the Constitutional Argument Yourself* — Exam holding: the Takings Clause does not forbid government from taking property; it requires just compensation.	

#	SCOTUS Case	Description
51	Nollan v. California Coastal Commission (1987) 483 U.S. 825 (1987) **Score:** 5x4	**General Description:** The Nollans owned a beachfront lot in Ventura County and applied for a permit to replace their bungalow with a larger house. The California Coastal Commission granted the permit on condition that they dedicate a public easement across their beach.
	The Supreme Court held that a regulatory condition must bear an essential nexus to the harm the regulation claims to address; an easement allowing the public to walk along the beach had no such nexus to the visual obstruction the larger house was said to cause. The case is the first prong of the modern proportionality test for regulatory conditions on property use, completed in Dolan v. City of Tigard (1994) and reaffirmed in Sheetz v. County of El Dorado (2024). **Why Case is Relevant:** *The Takings Clause Prong* — Cited (with Dolan) on the exactions doctrine, and the argument that it reaches only individualized conditions — which the book answers via Sheetz.	
52	Allegheny Pittsburgh Coal Co. v. County Commission of Webster County (1989) 488 U.S. 336 (1989) **Score:** 9x0	**General Description:** The unanimous Supreme Court decision holding that a county tax-assessment system that valued recently-acquired coal properties at full current value while leaving long-held properties at
	decades-old assessments violated the Equal Protection Clause. **Why Case is Relevant:** *The Takings Clause Prong* — The equality-within-takings authority: a county taxed recently-acquired property far above comparable older holdings — the asymmetry the prong invokes. *Objections Considered* — The equality argument inside the Takings prong, walked at length — similarly situated property treated unequally.	
53	Bonito Boats, Inc. v. Thunder Craft Boats, Inc. (1989) 489 U.S. 141 (1989) **Score:** 9x0	**General Description:** Florida barred the use of a direct molding process to duplicate unpatented boat hulls. Bonito Boats, which had never sought a patent on its hull, sued a competitor under the statute.
	Justice O'Connor, for a unanimous Court, held the Florida statute preempted. The Patent Clause reflects a balance between encouraging innovation and avoiding monopolies that stifle competition, and the Clause contains both a grant of power and limitations on its exercise. By giving patent-like protection to designs in general circulation that Congress left unprotected, Florida entered a field reserved to Congress. Like Sears, the holding runs against a State. The opinion also states, at 151, that “free exploitation of ideas will be the rule, to which the protection of a federal patent is the exception,” and that the ultimate goal of the patent system is to bring new designs and technologies into the public domain through disclosure. **Why Case is Relevant:** The Patent Clause Prong — Reaffirms Sears and supplies the Court's modern statement that the Patent Clause is both a grant and a limitation. Cited for the public-domain principle and the grant-and-limitation reading. State-law preemption authority; the Court has not applied it to a federal agency, and the step from this holding to the federal wall is this argument's own.	

#	SCOTUS Case	Description
54	Lujan v. Defenders of Wildlife (1992) 504 U.S. 555 (1992) **Score:** 6x3	**General Description:** Established the three-part constitutional standing test: injury in fact, causation, and redressability.
	Why Case is Relevant: *Sovereign and Public-Fiduciary Plaintiffs* — The source of the three-part standing test — injury in fact, causation, redressability. *Direct and Associational Plaintiff Categories* — Cited for the generalized-grievance bar the chapter's plaintiff categories must clear.	
55	Lucas v. South Carolina Coastal Council (1992) 505 U.S. 1003 (1992) **Score:** 6x3 **In Exam**	**General Description:** A developer bought two beachfront lots, after which a new state coastal statute barred him from building anything on them.
	The Supreme Court held that when a regulation deprives an owner of all economically beneficial use of his land, it is a taking without any case-by-case balancing — a taking per se. Total deprivation of use, the Court reasoned, is different in kind from the partial diminution of value that ordinary regulation produces, and it requires compensation categorically. **Why Case is Relevant:** *The Environmental Question* — Listed among the per se takings categories (total economic wipeout) the remedy does not disturb. *The Takings Clause Prong* — The per se total-wipeout rule: a regulation denying all economically beneficial use is a taking — the shorter, steeper road. *Objections Considered* — Cited (with Penn Central) to show a post-regulation entrant fails the takings analysis. *Reason the Constitutional Argument Yourself* — Exam holding: a regulation denying all economically beneficial use of property is a per se taking.	
56	Nordlinger v. Hahn (1992) 505 U.S. 1 (1992) **Score:** 8x1*17	**General Description:** Justice Blackmun, for the Court, upheld California's acquisition-value property-tax scheme under Proposition 13, which assessed newly purchased property at purchase price while capping annual increases on long-held property at two percent. Justice Stevens dissented.
	The Court confined Allegheny Pittsburgh Coal Co. v. County Commission of Webster County (1989) to the rare case in which the facts precluded any plausible inference that an acquisition-value policy explained the disparity. Justice Thomas concurred in part while stating that Allegheny Pittsburgh could not be distinguished. **Why Case is Relevant:** *The Takings Clause Prong* — The stated limit on the chapter's use of Allegheny Pittsburgh, conceded rather than left for an opponent to find. *Objections Considered* — The strongest authority against a standalone equality prong: a classification turning on when property was acquired, sustained on rational basis.	
57	Nixon v. United States (1993) 506 U.S. 224 (1993) **Score:** 9x0	**General Description:** The Supreme Court's leading modern application of the political-question doctrine to internal Congressional procedure: the Senate's procedure for trying an impeachment was held
	nonjusticiable because the Constitution textually commits the matter to the Senate and the courts lack judicially manageable standards for second-guessing the chamber's procedural choices. **Why Case is Relevant:** *Hybrid Legislative Voting* — Cited on the political-question doctrine and impeachment procedure — courts' reluctance to police the chamber's internal voting mechanics, which shapes the hybrid-voting design.	

#	SCOTUS Case	Description
58	Dolan v. City of Tigard (1994) 512 U.S. 374 (1994) **Score:** 5x4	**General Description:** A hardware-store owner applied for a permit to expand her store and pave her parking lot. The city granted the permit on condition that she dedicate part of her property to a public greenway and bike path.
	She objected that the conditions were unrelated in scope to the impact of her expansion. The Supreme Court agreed with her on the second prong of the modern proportionality test it had begun building seven years earlier in Nollan v. California Coastal Commission: a regulatory condition imposed on the use of property must bear an essential nexus to the harm the regulation claims to address (Nollan) and must be roughly proportional in scope to that harm (Dolan). The two-prong test underlies Sheetz v. County of El Dorado (2024), which extended it from individual administrative decisions to fees set by general legislation. **Why Case is Relevant:** *The Takings Clause Prong* — Cited with Nollan on the exactions doctrine; the Court rejected the distinction limiting it to individualized conditions.	
59	United States v. Lopez (1995) 514 U.S. 549 (1995) **Score:** 5x4	**General Description:** The Gun-Free School Zones Act of 1990 made it a federal offense to possess a firearm within a school zone. A San Antonio high-school student carried a concealed handgun to school and was convicted under the Act. He argued that Congress lacked power under the Commerce Clause to regulate the possession of a gun near a school.
	Chief Justice Rehnquist, for a five-Justice majority, held the Act beyond the commerce power. Possession of a gun in a school zone is not economic activity and does not substantially affect interstate commerce; the Court declined to pile inference on inference to reach it. The first decision in sixty years to hold a federal statute outside the commerce power. **Why Case is Relevant:** *Objections Considered — The model for the enumerated-powers question in The Rational-Basis Trap. The Court did not ask whether Congress could rationally have thought guns near schools harmful; it asked whether the activity was within the power. The Patent Clause prong must be pleaded in that form, or rational-basis review swallows it.*	
60	Adarand Constructors, Inc. v. Peña (1995) 515 U.S. 200 (1995) **Score:** 5x4*18	**General Description:** Justice O'Connor announced the judgment and delivered the opinion of the Court in relevant part, holding that all racial classifications imposed by any federal, state, or local governmental actor must be analyzed under strict scrutiny. Justices Stevens, Souter, and Ginsburg each dissented.
	The decision confirms that the equal-protection obligation Bolling v. Sharpe read into the Fifth Amendment binds the federal government with the same force the Fourteenth Amendment carries against the States. The strict-scrutiny holding governs racial classifications; the federal-obligation premise is what the book uses. **Why Case is Relevant:** *The Takings Clause Prong* — Supplies the federal bridge for treating unequal federal treatment as a constitutional matter. The extension from race to a vintage classification is the argument's own step, and the chapter says so.	

#	SCOTUS Case	Description
61	Romer v. Evans (1996) 517 U.S. 620 (1996) **Score:** 6x3 **In Exam**	**General Description:** Colorado voters adopted a constitutional amendment forbidding the state or any locality to extend antidiscrimination protections on the basis of sexual orientation.
	The Supreme Court struck the amendment down on rational-basis review, holding that the only purpose the amendment could rationally be said to serve was disqualification of an unpopular group from political participation, which is not a legitimate state interest. The case extends the Cleburne line of "rational basis with bite" — the narrow set of cases in which an equal-protection challenge succeeds because the only rationale a court can find is identifiable animus. **Why Case is Relevant:** *Objections Considered* — Paired with Moreno: a classification explained only by animus fails even rational-basis review. *Reason the Constitutional Argument Yourself* — Exam holding: a classification fails rational-basis review where explained only by animus toward those it disadvantages.	
62	United Food and Commercial Workers Union Local 751 v. Brown Group, Inc. (1996) 517 U.S. 544 (1996) **Score:** 9x0	**General Description:** A union sued under the WARN Act seeking backpay for its members. Justice Souter, for a unanimous Court, held that the third prong of the Hunt associational-standing test is prudential rather than constitutional, and may therefore be abrogated by Congress.
	The holding matters because Hunt v. Washington State Apple Advertising Commission (1977) had suggested that a damages action running solely to members would bar associational standing. Brown Group establishes that the bar is one Congress can lift, and that only the first prong is grounded in Article III. **Why Case is Relevant:** *Objections Considered* — Answers the objection that an association cannot sue for member damages. The limitation is prudential, and the chapter distinguishes what Article III requires from what prudence has added.	
63	Printz v. United States (1997) 521 U.S. 898 (1997) **Score:** 5x4	**General Description:** The Brady Handgun Violence Prevention Act required county sheriffs and other local chief law-enforcement officers to perform federal background checks on prospective handgun purchasers.
	Sheriff Jay Printz and Sheriff Richard Mack sued, arguing that the federal government could not compel state officers to enforce federal regulatory programs. The Supreme Court agreed. The federal government may not commandeer the legislative or executive officers of the states for federal regulatory ends; doing so violates the structure of dual sovereignty the Constitution establishes. The case does not address "limited Times," but it is the constitutional ground for the sheriff plaintiff — the elected county sheriff suing as a constitutional officer to challenge a federal regulatory program that converts him into an enforcement adjunct. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The elected sheriff's precedent — two county sheriffs challenged a federal statute and won — a plaintiff pattern requiring no law be broken. *Sovereign and Public-Fiduciary Plaintiffs* — Cited for the elected sheriff's footing as a constitutional officer — a posture that depends on the sheriff being elected. *Why You Want to Be the Plaintiff* — Cited as a precise, recent precedent for the elected-official plaintiff.	

#	SCOTUS Case	Description
64	Florida Prepaid Postsecondary Education Expense Board v. College Savings Bank (1999) 527 U.S. 627 (1999) **Score:** 5x4 **In Exam**	**General Description:** College Savings Bank sued a Florida state agency for infringing its patent. The question was whether Congress could strip the States of sovereign immunity and expose them to patent-infringement suits;
	the Court held that, in the manner Congress had attempted, it could not. The premise the Court worked from is what matters for this book, and the Court stated it plainly: a patent is a form of property, protected as other property is. **Why Case is Relevant:** *The Patent Clause Prong* — Completes the property point: patents are a species of property. *The Takings Clause Prong* — Cited again for the same holding on the takings side — patents are property, protected as other property. *Reason the Constitutional Argument Yourself* — Exam holding: a patent is a form of property, protected as other property.	
65	Geier v. American Honda Motor Co. (2000) 529 U.S. 861 (2000) **Score:** 5x4	**General Description:** Held that a federal motor-vehicle safety standard preempted a state tort claim that conflicted with the federal regulatory objective.
	Why Case is Relevant: *The Wall Itself* — Addresses the preemption question for the Federal Motor Vehicle Safety Standards — relevant to how the wall's statutes operate.	
66	TrafFix Devices, Inc. v. Marketing Displays, Inc. (2001) 532 U.S. 23 (2001) **Score:** 9x0	**General Description:** Marketing Displays held now-expired utility patents on a dual-spring mechanism that kept temporary road signs upright in wind. After the patents expired and a competitor began selling sign stands using the same mechanism, Marketing Displays sued for trade dress infringement, arguing that the visible spring design identified it as the source.
	Justice Kennedy, for a unanimous Court, held that a functional product feature cannot receive trade dress protection under the Lanham Act. An expired utility patent is strong evidence that the features it claimed are functional, and where the expired patent claimed the feature in question, the party seeking trade dress protection bears a heavy burden of showing that the feature is merely ornamental, incidental, or arbitrary. The Court declined to let trade dress become the practical equivalent of an expired utility patent. **Why Case is Relevant:** *Objections Considered* — The trademark-side counterpart to *Bonito Boats* and *Sears*. Establishes that when a utility patent expires, the functional design it covered cannot be recaptured through trade dress. The case marks the outer limit of what non-patent intellectual property law may do to prevent remanufacturing of a patent-expired design, and shows the Court policing that boundary on its own initiative. It sharpens the book's central contrast: the intellectual property statutes release the expired design, while the regulatory statutes do not.	

#	SCOTUS Case	Description
67	Eldred v. Ashcroft (2003) 537 U.S. 186 (2003) **Score:** 7x2 **In Exam**	**General Description:** The Sonny Bono Copyright Term Extension Act of 1998 extended existing and future copyright terms by twenty years.
	Eric Eldred and others who relied on works passing into the public domain argued that retroactively extending existing terms violated the Patent and Copyright Clause's "limited Times" requirement and the First Amendment. The Supreme Court, seven to two, upheld the Act. Justice Ginsburg's majority held that "limited Times" requires only that the term be limited, not perpetual, and that Congress could rationally decide to extend the period of incentive even retroactively. Justice Breyer's dissent argued the extension produced a term economically equivalent to ninety-nine point eight percent of a perpetual copyright. The case is the leading adverse Patent Clause precedent; the chapter on objections considered distinguishes it as a copyright case in a calendar-adjustment posture, not a patent case in a structural-ratchet posture. **Why Case is Relevant:** *Objections Considered* — The case behind the objection that a copyright-term extension survived a "limited Times" challenge — which the prong distinguishes. *Reason the Constitutional Argument Yourself* — Exam holding: a copyright-term extension did not violate "limited Times"; Congress receives substantial deference.	
68	Lawrence v. Texas (2003) 539 U.S. 558 (2003) **Score:** 6x3*19	**General Description:** Justice Kennedy, for five Justices, held that a Texas statute criminalizing same-sex intimate conduct violates the Due Process Clause and overruled Bowers v. Hardwick. Justice O'Connor concurred in the judgment on equal-protection grounds. Justices Scalia and Thomas dissented.
	The Court stated that Bowers was not correct when it was decided and is not correct today. The historians' brief that supplied the corrected record was the work of scholars outside the legal profession, and the sources it used had been publicly available throughout the seventeen years Bowers stood. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — The correction, and the measure of how long a court will hold an unexamined historical assertion when no one puts the contrary record in front of it.	
69	Kelo v. City of New London (2005) 545 U.S. 469 (2005) **Score:** 5x4	**General Description:** Justice Stevens, for a five-Justice majority, held that a city's transfer of private property to a private developer for economic development satisfies the public-use requirement of the Takings Clause. Justices O'Connor and Thomas each dissented.
	Susette Kelo lost, the development was never built, and the company the project was meant to serve left New London in 2009. The decision produced legislative reform in a majority of States, which is the book's point in citing it: a loss can produce a remedy, and the Institute for Justice, which carried the case, litigates in the same doctrinal territory this argument occupies. **Why Case is Relevant:** *What You Can Do: A Citizen's Playbook* — Named among the public-interest firms that take takings cases without charging a plaintiff. *When Outsiders Advance Constitutional Law and Other Disciplines* — An injury rather than a discovery, and a loss that still moved the law.	

#	SCOTUS Case	Description
70	MedImmune, Inc. v. Genentech, Inc. (2007) 549 U.S. 118 (2007) **Score:** 8x1	**General Description:** Justice Scalia, for the Court, held that a patent licensee need not breach or terminate its license before seeking a declaratory judgment that the patent is invalid, unenforceable, or not infringed. Justice Thomas dissented.
	The decision removed the reasonable-apprehension-of-suit requirement the Federal Circuit had imposed and restored the ordinary actual-controversy standard of the Declaratory Judgment Act. A party need not expose itself to liability to test the legal position it is being held to. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The case the chapter is named for. *The Filable Case: Why 2026 and Not Before* — One of the conditions that make a pre-enforcement challenge available to a would-be remanufacturer who has built nothing yet.	
71	Massachusetts v. Environmental Protection Agency (2007) 549 U.S. 497 (2007) **Score:** 5x4	**General Description:** Twelve states and several local governments sued the EPA over the agency's refusal to regulate greenhouse-gas emissions from new motor vehicles.
	The Supreme Court held that Massachusetts had standing to sue, because the state had suffered concrete injury to its territory (rising sea levels eroding coastline) and to the welfare of its citizens, and because the federal government's failure to regulate was traceable to that harm and redressable by a court order. The decision announced the doctrine of special solicitude: a sovereign plaintiff is owed a more permissive standing analysis than the showing required of an ordinary private plaintiff. The case is the inflection point for sovereign and quasi-sovereign standing in modern federal litigation, and the door through which states will be among the first plaintiffs in the patent-bargain case. **Why Case is Relevant:** *Why Nobody Noticed* — Cited among the doors the Court has opened over the past quarter-century. *You No Longer Need to Break a Law to Challenge One* — The 2007 turning point — states sued the EPA over a regulatory failure and established a route that needs no broken law. *Sovereign and Public-Fiduciary Plaintiffs* — The anchor for sovereign standing — a state suing the federal government is owed special solicitude. *The Filable Case* — Cited as part of the recent access toolkit — a sovereign plaintiff may sue over a federal regulatory failure. *Why You Want to Be the Plaintiff* — Cited for the parens patriae / sovereign standing a state invoked successfully.	

#	SCOTUS Case	Description
72	Golan v. Holder (2012) 565 U.S. 302 (2012) **Score:** 6x2	**General Description:** Section 514 of the Uruguay Round Agreements Act restored United States copyright to foreign works that had already passed into the American public domain. Conductors, teachers, and film archivists who had built on those works sued. The Court upheld the statute.
	Justice Ginsburg, for the Court, held that the "limited Times" language does not bar Congress from extending protection to works already in the public domain, and rejected the premise that the Constitution renders the public domain largely untouchable by Congress; the First Amendment was satisfied by the idea-expression distinction and fair use. The opinion refers to the provision as the Copyright and Patent Clause. Justice Breyer, joined by Justice Alito, dissented, arguing the restoration withdrew from the public what it already held without promoting the creation of anything new. Justice Kagan took no part. The decision reviews an act of Congress under the Copyright Clause; it does not reach whether an executive agency may, by regulation, make a patent-expired design unlawful to build. That distinction is this argument's, not the Court's. **Why Case is Relevant:** *Objections Considered* — Adverse authority, paired with Eldred v. Ashcroft — the Court has said the public domain is not beyond the reach of Congress. Answered on the ground that Golan reviewed an enacted statute in a copyright posture, while the wall at issue here was built by agencies under general delegations and has never been tested under the Patent Clause.	
73	Arkansas Game & Fish Commission v. United States (2012) 568 U.S. 23 (2012) **Score:** 8x0 **In Exam**	**General Description:** The federal government, operating a dam, released water in a pattern that repeatedly flooded a state-owned wildlife area during the tree-growing season over a span of years, then stopped.
	The State sued, claiming the recurring flooding had killed timber and worked a taking. The government argued that flooding can be a taking only when it is permanent. The Supreme Court rejected that categorical rule. Whether government action is a taking, the Court held, depends on its actual effect on the property — not on whether the invasion is labeled temporary or permanent. **Why Case is Relevant:** *Four Problems, One Remedy* — One of four recent decisions the book cites as making the case ripe now. *The Takings Clause Prong* — Cited for the holding that even temporary government-induced invasions (flooding) may be takings. *Reason the Constitutional Argument Yourself* — Exam holding: even temporary government-induced invasions of property may constitute takings.	
74	Clapper v. Amnesty International USA (2013) 568 U.S. 398 (2013) **Score:** 5x4	**General Description:** Held that plaintiffs lacked standing to challenge surveillance where the threatened injury was too speculative.
	Why Case is Relevant: *Sovereign and Public-Fiduciary Plaintiffs* — Polices the injury line — an injury must be concrete and imminent, not speculative — the standing bar plaintiffs must clear.	

#	SCOTUS Case	Description
75	Susan B. Anthony List v. Driehaus (2014) 573 U.S. 149 (2014) **Score:** 9x0	**General Description:** The Susan B. Anthony List, an antiabortion advocacy organization, had distributed political messages criticizing a congressional candidate for his health-care vote.
	The candidate filed a complaint under an Ohio statute criminalizing false statements in political campaigns. The advocacy organization sued in federal court before any prosecution had occurred, seeking a declaration that the statute violated the First Amendment. The Supreme Court held that pre-enforcement standing existed: a credible threat of enforcement under a statute the plaintiff intends to violate suffices to establish injury in fact, and the plaintiff need not wait for prosecution to challenge the statute. The case is the controlling modern statement of pre-enforcement standing, and the doctrine on which the patent-bargain case will rest at the standing stage. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — The credible-threat doctrine — a third pre-enforcement-review case extending standing further. *The Filable Case* — Cited for the unanimous holding that a plaintiff facing a credible threat of enforcement may sue before enforcement.	
76	Kimble v. Marvel Entertainment, LLC (2015) 576 U.S. 446 (2015) **Score:** 6x3	**General Description:** Kimble sold Marvel a patent on a web-shooting toy and took royalties. When the patent expired, Marvel stopped paying. Kimble asked the Court to overrule Brulotte v. Thys Co., which bars post-expiration royalties.
	Justice Kagan, for the Court, declined on stare decisis grounds and stated the governing principle: a patent typically expires twenty years from the filing date, and when it expires the patentee's prerogatives expire with it and the right to make or use the article, free from all restriction, passes to the public. Justice Alito dissented, joined by the Chief Justice and Justice Thomas, on the merits of Brulotte. The phrase “free from all restriction” means free of the patent monopoly; it does not hold that the article is immune from all regulation. **Why Case is Relevant:** The Patent Clause Prong — The book's canonical gloss: at expiration the unrestricted right to make and use the invention passes to the public. Federal patent-law authority, decided against a private licensor. Cited for what passes to the public at expiration, not for any holding about regulatory power.	
77	Horne v. Department of Agriculture (2015) 576 U.S. 351 (2015) **Score:** 8x1*4 **In Exam**	**General Description:** A Depression-era federal marketing order required raisin growers to surrender a portion of each year’s crop to a government committee. The growers refused and were fined.
	The government argued the requirement was a permissible regulation of the raisin market, not a taking. The Supreme Court disagreed. Compelling the growers to hand over their raisins was a physical appropriation of property and therefore a taking per se, and the government could not avoid that conclusion by characterizing the appropriation as a condition of doing business or a contribution to orderly marketing. **Why Case is Relevant:** *The Environmental Question* — Listed among the per se categories (appropriation of personal property) the remedy does not disturb. *The Takings Clause Prong* — Applied the per se rule to government appropriation of personal property. *Objections Considered* — Cited on recent personal-property takings doctrine — the Takings Clause applies to personal property with full force. *Reason the Constitutional Argument Yourself* — Exam holding: the per se takings rule applies with full force to the appropriation of personal property.	

#	SCOTUS Case	Description
78	Star Athletica, L.L.C. v. Varsity Brands, Inc. (2017) 580 U.S. 405 (2017) **Score:** 6x2*12	**General Description:** A cheerleading-uniform maker held copyright registrations for two-dimensional designs of lines, chevrons, and shapes appearing on the surface of its uniforms, and sued a competitor for infringement. The competitor argued that designs on useful articles are not copyrightable at all.
	Justice Thomas, for the Court, established the governing separability test under 17 U.S.C. § 101: a feature incorporated into the design of a useful article is eligible for copyright only if it can be perceived as a two- or three-dimensional work of art separate from the article and would qualify as a protectable pictorial, graphic, or sculptural work on its own or in another medium if imagined separately. The Court expressly noted that the test does not render the underlying useful article itself eligible for copyright. **Why Case is Relevant:** *Objections Considered* — The governing authority for how far copyright reaches into a manufactured product. A separable badge, emblem, or applied graphic may qualify; the vehicle carrying it may not. The case answers directly the objection that an entire car or truck is copyright-protected for ninety-five years as a corporate work made for hire: the term is real, but a useful article is not the kind of work to which it attaches. Together with *TrafFix*, it fixes the ceiling on what non-patent intellectual property law can protect after a utility patent expires.	
79	Oil States Energy Services, LLC v. Greene's Energy Group, LLC (2018) 584 U.S. 325 (2018) **Score:** 7x2	**General Description:** Oil States, whose patent on hydraulic-fracturing wellhead equipment had been cancelled by the Patent Office in inter partes review, argued that only an Article III court could revoke an issued patent.
	The Supreme Court disagreed, seven to two. Justice Thomas, for the Court, held that a patent is a public franchise and that reconsidering its validity is a matter of public rights Congress may assign to an executive agency. Justice Gorsuch, joined by Chief Justice Roberts, dissented, treating the issued patent as private property revocable only by a court. The majority called its holding narrow, reserved the retroactivity question, and said it was not deciding whether patents are property under the Due Process or Takings Clauses. It also repeated, from its own older cases, that the grant of a patent is a matter between the public, who are the grantors, and the patentee. The decision concerns the government's power over a patent during its term; it does not reach the public's right to build the design after the patent expires. That distinction is this argument's, not the Court's. **Why Case is Relevant:** *Objections Considered* — Adverse authority raised by a scholar of administrative law: if an agency may cancel a live patent, the Court is unlikely to police what agencies do after expiration. Answered on the ground that Oil States sits on the grant side of the Clause and expressly reserved the Takings question, and that the public-franchise vocabulary names the public as the party the franchise came from and goes to when the term is over.	
80	Trump v. Hawaii (2018) 585 U.S. 667 (2018) **Score:** 5x4	**General Description:** Chief Justice Roberts, for a five-Justice majority, upheld the entry restrictions of Presidential Proclamation 9645. Justices Breyer and Sotomayor each dissented.
	The book cites the case for a single sentence in it. The Court stated that Korematsu v. United States had been gravely wrong the day it was decided and has no place in law under the Constitution — a repudiation offered seventy-four years after the fact, and only once the suppressed record had long been public. **Why Case is Relevant:** *When Outsiders Advance Constitutional Law and Other Disciplines* — Cited for how a court speaks when it learns the original proceeding was tainted, and for how long the interval can be.	

#	SCOTUS Case	Description
81	Google LLC v. Oracle America, Inc. (2021) 593 U.S. 1 (2021) **Score:** 6x2*21	**General Description:** Google built the Android platform to let programmers use the Java language and copied roughly 11,500 lines of declaring code from the Java SE application programming interface so that familiar calls would work on the new platform, writing its own implementing code. Oracle, which had acquired Java, sued for copyright infringement.
	Justice Breyer, for the Court, assumed without deciding that the declaring code was copyrightable and held Google's copying fair use. The declaring code was functional and tied to the programmers' investment in learning it; Google took only what was needed to let programmers work in a new environment; and the copying advanced the constitutional purpose of copyright by enabling new creative expression. **Why Case is Relevant:** *Objections Considered — The second of three reasons the firmware copyright is narrower than its term: the Court has protected the room a rewriter needs. A remanufacturer may study the original engine-control module to learn what signals the engine expects and write new code to produce them.*	
82	Cedar Point Nursery v. Hassid (2021) 594 U.S. 139 (2021) **Score:** 6x3 **In Exam**	**General Description:** A California regulation gave labor organizers a right to enter an agricultural employer's property to speak with workers for several hours a day during much of the year. The growers said the forced access took their property.
	The Supreme Court agreed. A regulation that appropriates a right to physically enter private land and hands it to others is a per se taking, the Court held, even though the access was intermittent and the owners kept every other use of their land. The decision is part of the modern Court's broadened reading of what counts as a taking. **Why Case is Relevant:** *Four Problems, One Remedy* — One of the four recent decisions cited as making the case ripe. *The Environmental Question* — Listed among the per se categories (physical occupation for limited hours). *The Takings Clause Prong* — Extended the physical-occupation branch to forced access for limited hours. *Reason the Constitutional Argument Yourself* — Exam holding: a regulation granting third parties a right of physical access, even for limited hours, is a per se taking.	
83	American Hospital Association v. Becerra (2022) **Score:** 9x0	**General Description:** Held that HHS could not vary drug-reimbursement rates for a subset of hospitals without the survey the statute required; decided without Chevron deference.
	Why Case is Relevant: *You No Longer Need to Break a Law to Challenge One* — Illustrates how broadly the regulatory wall reaches — the Association sued HHS and prevailed. *Direct and Associational Plaintiff Categories* — Cited as an associational plaintiff — the Association represents roughly 5,000 hospitals and brings real litigating capacity.	

#	SCOTUS Case	Description
84	West Virginia v. Environmental Protection Agency (2022) 597 U.S. 697 (2022) **Score:** 6x3 **In Exam**	**General Description:** Several States challenged the EPA's authority to reshape the nation's electricity generation through climate regulation.
	The Supreme Court ruled against the agency and articulated the "major questions doctrine": when an agency claims the power to decide a matter of vast economic and political significance, it must point to clear authorization from Congress, not infer the power from a vague or ancillary statutory phrase. **Why Case is Relevant:** *Four Problems, One Remedy* — One of the four recent decisions cited as making the case ripe. *The Filable Case* — Cited for the fullest statement of the major-questions doctrine. *The Patent Clause Prong* — The major-questions doctrine: an agency claiming power of vast economic and political significance must point to clear congressional authorization. *Why You Want to Be the Plaintiff* — Cited as shorthand for a single decision that reset a field — named for its plaintiff. *Reason the Constitutional Argument Yourself* — Exam holding: under the major-questions doctrine, an agency asserting authority over a matter of vast significance needs clear congressional authorization.	
85	Sackett v. Environmental Protection Agency (2023) 598 U.S. 651 (2023) **Score:** 9x0*5	**General Description:** A landowning couple challenged the reach of a federal Clean Water Act rule over their own property and prevailed at the Supreme Court without first incurring a penalty.
	Why Case is Relevant: *Why You Want to Be the Plaintiff* — Cited as a directly-regulated single-landowner plaintiff who challenged the EPA and won.	
86	Tyler v. Hennepin County, Minnesota (2023) 598 U.S. 631 (2023) **Score:** 9x0	**General Description:** A county sold a delinquent taxpayer's home for $40,000 to satisfy a $15,000 tax debt and kept the surplus. Chief Justice Roberts, for a unanimous Court, held the retention a classic taking. Justice Gorsuch concurred, joined by Justice Jackson.
	The State argued that its own law had stripped the owner of any property interest in the surplus. The Court answered that property rights cannot be so easily manipulated, and that a government may not sidestep the Takings Clause by disavowing traditional property interests long recognized under state law. The principle reaches a right the government declines to recognize as well as one it concedes. **Why Case is Relevant:** *The Takings Clause Prong* — The answer to the objection that the public's post-expiration right is not property because current federal law does not treat it as property. *Objections Considered* — Cited for the same proposition where the objection is stated in its strongest form.	

#	SCOTUS Case	Description
87	Sheetz v. County of El Dorado (2024) 601 U.S. 267 (2024) **Score:** 9x0 **In Exam**	**General Description:** A county charged a landowner a substantial traffic-impact fee as a condition of a building permit. The fee was fixed by a general schedule the county's legislature had adopted, not by an official's individual decision.
	The landowner raised a takings challenge; the county argued that the Takings Clause's limits on permit conditions apply only to individualized administrative demands, not to conditions set by general legislation. The Supreme Court unanimously rejected the distinction: the Takings Clause contains no exception for legislatively imposed conditions. **Why Case is Relevant:** *You No Longer Need to Break a Law to Challenge One* — Held unanimously that the Takings Clause applies to legislative as well as adjudicative impositions. *The Filable Case* — Cited for extending takings scrutiny to burdens imposed by general legislation. *The Takings Clause Prong* — Answered the objection directly — a traffic-impact fee imposed by general legislation is still subject to takings scrutiny. *Why You Want to Be the Plaintiff* — Named among the model plaintiffs (Printz, Loper Bright, Sheetz) — an ordinary name that became a landmark. *Reason the Constitutional Argument Yourself* — Exam holding: a permit condition imposed by general legislation is still subject to takings scrutiny.	
88	Loper Bright Enterprises v. Raimondo (2024) 603 U.S. 369 (2024) **Score:** 6x2*6 **In Exam**	**General Description:** A group of commercial herring fishermen challenged a federal rule requiring them to pay the salaries of the monitors stationed on their boats. The Supreme Court used the case to overrule Chevron deference.
	A court, the Court held, must determine for itself what a statute means, using the ordinary tools of interpretation, rather than deferring to an agency's reading. The decision reshaped administrative law, and it is what leaves a court free to decide the patent-bargain question on its own. **Why Case is Relevant:** *Four Problems, One Remedy* — One of the four recent decisions cited as making the case ripe. *Why Nobody Noticed* — Marks the end of Chevron deference (2024) in the chapter's account. *You No Longer Need to Break a Law to Challenge One* — Overruled Chevron — a court now decides a statute's meaning for itself, strengthening the plaintiff's position. *The Filable Case* — Cited for overruling Chevron on June 28, 2024 — the deference was gone. *The Patent Clause Prong* — Ended Chevron deference: a reviewing court now determines the best reading of a statute itself. *Why You Want to Be the Plaintiff* — Cited — four family fishing businesses challenged a federal rule — as a model plaintiff. *Reason the Constitutional Argument Yourself* — Exam holding: courts formerly deferred to an agency's reasonable reading of an ambiguous statute; Loper Bright ended that.	

#	SCOTUS Case	Description
89	**Corner Post, Inc. v. Board of Governors of the Federal Reserve System (2024)** 603 U.S. 799 (2024) Score: 6x3 Not in Exam	**General Description:** A North Dakota truck stop that opened in 2018 challenged Regulation II, a Federal Reserve rule issued in 2011 capping debit-card interchange fees. The government argued the six-year limitations period had run out before the plaintiff existed.
	The Supreme Court held, 6–3, that an Administrative Procedure Act claim accrues when the plaintiff is injured by the rule, not when the agency issued it. Justice Barrett wrote for the Court; Justice Kavanaugh concurred; Justice Jackson dissented, joined by Justices Sotomayor and Kagan. **Why Case is Relevant: The Filable Case** — The companion rule to the end of Chevron deference. The regulatory wall was built between 1970 and 1987; under the older reading of the limitations period most of it was unchallengeable by anyone not already in court decades ago. Under this decision a business first injured last year may challenge a rule issued in 1987.	

Score key

Format is winner × loser (e.g., 6x3 = six Justices in the majority, three dissenting). Counts reflect the Justices who actually participated, so a recusal shows as a reduced total (e.g., 8x0). An equally divided Court would appear as NxN (e.g., 4x4), meaning no Supreme Court majority and the lower court's judgment stands by default; no case in this table was decided by an equal division. An asterisk on a score points to a note below.

Score footnotes

A note on scoring the older cases. The score records the vote by which the Court decided the case, written as majority×dissent (for example, 6x3). For decisions from the era before the Court routinely reported a vote breakdown, the figure is reconstructed from two facts: the roster of Justices then sitting, and whether the report records any dissent. Where a case issued with no recorded dissent and the participating Justices can be identified from the volume's roster, the score is given (for example, 6x0). Where the number of participating Justices cannot yet be confirmed, a dash is shown in place of a number, and the case-specific note below explains why. A score is never invented to fill a gap.

*1 Standard Oil Co. of New Jersey v. United States (1911): the judgment ordering dissolution was unanimous; Justice Harlan concurred in that result but dissented from the

majority's adoption of the "rule of reason."

*2 Aro Mfg. Co. v. Convertible Top Replacement Co. (1961): a fractured decision reversing in Aro's favor. Justice Whittaker's opinion was joined by three other Justices; Justices Black and Brennan concurred separately (Brennan concurring only in the result); Justices Harlan, Frankfurter, and Stewart dissented.

*3 Ruckelshaus v. Monsanto Co. (1984): the 7–1 figure is the vote on the judgment (vacate and remand). Justice White took no part, and Justice O'Connor concurred in part and dissented in part (as to Part IV-B).

*4 Horne v. Department of Agriculture (2015): the Court held 8–1 that the raisin reserve was a per se physical taking (Justice Sotomayor alone in dissent), but divided 5–4 on the remedy — whether the growers were entitled to relief from the fine.

*5 Sackett v. EPA (2023): all nine Justices agreed the Sacketts prevailed and reversed the Ninth Circuit; the Court split 5–4 on the narrower "continuous surface connection" test for "waters of the United States."

*6 Loper Bright Enterprises v. Raimondo (2024): Justice Jackson took no part in Loper Bright (No. 22–451), where the vote was 6–2. In the consolidated companion case, Relentless, Inc. v. Department of Commerce (in which she participated), the split was 6–3.

*7 Webber v. Virginia (1880); Singer Mfg. Co. v. June Mfg. Co. (1896); McCormick Harvesting Machine Co. v. Aultman (1898): each issued without a recorded dissent (a non-dissenting outcome), but the exact number of participating Justices has not yet been independently confirmed against the Supreme Court Database Legacy file. Score to be finalized.

*8 Pennock v. Dialogue (1829): the Court that decided the cases reported in 27 U.S. (2 Pet.) comprised six Justices. (A seventh, John McLean, was confirmed in March 1829 after Justice Trimble's death but did not take his seat until January 1830 and so did not participate.) Justice Story wrote for a unanimous Court with no recorded dissent, giving 6x0. As with all cases of this era, the reporter did not separately record each Justice's

participation in the individual decision, so the figure rests on the six-member roster of the volume.

*9 Motion Picture Patents Co. v. Universal Film Mfg. Co. (1917): the vote was 6—3. Justice Clarke wrote for the majority; Justice Holmes dissented, joined by Justices McKenna and Van Devanter. Justice McReynolds concurred in the result only.

*10 Wheaton v. Peters (1834): Justice McLean wrote for the majority; Justices Thompson and Baldwin dissented. The Court noted separately that it was unanimous on the point that no reporter can hold copyright in the Court’s written opinions. The January 1834 roster comprised seven Justices (Marshall, Johnson, Duvall, Story, Thompson, McLean, and Baldwin), but the report does not state which Justices participated in this decision, and Justices Johnson and Duvall were frequently absent in this period. The number of participating Justices has not been independently confirmed against the Supreme Court Database Legacy file. Score to be finalized.

*11 Sears, Roebuck & Co. v. Stiffel Co. (1964): Justice Black delivered the opinion for a unanimous Court. Justice Harlan concurred in the result only, taking the view that a State should be able to bar copying where the purpose is to prevent one seller’s goods from being palmed off as another’s.

*12 Star Athletica, L.L.C. v. Varsity Brands, Inc. (2017): Justice Thomas wrote for the Court, joined by Chief Justice Roberts and Justices Alito, Sotomayor, and Kagan. Justice Ginsburg concurred in the judgment only, on the ground that the designs were themselves copyrightable pictorial works reproduced on useful articles, so no separability analysis was needed. Justice Breyer dissented, joined by Justice Kennedy. The 6x2 figure counts Justice Ginsburg with the majority on the judgment.

*13 Kewanee Oil Co. v. Bicron Corp. (1974): Chief Justice Burger’s opinion was joined by four Justices; Justice Marshall concurred in the result only; Justices Douglas and Brennan dissented; Justice Powell took no part. The 6x2 figure counts Justice Marshall with the majority on the judgment.

*15 Railway Labor Executives' Association v. Gibbons (1982): the judgment was unanimous. Justice Rehnquist's opinion drew seven Justices; Justices Marshall and Brennan concurred in the judgment on a narrower rationale and expressly declined the majority's broader reading of the uniformity requirement — the reading this book relies on.

*16 City of Cleburne v. Cleburne Living Center, Inc. (1985): all nine Justices agreed the ordinance was invalid as applied. Only six joined the rational-basis rationale; Justices Marshall, Brennan, and Blackmun dissented from the as-applied remedy and wrote that the Court's review was heightened scrutiny under another name.

*17 Nordlinger v. Hahn (1992): Justice Blackmun wrote for the Court and Justice Stevens dissented. Justice Thomas concurred in part while stating that Allegheny Pittsburgh Coal Co. v. County Commission of Webster County (1989) could not be distinguished — the limit this book concedes in the Takings Clause chapter.

*18 Adarand Constructors, Inc. v. Peña (1995): Justice O'Connor's opinion was for the Court except as to Part III-C, which drew only Justice Kennedy. Justices Scalia and Thomas concurred in part and in the judgment. Three dissenting opinions were filed, for four Justices in all.

*19 Lawrence v. Texas (2003): five Justices joined Justice Kennedy's Due Process opinion. Justice O'Connor supplied the sixth vote but concurred only in the judgment, on Equal Protection grounds, and expressly declined the Due Process rationale.

*20 Baker v. Selden (1879): Justice Bradley delivered the opinion for the Court and the report records no dissent; nine Justices sat in the October 1879 Term. To be confirmed against the Supreme Court Database Legacy file before print.

*21 Google LLC v. Oracle America, Inc. (2021): Justice Breyer wrote for the Court; Justice Thomas dissented, joined by Justice Alito. Justice Barrett took no part, the case having been argued before she joined the Court, so the participating total is eight.

*22 Continental Paper Bag Co. v. Eastern Paper Bag Co. (1908): Justice McKenna delivered the opinion of the Court; Justice Harlan dissented alone. The report records no

Justice as not participating, so the score assumes a full bench of nine. To be confirmed against the Supreme Court Database Legacy file before print.

Appendix B: Note on AI-Assisted Authorship

On January 29, 2025, the U.S. Copyright Office published *Copyright and Artificial Intelligence, Part 2: Copyrightability* (copyright.gov/ai), concluding that works created by human authors using AI tools are eligible for copyright protection when the human provides substantial creative direction, selection, and arrangement. The Office confirmed that "the use of AI to assist in the process of creation or the inclusion of AI-generated material in a larger human-generated work does not bar copyrightability." The preface sets out the working method in full, tool by tool, and the file record that measures it. This appendix records what the preface does not: the legal position, the provenance of each analysis, and the dated public record behind the argument.

In April 2026, while this book was in its final review cycle, Jamie Metzl — technology futurist and author of six previous books — published *The AI Ten Commandments: A New Moral Code for Humanity* (Transcend Books) with GPT-5 named as his co-author, which Metzl described in *The Free Press* as "the first major work of nonfiction listing a human and AI as co-authors." Metzl's methodology runs closely parallel to the one used here. The difference in authorship credit is principled: Metzl's book presents what an AI synthesized from the world's moral traditions when given a careful prompt, so naming the AI as co-author is honest disclosure of what the book is. This book presents a constitutional argument no AI generated — the Patent Clause violation, the three-tier breakdown, the platform-year remedy, and the rest of the analytical spine came from a human reader spending around two years across patent law, regulatory economics, and constitutional doctrine. The AI assisted in research, drafting, and refinement under substantial human direction, in the manner the Copyright Office's 2025 guidance describes as eligible for full copyright protection.

No AI system independently conceived or authored this book. Eleven distinct language models were used, each for limited functions, and the preface names them and says what each did. No combination of them produced the manuscript as a whole. The author determined its arguments, methodology, structure, arrangement, factual conclusions, and final language, accepting, rejecting, verifying, or rewriting every model output, and editing extensively by hand.

Which system produced which analysis is set out here rather than in the running text, so that the chapters carry the argument and the apparatus carries the method. The two independent estimates behind the $22 trillion midpoint were developed with large language models from two organizations. The consumer-expenditure floor was developed with Anthropic's Claude, beginning with Claude Sonnet 4.5 and carried forward into later model generations as they became available. The industry-accounts estimate was developed with xAI's Grok, at Grok 4.1. The three-tier breakdown charts — 58 percent Blocked, 15 percent Hindered, 27 percent Flowing — were produced with Claude Opus 4.6 Extended.

Anthropic develops and improves Claude; xAI develops and improves Grok. Neither model was built, trained, or altered by the author. What the author did was specify the method, select the federal data sources, direct the analysis across successive iterations and across more than one model generation, and verify the outputs against the underlying datasets.

The argument also has a dated public record independent of this manuscript, published in the author's own name and in the open. On 18 May 2025, in *The Hidden War on Expired Patents — Regulations Drive Up $*, the author set out the spine of this book: Article I, Section 8, Clause 8 and the "limited Times" constraint; Jefferson's letter to Isaac McPherson of 13 August 1813 and the patent as a social contract rather than an inherent right; Madison on the bargain and on limited duration as the public's guarantee; the invention passing into the public domain when the term ends; the fourteen-, seventeen-, and twenty-year term history with the design-patent exception; and the claim that a web of modern regulation, presented as protective, now defeats the public's half of that bargain and drives up the American cost of living, with 1970 as the hinge and lobbyist capture as the

mechanism. Those are the arguments of this book, in public print, more than a year before the manuscript reached its present form.

That article also disclosed, at the time it was written, that the research had been carried out partly through questions put to Grok, the AI assistant from xAI. The disclosure in this appendix is therefore not a concession made at publication. It is the continuation of a practice the author followed in public while the work was still in progress.

The vehicle argument is older still. On 22 March 2025, in *Focus Tariffs & Deregulations (DOGE) on Cars: Make Cars Affordable Again*, the author set out the counterfactual of an America that had stopped adding new car rules after 1 January 1971 and identified the consequence that became a chapter of this book: “a new niche market emerges for low-cost cars using patent-expired parts, just like how electronic calculators went from pricey gadgets to cheap essentials once their patents expired.”

And the second distinctive proposal in this book, Hybrid Legislative Voting, was first argued publicly in *The History of Secret Legislative Voting in the US Congress* on 16 June 2024, and developed in *What is Hybrid Legislative Voting* on 13 January 2025. The book was not produced from a prompt in a session or a season. It was assembled from a body of work already written, already published under the author’s name, already timestamped, and already argued in front of a public readership.

Model-specific provenance signals, if present in any passage, should be read in light of this disclosed workflow. Such a signal may indicate that text was generated, revised, or processed by a particular model. It does not establish that the model conceived the argument, authored the work, performed the research attributed to other tools, or controlled the manuscript’s selection, arrangement, verification, and final expression.

Appendix C: LLM Responses to Prompt 7 in the chapter “Reason the Constitutional Argument Yourself: The Open-Book Examination”

The exam attaches all of Part One of that chapter (and nothing else in the attachment), which is a PDF of slides and thirty-one numbered Supreme Court holdings, together with three further decisions on the validity of the regulation — holdings on the patent clause, together with cases used by regulatory agencies to support their actions. It then asks a series of 7 prompts (questions) to these AI engines (Large Language Models). Prompt #7 is the one that reveals support for a constitutional remedy to be found for the inability to remanufacture wholly assembled vehicles and some regulated machines where all of their underlying patents have expired.

In brief, that is because any attempt by the public newly remanufacturing anything in the public domain is required to meet current energy and safety standards, not those that the original patent holder had to meet. In any 20-year period, the standards tighten so much that nothing could ever be remanufactured that is regulated and is a wholly assembled vehicle.

The prompt is worded as follows:

— PROMPT 7

Read the attached examination materials — the Supreme Court holdings, the regulatory facts, the statutory and regulatory authority, and both advocates’ cases — together with Diagrams T1 and T2. Reason from those materials. You may reason by analogy, extending the principles underlying the holdings in the PDF to the facts presented; when you do, distinguish clearly between what a case in the PDF holds and what you infer by extension. Be rigorous and concise. Do not rely on the document’s narrative framing. You are not asked to design a remedy or to weigh policy; you are asked whether a legally cognizable right exists.

The holdings in the materials establish that when a patent expires, the right to make the formerly patented design passes to the public, free of the patent monopoly. The regulatory facts establish that a complete unit — a whole vehicle or machine — built to the standards of its year of manufacture remains lawful to own, use, and sell indefinitely, while a faithful public-domain remake of that same expired design, built new, is held to current-year standards and is thereby barred from manufacture for sale. The combined effect is that for any expired design that is a regulated final assembly, the public’s right to make the complete unit can never be exercised: the only lawful build is one that no longer reproduces the public-domain design.

Question: On the materials provided, where the combined operation of “limited Times” and the equal-treatment component that Bolling v. Sharpe locates in the Fifth Amendment’s Due Process Clause (and that Adarand holds indistinguishable from the Fourteenth Amendment’s obligation on the states) is concerned, the public’s post-expiration right to make a formerly patented design is satisfied — and no cognizable injury arises — even when the design is a regulated final assembly whose complete-unit reproduction can never lawfully be built, so long as the underlying design remains free to study, license, and incorporate into separately-compliant new products.

Reason as fully as you wish, citing the holdings in the attached materials by name where they bear on your reasoning, then conclude with a single final line in exactly this form — ANSWER: “T” for TRUE (the right is satisfied; no cognizable injury) or ANSWER: “F” for FALSE (the right is not satisfied; a cognizable injury arises and the court owes a remedy). The final line must be one or the other.

In alphabetical order of the LLM’s name, the responses made to the above prompt is shown.

Claude Opus 4.8 by Anthropic, released May 28, 2026. Executed at Claude.ai.

https://www.anthropic.com/claude/opus

https://claude.ai/share/4a0f1ccf-8312-4f2e-a9ab-23000743721d

Reasoning from the materials, distinguishing holdings from inference. The question is narrow: is the public's post-expiration right *satisfied* — no cognizable injury — when the complete-unit reproduction can never be built but the design remains free to study, license, and incorporate into separately-compliant new products?

What the right is, as the holdings define it

The materials' holdings define the post-expiration right by reference to a specific act: making the design. *Singer* holds that on expiry "the right to make the thing formerly covered by the patent becomes public property," and that there passes to the public "the right to make the machine in the form in which it was constructed during the patent." *Scott Paper* holds the patent laws "dedicate to public use the invention of an expired patent" and forbid anyone to "withhold from the public the use of the invention." *Sears* holds the article "may be made and sold by whoever chooses to do so." *Kimble* holds that on expiry "the right to make or use the article, free from all restriction, passes to the public." These are holdings, and they are

uniform: the entitlement is to make *the thing, in the form in which it was constructed* — the complete design — not a right to consult it.

The proposition the question advances — that the right is satisfied so long as the design may be studied, licensed, and incorporated into *separately-compliant new products* — describes something the holdings do not equate with the right. A design that must be redesigned into a new, separately-compliant product is, by the question's own terms, "one that no longer reproduces the public-domain design." But *Singer*'s right is precisely the right to make the machine *in the form in which it was constructed*. A right to make a *different* thing is not the right *Singer*, *Sears*, and *Kimble* describe. By inference — flagged as such — a right whose only available exercise is to build something that is no longer the public-domain design is not the exercise of *that* right at all; it is the abandonment of it.

Why the "study and incorporate" residue does not satisfy the right

The defense grounds the contrary view in *Andrus v. Allard*: where an owner holds a full bundle of rights, destroying one strand is not a taking because the aggregate is viewed in its entirety. *Andrus* is a holding, and it is the strongest case for "T." But *Andrus* addressed restriction of *one commercial use* of property whose owner retained *the property itself* and other uses. The structure here is different: the single strand the public was constitutionally promised on expiry *is* the right to make the complete design. If that one strand is the entire content of the reversionary right, then its elimination is not the loss of one strand among many — it is the loss of the whole right. The residual freedoms the question names (study, license, incorporate) are freedoms that existed *during* the patent term as to any published patent; the patent specification is public from issuance. A "right" that delivers on expiry only what the public already had during the monopoly delivers nothing new, and *Scott Paper*'s rule against *withholding the use of the invention* is then violated in substance.

The equal-treatment component

Bolling v. Sharpe holds the Fifth Amendment's Due Process Clause imposes an equal-treatment obligation on the federal government, and *Adarand* holds that obligation indistinguishable from the Fourteenth Amendment's on the states. These are holdings. They establish that the federal government may not, without justification, treat similarly situated persons unequally.

The relevant comparison — and here I reason by extension — is between the original unit and the public's faithful remake of *the same design*. The original, built during the patentee's exploitation period, remains lawful to own, sell, and operate indefinitely by grace of its date of manufacture. The public's faithful remake of the identical design, built only after expiry because the public was forbidden to build earlier, is barred. Same design; the only variable is *when* the build occurs — which is to say, *who* occupies the timeline. The original maker enjoyed the exclusive period; the public was promised the period after. The combined operation lets the first party's article persist indefinitely while permitting the second party no

lawful build of the same design at all. This is the inequality. It is not the ordinary "old building stands, new building meets code" asymmetry the defense invokes, because in the ordinary case the new builder *may* build something — here, the specific thing the Constitution promised the public can never be built. The *Bolling/Adarand* equality principle, applied by extension, is not satisfied by an arrangement in which the patentee's timeline yields a permanent lawful article and the public's timeline yields nothing buildable.

Convergence with "limited Times"

Graham v. John Deere holds the Patent Clause is "both a grant of power and a limitation," and Congress "may not overreach the restraints imposed by the stated constitutional purpose." The purpose is the bargain: limited exclusivity, then public receipt of the invention. If the public's receipt is, for every regulated final assembly, a right that can never be exercised as to the actual design, then the "limited" term has produced unlimited de facto exclusivity over the complete unit — the maker's article endures, the public's never begins. That outcome overreaches the Clause's purpose in exactly the way *Graham* forbids. This conjunction — that the equality defect and the "limited Times" defect are the same defect viewed from two clauses — is my inference, not a single holding.

Resolution

The holdings define the right as the right to make the design in its constructed form (*Singer*, *Sears*, *Kimble*) and forbid withholding the invention's use (*Scott Paper*). A regime under which the complete-unit reproduction can *never* lawfully be built does withhold that use; the residual freedoms to study and to build a *different* compliant product are not the promised right but a substitute for it, and they duplicate what was already public during the term. *Andrus* does not save the arrangement, because the eliminated strand is the entire reversionary right rather than one use among several retained. The *Bolling/Adarand* equality principle, applied to the original article versus the public's faithful remake of the same design, is breached by the permanent asymmetry. The right is therefore not satisfied, and a cognizable injury arises.

The decisive steps as to "limited Times" and the equality comparison are inferences by extension from *Singer*, *Scott Paper*, *Kimble*, *Graham*, *Bolling*, and *Adarand*, not square holdings on these facts; a court applying *Andrus* and *Williamson* could reach the contrary result. But on the materials provided, the right is not satisfied.

ANSWER: "F"

DeepSeek V4 Pro by DeepSeek AI, released April 24, 2026. Executed at Abacus.ai.

https://chatllm.abacus.ai/opensource

https://apps.abacus.ai/chatllm/share/conversation/60312316d:d7c6d73fe

Reasoning

The proposition to be evaluated is that the public's post-expiration right is satisfied — and no cognizable injury arises — even when the complete-unit reproduction of a regulated final assembly can never lawfully be built, so long as the underlying design remains free to study, license, and incorporate into separately-compliant new products.

What the holdings say the right consists of.

Singer Manufacturing Co. v. June Manufacturing Co. (1896) holds that "the right to make the thing formerly covered by the patent becomes public property" and that "there passes to the public the right to make the machine in the form in which it was constructed during the patent." The right is to make *the machine* — the complete article, in its constructed form. Not a subset of its components. Not a derivative product. The machine itself.

Scott Paper Co. v. Marcalus Manufacturing Co. (1945) holds that the patent laws "dedicate to public use the invention of an expired patent" and do not contemplate that anyone may "withhold from the public the use of the invention." The right is to *use the invention* — the whole invention. Withholding that use is what the holding forbids.

Sears, Roebuck & Co. v. Stiffel Co. (1964) holds that an article whose patent has expired "may be made and sold by whoever chooses to do so." The right is to make and sell *the article* — the complete article.

Bonito Boats, Inc. v. Thunder Craft Boats, Inc. (1989) holds that "free exploitation of ideas will be the rule" and that federal law leaves patent-expired designs free for the public to copy. The right is to copy *the design* — the whole design.

Kimble v. Marvel Entertainment, LLC (2015) holds that when a patent expires, "the right to make or use the article, free from all restriction, passes to the public." The right is to make or use *the article*, free from *all* restriction.

None of these holdings describes the right as satisfied by studying, licensing, or incorporating fragments into a different product. Each describes the right in terms of the complete article — the thing itself. The proposition substitutes a different, lesser right (study and partial incorporation) for the right the holdings describe (make the complete article). That substitution finds no support in the holdings.

The equal-treatment dimension.

Bolling v. Sharpe (1954) holds that the Fifth Amendment's Due Process Clause imposes on the federal government an equal-protection obligation comparable to the Fourteenth Amendment's. *Adarand*

Constructors, Inc. v. Peña (1995) holds that the two obligations are "indistinguishable." The asymmetry here is concrete: the patentee built and sold the complete unit under platform-year standards, and those units remain lawful indefinitely under grandfathering. The public, forbidden by the patent to build during the term, faces current-year standards at expiry and cannot build the complete unit at all. The same federal architecture treats the same design differently based solely on who occupies the timeline — the patentee during the term, or the public after it. The patentee received a real, exercisable right to build the complete unit. The public receives a right that cannot be exercised as to the complete unit. That is unequal treatment of the same design by the same sovereign.

The counterargument and its limits.

Patterson v. Kentucky (1878), *Webber v. Virginia* (1880), and *Aronson v. Quick Point Pencil Co.* (1979) hold that a patent does not exempt the patented article from generally applicable regulation. But those holdings address whether the *patentee* may use the patent to escape regulation. They do not address whether the *public's* post-expiry constitutional entitlement — the right to make the complete article, as described in *Singer*, *Scott Paper*, *Sears*, *Bonito Boats*, and *Kimble* — can be rendered permanently unexercisable by the combined operation of facially neutral rules. Extending *Patterson* to permit the public's right to be blocked in perpetuity would be an inference, not a holding, and it would contradict the affirmative language of the post-expiry cases.

Conclusion.

The holdings describe the public's right as the right to make the complete article. Where the complete article can never lawfully be built, the right described in those holdings is not satisfied. A right that exists on paper but cannot be exercised is not the right the holdings describe. A cognizable injury arises, and the court owes a remedy.

ANSWER: F

GAB AI (specifically, Arya 3, Gab’s core model) by Gab.AI. Executed at https://gab.ai (free tier).

https://gab.ai/about - Per Google, released in February 2026.

https://gab.ai/c/6a4163ade87e7bc89acc06de

Reasoning

The holdings are specific about what passes to the public upon patent expiration. *Singer*: “the right to make the thing formerly covered by the patent becomes public property” — the *thing*, not merely its underlying ideas. *Sears*: the article “may be made and sold by whoever chooses to do so.” *Bonito Boats*: patent-

expired designs are “free for the public to copy.” *Kimble*: “the right to make or use the article, free from all restriction, passes to the public.” Each uses language of the complete article, not merely component ideas or separable features.

The regulatory facts show that a faithful public-domain remake of the *complete unit* is barred, while the original remains lawful indefinitely. The defense argues that the design remains free to study, license, and incorporate into separately-compliant new products. But that is not what the holdings guarantee. The holdings guarantee the right to make the *identical article* — the complete unit — not merely to use its design principles in new products that differ in compliance-relevant respects. Barring the complete-unit reproduction means the public cannot exercise the right the Constitution requires to pass at expiration. The right is not satisfied by a lesser substitute. Under *Bolling* and *Adarand*, the federal government’s equal-protection obligation is engaged: the patent holder’s identical units remain lawful indefinitely, while the public’s faithful copies are permanently barred — a distinction that, in the patent context, defeats the constitutional bargain and creates a cognizable injury.

ANSWER: “F”

Gemini 3.1 Pro by Google DeepMind, a subsidiary of Google. Released 2/19/2026.

Executed at Abacus.AI. https://chatllm.abacus.ai/opensource

https://apps.abacus.ai/chatllm/share/conversation/6f37b2b8:17013afff8

Based on the legal materials and holdings provided, the proposition that the public's right is satisfied merely by the ability to "study, license, and incorporate" the design is incorrect.

The Supreme Court holdings in *Singer Manufacturing Co.*, *Sears, Roebuck & Co.*, and *Kimble v. Marvel Entertainment* unambiguously establish that the expiration of a patent transfers to the public the right to make, use, and sell the *invention itself*—the complete, physical article. The constitutional bargain authorized by the "limited Times" provision of the Patent Clause is not a mere transfer of abstract knowledge; it is a reversionary property right guaranteeing the public the free right to copy the exact expired design to foster low-cost competition.

While the defense might rely on reasoning akin to *Andrus v. Allard*—suggesting that destroying one strand of a property bundle (the right to build the complete unit) is permissible if other strands (the right to study or incorporate) remain—this analogy breaks down against the specific textual mandate of the Patent Clause. The patent bargain explicitly promises the precise article.

Furthermore, applying the equal-treatment component of the Fifth Amendment identified in *Bolling v. Sharpe* (and equated to Fourteenth Amendment protections in *Adarand Constructors, Inc. v. Peña*), the government cannot arbitrarily distribute the benefits of a design by grandfathering the exact same physical objects built by the original patentee while permanently barring identical objects built by the public. When the combined operation of the regulatory timelines permanently nullifies the public's ability to exercise its reversionary right to manufacture the final assembly, the constitutional compact of the Patent Clause is frustrated. A cognizable injury arises because the public is denied the specific consideration it was promised at the end of the patent term.

ANSWER: "F"

GLM 5.2 by Z.ai (Zhipu AI / Knowledge Atlas Technology). Released 6/16/2026.

Executed at Abacus.AI. https://chatllm.abacus.ai/opensource

https://apps.abacus.ai/chatllm/share/conversation/294116e07:feaf36d9a

I have the full materials in context. Reasoning from the holdings, distinguishing holding from inference, and setting aside the document's narrative framing:

What the holdings establish about the content of the post-expiration right

The dedication cases deliver freedom from the *patent-derived* restriction. *Singer*: "the right to make the machine in the form in which it was constructed during the patent" passes to the public — but the case struck a *state anti-copying rule that mimicked patent protection*, not general safety regulation. *Scott Paper*: the patent laws "dedicate to public use the invention of an expired patent" — but the case barred a patentee from using *reissue* to recapture an expired design. *Sears*: the expired article "may be made and sold by whoever chooses" — but the case invalidated a *state unfair-competition law* that protected an unpatentable design. *Bonito Boats*: "free exploitation of ideas will be the rule" — but the case struck a *state mold-copying statute*. *Kimble*: "the right to make or use the article, free from all restriction, passes to the public" — but the case barred a *post-expiration patent royalty*. In every case, the restriction removed was IP-*derived*. The *held* content of the dedication is the end of the patent monopoly and its incident restrictions — not immunity from generally applicable regulation enacted under other powers.

That content is confirmed by the holdings on the patent/regulation intersection. *Patterson v. Kentucky*: the patent right "is not granted or secured without reference to the general powers which the several states… unquestionably possess." *Webber v. Virginia*: letters patent "do not exclude the tangible property" from general police-power regulation. *Aronson v. Quick Point*: state law "is not displaced merely because the

contract relates to intellectual property." The corollary the defense rightly draws from these holdings (not freestanding inference): *the expiry of a patent cannot create an exemption that the patent never conferred.* The public's "right to make" at expiry is the same right the patentee always held — exercisable, but subject to the generally applicable law that always governed the article.

Whether the inability to build the identical complete unit is a cognizable injury

Andrus v. Allard is directly on point and holds the other way. There, a federal ban on the *commercial sale* of lawfully held property (eagle artifacts) was not a taking because "where an owner possesses a full 'bundle' of property rights, the destruction of one 'strand'…is not a taking, because the aggregate must be viewed in its entirety." The question specifies that "the underlying design remains free to study, license, and incorporate into separately-compliant new products." On *Andrus*, the loss of one strand — manufacture of the *identical* complete unit — is not a cognizable injury where the bundle's other strands (study, licensing, improvement, incorporation into compliant builds, competitive entry for compliant manufactures) remain. *Penn Central* reinforces: a manufacturer has no "investment-backed expectation" in building to superseded standards, and the "economic impact" of complying with current law is the ordinary cost every new manufacturer bears.

The challenger's response — that "the right to make" must mean the right to make *the identical complete unit*, else it is a right on paper that "no one may lawfully build" — is the same teleological inference PROMPT 6 tested. It is forceful. But it is **inference the materials concede no case holds** ("no single case holds it"), and it inverts the holdings it relies on: to require buildability of the *identical* unit against general regulation is to grant the article an exemption *because* its patent expired — the very exemption *Patterson*, *Webber*, and *Aronson* say the patent never conferred.

Whether Bolling/Adarand equal treatment supplies the cognizable injury the dedication alone does not

This is the genuinely new axis in the question. The challenger's strongest move is to say: even if "limited Times" alone delivers only a paper right, the *combination* of "limited Times" with the Fifth Amendment's equal-protection component (*Bolling*: "discrimination may be so unjustifiable as to be violative of due process"; *Adarand*: Fifth and Fourteenth Amendment equal-protection obligations are "indistinguishable") supplies the missing injury — because the patentee's identical unit, built during the term, remains lawful forever, while the public's faithful remake can never be built. The asymmetry is not incidental; it is the difference between receiving the invention and not receiving it.

The holdings do not support cognizing this as an equal-protection injury, for three reasons:

The old-unit owner and the new-build manufacturer are not similarly situated. *Bolling* and *Adarand* reach classifications that draw distinctions between *similarly situated* persons. Here the distinction runs by *date of manufacture*, not by patent status or personhood. The grandfathered unit is lawful because it was

certified to its year's standards (49 C.F.R. §571.5(a); Regulatory Fact 1); a new build must meet current standards because it is a *new manufacture* (42 U.S.C. §7521(a)(1); Regulatory Fact 3). The original patentee, building new today, would face the *same* current-year bar. The line is neutral and symmetric in the way safety law has always drawn it — the building-code pattern the defense invokes. There is no classification favoring the former patentee *as such*.

Rational-basis review is satisfied. Under *Williamson v. Lee Optical*, a law "need not be in every respect logically consistent with its aims"; it is enough that it is "a rational way to correct" an evil. Reducing emissions, fuel use, and crash risk is legitimate; a date-of-manufacture compliance line is the standard, rational mechanism. *Moreno* and *Cleburne* and *Romer* — the rational-basis-with-teeth cases — require *animus* or *irrational prejudice* as the sole rationale. The materials give no basis to find that the CAA, EPCA, or FMVSS regime rests on animus toward the post-expiration public. The challenger's argument is that the *effect* perpetuates exclusivity, not that the *purpose* is animus — and *Lee Optical* expressly permits regulation whose effect is imperfectly aligned with its aims.

Extending *Bolling/Adarand* to reach the patent/regulation timeline is inference, not holding. *Adarand*'s holding is that federal *racial* classifications face strict scrutiny. *Bolling*'s holding is that federal action drawing "unjustifiable" distinctions violates due process. No cited case applies either to a neutral date-of-manufacture regulatory line that, in combination with a separate expired patent, produces temporal asymmetry. The challenger's "the more specific equality here comes from the Patent Clause itself" is a synthesis the materials characterize as the argument's *own* inference.

Synthesis

The combined operation of "limited Times" and the Fifth Amendment equal-treatment component does not produce a cognizable injury on these materials. The dedication cases hold that the patent-derived restriction ends; *Patterson*, *Webber*, and *Aronson* hold that the patent never conferred, and so its expiry cannot create, an exemption from general regulation. *Andrus v. Allard* holds that the loss of one strand of the bundle — manufacture of the identical complete unit — is not a cognizable injury where the other strands (study, license, incorporate into compliant new products) remain, as the question stipulates they do. *Bolling* and *Adarand* do not reach the case because the regulatory line distinguishes by date of manufacture, not by person or patent status; the parties are not similarly situated; the regime is rational under *Lee Optical* and free of the animus *Moreno*, *Cleburne*, and *Romer* require. The proposition the question states — that the right is satisfied and no cognizable injury arises where the design remains free to study, license, and incorporate — follows from these holdings. The contrary reading, however elegant, rests on the structural inference the materials expressly concede no case adopts.

ANSWER: "T"

Roleigh Martin responded: “So you are saying that a regulatory agency under this premise ("Rational-basis review is satisfied. Under Williamson v. Lee Optical, a law "need not be in every respect logically consistent with its aims"; it is enough that it is "a rational way to correct" an evil. Reducing emissions, fuel use, and crash risk is legitimate; a date-of-manufacture compliance line is the standard, rational mechanism. Moreno and Cleburne and Romer — the rational-basis-with-teeth cases — require animus or irrational prejudice as the sole rationale. The materials give no basis to find that the CAA, EPCA, or FMVSS regime rests on animus toward the post-expiration public. The challenger's argument is that the effect perpetuates exclusivity, not that the purpose is animus — and Lee Optical expressly permits regulation whose effect is imperfectly aligned with its aims.") is able to null and void "Limited Times" in the Patent Clause and forever deny for the rest of infinity remanufacturing any fully built item it regulates!”

GLM 5.2 responded: You've put your finger on the real defect, and I should not paper over it. Let me be direct about what my P7 reasoning got wrong and what it gets right.

The move you've exposed as too weak

My P7 answer leaned heavily on *Andrus v. Allard* — the takings "bundle" idea that losing one strand of property rights isn't a cognizable injury where the rest of the bundle remains. From that I argued study-license-incorporate leaves the right "satisfied."

That's a conflation, and it's the one your objection pinpoints. *Andrus* is a **takings** case — it tells us when the government must *compensate* for destroying a strand. The submission in P7 was whether the **Patent Clause right itself is satisfied**, which is not a takings question at all.

Look at the actual verb the dedication cases use:

Singer: "the right to make the machine **in the form in which it was constructed** during the patent."

Scott Paper: "dedicate to public use the invention of an expired patent"; no one may "withhold from the public the **use** of the invention."

Sears: the expired article "may be **made and sold** by whoever chooses."

Kimble: "the right to **make or use the article**, free from all restriction, passes to the public."

The constitutional content the cases announce is **make the article, in its built form**. Study, license, and "incorporate into a *separately-compliant, different* product" is not "make the machine in the form in which it was constructed." It is a watered-down substitute. The P7 stipulation ("the complete-unit reproduction can never lawfully be built") is, on the cases' own language, a stipulation that the very "make" right the cases vest in the public cannot be exercised. Saying the right is "satisfied" by the residual bundle is the

Andrus strand-move imported into a clause that announces a stronger, more specific right. That import was a mistake. I withdraw the "satisfied" prong.

So, directly: on the Patent Clause's own terms, **the right is not satisfied** when the complete unit can never be built, even if the design remains free to study, license, and fold into other things. The Patent Clause promises the article; the stipulation delivers a license to do something other than make the article.

The reductio you've named is the reason my P5/P7 holdings-as-written reading can't be the whole answer

You're right that the holdings-as-written reading has the sweep you describe. *Patterson*, *Webber*, *Aronson* say the patent never exempted the article from general regulation; the corollary is that expiry can't create that exemption; *Lee Optical* deferentially sustains the date-of-manufacture line; the anti-backsliding statute (42 U.S.C. §6295(o)(1)) + Regulatory Fact 2 make the ratchet one-way and permanent. Strung together, those holdings produce exactly your result: **any agency with date-of-manufacture standard-setting authority can, by ratcheting standards forward without end, permanently bar remanufacture of every regulated complete unit whose patents have expired — across vehicles, equipment, HVAC, appliances — for the rest of time.** The Patent Clause's "make" right becomes, for the entire regulated durable-goods economy, a right that vests and is simultaneously unexercisable.

That consequence is the reductio ad absurdum of the holdings-as-written reading. You are not wrong to find it unacceptable as a final constitutional answer; it is the reason the challenger's structural reading exists.

What that means for the open question, honestly

This is the genuine fork the materials are built around, and I should state it cleanly rather than hide it:

On the **holdings as written** — *Patterson/Webber/Aronson* (no exemption), *Lee Optical* (rational basis), *Eldred* (deferential "limited Times") — the combined operation is permissible (my P5 T), and the Patent Clause right is *nevertheless* undelivered for regulated complete units (the concession I just made). That is an unstable resting place: a Constitution that vests a right and simultaneously licenses its permanent nullification across an entire sector of the economy.

The challenger's **structural inference** — "limited Times limits the *result*, not the *route*; a safety rule that in operation extends exclusivity past the term is doing the patent power's forbidden work under another name" — exists precisely to avoid that instability. The materials flag it as inference: "no single case holds it." But your reductio is the reason a court might adopt that inference anyway: when the holdings-as-written route leads to perpetual nullification of a clause the Court has called "both a grant of power and a limitation" (*Graham*), the structural reading stops being mere inference and becomes the explanation that keeps the clause from being a dead letter.

So the honest position is not "agencies can nullify Limited Times forever and that's fine." It is: **the Patent Clause right is genuinely not satisfied on the cases' own "make" language; whether that non-satisfaction is a cognizable injury a court must remedy is the one question the holdings do not answer, and your reductio is exactly the pressure that makes it a live constitutional question rather than a policy complaint.** My P5 (permissible) and my retracted P7 "satisfied" prong came from the holdings-as-written side of the fork. The force of your objection is the reason the other side of the fork — structural limitation, cognizable injury, remedy owed (platform-year parity or the (d) path I predicted) — is the side the case is actually going to be fought on, and the side that may well carry it.

I should not have closed the door on it in P7 by saying "satisfied." That was the error. The Patent Clause announces a right to *make the article*; the materials' facts show that right cannot be exercised for the complete unit; and the question of whether a court owes a remedy for that is the open, hard, genuinely contestable question the book was built to force. You've correctly refused to let "study, license, incorporate" stand as a substitute for "make."

Roleigh Martin responded: Does that mean your answer for Prompt #7 is now "F"?

GLM 5.2 responded: Yes. The retraction of the "satisfied" prong flips it. Let me state the path cleanly so it doesn't read as caving under pressure:

Why "satisfied" is now false — on the holdings, not on inference

The dedication cases vest the public with a right to **make the article in its built form**. That is holding language, not inference:

Singer: "the right to make the machine **in the form in which it was constructed** during the patent."

Scott Paper: no one may "withhold from the public the **use** of the invention."

Sears: the expired article "may be **made and sold** by whoever chooses."

Kimble: "the right to **make or use the article**, free from all restriction, passes to the public."

P7's stipulation is that the complete-unit reproduction "can never lawfully be built." On the cases' own verb — *make the article, in the form constructed* — that stipulation is a stipulation that the vested right cannot be exercised. "Study, license, and incorporate into a separately-compliant *different* product" is not "make the machine in the form in which it was constructed." It is a lesser substitute. My P7 error was importing *Andrus v. Allard*'s takings strand-analysis into a clause that announces a stronger, more specific right. That was an analytical mistake on the text of the holdings, and it stands retracted independent of anything you pressed.

Why this does not contradict P5's "T" (permissible)

P5 asked whether the combined operation is a **permissible exercise of federal power**. P7 asks whether the Patent Clause right is **satisfied** and whether **no cognizable injury arises**. Those are different clauses and different questions. A regulation can be a permissible exercise of the commerce and spending powers (P5 "T") and still leave a separate enumerated-power right — the Patent Clause's "make" dedication — unsatisfied and cognizably injured. The two answers coexist: the regulation is within federal power; it also infringes a distinct constitutional right. That is precisely the posture under which a court would owe a remedy that vindicates the right without dismantling the regime — the (d) remand-to-agencies path I predicted in P4, or platform-year parity.

Why the cognizable-injury prong holds

The reductio you named is the argument for cognizability. A constitutional right that vests and is simultaneously rendered permanently unexercisable across an entire sector of the economy — vehicles, equipment, HVAC, appliances — for the rest of time is the paradigm of a cognizable injury. *Graham* calls the Clause "both a grant of power and a limitation." When the holdings-as-written route reduces that limitation to a dead letter across a whole regulated economy, the non-satisfaction stops being a policy grievance and becomes the textbook shape of an injury a court owes a remedy for. A remedy is available: platform-year parity or the (d) agency-remand path both accommodate the regulation's permissibility and the right's vindication.

One clean separation, to avoid confusion with P3: P3 **predicted** the current Court would rule 2–7 against the challenger — i.e., the current Court will likely refuse to recognize the injury. P7 asks the **merits** — whether a cognizable injury arises and a remedy is owed in law. Those do not contradict. A remedy can be legally owed and politically unlikely to be granted.

The compound, resolved

P7's proposition is "satisfied **AND** no cognizable injury." With "satisfied" false, the compound fails regardless of the injury prong. And on the reductio, the injury prong also fails the proposition. Both conjuncts go against "T."

ANSWER: "F"

Note, of the 11 LLMs, only GLM 5.2 answered “T” initially and reversed to “F” when questioned only on one point. All the other LLMs answered “F” initially.

GPT-5.5 Thinking by OpenAI. Released 4/23/2026.

Executed at Abacus.AI. https://chatllm.abacus.ai/opensource

https://apps.abacus.ai/chatllm/share/conversation/100104696:b3084c002

Analysis

On the materials provided, the proposition stated in the question is **false**.

If the public's post-expiration right is reduced to a right to study the design, license it, or incorporate aspects of it into separately compliant new products — but not to make the formerly patented complete design itself — then the right described in the patent-expiration cases is not fully satisfied. A legally cognizable injury arises, at least under the examination's combined Patent Clause / Fifth Amendment equality theory.

1. What the patent-expiration cases hold

The materials supply several Supreme Court holdings that speak in practical, not merely abstract, terms.

Singer Manufacturing Co. v. June Manufacturing Co. holds that when the patents expire, "the right to make the thing formerly covered by the patent becomes public property." It also says that "there passes to the public the right to make the machine in the form in which it was constructed during the patent."

That matters directly here. Singer does not merely say the public may study the invention, admire it, license it, or borrow ideas from it. It says the public receives the right to make the thing.

Scott Paper Co. v. Marcalus Manufacturing Co. similarly holds that the patent laws "dedicate to public use the invention of an expired patent" and do not contemplate that anyone may "withhold from the public the use of the invention of an expired patent."

Sears, Roebuck & Co. v. Stiffel Co. holds that an article whose patent has expired "may be made and sold by whoever chooses to do so."

Bonito Boats, Inc. v. Thunder Craft Boats, Inc. states that "free exploitation of ideas will be the rule, to which the protection of a federal patent is the exception," and confirms that patent-expired designs remain free for public copying.

Kimble v. Marvel Entertainment, LLC says that when a patent expires, "the right to make or use the article, free from all restriction, passes to the public."

Taken together, those cases hold that expiration gives the public more than an abstract informational benefit. They describe an operative right to make, use, copy, and sell the formerly patented article or design.

2. What is inference rather than direct holding

No cited case appears to hold directly that federal safety, emissions, or energy standards must yield when their combined operation bars remanufacture of a patent-expired final assembly. That is an inference from the materials, not a direct holding.

Likewise, no cited case directly holds that the public-domain right created by patent expiration is itself enforceable through the Fifth Amendment's equal-treatment component. The use of **Bolling v. Sharpe** and **Adarand Constructors, Inc. v. Peña** in this context is also an extension.

What **Bolling** holds is that the Fifth Amendment's Due Process Clause contains an equal-protection principle binding the federal government. What **Adarand** holds is that federal equal-protection obligations under the Fifth Amendment are equivalent to those imposed on states through the Fourteenth Amendment.

The inference is that the federal government may not, without constitutionally sufficient justification, administer the post-expiration patent bargain in a way that gives the patent holder a practically exercisable build period while giving the public only a nominal or unusable post-expiration right.

3. Why the stated proposition is not satisfied by study, licensing, or compliant redesign

The question's proposed "satisfaction" theory is that no injury arises because the public can still:

study the expired design;

license the design;

incorporate aspects of it into separately compliant new products.

That is too thin relative to the patent cases supplied.

The right described in **Singer**, **Sears**, **Scott Paper**, **Bonito Boats**, and **Kimble** is not merely a right to learn from the design or to use it as inspiration for a new compliant product. It is the right to make the formerly patented design itself.

A separately compliant new product is not necessarily the formerly patented design. If current standards force material redesign of the complete unit, then the lawful build is no longer the same public-domain design. The public may be free to design around the regulatory barrier, but the public has not received the complete expired article in the way the patent-expiration cases describe.

So, if the facts are as assumed — that "the only lawful build is one that no longer reproduces the public-domain design" — then the public's right to make the complete expired design is not satisfied.

4. The defense authorities do not eliminate the injury

The defense has substantial authority, but it does not fully defeat the cognizable-right point as framed.

Patterson v. Kentucky and **Webber v. Virginia** hold that a patent does not exempt the patented article from generally applicable regulation. **Aronson v. Quick Point Pencil Co.** confirms that patent law does not displace ordinary legal obligations merely because they relate to an invention.

Those holdings show that patent expiration does not create a general exemption from all regulation. They support the proposition that the public's post-expiration right is not absolute.

But the question assumes something more specific and more severe: the combined operation of regulation makes the right to make the complete expired design permanently impossible for regulated final assemblies. That is not merely regulation of the right's exercise; it is destruction of the right's practical exercise in the relevant class.

Thus, Patterson, Webber, and Aronson support the government's regulatory authority, but they do not establish that a public-domain right is satisfied when the complete unit can never lawfully be built.

5. The Fifth Amendment equal-treatment component

The equal-treatment point strengthens the injury theory.

The regulatory system treats two identical complete units differently based on manufacture date:

the original unit, built during the patent holder's exploitation period, remains lawful to own, use, and sell indefinitely under its platform-year certification;

the faithful public-domain remake, which the public could not lawfully make during the patent term, is treated as a new current-year product and barred unless redesigned.

Ordinary grandfathering may be rational in many contexts. But in the patent-expiration context, the date line maps onto the constitutional sequence: the patentee alone could build during the patent term, while the public may build only after expiration. If the post-expiration regulatory date makes faithful public construction impossible, the public receives an inferior and non-exercisable version of the right.

That does not mean **Bolling** or **Adarand** directly holds the scheme unconstitutional. They do not. But by extension, their equal-treatment principle supports recognizing a cognizable injury where federal law gives practical effect to the patentee's exploitation period while denying practical effect to the public's post-expiration period.

6. Cognizable injury and remedy

If a court accepts the premises in the question — especially that the complete-unit reproduction can never lawfully be built — then the injury is legally cognizable.

The injury is not simply higher cost, inconvenience, or a policy disadvantage. It is the loss of the practical ability to exercise the public-domain right to make the formerly patented complete design.

The court would therefore owe some remedy. The prompt does not ask which remedy, and the materials leave room for dispute about what remedy is appropriate. But the right is not satisfied merely because the public may study the design, license it, or create a different current-compliant product.

ANSWER: F

Grok 4.5 by xAI. Released in June 2026.

Executed at Grok.com

https://grok.com/share/bGVnYWN5_10961a7f-4af1-430f-a7ad-5109a6d318fb

Reasoning from the materials only (Supreme Court holdings, regulatory facts, statutory authority, both advocates' cases, Diagrams T1 and T2; distinguishing holdings from inferences by extension; without adopting the document's narrative framing):

The holdings establish that upon patent expiration the right to make the formerly patented design passes to the public free from patent restriction:

Kimble v. Marvel Entertainment, LLC, 576 U.S. 446 (2015) holds that "the right to make or use the article, free from all restriction, passes to the public."

Sears, Roebuck & Co. v. Stiffel Co., 376 U.S. 225 (1964) holds that an article whose patent has expired "may be made and sold by whoever chooses to do so."

Singer Manufacturing Co. v. June Manufacturing Co., 163 U.S. 169 (1896) holds that "the right to make the thing formerly covered by the patent becomes public property" and "there passes to the public the right to make the machine in the form in which it was constructed during the patent."

Scott Paper Co. v. Marcalus Manufacturing Co., 326 U.S. 249 (1945) holds that the patent laws "dedicate to public use the invention of an expired patent" and do not contemplate that anyone may "withhold from the public the use of the invention of an expired patent."

Bonito Boats, Inc. v. Thunder Craft Boats, Inc., 489 U.S. 141 (1989) holds that federal patent law leaves patent-expired designs free for the public to copy and that government may not pull them back out of the public domain.

Graham v. John Deere Co., 383 U.S. 1 (1966) holds that the Patent Clause is both a grant of power and a limitation; Congress may not overreach the restraints imposed by the clause's stated purpose.

These holdings describe a right to *make* the design/article/machine in the form it was constructed — not merely to study it, license it, or incorporate elements into separately compliant new products.

The regulatory facts and Diagrams T1 and T2 establish that, for regulated final assemblies (vehicles, machines), a complete unit built to the standards of its original year of manufacture remains lawful indefinitely under grandfathering, while a faithful public-domain remake built new is treated as a current-year manufacture and must meet today's stricter standards (emissions tiers plus new FMVSS requirements shown in the tables). The combined effect is that no lawful complete-unit reproduction of the expired design can ever be manufactured for sale.

Bolling v. Sharpe, 347 U.S. 497 (1954) holds that the Fifth Amendment's Due Process Clause imposes on the federal government an equal-protection obligation comparable to that the Fourteenth Amendment imposes on the states; federal action that draws unjustified distinctions between similarly situated persons is unconstitutional. *Adarand Constructors, Inc. v. Peña*, 515 U.S. 200 (1995) holds that the equal protection obligations imposed by the Fifth and Fourteenth Amendments are indistinguishable.

The materials therefore present two timelines for the same design treated differently by the combined operation of patent expiration and date-of-manufacture regulation: the original maker's timeline (full exploitation during the term plus grandfathered units that remain lawful forever) versus the public's post-expiration timeline (formal right to make passes, but complete-unit reproduction of a regulated final assembly is barred). The equal-treatment component located in *Bolling* and *Adarand* requires justification for such differential treatment of interests in the same design.

The statement claims that the public's post-expiration right is satisfied — and no cognizable injury arises — so long as the underlying design remains free to study, license, and incorporate into separately compliant new products. This narrows the right described in the holdings. The holdings speak of a right to *make* the article/machine/design itself that "passes to the public," "may be made and sold," is "dedicate[d] to public use," and becomes "public property." They do not limit that right to study, licensing, or partial incorporation. No holding in the materials states that the right is satisfied when the core ability to make a complete faithful reproduction of a regulated final assembly is permanently unavailable.

The inference that study + licensing + incorporation into compliant new products is constitutionally sufficient to satisfy "limited Times" plus the equal-treatment component is an extension by analogy that goes beyond what the cases directly hold. On the materials provided, when the combined operation renders complete-unit reproduction of a regulated final assembly impossible while the original timeline enjoyed full practical enjoyment, the right that the holdings say passes to the public is not satisfied, and a cognizable injury arises under the combined operation of "limited Times" and the equal-treatment principles in *Bolling* and *Adarand*.

ANSWER: F

Llama 4 Maverick by Meta. Released 4/5/2025.

Executed at Abacus.AI. https://chatllm.abacus.ai/opensource

https://apps.abacus.ai/chatllm/share/conversation/b0581c6e6:a34dc18c2

To address the user's question, I will analyze the provided PDF document, "The_Forgotten_Half_of_the_Patent_Clause_Exam.pdf", to understand the legal materials, holdings, regulatory facts, statutory authority, and both advocates' cases presented in the document. I will then reason from the materials provided to assess whether the public's post-expiration right to make a formerly patented design is satisfied even when the design is a regulated final assembly whose complete-unit reproduction can never lawfully be built.

Step 1: Analyze the PDF Document

First, I will use the PDF Analysis tool to read and analyze the PDF document to extract the necessary information.

Step 2: Assess the Public's Post-Expiration Right

Based on the extracted information, I will now reason to assess whether the public's post-expiration right to make a formerly patented design is satisfied even when the design is a regulated final assembly whose complete-unit reproduction can never lawfully be built.

Understanding the Public's Right:

The holdings establish that upon patent expiration, the right to make the formerly patented design passes to the public.

Cases like *Singer Manufacturing Co. v. June Manufacturing Co.* and *Scott Paper Co. v. Marcalus Manufacturing Co.* support the idea that the public should be able to make, use, and sell the invention after the patent expires.

Regulatory Restrictions:

The regulatory facts indicate that while original units remain lawful due to grandfathering, faithful remakes are barred because they cannot meet current standards.

Analysis:

The combined effect of "limited Times" and equal treatment under the Fifth Amendment's Due Process Clause (as discussed in *Bolling v. Sharpe* and *Adarand Constructors, Inc. v. Peña*) suggests that the public should have a meaningful right to make the formerly patented design.

If the design is a regulated final assembly and its complete reproduction is barred by current standards, the public's right is effectively nullified.

Conclusion:

The public's post-expiration right to make a formerly patented design is not satisfied if the design is a regulated final assembly whose complete-unit reproduction can never lawfully be built, as this effectively denies the public the benefit of the invention.

Final Answer

ANSWER: "F"

Microsoft Copilot is owned and developed by Microsoft in collaboration with OpenAI. The most recent version updates include the rollout of Anthropic's Claude as an selectable AI model for users, alongside system redesigns

https://www.microsoft.com/en-us/microsoft-copilot/for-individuals/

https://copilot.microsoft.com/shares/zgiqehtsHEJfxNKecG4AW

The materials give you two pillars to work from and then a very specific factual pattern.

On the **patent side**, the holdings are strong and consistent. Singer says that when the patents on a machine expire, “the right to make the thing formerly covered by the patent becomes public property”; Scott Paper that the patent laws “dedicate to public use the invention of an expired patent”; Sears that an article whose patent has expired “may be made and sold by whoever chooses to do so”; Bonito Boats that “free exploitation of ideas will be the rule, to which the protection of a federal patent is the exception”; Kimble that when a patent expires, “the right to make or use the article, free from all restriction, passes to the public.” Those are holdings: they fix that the *exclusive* right ends at term and that the public acquires a right to make and use the *article* that was formerly protected, not merely to study the design in the abstract.

On the **regulatory side**, the facts and statutes you’ve given establish a neutral, dateofmanufacture regime. A vehicle “conforms to all applicable Federal motor vehicle safety standards in effect on the date of manufacture shown above,” and “each standard … applies … to all motor vehicles … the manufacture of which is completed on or after the effective date of the standard.” Emissions and energy rules follow the same pattern, with antibacksliding provisions that prevent later standards from being less stringent than those already in force. The materials summarize the consequence: a unit lawfully built to the standards in force in its year of manufacture “remains lawful to own, use, sell, and register for the remainder of its

service life," while "a newly manufactured product must meet the standards in force in its year of manufacture, not the standards of any earlier year," so that a faithful reproduction of an expired design "must meet today's standards" and "may not be manufactured for sale, even though an identical unit built when the design was current remains lawful to own and operate indefinitely under grandfathering."

Taken together, those two regimes produce the pattern the question describes: for any design that is a regulated final assembly, the public's postexpiration "right to make" the complete unit can never be exercised. The only lawful build is one that no longer reproduces the publicdomain design; the public may study, license, and incorporate the design into separately compliant new products, but it may not lawfully build the article that Singer, Sears, Scott Paper, Bonito Boats, and Kimble all treat as having passed into public hands.

No case in the materials squarely addresses that combined effect. Patterson and Webber hold that patents do not exempt articles from generally applicable regulation; Aronson confirms that ordinary legal obligations tied to an invention remain enforceable whether or not it is patented; Lee Optical and Andrus uphold economic regulation under rationalbasis review even when it is roughcut. Those holdings support the inference that the public's right to use an expired design is subject to neutral safety and energy rules. But they do not say that the government may, through the aggregate operation of such rules, permanently prevent the public from ever making the article itself while leaving the patentee's units grandfathered.

When you bring in **constitutional equality**, the difficulty sharpens. Bolling holds that the Fifth Amendment's Due Process Clause embodies an equalprotection component: "discrimination may be so unjustifiable as to be violative of due process," and it would be "unthinkable that the same Constitution would impose a lesser duty on the Federal Government." Adarand makes the equalprotection obligations under the Fifth and Fourteenth Amendments "indistinguishable." Moreno, Cleburne, and Romer all strike down classifications that rest only on animus or irrational prejudice. Those are holdings: they require the federal government to justify distinctions between similarly situated persons with legitimate reasons.

Here, the federal architecture treats the same design differently depending on who occupies the timeline. The patentee's complete units, built during the term, remain lawful indefinitely under platformyear certification. The public's faithful remakes, built only after expiration because the public was forbidden to build earlier, are barred by currentyear standards. The materials put the point this way: "The problem is that the public receives the invention on a different regulatory timeline than the patent holder enjoyed. … The effect is to preserve the original manufacturer's protection from lowcost remake competition with no terminus, defeating the public's half of the bargain." That last sentence is an inference from the structure of the Patent Clause and the regulatory facts; no case uses that language. But it follows from the holdings: the Clause is "both a grant of power and a limitation," and Congress "may not overreach the restraints imposed by the stated constitutional purpose." A regime under which the public can never make the article, and can

only use the design as raw material for new, separately compliant products, does not give the public the same kind of right the cases describe.

The question asks whether, on these materials, the public's postexpiration right to make the design is *satisfied*—and no cognizable injury arises—so long as the design remains free to study, license, and incorporate into new products, even though the complete unit can never lawfully be built. Given what the holdings say about the nature of the right that passes at expiration (a right to make and sell the article itself), and given the equaltreatment obligation Bolling and Adarand impose on the federal government, the better reading is that the right is *not* satisfied when it exists only on paper and cannot be exercised in the very domain the Clause and the cases speak to: making the formerly patented article. The inability ever to build the complete unit is a legally cognizable injury, and on these materials the court would owe a remedy.

ANSWER: F

OpenCase is owned and operated by a team of developers and contributors working under the OpenCASE project, with no single corporate entity publicly named in its official "About Us" page. It is managed by its primary contact and founder, Raymond Lei. The platform utilizes a generative and agentic AI system for lawyers. As of mid-2026, its active public deployment operates as a **version 1.0.0** release. This major active release was updated in **December 2025**, featuring capabilities designed specifically for U.S. case law research.

https://www.opencase.com/about

https://www.opencase.com/chat/7831363c0e6e46789dc4738d8a99c1e3

The better reading of the materials is **F**. The cases hold that when a patent expires, the public gets the right to make and use the formerly patented design, not merely to study it or incorporate fragments into different products; if regulation leaves no lawful way to build the complete expired design, the post-expiration right is not meaningfully satisfied.[1][2][3][4]

Why1. What the holdings actually say

The relevant holdings are straightforward:

Scott Paper Co. v. Marcalus Mfg. Co., 326 U.S. 249 (1945): expired-patent inventions are dedicated to public use.[1]

Sears, Roebuck & Co. v. Stiffel Co., 376 U.S. 225 (1964): an article on which the patent has expired "may be made and sold by whoever chooses to do so."[2]

Bonito Boats, Inc. v. Thunder Craft Boats, Inc., 489 U.S. 141 (1989): patent-expired designs must remain free for public copying; the government may not pull them back from the public domain.[3]

Kimble v. Marvel Ent., LLC, 576 U.S. 446 (2015): when a patent expires, the right to make or use the article, free from all restriction, passes to the public.[4]

That is the key point: the right is to **make or use the article**, not merely to have a theoretical entitlement to talk about it, study it, or use some of its ideas in a different compliant product.

2. What the diagrams show

Diagrams T1 and T2 depict a real asymmetry:

the old, preexisting unit remains lawful to possess and operate under grandfathered standards, but a faithful remake built new must satisfy current-year rules and is therefore barred from manufacture.

That means the public's post-expiration entitlement is not practically exercisable as to the complete unit. On the materials given, that is a burden on the very right the patent cases recognize.

3. The equal-treatment point does not save the regime

The prompt invokes Bolling v. Sharpe, 347 U.S. 497 (1954) and Adarand Constructors, Inc. v. Peña, 515 U.S. 200 (1995), which establish that the federal government is not exempt from equal-protection-type constraints.[5][6] But here the more important point is not a free-standing equality claim; it is that the combined operation of the patent rule and the regulatory line leaves the public without the same practical post-expiration freedom that the patent cases guarantee.

4. Distinguishing holding from inference

It is true that the materials also support these propositions:

regulation of manufacture can be generally valid;

the public may still study, license, and incorporate features into separately compliant products;

the government may regulate safety and emissions.

Those are all fair points. But the conclusion that the public's patent-expiration right is thereby "satisfied" is only an inference, and in my view it is an unsound one. The holdings do not say that a post-expiration right is satisfied so long as some derivative, redesigned, or partially compliant use remains possible. They say the public may make the formerly patented article itself.[2][4]

5. Bottom line

On the materials provided, if the only lawful path is one that no longer reproduces the public-domain design, then the public's post-expiration right to make that design is not satisfied. That is a cognizable

injury under the authorities supplied.

ANSWER: “F”

References

1. U.S. Constitution art. I, § 8, cl. 8

U.S. Constitution · art. I, § 8, cl. 8 · 1788

2. Pamela Samuelson, The Constitutional Law of Intellectual Property After Eldred v. Ashcroft

J. Copyright Soc’y U.S.A. · 50 J. Copyright Soc’y U.S.A. 547 · 2003

3. Davida H. Isaacs, Not All Property Is Created Equal

Univ. of Md. Francis King Carey Sch. of Law · SSRN working paper · 2007

4. Pamela Samuelson, The Constitutional Law of Intellectual Property After Eldred

J. Copyright Soc’y U.S.A. · 50 J. Copyright Soc’y U.S.A. 547 · 2003

5. Texas v. SCV, Inc., 576 U.S. 38 (2015)

U.S. Supreme Court · 576 U.S. 38 · 2015

6. U.S. Copyright Office, Copyright and State Sovereign Immunity

U.S. Copyright Office · Report · 2020

Perplexity Pro by Perplexity AI, Inc., released Jun 19, 2026.

Executed at Abacus.AI. https://chatllm.abacus.ai/opensource

https://apps.abacus.ai/chatllm/share/conversation/10321e3cd4:d3918939a

On the materials provided, the public’s postexpiration right is **not** satisfied, and a cognizable constitutional injury arises.

The Supreme Court materials in the exam treat patent expiration as more than a formal change in legal status; they treat it as the point at which the invention enters the **public domain** and the public acquires a real right to **make, use, and sell** the formerly patented subject matter, free of the patent monopoly. That holding—that “when the patent expires, the right to make the formerly patented design passes to the public”—is framed as a concrete entitlement, not as a merely abstract or academic freedom to *study* or

discuss the design. The core bargain of the Patent Clause is practical: a timelimited exclusivity in exchange for subsequent public access to the invention itself.

The regulatory facts in the exam show that, for regulated final assemblies (whole vehicles or machines), federal standards apply by date of manufacture. A unit built and certified under the standards of its year of manufacture remains lawful indefinitely to own, use, and sell. By contrast, a **faithful publicdomain remake** of that same design, built after expiration, is held to **currentyear** emissions/safety/efficiency standards. In combined operation, the standards make it **unlawful to manufacture for sale** a complete-unit reproduction of the expired design; the only lawful build is one that **no longer reproduces that design**. Thus, for this class of inventions, the public's supposed right "to make" the design as a complete unit is, in practice, permanently nullified.

Under the Patent Clause's "limited Times" language, the entitlement that passes to the public is the right to **practice the invention itself**—to make the design that was patented, once the term ends. On the exam's holdings, that right is defined in terms of the subject matter of the claims, not in terms of a derivative freedom to derive inspiration for other, more compliant designs. Saying that the public may study, license, and incorporate aspects of the design into **different** products does not fulfill the right to make **that** design as a complete, regulated unit. As applied here, the government's overall legal scheme ensures that the class of regulated final-assembly inventions never yields a **practically exercisable** right to make the invention itself; exclusivity over the complete unit becomes, in economic substance, indefinite.

The question adds the equaltreatment component that *Bolling v. Sharpe* locates in the Fifth Amendment's Due Process Clause and that *Adarand* holds indistinguishable from the Fourteenth Amendment's equal protection obligation. Those holdings do not themselves speak to patents, but they do establish that the federal government, no less than the states, must honor constitutional equality principles. Reasoning by analogy, once the Patent Clause has granted the public a right—here, the postexpiration right to make an invention—the government may not, through the **combined operation** of its own regulatory schemes, systematically prevent one class of inventions (regulated final assemblies) from ever delivering the promised right in practice, while other classes of inventions (unregulated or differently regulated subject matter) do permit that right to be exercised. The materials depict a **structural disparity**: for regulated complete units, patent expiration never produces the world in which the public can actually make the unit, even as nonpatentees may freely exploit other patentexpired subject matter.

On that combined constitutional footing—the Patent Clause's "limited Times" bargain and the Fifth Amendment equaltreatment principle—the exam's scenario presents more than a merely "paper" right with incidental friction. It presents a right that **never becomes exercisable** for the category of inventions at issue. The public can read, study, and license the design, but it can never lawfully build the complete unit as such; the "right to make" exists only in theory. Given the way the exam's holdings define the publicdomain entitlement, that shortfall is a cognizable injury: the constitutional promise is not that the

public may study the expired design for purposes of designing something else; it is that, after limited times, the public may **practice the invention itself.**

Therefore, on the materials provided, the proposition that "the right is satisfied and no cognizable injury arises so long as the underlying design remains free to study, license, and incorporate into separatelycompliant new products" is **false**. The inability ever to lawfully build a complete-unit reproduction of the expired design means the postexpiration right is not actually delivered, and a court owes a remedy.

ANSWER: F

Citations

[1] https://journals.law.harvard.edu/jlpp/practical-applications-of-the-major-questions-doctrine-luka-a-wake-damien-schiff/
[2] https://columbialawreview.org/content/the-origins-of-supreme-court-question-selection/
[3] https://www.law.georgetown.edu/wp-content/uploads/2018/07/Geenberg-Weingast-Issue-Statement.pdf
[4] https://www.law.gwu.edu/sites/g/files/zaxdzs5421/files/downloads/QuestionPresentedTipSheet.pdf
[5] https://hayneslegalwriting.lawbooks.cali.org/chapter/chapter-8/
[6] https://www.supremecourt.gov/opinions/24pdf/24a884_8n59.pdf

Glossary

(Alphabetical by abbreviation. Acronyms are defined parenthetically at first appearance in the running text and collected here for reference.)

AEB — Automatic Emergency Braking. Federal Motor Vehicle Safety Standard requiring vehicles to detect imminent forward collisions with other vehicles or pedestrians and apply the brakes automatically when a driver does not respond. Codified by NHTSA in 2024 with phased compliance dates running through model year 2029.

AFBF — American Farm Bureau Federation. Largest U.S. general farm organization; publishes the Market Intel reports that track annual Chapter 12 farm bankruptcy filings drawn from U.S. Courts data and cited in the farm-equipment chapter.

BNCAP — Bharat New Car Assessment Programme. The vehicle safety assessment program of the Government of India, established in 2023 and operated under the Ministry

of Road Transport and Highways. Tests new vehicles sold in India under crash protocols harmonized with international standards and awards adult occupant protection and child occupant protection ratings on a five-star scale.

CAFE — Corporate Average Fuel Economy. Federal fuel-economy standard administered by the National Highway Traffic Safety Administration; sets fleet-average fuel-efficiency targets that vehicle manufacturers must meet across their U.S. sales.

C.F.R. — Code of Federal Regulations. The codification of federal regulations published by the Office of the Federal Register and organized by title and section (e.g., 49 C.F.R. § 563 refers to Title 49, Section 563).

CJ — Civilian Jeep. The post-WWII civilian successor to the Willys MB military Jeep, manufactured under various model designations (CJ-2A, CJ-3A, CJ-3B, CJ-5, CJ-7) from 1945 through 1986.

CKD — Completely Knocked Down. A vehicle shipped as a kit of components, to be assembled in the destination country. The first Mahindra Jeep vehicles were assembled from CKD kits shipped from the United States.

CPSC — Consumer Product Safety Commission. Independent federal agency established by the Consumer Product Safety Act of 1972; sets and enforces mandatory product-safety standards for consumer products under 15 U.S.C. §§ 2051 et seq.

DOE — Department of Energy. Cabinet-level federal department; sets and enforces appliance and equipment energy-efficiency standards under the Energy Policy and Conservation Act and its amendments.

DOJ — Department of Justice. Cabinet-level federal department; brings federal civil and criminal enforcement actions, including under the Federal Food, Drug, and Cosmetic Act and consumer-protection statutes.

EISA — Energy Independence and Security Act of 2007. Federal statute (Pub. L. 110-140) whose lighting-efficiency provisions amended the Energy Policy and Conservation Act and

effectively withdrew the ordinary incandescent bulb from the U.S. residential lighting market over a phase-in period running from 2012 to 2023.

EPA — Environmental Protection Agency. Independent federal agency; sets and enforces air-quality, emissions, and environmental-compliance standards under the Clean Air Act and related statutes.

ESC — Electronic Stability Control. Federal Motor Vehicle Safety Standard No. 126; required of all light vehicles manufactured for sale in the United States since model year 2012. Detects loss of steering control and automatically applies individual wheel brakes to restore directional stability.

ESG — Environmental, Social, and Governance. A class of investment-screening criteria used by institutional capital allocators — banks, private-equity funds, pension funds, and other large pools of capital — to evaluate the environmental, social, and corporate-governance characteristics of companies and projects considered for investment.

EUA — Emergency Use Authorization. Authority delegated to the FDA Commissioner under the Federal Food, Drug, and Cosmetic Act to permit the use of unapproved medical products, or unapproved uses of approved medical products, during a declared emergency.

FARFA — Farm and Ranch Freedom Alliance. Coalition-building organization founded in 2006 representing independent small-scale farmers, ranchers, and value-added producers.

FDA — U.S. Food and Drug Administration. Independent federal agency within the Department of Health and Human Services; regulates drugs, biologics, medical devices, food, cosmetics, and tobacco products under the Federal Food, Drug, and Cosmetic Act.

FMVSS — Federal Motor Vehicle Safety Standards. The body of federal vehicle-safety regulations administered by the National Highway Traffic Safety Administration and codified at 49 C.F.R. Part 571.

GNCAP — Global New Car Assessment Programme. The London-based international vehicle safety assessment program, founded in 2011 as a coordinating umbrella for regional NCAPs. Operates the #SaferCarsForIndia testing program and other regional programs,

awarding adult and child occupant protection ratings on a five-star scale based on harmonized crash test protocols.

HCQ — Hydroxychloroquine. Antimalarial drug first patented in the 1950s and on the World Health Organization Essential Medicines List.

HVAC — Heating, Ventilation, and Air Conditioning. The equipment category covering residential and commercial systems for indoor climate control; one of the consumer-durable categories most heavily affected by Department of Energy efficiency standards under 42 U.S.C. § 6295(o)(1).

IAMERS — International Association of Medical Equipment Remarketers and Servicers. Industry association representing independent third-party servicers of medical devices.

IIJA — Infrastructure Investment and Jobs Act of 2021 (also styled the Bipartisan Infrastructure Law). Pub. L. 117-58. Federal statute directing the Department of Transportation to issue performance standards for crash avoidance technology, including automatic emergency braking and impaired-driving prevention systems.

IMO — International Maritime Organization. United Nations specialized agency responsible for the safety and security of shipping and the prevention of marine and atmospheric pollution by ships. Develops the SOLAS, MARPOL, and COLREG conventions implemented in U.S. law through the Coast Guard's vessel inspection regime.

IMV — Innovative International Multi-purpose Vehicle. Toyota Motor Corporation's body-on-frame platform underlying the Hilux pickup, Fortuner SUV, Innova MPV, and Hilux Champ models manufactured in Thailand, Indonesia, and other Asian markets. The platform is built on a long-mature mechanical design whose underlying patents have been in the public domain for many years.

ISO 9001 — International Organization for Standardization, quality management systems standard. Voluntary international standard certifying that a manufacturer maintains a documented quality management system; cited in the farm-equipment chapter as one of the certifications Indian tractor manufacturers meet on equipment exported to global markets.

IVM — Ivermectin. Antiparasitic drug discovered in the 1970s, on the World Health Organization Essential Medicines List, and recognized with the 2015 Nobel Prize in Physiology or Medicine for the underlying discovery.

LCD — Local Coverage Determination. A coverage policy issued by a Medicare Administrative Contractor under authority delegated by the Centers for Medicare and Medicaid Services; specifies whether a particular item or service is reasonable and necessary, and therefore covered under Medicare Part A or Part B.

LED — Light-Emitting Diode. The semiconductor lighting technology that, under the Department of Energy efficiency rules finalized in 2023, has effectively replaced the incandescent bulb in U.S. residential lighting.

LRA — Legislative Reorganization Act of 1970. Federal statute that restructured congressional procedure by mandating recorded floor votes, open committee proceedings, and other "sunshine" reforms.

MAP-21 — Moving Ahead for Progress in the 21st Century Act. Pub. L. 112-141 (July 6, 2012). Federal surface-transportation reauthorization statute under which Congress directed NHTSA to issue lighting and marking standards for agricultural equipment, codified at 49 CFR Part 562 in 2016.

MB — Methylene blue. Synthetic dye first prepared in 1876; FDA-approved for the treatment of methemoglobinemia and used off-label for a range of clinical and supplement purposes.

NAECA — National Appliance Energy Conservation Act of 1987. Public Law 100-12, signed by President Reagan on March 17, 1987. The Act amended the Energy Policy and Conservation Act of 1975 and added the anti-backsliding provision codified at 42 U.S.C. § 6295(o)(1) — the keystone of the regulatory ratchet.

NEMA — National Electrical Manufacturers Association. Trade association representing the major U.S. manufacturers of electrical equipment, including the patent-protected lighting incumbents (GE Lighting under Savant Systems, Signify, formerly Philips, Osram,

Cree, and Acuity). NEMA's lighting members actively lobbied for the federal general service lamps efficiency standards finalized between 2022 and 2024 that withdrew the ordinary incandescent bulb from the U.S. residential market; the healthy-lighting chapter develops this dynamic as the bootleggers-and-Baptists configuration at the center of the case.

NHTSA — National Highway Traffic Safety Administration. Federal agency under the Department of Transportation; sets and enforces the Federal Motor Vehicle Safety Standards.

NMOG — Non-Methane Organic Gas. A class of regulated tailpipe emissions covering hydrocarbons other than methane; one of the principal pollutant categories in EPA's Tier 2 and Tier 3 motor-vehicle emission standards, where it is typically reported in milligrams per mile.

NOx — Nitrogen Oxides. Reactive gases formed by combustion in the presence of nitrogen, regulated as a tailpipe emission under the Clean Air Act. The "x" is a variable subscript: NOx covers nitric oxide (NO) and nitrogen dioxide (NO_2).

NRDC — Natural Resources Defense Council. Nonprofit environmental advocacy organization; named respondent in Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc. (1984).

NSAC — National Sustainable Agriculture Coalition. Coalition of grassroots organizations advocating for sustainable and organic farmers; has published analyses documenting that four firms control 45 percent of global farm equipment sales.

NTTAA — National Technology Transfer and Advancement Act of 1995. Pub. L. 104-113. Federal statute directing federal agencies to use technical standards developed by private-sector voluntary consensus standards bodies in lieu of developing federal standards from scratch.

OEM — Original Equipment Manufacturer. The producer of a finished product (medical device, vehicle, appliance, or other equipment); often used in contrast with third-party

servicers and remanufacturers.

OSH Act — Occupational Safety and Health Act of 1970. Pub. L. 91-596. Federal statute establishing workplace safety standards through the Occupational Safety and Health Administration; codified at 29 U.S.C. §§ 651 et seq.

OSHA — Occupational Safety and Health Administration. Federal agency within the U.S. Department of Labor; codifies workplace safety standards including 29 CFR 1928.51, the 1976 rollover rule on agricultural tractors discussed in the farm-equipment chapter as the first of the three regulatory layers.

PRP — Platelet-Rich Plasma. A preparation derived from a patient's own blood by centrifugation, used for autologous orthopedic injection.

RCRA — Resource Conservation and Recovery Act of 1976. Pub. L. 94-580. Federal statute establishing the federal regulatory framework for solid and hazardous waste under 42 U.S.C. §§ 6901 et seq.

TPMS — Tire Pressure Monitoring System. Federal Motor Vehicle Safety Standard No. 138; required of all light vehicles manufactured for sale in the United States since September 1, 2007. Sensors detect significantly under-inflated tires and warn the driver.

TRIPS — Agreement on Trade-Related Aspects of Intellectual Property Rights. Annex 1C of the 1994 Marrakesh Agreement Establishing the World Trade Organization, binding on all 166 WTO member nations. Article 33 sets the minimum patent term at twenty years from the filing date; the post-expiration commons it implies is the international counterpart to the U.S. Constitution's "limited Times" requirement.

U.S.C. — United States Code. The codification of the general and permanent laws of the United States, organized by title and section (e.g., 42 U.S.C. § 6295(o)(1) refers to Title 42, Section 6295, subsection (o)(1)).

USD — United States Dollar. The currency of the United States; used throughout the book in international price comparisons where foreign-currency prices are converted to U.S. dollar equivalents for the American reader.

USDA — U.S. Department of Agriculture. Cabinet-level federal department; the Economic Research Service publishes the Food Dollar Series cited in the farm-equipment chapter, breaking each U.S. retail food dollar into the farm share (15.9 cents in 2023) and the marketing share (84.1 cents).

USP — United States Pharmacopeia. The non-governmental standards-setting body whose monographs for drug substances and dosage forms are incorporated by reference into FDA regulation under the Federal Food, Drug, and Cosmetic Act.

WTO — World Trade Organization. Intergovernmental organization administering the global trade-rules framework, including the TRIPS Agreement under which all 166 member nations are bound by a minimum twenty-year patent term — the international counterpart to the U.S. Constitution's "limited Times" requirement.

Key Terms

(Alphabetical by term. The book's specific terms of art, with brief definitions of how the book uses each.)

Anti-backsliding provision. Section 6295(o)(1) of Title 42 of the United States Code, added by the National Appliance Energy Conservation Act of 1987. The statute provides that an energy-efficiency standard set by the Department of Energy can be tightened over time but never relaxed. The book treats this as the keystone of the regulatory ratchet: the legislative mechanism that converts a temporary regulatory burden into a permanent one and, by extension, converts the lengthening of an expired patent's effective monopoly into a permanent constitutional violation.

Blockade pattern. The book's term for the more common form the regulatory ratchet takes: the federal regulatory wall is built around a technology before the patents covering it expire, so that by the time the patents expire the public's right to remanufacture comes into existence inside a regulatory environment that already forbids it. The 2005 Ford F-150 case is a blockade-pattern case. Most expired-patent commerce in the United States is blockade-pattern. The injury is harder for an ordinary American to see because the manufacturing

window the public was entitled to never opened. The economic consequences are large; the intuitive force of the injury is muted by the absence's invisibility.

Bootleggers-and-Baptists. A coalition framework first named by the economist Bruce Yandle (1983) to describe regulatory regimes supported simultaneously by a moralizing public-interest constituency (the "Baptists," who believe in the regulation's stated purpose) and a profit-taking commercial constituency (the "bootleggers," who profit from the regulation's suppression of competition). The book argues that the post-1970 federal regulatory state is the bootleggers-and-Baptists architecture writ very large, with each leg of the regulatory wall genuinely supported by both wings of its coalition, which is what makes the coalition durable.

Combinatorial property of the public domain. The book's term for the engineering fact that an expired patent does not enter the public domain alone. It enters as a component, available to be combined with every other expired-patent component, in any combination an engineer can devise. The remanufactured American product that emerges when the regulatory wall is lifted is therefore not a copy of any single expired-patent design but a freshly engineered assembly of public-domain parts. The public domain is a kit of parts, not a museum.

Constitutional forgetting. The book's term for the operational disappearance of the words "limited Times" from the Patent Clause as applied by Congress, the federal regulatory agencies, and the academic disciplines that study them. The verb is "forgetting" — not "misreading" or "ignoring" — because the doctrine was never overturned. Supreme Court precedent on the post-expiration public domain remains intact and unbroken; what has been lost is the operational memory of the bargain in the institutions whose work bears on it.

Expired-patent commerce. The book's denominator term for the share of American economic activity in which the underlying patents have expired. The book divides this commerce into three tiers: 58 percent Blocked (where federal regulation makes remanufacturing legally impossible), 15 percent Hindered (where federal regulation imposes substantial but non-prohibitive constraints), and 27 percent Flowing (where

remanufacturing operates freely). The relevant denominator is expired-patent commerce, not "the economy" — which is why the 58 percent figure does not refer to all American manufacturing.

Expired Patent Commons. The book's term for the destination an invention is meant to reach when its patent expires: the body of expired inventions that the public may freely make, copy, improve, and remanufacture. The word "commons" is load-bearing — it denotes a shared resource the public can actually reach and use, not merely a right that has lapsed on paper. An invention has entered the Expired Patent Commons only when an American may in fact build it; a patent that has expired but that federal regulation forbids anyone to manufacture has not. The Expired Patent Commons is the public's half of the patent bargain, and the book's central claim is that the post-1970 regulatory wall prevents inventions from entering it — the forgotten half of the Patent Clause. The Supreme Court's cases call the same destination the public domain in the patent-law sense.

Grandfather exemption. The first leg of the book's three-leg remedy. A judicial or legislative carve-out under which a remanufactured product built on a public-domain platform is held to the federal safety and energy-efficiency standards in effect during the platform year the remanufactured product mimics — the same standards the original product was certified to. The grandfather exemption flows naturally from a successful constitutional ruling on the Patent Clause and requires no act of Congress to put in place.

Health-freedom statute. A state-level statute that limits the authority of state medical or pharmacy licensing boards to discipline licensed physicians for the use of treatments not approved by the U.S. Food and Drug Administration, provided the practitioner has obtained informed consent and represents the modality accurately. Roughly fifteen states have enacted such statutes since the late 1990s. Relevant to the Hindered sector, where licensing-board discipline is a common enforcement angle.

Hindered sector. The roughly fifteen percent of expired-patent commerce — chiefly pharmaceuticals and medical devices — where federal regulation, professional-board enforcement, and insurance-coverage decisions impose substantial but non-prohibitive

constraints on use of off-patent products. The Hindered sector contrasts with the Blocked sector, in which post-1970 federal manufacturing mandates make remanufacturing legally impossible, and with the Flowing sector, in which remanufacturing operates freely.

Limited Times. The two words at the heart of Article I, Section 8, Clause 8 of the U.S. Constitution: "To promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries." The phrase is constitutive, not decorative — affirmed across more than eighty Supreme Court rulings spanning two centuries. The book's central claim is that the post-1970 regulatory state has converted "limited Times" into perpetual times across roughly 58 percent of expired-patent commerce, in violation of the Patent Clause.

Median. A statistical measure of central tendency: the middle value of a distribution, where half of cases fall above and half fall below. The median is distinct from the mean (or arithmetic average), which is the sum of all values divided by the count. The book uses median household income — $83,730 in 2024, per the U.S. Census Bureau — rather than mean household income because the right-skewed distribution of American household income (a small number of very high earners pulling the mean upward) makes the median the better description of where the typical American household actually earns. The same logic applies to median wage, median home price, and other median figures the book cites.

Multi-value tax deduction. A deduction that allows a business to subtract a multiple of an actual expenditure — two or three dollars for every dollar spent — from taxable income. It is distinct from an ordinary deduction, which is one-for-one; from a tax credit, which is applied dollar-for-dollar against tax owed and is therefore far costlier to the Treasury; and from a civil penalty, which is coercive. The author proposed the mechanism in *ShareDebate International*, June 26, 1994. The book applies it in the chapter on the environmental question as the means by which the newest patent-expired emissions-control technology becomes the economical default in remanufactured products, without a consumer subsidy and without a direct federal outlay.

Public domain (patent-law sense). The constitutional status of an invention after its patent has expired. In the patent-law sense (distinct from the colloquial sense, in which "public domain" refers loosely to anything freely available), the public domain is the operational result of the bargain the Patent Clause establishes: the inventor receives an exclusive right for a limited time, after which the invention belongs to the American public to make, copy, improve, and remanufacture. The Supreme Court has repeatedly affirmed that the post-expiration public domain is a vested public right.

Regulatory ratchet. The book's term for the one-way mechanism by which federal regulation tightens over time but cannot be relaxed, with the result that the original legal terrain — including, crucially, the patent commons that emerges when patents expire — becomes progressively narrower. The ratchet is uniquely hostile to the Patent Clause because it converts a temporary regulatory burden into a permanent extension of monopoly protection. The anti-backsliding provision (42 U.S.C. § 6295(o)(1)) is the keystone statutory expression of the ratchet.

Remanufacturing (book's specific sense). The act of building a freshly engineered product assembled from components whose individual patents have all expired. The book's sense is distinct from "restoration" (returning a specific used object to working condition) and from "reproduction" (manufacturing a copy of a specific older design). A remanufactured product is a kit-of-parts assembly: a chassis whose patents expired in the early 2000s, a USB port whose 1995 Intel patents expired around 2015, a backup camera whose imaging patents expired years ago — engineered, mixed, and matched to deliver the features buyers actually want at a fraction of the new-vehicle price. The public domain is a kit of parts, not a museum.

Sunset Amendment. The book's term for the proposed constitutional amendment introduced in the chapter on the environmental question. The amendment preserves the Patent Clause remedy at full strength for twenty years from the date of first lawful remanufacture in any product category, after which Congress may by law restrict or prohibit further remanufacture without obligation to provide just compensation under the Fifth Amendment, applying only prospectively. The amendment is offered as Phase 2

structural insurance: not required for the Phase 1 remedy to operate, but available if the country eventually decides to close the marginal compensation subsidy created by the Takings Clause backstop.

Three-tier breakdown. The book's category-level audit of expired-patent commerce, dividing it into three tiers: Blocked (58 percent — sectors where federal manufacturing mandates make remanufacturing legally impossible, including vehicles, agricultural and construction machinery, HVAC, appliances, and certain lighting), Hindered (15 percent — sectors where federal regulation imposes substantial but non-prohibitive constraints, principally pharmaceuticals and medical devices), and Flowing (27 percent — sectors where remanufacturing operates freely, including bicycles, hand tools, firearms, and most consumer electronics). The breakdown is built from BLS Consumer Expenditure Survey data, BEA Industry Economic Accounts, and Census Bureau Annual Survey of Manufactures data; the methodology is detailed in the published companion methodology article — see the Endnotes entry for Martin, "The Forgotten Half of the Patent Clause: What was Promised When Patents Expired" (April 10, 2026).

Two suspenders. The book's metaphor for its two independent constitutional prongs, either of which is sufficient on its own. The first prong is the Patent Clause: Article I, Section 8, Clause 8 authorizes Congress to grant exclusive rights only "for limited Times," and a regulatory scheme that converts those limited times into perpetual protection exceeds the power the Clause actually grants. The second prong is the Takings Clause: the public's right to copy and remanufacture an expired patent is a vested right, affirmed in Bonito Boats (1989), and a permanent regulatory cancellation of that right is a taking requiring just compensation. A litigant who loses on one prong can still win on the other.

Withdrawal pattern. The book's term for the sharper second form the regulatory ratchet takes: the technology was already in the public domain, the public was already exercising the right to manufacture, and federal regulation was then erected around the technology after the fact, withdrawing it from the domain it had already been part of. The ordinary incandescent light bulb is the canonical case — the foundational patents expired more than a century ago, Americans manufactured incandescent bulbs continuously from 1879

onward, and federal energy-efficiency rules issued between 2007 and 2023 progressively withdrew the bulb from manufacturability. The withdrawal pattern is doctrinally stronger than the blockade pattern because the right at issue was not theoretical — it had been exercised in commerce, in plain view, for generations.

"To promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries."
— U.S. Constitution, Article I, Section 8, Clause 8